# Digital Camera Z 8

## USB Still Image Capture Device Media Transfer Protocol (MTP) Specifications

Rev. 2.00

Nikon Corporation
Imaging Division
Development Management Department

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# 1 Introduction

## 1.1 Application

These specifications describe the operations of the Z 8 (hereinafter referred to as the camera) as the USB Still Image Capture Device.

The specifications of the USB Still Image Capture Device are defined by “MTP Specification v1.0 - Media Transfer Protocol (MTP)” and the camera is based on it.

The camera conforms to the USB-related specifications below. For the details of each specification, refer to the related specifications manual.

| Item | Contents |
|---|---|
| USB specifications | Revision 3.0 |
| Class | Image Interface |
| Subclass | Still Image Capture Device |
| Protocol | Bulk-Only Transport Protocol |

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# 2 Outline

## 2.1 PC Connection Mode

When the camera is connected to the PC, the camera is switched to the PC connection mode.

## 2.2 PC Connection Mode and Camera Operations

The operations of the camera in the PC connection mode differ from those of the camera alone in the following points.

| No. | Description |
|---|---|
| 1 | When the remote mode is set (the camera is controlled by the PC), the operations by the dials and buttons of the camera body are locked with some exceptions. (Refer to section 2.4.) |
| 2 | The captured images are recorded either in the card or in the SDRAM. (Refer to section 2.6.) |
| 3 | The image playback cannot be performed. The image deletion by operating the camera body also cannot be performed. (Except in the application mode) |
| 4 | The Auto meter-off delay is set to “No limit”. |

## 2.3 PC Camera Mode and Remote Mode

The PC connection mode has the PC camera mode and the remote mode, and the camera is set to the PC camera mode when it is connected to the PC. Switching between the PC camera mode and the remote mode is performed by the command processing routine in the camera automatically for each command sent from the host or by the ChangeCameraMode command defined as a vendor command.

If a command for which changing the PC camera mode to the remote mode is necessary is issued while the remote mode is specified by the ChangeCameraMode command, the remote mode is retained even when the command processing is terminated.

During switching to the remote mode by the ChangeCameraMode command, the operation of each dial or button is ignored and the value set by the host is valid. When switching to the PC camera mode, the values set by the host in the remote mode are canceled and those set by each dial or button are used. For the setting value of each dial or button when switching to the remote mode, the value set by each dial or button becomes the initial value.

## 2.4 Application Mode

The camera is switched to the application mode by the ChangeApplicationMode property. At this time, the StoreRemoved event is passed.

In the application mode by the PC camera mode, the image playback, deletion and the video recording can be performed on the camera.

The host, which acquires the events by using the GetEventEx command, can acquire all the events generated in the application mode.

The host, which acquires the events by using the Interrupt IN transfer, can acquire only the DevicePropChanged event of the standard property.

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# 2.5 Photo Mode and Video Mode

In the case of PC camera mode, you can switch between photo mode and video mode with the photo/video selector (switching lever) of the camera. Also, the status can be checked from the host using the LiveViewSelector property (section 6.5.6.1).

When in remote mode, you can switch using the LiveViewSelector property.

# 2.6 Recording Destination

The camera supports the following three methods for the recording destinations of the image data when the images are acquired by using the shutter-release button of the camera or the release request command from the host during the USB connection.

| No. | Description |
|---|---|
| 1 | Records in the card (default). |
| 2 | Transfers to the host (records in the SDRAM temporarily). |
| 3 | Records in the card and transfers to the PC simultaneously. |

The camera has the RecordingMedia property to save the recording destination setting. The RecordingMedia property can be changed only by issuing the command from the host.

| RecordingMedia property | Operations |
|---|---|
| Card | The captured data are recorded in the card. |
| SDRAM | The captured data are recorded in the SDRAM temporarily and transferred to the host. |
| Card and SDRAM | Operation with the recording destination set to the card and that with the recording destination set to the SDRAM are both performed. |

The operation that changes the recording destination depending on the RecordingMedia property is as follows.

- Shutter release button of the camera
- Interval timer shooting
- Focus shift shooting

Shooting with the following commands is not affected by the RecordingMedia property, and the recording destination differs depending on the requested command.

| RecordingMedia property | Operations |
|---|---|
| InitiateCapture | The captured data are recorded in the card. |
| InitiateCaptureRecInSdram | The captured data are recorded in the SDRAM temporarily and transferred to the host. |
| AfAndCaptureRecInSdram | |
| InitiateCaptureRecInMedia | The recording destination is selected with specifying the parameter. |

If the host is incapable of image acquisition operation by the SDRAM recording shown in section 2.8, the value of the RecordingMedia property must not be changed.

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## 2.6.1 Access to the Card

During the USB connection, the insertion/ejection, existence/nonexistence, capacity, and the type of the card are detected by the camera. For the access to the card, the file system of the camera is always used and that of the host is not used. Therefore the host can access the card only after the USB connection is performed and the initialization of the information in the card is completed. During the USB connection, even if the card does not exist and then it is inserted, the host cannot access the data in the card until the initialization of the card information is completed.

The file system conforms to the DCF. The DPOF specification is also supported. Therefore the host cannot access the card with the directory structure or the file structure that does not conform to the DCF or the DPOF specification.

The host can read the data in the card, but cannot write the data to the card.

The camera can make the deletion of data in the card and perform card formatting by the command from the host.
When the captured images are being recorded on the card, the data deletion and the card formatting cannot be performed until all the images are recorded on the card. The release operation by the shutter-release button of the camera and the shooting request command cannot be executed during the data deletion and the card formatting.

## 2.6.2 Access to the SDRAM

When accessing the image data recorded in the SDRAM, accessing can be performed only to the one image data that is ready to be taken in and to the oldest image data in order of storage in the SDRAM. The image data in the SDRAM cannot be accessed randomly. (Refer to section 8.2).
The image data sent to the host completely is erased from the SDRAM.

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# 2.7 Sending the Event

When the status in the camera is changed, the camera sends an event by the Interrupt transfer to notify the host of the contents of the change. However, the host may not be able to get the event sent by the camera depending on the OS type of the PC (host). Therefore the camera provides two methods of getting the event.

All the generated events can be acquired in order of generation by either of the two methods. The methods are shown below.

| No. | Description |
|---|---|
| 1 | Sending the event by the Interrupt transfer (based on the PTP specifications) |
| 2 | Sending the event by the GetEventEx command (vendor-defined) |

The camera stores the event generated in the camera in order in the queue buffer and retains the queue buffer status until the event is acquired by the host. The camera is provided with two queue buffers for the two methods of getting the event.

It is necessary for the host to use only one of the two methods of acquiring the event to perform the event processing. The camera does not manage the coordination of the two queue buffers.

Up to 2048 events can be stored in the queue buffer. When the queue buffer becomes full, the camera deletes the old event from the queue buffer and stores the new event in the queue buffer.

The event of the GetEventEx command is stored in the queue buffer only when the session is open, and the information in the queue buffer is cleared when the session is closed.

For sending the event in the application mode, refer to “Application Mode” (section 2.4).

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# 2.8 Sequence of Shooting Operation and Image Data Acquisition in the PC Connection Mode

While the session is open in the PC connection mode, the camera performs the shooting operation by the command from the host or the shutter-release button of the camera. The captured images are stored in the card or the SDRAM by the command from the host or the setting value of the recording destination property (refer to section 2.6).

The host can acquire the saved image data by the GetObject or the GetPartialObject command.

The command sequence of each operation is described below.

## 2.8.1 Recording in the Card

The command sequence leading from the recording of the captured image data in the card to the data acquisition by the host is shown below.

| No. | Description |
|---|---|
| 1 | When acquiring the event by the command, the host issues the GetEventEx command regularly to acquire the event. |
| 2 | The host issues the InitiateCapture command or the InitiateCaptureRecInMedia command to capture the image data. The camera performs the single shot operation or the continuous shot operation according to the shooting mode setting. |
| 3 | When the captured image data is saved in the card, the camera issues the ObjectAdded event. The object handle indicating the saved image data is added to the ObjectAdded event. |
| 4 | If the continuous shot operation is set, the camera repeats the procedure in (3) as many times as the number of shooting. |
| 5 | The camera issues the CaptureComplete event when all the image data captured by the InitiateCapture command or the InitiateCaptureRecInMedia command are saved completely. |
| 6 | The host acquires the image data information by issuing the GetObjectInfo command. |
| 7 | The host acquires the thumbnail data by issuing the GetThumb and the GetLargeThumb commands, if necessary. |
| 8 | The host acquires the image data by issuing the GetObject or GetPartialObject command. |
| 9 | When two or more image data are saved by the continuous shot operation, the host repeats the procedures in (6) to (8) to acquire all the image data. |

## 2.8.2 Recording in the SDRAM

The command sequence leading from the recording of the captured image data in the SDRAM to the data acquisition by the host is shown below.

| No. | Description |
|---|---|
| 1 | When acquiring the event by the command, the host issues the GetEventEx command regularly to acquire the event. |
| 2 | The host issues the InitiateCaptureRecInSdram, the AfAndCaptureRecInSdram, or the InitiateCaptureRecInMedia command to capture the image data. The camera performs the single shot operation or the continuous shot operation according to the shooting mode setting. |
| 3 | The host issues the DeviceReady command repeatedly while executing the operations in (4) and after. |
| 4 | The host waits for the camera to issue the ObjectAddedInSdram event. |
| 5 | The camera saves the captured image data in the SDRAM in order, and issues the ObjectAddedInSdram event in sequence when the sending of image data to the host becomes enabled. |
| 6 | The host acquires the image data information by issuing the GetObjectInfo command. |
| 7 | The host acquires the thumbnail data by issuing the GetThumb and the GetLargeThumb commands, if necessary. |
| 8 | The host acquires the image data by issuing the GetObject or GetPartialObject command. |
| 9 | The host repeats the procedures in (6) to (8) as many times as the number of received ObjectAddedInSdram events. |
| 10 | When all the image data captured by one InitiateCaptureRecInSdram, the AfAndCaptureRecInSdram, or the InitiateCaptureRecInMedia command are sent completely, the camera sends the CaptureCompleteRecInSdram event. |

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## 2.8.3 Recording by the Shutter-Release Button of the Camera

The command sequence leading from the capture of the image data by the shutter-release button of the camera to the acquisition of the recorded image data is shown below.

| No. | Description |
|---|---|
| 1 | When acquiring the event by the command, the host issues the GetEventEx command regularly to acquire the event. |
| 2 | The shooting operation is performed by the shutter-release button of the camera. The camera performs the shooting operation (single shot operation, the continuous shot operation, or the interval timer shooting) according to the shooting mode setting. |
| 3 | The camera saves the images in the card, the SDRAM, or the card and the SDRAM according to the setting of the recording destination property. |
| 4 | When the images are saved in the card, the procedures in (5) to (10) should be performed. When the images are saved in the SDRAM, the procedures in (11) to (15) should be performed. |
| 5 | When the captured image data is saved in the card, the camera issues the ObjectAdded event. The object handle indicating the saved image data is added to the ObjectAdded event. |
| 6 | If the continuous shot operation is set, the camera repeats the procedure in (5) as many times as the number of shooting. |
| 7 | The host acquires the image data information by issuing the GetObjectInfo command. |
| 8 | The host acquires the thumbnail data by issuing the GetThumb and the GetLargeThumb commands, if necessary. |
| 9 | The host acquires the image data by issuing the GetObject or GetPartialObject command. |
| 10 | When two or more image data are saved by the continuous shot operation, the host repeats the procedures in (11) to (14) to acquire all the image data. |
| 11 | The camera saves the captured image data in the SDRAM in order, and issues the ObjectAddedInSdram event in sequence when the sending of image data to the host becomes enabled. |
| 12 | The host acquires the image data information by issuing the GetObjectInfo command. |
| 13 | The host acquires the thumbnail data by issuing the GetThumb and the GetLargeThumb commands, if necessary. |
| 14 | The host acquires the image data by issuing the GetObject or GetPartialObject command. |
| 15 | The host repeats the procedures in (12) to (15) as many times as the number of received ObjectAddedInSdram events. |
| 16 | The camera issues CaptureCompleteRecInSdram event when the transmission of image data is complete. |

## 2.8.4 Bulb Shooting

The command sequence to acquire the image data recorded by Bulb shooting is shown below.

| No. | Description |
|---|---|
| 1 | The host changes the camera to the remote mode by using the ChangeCameraMode command. |
| 2 | The host sets “M” by using the ExposureProgramMode property. |
| 3 | The host sets Bulb shooting (0xFFFFFFFF) by using the ExposureTime property and the ShutterSpeed property. |
| 4 | The host issues the InitiateCaptureRecInMedia command to start shooting. |
| 5 | The host issues the DeviceReady command and confirms that the response code is Bulb_Release_Busy. |
| 6 | The host issues the TerminateCapture command to terminate Bulb shooting. |
| 7 | The camera records the created images in the card, the SDRAM, or the card and the SDRAM after completion of the shooting and issues the ObjectAdded event or the ObjectAddedInSdram event to the host. |
| 8 | The host acquires the ObjectAdded event or the ObjectAddedInSdram event by the GetEventEx command to acquire the ObjectHandle of the created images. |
| 9 | The host acquires the image information by the GetObjectInfo command. |
| 10 | The host acquires the thumbnail image by the GetThumb and the GetLargeThumb commands, if necessary. |
| 11 | The host acquires the image data by issuing the GetObject or GetPartialObject command. |
| 12 | The camera issues the CaptureComplete event or the CaptureCompleteRecInSdram event. |
| 13 | The host acquires the CaptureComplete event or the CaptureCompleteRecInSdram event by the GetEventEx command. |
| 14 | The host changes the camera to the PC camera mode by using the ChangeCameraMode command. |

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## 2.8.5 Interval Timer Shooting

The command sequence to acquire the image data recorded by interval timer shooting is shown below.

| No. | Description |
|---|---|
| 1 | The host acquires information about interval timer shooting by issuing the GetOpenCaptureInfo commands, if necessary. |
| 2 | The host requests the start of the interval timer shooting by issuing InitiateOpenCaptureV command. |
| 3 | The specialShootingMode property of the camera is turned on, and the shooting operation is performed according to the interval timer shooting setting. |
| 4 | The camera saves the images in the card, the SDRAM, or the card and the SDRAM according to the setting of the recording destination property. |
| 5 | When saving an image on a card, perform operations (6) to (8), and when saving an image on an SDRAM, perform operations (9) to (12). |
| 6 | When the captured image data is saved in the card, the camera issues the ObjectAdded event. |
| 7 | The host acquires the image data.<br>The acquisition method is the same as the sequence in section 2.8.1. |
| 8 | Multiple image data is saved depending on the interval timer shooting setting. The host repeats (7) to acquire all the image data. |
| 9 | The camera sequentially saves the captured image data in SDRAM.<br>The camera issues ObjectAddedInSdram events in sequence when it is ready to send to the host. |
| 10 | The host acquires the SDRAM image data.<br>The acquisition method is the same as the sequence in section 2.8.2. |
| 11 | The host repeats (10) the number of times the ObjectAddedInSdram event is received. |
| 12 | The camera issues a CaptureCompleteRecInSdram event when the transmission of SDRAM image data is completed.<br>The camera issues a CaptureCompleteRecInSdram event when all the SDRAM images taken up to that point are transmitted even during the interval timer shooting. |
| 13 | When the interval timer shooting can be skipped, the camera issues a CaptureSkip event if shooting is skipped. |
| 14 | If the host needs to stop the interval timer shooting in the middle, it requests the stop with the TerminateOpenCaptureV command. |
| 15 | If the time-lapse video option for interval timer shooting is enabled, the camera writes the time-lapse video file to the card and issues an ObjectAdded event. |
| 16 | The host gets the video file.<br>The acquisition method is the same as the video recording sequence (section 2.9.2, section 2.9.2.2). |
| 17 | The camera issues an OpenCaptureComplete event when the interval timer shooting started by the InitiateOpenCaptureV command is completed (including when stopped in the middle of shooting). |
| 18 | The camera turns off the SpecialShootingMode property when interval timer shooting is complete. |

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## 2.8.6 Focus Shift Shooting

The command sequence to acquire the image data recorded by focus shift shooting is shown below.

| No. | Description |
|---|---|
| 1 | The host acquires information about focus shift shooting by issuing the GetOpenCaptureInfo commands, if necessary. |
| 2 | The host requests the start of the focus shift shooting by issuing InitiateOpenCaptureV command. |
| 3 | The specialShootingMode property of the camera is turned on, and the shooting operation is performed according to the focus shift shooting setting. |
| 4 | The camera saves the images in the card, the SDRAM, or the card and the SDRAM according to the setting of the recording destination property. |
| 5 | When saving an image on a card, perform operations (6) to (8), and when saving an image on an SDRAM, perform operations (9) to (12). |
| 6 | When the captured image data is saved in the card, the camera issues the ObjectAdded event. |
| 7 | The host acquires the image data.<br>The acquisition method is the same as the sequence in section 2.8.1. |
| 8 | Multiple image data is saved depending on the focus shift shooting setting. The host repeats (7) to acquire all the image data. |
| 9 | The camera sequentially saves the captured image data in SDRAM.<br>The camera issues ObjectAddedInSdram events in sequence when it is ready to send to the host. |
| 10 | The host acquires the SDRAM image data.<br>The acquisition method is the same as the sequence in section 2.8.2. |
| 11 | The host repeats (10) the number of times the ObjectAddedInSdram event is received. |
| 12 | The camera issues a CaptureCompleteRecInSdram event when the transmission of SDRAM image data is completed.<br>The camera issues a CaptureCompleteRecInSdram event when all the SDRAM images taken up to that point are transmitted even during the focus shift shooting. |
| 14 | If the host needs to stop the focus shift shooting in the middle, it requests the stop with the TerminateOpenCaptureV command. |
| 15 | The camera issues an OpenCaptureComplete event when the focus shift shooting started by the InitiateOpenCaptureV command is completed (including when stopped in the middle of shooting). |
| 16 | The camera turns off the SpecialShootingMode property when focus shift shooting is complete. |

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# 2.9 Sequence of Live View Image and Data Acquisition in the PC Connection Mode

When the session is open in the PC connection mode, the camera starts the live view by the request from the host.

During the remote live view after the host executed the StartLiveView command, the live view images can be acquired by the GetLiveViewImageEx command.

The command sequence of each operation is described below.

## 2.9.1 Live View

The command sequence leading from the start of live view by the request from the host to the acquisition of the live view image is shown below.

| No. | Description |
|---|---|
| 1 | The host issues the StartLiveView command to request the camera to start the remote live view. |
| 2 | After confirming the normal termination of the StartLiveView command, the host issues the DeviceReady command repeatedly until the DeviceReady command response other than Device_Busy appears. |
| 3 | The camera returns the normal termination response to the DeviceReady command when the live view image acquisition becomes enabled. If the live view image acquisition is disabled for some reason, the camera returns an error response to the DeviceReady command to terminate the live view. |
| 4 | After confirming the normal termination of the DeviceReady command response, the host issues the GetLiveViewImageEx command to acquire the live view image. |
| 5 | The host issues the GetLiveViewImageEx command repeatedly while the live view continues. At this time, the AF driving (AfDrive command), the MF driving (MfDrive command), the AF area change (ChangeAfArea command), etc. can be performed optionally. |
| 6 | When the remote live view is finished, the host issues the EndLiveView command. |

## 2.9.2 Video Recording

The sequence leading from the recording of the video in the card by the MovieRec command to the acquisition of the video file is shown below.

### 2.9.2.1 Video Recording Transfer

| No. | Description |
|---|---|
| 1 | The host issues MovieRecProhibitionCondition in video mode to confirm that video recording can be performed. |
| 2 | The host issues the StartMovieRecord command to start the video recording. |
| 3 | The camera stops the video recording or the host issues the EndMovieRecord command to stop the video recording. |
| 4 | The camera writes the captured video file in the card and issues the ObjectAdded event. |
| 5 | The camera issues the MovieRecordComplete event after stopping video recording. |
| 6 | The host acquires the MovieRecordComplete event by the GetEventEx command. |
| 7 | The host acquires the ObjectAdded event by the GetEventEx command to acquire the ObjectHandle of the video file. |
| 8 | The host acquires the file size of the video file by the GetObjectInfo command. |
| 9 | The host divides and acquires the video file by the GetPartialObject command.<br>When the video file is acquired by the GetObject command, the control from the host cannot be performed until the file transfer is finished because the file size of the video file is large. Therefore, use the GetPartialObject command. |

Do Not Copy

### 2.9.2.2 Over 4GB Video Transfer

The command sequence to transfer the video data over 4GB is shown below.

| No. | Description |
|---|---|
| 1 - 7 | Same as normal video recording transfer. |
| 8 | The host acquires the file size of the video file by the GetObjectInfo command.<br>When the file size of the GetObjectInfo command is 0xFFFFFFFF, use the following method to get the file size.<br><br>PTP: Use GetObjectSize command to get the file size.<br>MTP: Use ObjectSize of the GetObjectPropValue command to get the file size. |
| 9 | The host divides and acquires the video file by the GetPartialObjectHighSpeed or GetPartialObjectEx command.<br>When the video file is acquired by the GetObject command, the control from the host cannot be performed until the file transfer is finished because the file size of the video file is large. Therefore, use the GetPartialObjectHighSpeed or GetPartialObjectEx command. |

### 2.9.2.3 RAW Video Transfer

When video recording is performed with N-RAW12-bit(NEV)/ProResRAW HQ12bit(MOV), a video of the parent NEV (MOV) and the child MP4 is created.

The sequence to get this parent-child video is as follows.

| No. | Description |
|---|---|
| 1 - 3 | Same as normal video recording transfer. |
| 4 | The camera writes the captured video file of NEV(MOV) in the card and issues the ObjectAdded event. |
| 5 | The camera writes the captured video file of MP4 in the card and issues the ObjectAdded event. |
| 6 | The camera issues the MovieRecordComplete event after stopping video recording. |
| 7 | The host acquires the MovieRecordComplete event by the GetEventEx command. |
| 8 | The host acquires the ObjectAdded event of NEV(MOV) by the GetEventEx command to acquire the ObjectHandle of the video file. |
| 9 | The host acquires the file size of the video file of NEV(MOV) by the GetObjectInfo command.<br>When the file size of the GetObjectInfo command is 0xFFFFFFFF, use the following method to get the file size.<br>PTP: Use GetObjectSize command to get the file size.<br>MTP: Use ObjectSize of the GetObjectPropValue command to get the file size. |
| 10 | The host divides and acquires the video file of NEV(MOV) by the GetPartialObjectHighSpeed or GetPartialObjectEx command. |
| 11 | Perform No.8 to No.10 for MP4 files. |

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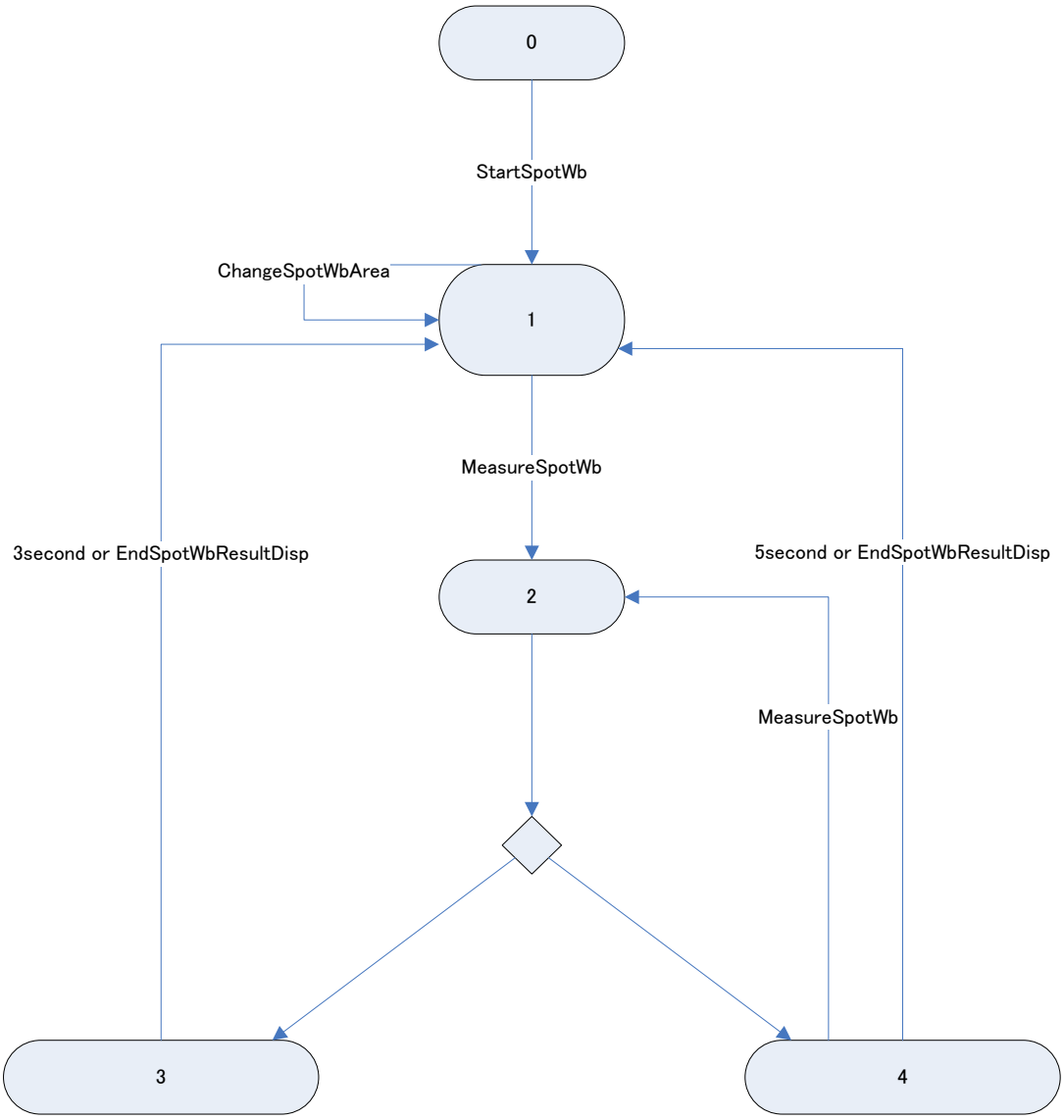
## 2.9.3 Spot WB

The command sequence to acquire the spot WB by the MeasureSpotWB command and set it to preset manual is shown below.

| No. | Description |
|---|---|
| 1 | The host issues the StartLiveView command to request the camera to start the remote live view. |
| 2 | The host changes the camera to the remote mode by using the ChangeCameraMode command. |
| 3 | The host sets WhiteBalance to Preset manual. |
| 4 | The host sets WbPresetDataNo in which the spot WB is saved. At this time, a WbPresetDataNo in which Preset manual is protected must not be set. |
| 5 | The host issues the StartSpotWb command to request the camera to start the spot WB acquisition standby state. Confirm the spot WB condition of LiveViewObject that can be acquired by the GetLiveViewImageEx command to check whether the spot WB acquisition standby state is started. |
| 6 | The host sets the area to be acquired by ChangeSpotWbArea. |
| 7 | The host issues the MeasureSpotWb command to acquire the spot WB. Confirm the spot WB condition of LiveViewObject to check the acquisition result. |
| 8 | If the acquisition succeeds, the camera cancels the spot WB acquisition standby state and sets the spot WB condition of LiveViewObject to “On standby” (acquisition success) for three seconds. |
| 9 | If the acquisition fails, the camera remains in the spot WB acquisition standby state and sets the spot WB condition of LiveViewObject to “On standby” (acquisition failure) for five seconds. |
| 10 | The host issues the EndSpotWbResultDisp command to cancel the spot WB acquisition result display quickly. |
| 11 | The host issues the EndSpotWb command to terminate the spot WB acquisition standby state. |
| 12 | The host changes the camera to the PC camera mode by using the ChangeCameraMode command. |

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The transition of the spot WB condition that can be acquired by GetLiveViewImageEx command is shown below.




The figures in the diagram above show “Spot WB condition” of LiveViewObject acquired by the GetLiveViewImageEx command.

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## 2.9.4 Tracking Operation

The command sequence from tracking start to tracking end is shown below.

| No. | Description |
|---|---|
| 1 | The host issues the StartLiveView command to request the camera to start the remote live view. |
| 2 | The host sets the AF area mode to Auto-area AF. |
| 3 | The host can execute the following commands to change the tracking state.<br><br>- StartTracking command<br>The host issues the StartTracking command with the area where you want to start tracking as the parameters.<br><br>When the subject recognition target is not detected at the specified coordinates, the camera sets the tracking state of LiveViewObject to "On tracking" and starts tracking operation at the specified coordinates.<br><br>When the subject recognition target is detected at the specified coordinates, the camera updates the selected face index.<br>At this time, the tracking state changes to the standby state.<br><br>When the StartTracking command is executed with "On tracking" in the tracking state, the camera starts tracking operation at the specified coordinates with or without detected subject recognition target. |
| 5 | The host issues the EndTracking command to end tracking action. |

*1 Whether the tracking state has changed or not is confirmed by checking the tracking state of LiveViewObject acquired by the GetLiveViewImageEx command.

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## 2.10 Reacquisition of the Image Data

When one of the following occurs during the reading sequence of the image data recorded in the SDRAM in sections 2.8.2 and 2.8.3, the host can redo the image data acquisition by issuing the GetObjectInfo command.

| No. | Description |
|---|---|
| 1 | The USB connection is cut. (Such as the case that the USB cable is extracted.) |
| 2 | The camera returns an error to the GetObject or the GetPartialObject command. |
| 3 | It is desirable for the host to redo the acquisition of the SDRAM image. |

When the image data that is not sent yet exists in the SDRAM, the camera must retain it even if the USB connection is cut.

If the shooting has been performed with “RAW+JPEG” and either of the image data has been sent normally, only the other image data should be sent again.

## 2.11 Operation with the Empty Battery

When the battery is exhausted, the camera cannot detect the card and notify the host of no card status even if it is inserted.

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# 3 Device Requests

## 3.1 Standard Device Requests

| Request | | wIndex | |
|---|---|---|---|
| | | Data stage | Status stage |
| Get Status | Device | The current device power status and the setting of REMOTE_WAKEUP function | Receive NULL data |
| | Endpoint: Endpoint number that is supported | The current ENDPOINT_STALL status | Receive NULL data |
| | Endpoint: Endpoint number that is not supported | STALL response | - |
| Clear Feature | Device: DEVICE_REMOTE_WAKEUP | - | Return NULL data |
| | Endpoint: END_POINT_HALT<br>Endpoint number that is supported | - | Return NULL data |
| | Endpoint: END_POINT_HALT<br>Endpoint number that is not supported | - | STALL response |
| Set Feature | Device: DEVICE_REMOTE_WAKEUP | - | Return NULL data |
| | Endpoint: END_POINT_HALT<br>Endpoint number that is supported | - | Return NULL data |
| | Endpoint: END_POINT_HALT<br>Endpoint number that is not supported | - | STALL response |
| Set Address | | - | Return NULL data |
| Get Descriptor | | The specified descriptor | Receive NULL data |
| Set Descriptor | | STALL response | - |
| Get Configuration | | The current configuration value | Return NULL data |
| Set Configuration | Configuration number that is supported | - | Return NULL data |
| | Configuration number that is not supported | - | STALL response |
| Set Interface | Interface number and alternate number that are supported | - | Return NULL data |
| | Interface number and alternate number that are not supported | - | STALL response |
| Get Interface | Interface number that is supported | The current alternate value | Return NULL data |
| | Interface number that is not supported | STALL response | - |
| Synch Frame | | STALL response | - |

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## 3.2 Class-Specific Requests

The camera supports the class-specific requests below.

### 3.2.1 Cancel Request

This request is used for the host to cancel the data transfer.

| bmRequestType | bRequest | wValue | wIndex | wLength |
|---|---|---|---|---|
| 00100001b | 01100100b | 0000h | 0000h | 06h |

The camera receives the Cancel request data according to the following format.

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | Cancellation Code | 2 | Code | 0x4001 |
| 2 | TransactionID | 4 | Number | TransactionID |

The camera cancels the command processing that corresponds to the TransactionID.

### 3.2.2 DeviceReset Request

This request is sent from the host to the camera in order to make the device become in the idle status when the Bulk Pipe is stalled.

| bmRequestType | bRequest | wValue | wIndex | wLength |
|---|---|---|---|---|
| 00100001b | 01100110 | 0000h | 0000h | 0 |

### 3.2.3 GetDeviceStatus Request

This request is used for the host to acquire the device information for the recovery of the endpoint that is in the halt status.

| bmRequestType | bRequest | wValue | wIndex | wLength |
|---|---|---|---|---|
| 10100001b | 01100110 | 0000h | 0000h | 0 |

The camera sends the GetDeviceStatus request data according to the following format.

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | WLength | 2 | Number | 4 |
| 2 | Code | 2 | Code | 0x2001: Status OK<br>0x2019: DeviceBusy |

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# 4 Descriptors

## 4.1 Standard Descriptors

The camera has the following standard descriptors.

### 4.1.1 Device Descriptor

#### 4.1.1.1 HIGH-SPEED

| USB 2.x | | | | |
|---|---|---|---|---|
| Offset | Field | Size | Value | Description |
| 0 | bLength | 1 | 12h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 01h | Type of the descriptor (Device descriptor) |
| 2 | bcdUSB | 2 | 0210h | USB specification number (0210h=Revision 2.10) |
| 4 | bDeviceClass | 1 | 00h | Class (specified by the interface descriptor) |
| 5 | bDeviceSubClass | 1 | 00h | Subclass (specified by the interface descriptor) |
| 6 | bDeviceProtocol | 1 | 00h | Protocol (specified by the interface descriptor) |
| 7 | bMaxPacketSize0 | 1 | 40h | Maximum packet size of endpoint 0 |
| 8 | idVendor | 2 | 04B0h | Vendor ID (“NIKON”) |
| 10 | idProduct | 2 | 0451h | Product ID |
| 12 | bcdDevice | 2 | 0100h | Device release number (0100h=1.00) |
| 14 | iManufacture | 1 | 01h | Index of the string descriptor describing the manufacturer name |
| 15 | iProduct | 1 | 02h | Index of the string descriptor describing the product name |
| 16 | iSerialNumber | 1 | 03h | Index of the string descriptor describing the serial number |
| 17 | bNumConfigurations | 1 | 01h | The number of configurations |

| USB 3.x | | | | |
|---|---|---|---|---|
| Offset | Field | Size | Value | Description |
| 0 | bLength | 1 | 12h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 01h | Type of the descriptor (Device descriptor) |
| 2 | bcdUSB | 2 | 0320h | USB specification number (0320h=Revision 3.20) |
| 4 | bDeviceClass | 1 | 00h | Class (specified by the interface descriptor) |
| 5 | bDeviceSubClass | 1 | 00h | Subclass (specified by the interface descriptor) |
| 6 | bDeviceProtocol | 1 | 00h | Protocol (specified by the interface descriptor) |
| 7 | bMaxPacketSize0 | 1 | 09h | Maximum packet size of endpoint 0 |
| 8 | idVendor | 2 | 04B0h | Vendor ID (“NIKON”) |
| 10 | idProduct | 2 | 0451h | Product ID |
| 12 | bcdDevice | 2 | 0100h | Device release number (0100h=1.00) |
| 14 | iManufacture | 1 | 01h | Index of the string descriptor describing the manufacturer name |
| 15 | iProduct | 1 | 02h | Index of the string descriptor describing the product name |
| 16 | iSerialNumber | 1 | 03h | Index of the string descriptor describing the serial number |
| 17 | bNumConfigurations | 1 | 01h | The number of configurations |

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#### 4.1.1.2 FULL-SPEED

| USB 2.x | | | | |
|---|---|---|---|---|
| Offset | Field | Size | Value | Description |
| 0 | bLength | 1 | 12h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 01h | Type of the descriptor (Device descriptor) |
| 2 | bcdUSB | 2 | 0210h | USB specification number (0210h=Revision 2.10) |
| 4 | bDeviceClass | 1 | 00h | Class (specified by the interface descriptor) |
| 5 | bDeviceSubClass | 1 | 00h | Subclass (specified by the interface descriptor) |
| 6 | bDeviceProtocol | 1 | 00h | Protocol (specified by the interface descriptor) |
| 7 | bMaxPacketSize0 | 1 | 40h | Maximum packet size of endpoint 0 |
| 8 | idVendor | 2 | 04B0h | Vendor ID (“NIKON”) |
| 10 | idProduct | 2 | 0451h | Product ID |
| 12 | bcdDevice | 2 | 0100h | Device release number (0100h=1.00) |
| 14 | iManufacture | 1 | 01h | Index of the string descriptor describing the manufacturer name |
| 15 | iProduct | 1 | 02h | Index of the string descriptor describing the product name |
| 16 | iSerialNumber | 1 | 03h | Index of the string descriptor describing the serial number |
| 17 | bNumConfigurations | 1 | 01h | The number of configurations |

| USB 3.x | | | | |
|---|---|---|---|---|
| Offset | Field | Size | Value | Description |
| 0 | bLength | 1 | 12h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 01h | Type of the descriptor (Device descriptor) |
| 2 | bcdUSB | 2 | 0320h | USB specification number (0320h=Revision 3.20) |
| 4 | bDeviceClass | 1 | 00h | Class (specified by the interface descriptor) |
| 5 | bDeviceSubClass | 1 | 00h | Subclass (specified by the interface descriptor) |
| 6 | bDeviceProtocol | 1 | 00h | Protocol (specified by the interface descriptor) |
| 7 | bMaxPacketSize0 | 1 | 09h | Maximum packet size of endpoint 0 |
| 8 | idVendor | 2 | 04B0h | Vendor ID (“NIKON”) |
| 10 | idProduct | 2 | 0451h | Product ID |
| 12 | bcdDevice | 2 | 0100h | Device release number (0100h=1.00) |
| 14 | iManufacture | 1 | 01h | Index of the string descriptor describing the manufacturer name |
| 15 | iProduct | 1 | 02h | Index of the string descriptor describing the product name |
| 16 | iSerialNumber | 1 | 03h | Index of the string descriptor describing the serial number |
| 17 | bNumConfigurations | 1 | 01h | The number of configurations |

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## 4.1.2 Device_Qualifier Descriptor

| USB 2.x | | | | |
|---|---|---|---|---|
| Offset | Field | Size | Value | Description |
| 0 | bLength | 1 | 0Ah | Size of the descriptor |
| 1 | bDescriptorType | 1 | 06h | Type of the descriptor (Device_Qualifier descriptor) |
| 2 | bcdUSB | 2 | 0210h | USB specification number (0210h=Revision 2.10) |
| 4 | bDeviceClass | 1 | 00h | Class (specified by the interface descriptor) |
| 5 | bDeviceSubClass | 1 | 00h | Subclass (specified by the interface descriptor) |
| 6 | bDeviceProtocol | 1 | 00h | Protocol (specified by the interface descriptor) |
| 7 | bMaxPacketSize0 | 1 | 40h | Maximum packet size of endpoint 0 |
| 8 | bNumComfigurations | 1 | 01h | The number of configurations other than USB2.0 |
| 9 | bReserved | 1 | 00h | Reserved |

| USB 3.x | | | | |
|---|---|---|---|---|
| Offset | Field | Size | Value | Description |
| 0 | bLength | 1 | 0Ah | Size of the descriptor |
| 1 | bDescriptorType | 1 | 06h | Type of the descriptor (Device_Qualifier descriptor) |
| 2 | bcdUSB | 2 | 0210h | USB specification number (0210h=Revision 2.10) |
| 4 | bDeviceClass | 1 | 00h | Class (specified by the interface descriptor) |
| 5 | bDeviceSubClass | 1 | 00h | Subclass (specified by the interface descriptor) |
| 6 | bDeviceProtocol | 1 | 00h | Protocol (specified by the interface descriptor) |
| 7 | bMaxPacketSize0 | 1 | 40h | Maximum packet size of endpoint 0 |
| 8 | bNumComfigurations | 1 | 01h | The number of configurations other than USB2.0 |
| 9 | bReserved | 1 | 00h | Reserved |

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## 4.1.3 Configuration Descriptor

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 09h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 02h | Type of the descriptor (Configuration descriptor) | |
| 2 | wTotalLength | 2 | 0027h | The total length of the data returned for this configuration<br>All the descriptors are included (configuration, interface, endpoint, and class-specific).<br>Total length is 0039h when the SuperSpeed is used. | |
| 4 | bNumInterfaces | 1 | 01h | The number of interfaces | |
| 5 | bConfiguration Value | 1 | 01h | The value used as an argument to Set Configuration Request for selecting this configuration | |
| 6 | iConfiguration | 1 | 00h | Index of the string descriptor | |
| 7 | bmAttributes | 1 | C0h | Configuration characteristics | |
| | | | | Bit7 | Reserved (1) |
| | | | | Bit6 | Self-powered |
| | | | | Bit5 | Remote Wakeup |
| | | | | Bit4…0 | Reserved (0) |
| 8 | MaxPower | 1 | FAh | Maximum power consumption supplied from the bus to the USB device (2 mA).<br>003fh (8 mA) for SuperSpeed. | |

## 4.1.4 Other_Speed_Configuration Descriptor

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 09h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 07h | Type of the descriptor (OtherSpeedConfiguration descriptor) | |
| 2 | wTotalLength | 2 | 0027h | The total length of the data returned for this configuration<br>All the descriptors are included (configuration, interface, endpoint, and class-specific). | |
| 4 | bNumInterfaces | 1 | 01h | The number of interfaces | |
| 5 | bConfiguration Value | 1 | 01h | The value used as an argument to Set Configuration Request for selecting this configuration | |
| 6 | iConfiguration | 1 | 00h | Index of the string descriptor | |
| 7 | bmAttributes | 1 | C0h | Configuration characteristics | |
| | | | | Bit7 | Reserved (1) |
| | | | | Bit6 | Self-powered |
| | | | | Bit5 | Remote Wakeup |
| | | | | Bit4…0 | Reserved (0) |
| 8 | MaxPower | 1 | FAh | Maximum power consumption supplied from the bus to the USB device (2 mA) | |

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### 4.1.5 Interface Descriptor

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 09h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 04h | Type of the descriptor (Interface descriptor) |
| 2 | bInterfaceNumber | 1 | 00h | Interface number (0 is the standard.) |
| 3 | bAlternatingSetting | 1 | 00h | The value used for selecting the interface |
| 4 | bNumEndpoints | 1 | 03h | The number of endpoints |
| 5 | bInterfaceClass | 1 | 06h | Class code (06=ImageInterface) |
| 6 | bInterfaceSubClass | 1 | 01h | Subclass code (01=Still Image Capture Device) |
| 7 | bInterfaceProtocol | 1 | 01h | Protocol (01h=Bulk-Only Transport) |
| 8 | iInterface | 1 | 00h | Index of the string descriptor describing this interface |

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## 4.1.6 Endpoint Descriptor

### 4.1.6.1 Bulk-Out Endpoint

For SUPER-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 02h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 02h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control, 01 = Isochronous, 10 = Bulk, 11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0400h | Maximum packet size of this endpoint (0400h=1024Byte) | |
| 6 | bInterval | 1 | 00h | Polling interval (invalid for the Bulk and the Control endpoints) | |

For HIGH-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 02h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 02h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control, 01 = Isochronous, 10 = Bulk, 11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0200h | Maximum packet size of this endpoint (0200h=512Byte) | |
| 6 | bInterval | 1 | 00h | Polling interval (invalid for the Bulk and the Control endpoints) | |

For FULL-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 02h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 02h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control, 01 = Isochronous, 10 = Bulk, 11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0040h | Maximum packet size of this endpoint (0040h=64Byte) | |
| 6 | bInterval | 1 | 00h | Polling interval (invalid for the Bulk and the Control endpoints) | |

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### 4.1.6.2 Bulk-In Endpoint

For SUPER-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 81h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 02h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control、01 = Isochronous、10 = Bulk、11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0400h | Maximum packet size of this endpoint (0400h=1024Byte) | |
| 6 | bInterval | 1 | 00h | Polling interval (invalid for the Bulk and the Control endpoints) | |

For HIGH-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 81h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 02h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control、01 = Isochronous、10 = Bulk、11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0200h | Maximum packet size of this endpoint (0200h=512Byte) | |
| 6 | bInterval | 1 | 00h | Polling interval (invalid for the Bulk and the Control endpoints) | |

For FULL-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 81h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 02h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control、01 = Isochronous、10 = Bulk、11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0040h | Maximum packet size of this endpoint (0040h=64Byte) | |
| 6 | bInterval | 1 | 00h | Polling interval (invalid for the Bulk and the Control endpoints) | |

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### 4.1.6.3 Interrupt Endpoint

For SUPER-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 83h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 03h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control, 01 = Isochronous, 10 = Bulk, 11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0018h | Maximum packet size of this endpoint (0018h=24Byte) | |
| 6 | bInterval | 1 | 0Ah | Polling interval (0Ah = 10ms) | |

For HIGH-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 83h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 03h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control, 01 = Isochronous, 10 = Bulk, 11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0008h | Maximum packet size of this endpoint (0008h=8Byte) | |
| 6 | bInterval | 1 | 09h | Polling interval | |

For FULL-SPEED

| Offset | Field | Size | Value | Description | |
|---|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor | |
| 1 | bDescriptorType | 1 | 05h | Type of the descriptor (Endpoint descriptor) | |
| 2 | bEndpointAddress | 1 | 83h | Address of the endpoint | |
| | | | | Bit7 | Transfer direction (0=OUT, 1=IN) |
| | | | | Bit6…4 | Reserved (0) |
| | | | | Bit3…0 | Endpoint number |
| 3 | bmAttributes | 1 | 03h | Attributes of the endpoint | |
| | | | | Bit1…0 | Transfer type |
| | | | | (00 = Control, 01 = Isochronous, 10 = Bulk, 11 = Interrupt) | |
| 4 | wMaxPacketSize | 2 | 0008h | Maximum packet size of this endpoint (0008h=8Byte) | |
| 6 | bInterval | 1 | 09h | Polling interval | |

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## 4.1.7 Super Speed Endpoint Companion Descriptor

### 4.1.7.1 Bulk-Out Endpoint

For SUPER-SPEED

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 06h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 30h | Type of the descriptor (SuperSpeed Endpoint Companion Descriptor) |
| 2 | bMaxBurst | 1 | 0Fh | Maximum number of packets that can be transmitted and received by burst transfer.<br>(Number of burst transfers -1) should be set. The maximum value is 15 |
| 3 | bmAttributes | 1 | 00h | Maximum number of streams for Bulk-Endpoint. 0 for no stream definition.<br>In the case of an isochronous end point, the maximum number of packets in the service interval. |
| 4 | wBytesPerInterval | 2 | 0000h | Maximum number of bytes at which a fixed interval interrupt or isochronous endpoint will transfer per service interval. |

### 4.1.7.2 Bulk-In Endpoint

For SUPER-SPEED

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 06h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 30h | Type of the descriptor (SuperSpeed Endpoint Companion Descriptor) |
| 2 | bMaxBurst | 1 | 0Fh | Maximum number of packets that can be transmitted and received by burst transfer.<br>(Number of burst transfers -1) should be set. The maximum value is 15 |
| 3 | bmAttributes | 1 | 00h | Maximum number of streams for Bulk-Endpoint. 0 for no stream definition.<br>In the case of an isochronous end point, the maximum number of packets in the service interval. |
| 4 | wBytesPerInterval | 2 | 0000h | Maximum number of bytes at which a fixed interval interrupt or isochronous endpoint will transfer per service interval. |

### 4.1.7.3 Interrupt Endpoint

For SUPER-SPEED

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 06h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 30h | Type of the descriptor (SuperSpeed Endpoint Companion Descriptor) |
| 2 | bMaxBurst | 1 | 0Fh | Maximum number of packets that can be transmitted and received by burst transfer.<br>(Number of burst transfers -1) should be set. The maximum value is 15 |
| 3 | bmAttributes | 1 | 00h | Maximum number of streams for Bulk-Endpoint. 0 for no stream definition.<br>In the case of an isochronous end point, the maximum number of packets in the service interval. |
| 4 | wBytesPerInterval | 2 | 0018h | Maximum number of bytes at which a fixed interval interrupt or isochronous endpoint will transfer per service interval. |

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## 4.1.8 String Descriptor

### 4.1.8.1 Index1 (iManufacture)

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 0Ch | Size of the descriptor |
| 1 | bDescriptorType | 1 | 03h | Type of the descriptor (String descriptor) |
| 2 | bString | 10 | 4E00h<br>4900h<br>4B00h<br>4F00h<br>4E00h | Unicode character string<br>“NIKON” |

### 4.1.8.2 Index2 (iProduct)

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 1Ch | Size of the descriptor |
| 1 | bDescriptorType | 1 | 03h | Type of the descriptor (String descriptor) |
| 2 | bString | 26 | 4E00h<br>4900h<br>4B00h<br>4F00h<br>4E00h<br>2000h<br>4400h<br>5300h<br>4300h<br>2000h<br>5A00h<br>2000h<br>3800h | Unicode character string<br>“NIKON DSC Z 8 |

### 4.1.8.3 Index3 (iSerialNumber)

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 42h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 03h | Type of the descriptor (String descriptor) |
| 2 | bString | 64 | XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h<br>XX00h | Unicode character string<br>“XXXXXXXXXXXX”<br><br>Currently 0000h is set after the twelfth character. |

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### 4.1.9 BOS Descriptor

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 05h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 0Fh | Type of the descriptor (BOS descriptor) |
| 2 | wTotalLength | 2 | 0016h | Length of this descriptor and all of its sub descriptors |
| 4 | bString | 1 | 02h | The number of separate device capability descriptors in the BOS |

### 4.1.10 USB 2.0 Extension Descriptor

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 07h | Size of the descriptor |
| 1 | bDescriptorType | 1 | 10h | Type of the descriptor (DEVICE CAPABILITY descriptor) |
| 2 | bDevCapabilityType | 1 | 02h | Capability type (USB 2.0 Extension) |
| 3 | bmAttributes | 4 | 00000006h | Bit0 : Reserved<br>Bit1 : LPM (Link Power Management protocol)<br>Bit2 : BESL & Altermate HIRD definitions supported<br>Bit3 to Bit31 : Reserved |

### 4.1.11 Super Speed Device Capabilities Descriptor

| Offset | Field | Size | Value | Description |
|---|---|---|---|---|
| 0 | bLength | 1 | 0Ah | Size of the descriptor |
| 1 | bDescriptorType | 1 | 10h | Type of the descriptor (DEVICE CAPABILITY descriptor) |
| 2 | bDevCapabilityType | 1 | 03h | Capability type (SUPERSPEED_USB) |
| 3 | bmAttributes | 1 | 00h | Bit0 : Reserved<br>Bit1 : LTM Capable<br>Bit2 to Bit31 : Reserved |
| 4 | wSpeedsSupported | 2 | 000Eh | Bit0 : Reserved<br>Bit1 : full-speed supported<br>Bit2 : high-speed supported<br>Bit3 : 5Gbs supported<br>Bit4 to Bit15 : Reserved |
| 6 | bFunctionalitySupport | 1 | 01h | |
| 7 | bU1DevExitLat | 1 | 01h | U1 Device Exit Latency.<br>01h : Less than 1μs |
| 8 | wU1DevExitLat | 2 | 0065h | U2 Device Exit Latency.<br>0065h : Less than 101μs |

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## 4.2 Class-Specific Descriptor

The camera has no class-specific descriptor.

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# 5 Protocol

The camera supports the Bulk-Only Transport protocol. In the Bulk-Only protocol, all transmission/reception of the command, data, and response are performed by the bulk transfer. When an asynchronous event is generated in the camera, the information is sent by the Interrupt transfer.

The protocol processing is composed of three phases (command phase, data phase, and response phase). All the commands, data, and responses are stored in the Generic Container Structure and transferred between the host and the camera. The processing starts by sending the command (Bulk-Out transfer) from the host to the camera first (command phase). Then the data is transmitted or received by the bulk transfer if the command needs the data transfer (data phase). The processing is completed when the device transmits the command response to the host (Bulk-In transfer) last (response phase). The command and the response phases are always present.

When sending the event to the host, the contents of the event to be sent are stored according to the asynchronous event interrupt data format and sent as the asynchronous event (Interrupt transfer).

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# 5.1 Generic Container Structure

Each field data of the Generic Container Structure is transferred in order of the LSB first (little endian). The Container Type and the contents of the Payload differ according to the phase difference.

| Offset | Size | Field | Description |
|---|---|---|---|
| 0 | 4 | Container Length | The number of bytes in the unsigned integer of this container is coded.<br>The still image capture device decides the container size by using this field. |
| 4 | 2 | Container Type | This field shows the container type.<br>0: Not defined<br>1: Command block<br>2: Data block<br>3: Response block<br>4: Event block |
| 6 | 2 | Code | This field conforms to MTP Specification v1.0. (OperationCode, ResponseCode, or EventCode)<br>For the data block, the OperationCode of the command block is used. |
| 8 | 4 | TransactionID | This is the number of the 32-bit unsigned integer created by the host related to all the phases.<br>It starts with 0x00000001 and is incremented along with the command block issue. When the OperationCode is OpenSession, 0x00000000 is set and 0x00000001 is given by the next command block.<br>When the number reaches the maximum value (0xFFFFFFFE), the next TransactionID becomes 0x00000001. |
| 12 | ?? | Payload | Differs depending on the phase. |

# 5.2 Asynchronous Event Interrupt Data Format

When an event is generated in the camera, the information is transferred to the host according to the following format. Each field data is transferred in order of the LSB first (little endian).

| Offset | Size | Field | Description |
|---|---|---|---|
| 0 | 4 | Container Length | The number of bytes in the unsigned integer of the length of this container is coded. The value is 0x00000010. |
| 4 | 2 | Container Type | Container Type = 0x0004 (Event) |
| 6 | 2 | Code | EventCode |
| 8 | 4 | TransactionID | TransactionID = 0xFFFFFFFF |
| 12 | ?? | Payload | Differs depending on the event. |

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# 5.3 Phases

The communication between the camera and the host is composed of the three phases; command phase, data phase, and response phase. The details of each phase are shown below.

## 5.3.1 Command Phase

In the command phase, the host sends the Generic Container Structure of the command block to the camera. The processing is started by sending the command block from the host in the command phase. The camera performs the processing according to the OperationCode sent in the command block. For the OperationCode and its processing, refer to section 6.2. The Generic Container Structure field data set in the command phase is shown below.

| Offset | Size | Field | Description |
|---|---|---|---|
| 0 | 4 | Container Length | Indicates the length of this container.<br>Each code has its own container length. |
| 4 | 2 | Container Type | 1: Command block |
| 6 | 2 | Code | Operation Code |
| 8 | 4 | TransactionID | This is the number given by the 32-bit unsigned integer created by the host related to all the phases. It starts with 0x00000001 and is incremented by 1 along with the command block issue. When the OperationCode is OpenSession, 0x00000000 is set and 0x00000001 is given by the next command block. When the number reaches the maximum value of the field (0xFFFFFFFE), the next TransactionID becomes 0x00000001. |
| 12 | 4 | Parameter 1 | This field includes the operation parameter. The format and the meaning of the parameter differ depending on the OperationCode. |
| 16 | 4 | Parameter 2 | |
| 20 | 4 | Parameter 3 | |

## 5.3.2 Data Phase

The data phase is an optional phase used to transfer the data that is larger than what can fit in the data sets of the command or the response block. According to the OperationCode specified by the command block, the data is transferred from the host to the camera, from the camera to the host, or not transferred at all. For the OperationCode and the corresponding data contents, refer to section 6.2. The Generic Container Structure field data set in the data phase is shown below.

| Offset | Size | Field | Description |
|---|---|---|---|
| 0 | 4 | Container Length | Indicates the length of this container.<br>The container length differs depending on the size of the Payload. |
| 4 | 2 | Container Type | 2: Data block |
| 6 | 2 | Code | The corresponding OperationCode sent in the command phase is set. |
| 8 | 4 | TransactionID | This is the number given by the 32-bit unsigned integer created by the host related to all the phases. It starts with 0x00000001 and is incremented by 1 along with the command block issue. When the OperationCode is OpenSession, 0x00000000 is set and 0x00000001 is given by the next command block. When the number reaches the maximum value of the field (0xFFFFFFFE), the next TransactionID becomes 0x00000001. |
| 12 | ?? | Payload | The contents of this field differ depending on the OperationCode. |

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### 5.3.3 Response Phase

In the response phase, the Generic Container Structure of the response block is sent from the camera to the host in order to indicate whether the command sent in the command phase succeeds or fails.

The contents of the response can be identified by the ResponseCode stored in the Code field. For the ResponseCode, refer to section 6.3.

The Generic Container Structure field data set in the response phase is shown below.

| Offset | Size | Field | Description |
|---|---|---|---|
| 0 | 4 | Container Length | Indicates the length of this container.<br>Each code has its own container length. |
| 4 | 2 | Container Type | 3: Response block |
| 6 | 2 | Code | The ResponseCode is set in order to indicate whether the processing corresponding to the OperationCode sent in the command phase succeeds or fails. |
| 8 | 4 | TransactionID | This is the number given by the 32-bit unsigned integer created by the host related to all the phases. It starts with 0x00000001 and is incremented by 1 along with the command block issue. When the OperationCode is OpenSession, 0x00000000 is set and 0x00000001 is given by the next command block. When the number reaches the maximum value of the field (0xFFFFFFFE), the next TransactionID becomes 0x00000001. |
| 12 | 4 | Response Parameter | This field includes the response parameter. The format and the meaning of the parameter differ depending on the OperationCode and the ResponseCode. |

## 5.4 Error Handling

If the following state is detected, the camera returns to the command phase status.

| No. | Description |
|---|---|
| 1 | Bus reset |
| 2 | Reset recovery |

### 5.4.1 Command Block Reception Failure

When the command block reception fails and the reset recovery and the bus reset are not detected, the camera stalls the Bulk-In and the Bulk-Out endpoints and returns to the command phase status.

### 5.4.2 Command Block Invalidity

When the following error is detected after the command block reception succeeds and the reset recovery and the bus reset are not detected, the camera stalls the Bulk-In and the Bulk-Out endpoints.

| No. | Description |
|---|---|
| 1 | Container Type of Generic Container Structure is other than the command block. |

### 5.4.3 Command Execution Error

When the command execution error is detected after a valid command block is received, the phase is changed to the response phase and the response code corresponding to the error is set in the Code field of the Generic Container Structure and sent. Stalling is not performed.

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# 6 Code

The codes supported by the camera are described.

## 6.1 ObjectFormatCode

The ObjectFormatCode indicates the format of the objects in the card inserted in the camera (image file and script file) and the related objects (corresponding to the directories and the virtual association representing the relation between the images that conform to the DCF standards and the DCF objects in the camera). The following table represents the ObjectFormatCodes supported by the camera.

| ObjectFormatCode | Format | Description |
|---|---|---|
| 0x3000 | Undefined | NDF (dust reference image)<br>NEF (when MTP is not supported)<br>(In the definition of MTP Specification v1.0, it is defined as “Format not defined”.) |
| 0x3001 | Association | Association<br>(Indicates the directories or the virtual association representing the relation between the images that conform to the DCF standards and the DCF objects.) |
| 0x3002 | Script | Script (only the virtual script file is the target) |
| 0x3008 | WAV | AudioClip |
| 0x3800 | Unknown Image Object | NEF (when MTP is supported) |
| 0x3801 | EXIF/JPEG | JEIDA Standard |
| 0x3808 | JFIF | JPEG File Interchange Format (represents the thumbnail format.) |
| 0x300D | MOV | Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF (RGB) | Tag Image File Format |
| 0xB200 | HEIF | High Efficiency Image File Format |
| 0xB201 | NEV | N-RAW |
| 0xB982 | MP4 Container | ISO 14496-1 |

The ObjectFormatCode may be used as one of the parameters in the command phase.
It is also used in the ObjectInfo data set.

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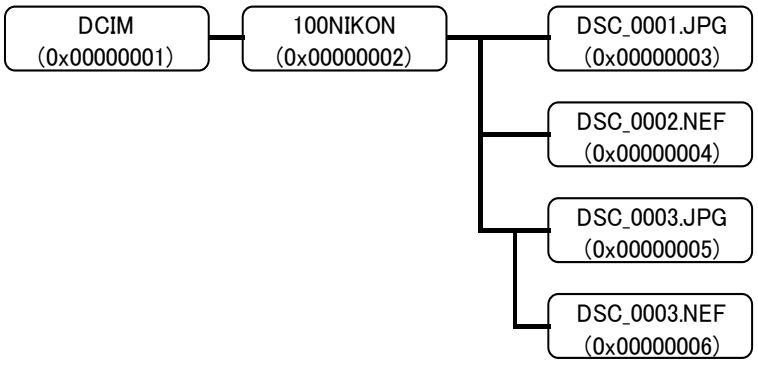
## 6.1.1 Association Type

There are various Types in the Association defined by ObjectFormatCode. Association is for representing the folders and the file system. All the objects that belong to the Association correspond to the branches of the tree structure under the Association. The associations to which the objects belong are specified in the ParentObject field of the ObjectInfo data set for each object. Type is specified in the AssociationType field of the ObjectInfo data set for the associations object. The Association Type used in the camera is shown below.

| Association Code | Association Type | Description |
|---|---|---|
| 0x0001 | GenericFolder | Indicates the general directory. For the camera, it indicates the DCF image directory and DCF directory. |

An example of how the Association is used in the DCF file system for the camera is shown below.
Note, however, that the directory name, the file name, and the ObjectHandle value may differ from the ObjectHandle that is actually used in the camera to simplify the explanation.

| ObjectHandle | Description | ObjectFormatCode | ParentObject |
|---|---|---|---|
| 0x00000001 | ¥DCIM folder | 0x3001 | 0x00000000 |
| 0x00000002 | ¥DCIM¥100NIKON | 0x3001 | 0x00000001 |
| 0x00000003 | ¥DCIM¥100NIKON¥DSC_0001.JPG | 0x3801 | 0x00000002 |
| 0x00000004 | ¥DCIM¥100NIKON¥DSC_0002.NEF | 0x3000 | 0x00000002 |
| 0x00000005 | ¥DCIM¥100NIKON¥DSC_0003.JPG | 0x3801 | 0x00000002 |
| 0x00000006 | ¥DCIM¥100NIKON¥DSC_0003.NEF | 0x3000 | 0x00000002 |




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# 6.2 Operation Code

The OperationCode is the command that is used by the host to request the operation of the camera in the command phase. The OperationCode is sent as a part of the command block data set.

OperationDataNone, OperationReceiveData and OperationSendData can execute 4Bytes OperationCodes.

Operation Codes that cannot be executed in photo mode, video mode, or video recording are described in the table below. Access_Denied response is sent when execution is disabled.

The OperationCodes supported by the camera are shown below.

| Operation Code | Description | Reference item | Photo mode | Video mode | During video recording |
|---|---|---|---|---|---|
| 0x1001 | GetDeviceInfo | 6.2.1.1 | ✓ | ✓ | ✓ |
| 0x1002 | OpenSession | 6.2.1.2 | ✓ | ✓ | - |
| 0x1003 | CloseSession | 6.2.1.3 | ✓ | ✓ | ✓ |
| 0x1004 | GetStorageIDs | 6.2.1.4 | ✓ | ✓ | ✓ |
| 0x1005 | GetStorageInfo | 6.2.1.5 | ✓ | ✓ | ✓ |
| 0x1006 | GetNumObjects | 6.2.1.6 | ✓ | ✓ | ✓ |
| 0x1007 | GetObjectHandles | 6.2.1.7 | ✓ | ✓ | ✓ |
| 0x1008 | GetObjectInfo | 6.2.1.8 | ✓ | ✓ | - |
| 0x1009 | GetObject | 6.2.1.9 | ✓ | ✓ | - |
| 0x100A | GetThumb | 6.2.1.10 | ✓ | ✓ | - |
| 0x100B | DeleteObject | 6.2.1.11 | ✓ | ✓ | - |
| 0x100C | SendObjectInfo | 6.2.1.12 | ✓ | ✓ | - |
| 0x100D | SendObject | 6.2.1.13 | ✓ | ✓ | - |
| 0x100E | InitiateCapture | 6.2.1.14 | ✓ | - | - |
| 0x100F | FormatStore | 6.2.1.15 | ✓ | ✓ | - |
| 0x1014 | GetDevicePropDesc | 6.2.1.16 | ✓ | ✓ | ✓ |
| 0x1015 | GetDevicePropValue | 6.2.1.17 | ✓ | ✓ | ✓ |
| 0x1016 | SetDevicePropValue | 6.2.1.18 | ✓ | ✓ | ✓ |
| 0x101B | GetPartialObject | 6.2.1.19 | ✓ | ✓ | - |
| 0x90C0 | InitiateCaptureRecInSdram | 6.2.4.2 | ✓ | - | - |
| 0x90C1 | AfDrive | 6.2.5.1 | ✓ | ✓ | ✓ |
| 0x90C2 | ChangeCameraMode | 6.2.2.15 | ✓ | ✓ | - |
| 0x90C3 | DeleteImagesInSdram | 6.2.3.9 | ✓ | ✓ | - |
| 0x90C4 | GetLargeThumb | 6.2.3.6 | ✓ | ✓ | - |
| 0x90C7 | GetEvent | 6.2.2.12 | ✓ | ✓ | ✓ |
| 0x90C8 | DeviceReady | 6.2.2.11 | ✓ | ✓ | ✓ |
| 0x90C9 | SetPreWbData | 6.2.8.1 | ✓ | ✓ | - |
| 0x90CB | AfAndCaptureRecInSdram | 6.2.4.3 | ✓ | - | - |
| 0x90CC | GetPicCtrlData | 6.2.9.1 | ✓ | ✓ | ✓ |
| 0x90CD | SetPicCtrlData | 6.2.9.3 | ✓ | ✓ | - |
| 0x90CE | DeleteCustomPicCtrl | 6.2.9.4 | ✓ | ✓ | - |
| 0x90CF | GetPicCtrlCapability | 6.2.9.5 | ✓ | ✓ | ✓ |
| 0x9201 | StartLiveView | 6.2.6.1 | ✓ | ✓ | ✓ |
| 0x9202 | EndLiveView | 6.2.6.2 | ✓ | ✓ | ✓ |
| 0x9203 | GetLiveViewImage | 6.2.6.3 | ✓ | ✓ | ✓ |

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| | | | | | |
|---|---|---|---|---|---|
| 0x9204 | MfDrive | 6.2.5.3 | ✓ | ✓ | ✓ |
| 0x9205 | ChangeAfArea | 6.2.6.6 | ✓ | ✓ | ✓ |
| 0x9206 | AfDriveCancel | 6.2.5.2 | ✓ | ✓ | ✓ |
| 0x9207 | InitiateCaptureRecInMedia | 6.2.4.1 | ✓ | - | - |
| 0x9209 | GetVendorStorageIDs | 6.2.3.1 | ✓ | ✓ | ✓ |
| 0x920A | StartMovieRecord | 6.2.7.1 | - | ✓ | - |
| 0x920B | EndMovieRecord | 6.2.7.2 | ✓ | ✓ | ✓ |
| 0x920C | TerminateCapture | 6.2.4.4 | ✓ | ✓ | - |
| 0x920F | GetFhdPicture | 6.2.3.7 | ✓ | ✓ | - |
| 0x9400 | GetPartialObjectHighSpeed | 6.2.3.5 | ✓ | ✓ | - |
| 0x9401 | GetIPTCInfo | 6.2.10.1 | ✓ | ✓ | ✓ |
| 0x9402 | StartSpotWb | 6.2.8.2 | - | ✓ | - |
| 0x9403 | EndSpotWb | 6.2.8.3 | ✓ | ✓ | - |
| 0x9404 | ChangeSpotWbArea | 6.2.8.4 | ✓ | ✓ | - |
| 0x9405 | MeasureSpotWb | 6.2.8.5 | ✓ | ✓ | - |
| 0x9406 | EndSpotWbResultDisp | 6.2.8.6 | ✓ | ✓ | - |
| 0x940C | CancelImagesInSdram | 6.2.3.8 | ✓ | ✓ | - |
| 0x9412 | GetIPTCPresetInfo | 6.2.10.2 | ✓ | ✓ | ✓ |
| 0x9413 | SetIPTCPresetInfo | 6.2.10.3 | ✓ | ✓ | - |
| 0x9414 | GetSBHandles | 6.2.11.1 | ✓ | ✓ | ✓ |
| 0x9415 | GetSBAttrDesc | 6.2.11.2 | ✓ | ✓ | ✓ |
| 0x9416 | GetSBAttrValue | 6.2.11.3 | ✓ | ✓ | ✓ |
| 0x9417 | SetSBAttrValue | 6.2.11.4 | ✓ | ✓ | - |
| 0x9418 | GetSBGroupAttrDesc | 6.2.11.5 | ✓ | ✓ | ✓ |
| 0x9419 | GetSBGroupAttrValue | 6.2.11.6 | ✓ | ✓ | ✓ |
| 0x941A | SetSBGroupAttrValue | 6.2.11.7 | ✓ | ✓ | - |
| 0x941B | TestFlash | 6.2.11.8 | ✓ | ✓ | ✓ |
| 0x941C | GetEventEx | 6.2.2.13 | ✓ | ✓ | ✓ |
| 0x9420 | SaveCameraSetting | 6.2.2.20 | ✓ | ✓ | - |
| 0x9421 | GetObjectSize | 6.2.3.2 | ✓ | ✓ | - |
| 0x9422 | ChangeMonitorOff | 6.2.2.17 | ✓ | ✓ | ✓ |
| 0x9423 | GetLiveViewCompressedSize | 6.2.6.5 | ✓ | ✓ | ✓ |
| 0x9424 | StartTracking | 6.2.6.7 | ✓ | ✓ | ✓ |
| 0x9425 | EndTracking | 6.2.6.8 | ✓ | ✓ | ✓ |
| 0x9426 | ChangeAELock | 6.2.4.5 | ✓ | ✓ | ✓ |
| 0x9428 | GetLiveViewImageEx | 6.2.6.4 | ✓ | ✓ | ✓ |
| 0x942E | GetHLGQualityData | 6.2.7.3 | ✓ | ✓ | ✓ |
| 0x942F | SetHLGQualityData | 6.2.7.4 | ✓ | ✓ | - |
| 0x9431 | GetPartialObjectEx | 6.2.3.4 | ✓ | ✓ | - |
| 0x9433 | InitiatePixelMapping | 6.2.4.6 | ✓ | ✓ | - |
| 0x9434 | GetObjectsMetaData | 6.2.3.3 | ✓ | ✓ | - |
| 0x9435 | ChangeApplicationMode | 6.2.2.16 | ✓ | ✓ | - |
| 0x9436 | ResetMenu | 6.2.2.21 | ✓ | ✓ | - |
| 0x9439 | GetVendorCodes | 6.2.2.1 | ✓ | ✓ | ✓ |
| 0x943A | GetDevicePropDescEx | 6.2.2.5 | ✓ | ✓ | ✓ |
| 0x943B | GetDevicePropValueEx | 6.2.2.6 | ✓ | ✓ | ✓ |
| 0x943C | SetDevicePropValueEx | 6.2.2.7 | ✓ | ✓ | ✓ |
| 0x943D | RegisterDevicePropGroup | 6.2.2.8 | ✓ | ✓ | ✓ |

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| | | | | | |
|---|---|---|---|---|---|
| 0x943E | GetDevicePropDescList | 6.2.2.9 | ✓ | ✓ | ✓ |
| 0x943F | GetDevicePropValueList | 6.2.2.10 | ✓ | ✓ | ✓ |
| 0x9440 | OperationDataNone | 6.2.2.2 | ✓ | ✓ | ✓ |
| 0x9441 | OperationReceiveData | 6.2.2.3 | ✓ | ✓ | ✓ |
| 0x9442 | OperationSendData | 6.2.2.4 | ✓ | ✓ | ✓ |
| 0x9443 | GetPicCtrlDataList | 6.2.9.2 | ✓ | ✓ | ✓ |
| 0x9444 | GetPicCtrlCapabilityList | 6.2.9.6 | ✓ | ✓ | ✓ |
| 0x9445 | InitiateOpenCaptureV | 6.2.4.7 | ✓ | - | - |
| 0x9446 | TerminateOpenCaptureV | 6.2.4.8 | ✓ | - | - |
| 0x9447 | GetOpenCaptureInfo | 6.2.4.9 | ✓ | ✓ | ✓ |
| 0x944C | GetCommandFeature | 6.2.2.24 | ✓ | ✓ | ✓ |
| 0x944D | GetPicCtrlHlgDataList | 6.2.9.7 | ✓ | ✓ | ✓ |
| 0x944E | SetPicCtrlHlgData | 6.2.9.8 | ✓ | ✓ | - |
| 0x944F | GetPicCtrlHlgCapabilityList | 6.2.9.9 | ✓ | ✓ | ✓ |
| 0x9453 | SetPortraitImpressionBalanceData | 6.2.2.23 | ✓ | ✓ | - |
| 0x9454 | GetPortraitImpressionBalanceDataList | 6.2.2.22 | ✓ | ✓ | ✓ |
| 0x9461 | ClearEvent | 6.2.2.14 | ✓ | ✓ | ✓ |
| 0x9463 | SetFlickerReductionShutterSpeed | 6.2.4.10 | ✓ | ✓ | ✓ |
| 0x9464 | PowerZoomDrive | 6.2.5.4 | ✓ | ✓ | ✓ |
| 0x946D | GetCameraSettingData | 6.2.2.18 | ✓ | ✓ | - |
| 0x946E | SendCameraSettingData | 6.2.2.19 | ✓ | ✓ | - |
| 0x9470 | GetManualSettingLensData2 | 6.2.5.5 | ✓ | ✓ | ✓ |
| 0x9801 | GetObjectPropsSupported | 6.2.12.1 | ✓ | ✓ | ✓ |
| 0x9802 | GetObjectPropDesc | 6.2.12.2 | ✓ | ✓ | ✓ |
| 0x9803 | GetObjectPropValue | 6.2.12.3 | ✓ | ✓ | - |
| 0x9805 | GetObjectPropList | 6.2.12.4 | ✓ | ✓ | - |

Do Not Copy

## 6.2.1 Standard

### 6.2.1.1 GetDeviceInfo

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1001 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DeviceInfo data set |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns the information of the camera (DeviceInfo data set).
The DeviceInfo data set includes information such as the camera version information and the codes supported by the camera.

**○ Command Details**

This operation is the only operation that may be issued inside or outside of a session.
The contents of the DeviceInfo data set sent by the camera are shown in section 9.1.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.1.2 OpenSession

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1002 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode starts the logical connection (session) between the camera and the host.

The SessionID is specified optionally by the host and retained during the session.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | SessionID |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Invalid_Parameter | The value of the parameter1 is 0x00000000. |
| Session_Already_Open | The session between the camera and the host has been already started. |

Do Not Copy

### 6.2.1.3 CloseSession

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1003 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode closes the logical connection (session) between the camera and the host.

When this operation is performed, the operation being run is terminated. Besides, the camera state changed after OpenSession returns to the default.

| State | Contents of change |
|---|---|
| During shooting operation | Only the shooting operation currently being performed is executed. For the continuous shooting, it is not continued. |
| During AF operation | The AF operation by command stops. |
| During power zoom operation | The power zoom operation by command stops. |
| AE locked | Release AE lock (If AE lock is not assigned to the camera button) |
| During remote live view | The camera live view is executed. |
| In the remote mode | The camera enters the PC camera mode. |
| The recording destination is “SDRAM” or “Card and SDRAM”. | The recording destination is set to “Card”. |
| In the application mode | Set to OFF. |
| During forcibly monitor-off | Forcibly monitor-off is released. |
| During AE lock | AE lock is released (when the AE lock is not assigned in the camera button). |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Parameter_Not_Supported | An unsupported parameter is specified. |

Do Not Copy

### 6.2.1.4 GetStorageIDs

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1004 |
| Operation Parameter | None |
| Data | StorageIDArray |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns a list of the currently valid StorageIDs.
For the application mode, the StorageID with the card not inserted can be acquired by this command. The valid StorageID can be known by using the GetVendorStorageIDs.

**○ Command Details**

The camera returns the StorageID of each slot.
The StorageID of each slot takes the following values.

| StorageID | Details |
|---|---|
| 0x00010001 | When the card is inserted in the slot1 |
| 0x00010000 | When the card is not inserted in the slot1 |
| | When the card in the slot1 is being formatted |
| | When the battery is exhausted |
| 0x00020001 | When the card is inserted in the slot2 |
| 0x00020000 | When the card is not inserted in the slot2 |
| | When the card in the slot2 is being formatted |
| | When the battery is exhausted |

The format of the StorageIDArray that is sent by the camera is shown below.
Each field data is stored in the little-endian format.

| Field | Size (Byte) | Data |
|---|---|---|
| NumElement | 4 | 0x00000002 (Two elements for the array) |
| ArrayEntry1 | 4 | StorageID (slot1) |
| ArrayEntry2 | 4 | StorageID (slot2) |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.1.5 GetStorageInfo

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1005 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | StorageInfo |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode obtains the information of the card inserted in the camera.

**○ Command Details**

The camera returns the StorageInfo data set of the card specified by the StorageID.
The StorageIDs supported by the camera are shown in section 6.2.1.4.
The StorageInfo data set sent by the camera is described in section 9.2.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | StorageID |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_StorageID | The StorageID sent by the camera differs from the StorageID specified by the host. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |

Do Not Copy

### 6.2.1.6 GetNumObjects

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x1006 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | NumObjects |

#### ○ Command Outline

The operation by this OperationCode obtains the number of objects that are present in the card. The number of objects recorded in the SDRAM is not included.

#### ○ Command Details

If the new object is being recorded in the card when this command is received, the processing starts after the acquisition of all the data is completed.

The camera returns the number of objects in the card specified by the StorageID. In addition, Parameter2 and Parameter3, which are optional, can be combined.

| Operation Parameter | Value | Details |
|---|---|---|
| Parameter1 | 0xFFFFFFFF | All the cards should be the targets. |
| | StorageID | The specified card should be the target. Refer to section 6.2.1.4. |
| Parameter2 (Option) | 0x00000000 | All the object formats should be the targets. |
| | No value | |
| | 0xFFFFFFFF | All the image formats in the card corresponding to the specified StorageID should be the targets. |
| | ObjectFormatCode | The specified object format should be the target. Refer to section 6.1. |
| Parameter3 (Option) | 0x00000000 | All the directories should be the targets. |
| | No value | |
| | 0xFFFFFFFF | The objects directly under the root should be the targets. |
| | ObjectHandle | The objects directly under the specified directory should be the targets. |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | StorageID |
| Parameter2 | [ObjectFormatCode] |
| Parameter3 | [ObjectHandle of the directory] |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_StorageID | The StorageID sent by the camera differs from the StorageID specified by the host. |
| Invalid_Object_Handle | For an invalid object handle |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Specification_By_Format_Unsupported | The specified ObjectFormatCode is not supported. |

Do Not Copy

| | |
|---|---|
| Invalid_Parent_Object | An ObjectHandle other than that indicating the directory in the card was specified for ObjectHandle (Parameter3) of the specified directory. |
| | The specified directory does not exist. |

Do Not Copy

### 6.2.1.7 GetObjectHandles

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1007 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectHandleArray |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode obtains the handle of the object in the card. The ObjectHandle of the object recorded in the SDRAM is not included.

**○ Command Details**

If the new object is being recorded in the card when this command is received, the processing starts after the acquisition of all the data is completed.

The camera returns the ObjectHandle in the card specified by the StorageID. In addition, Parameter2 and Parameter3, which are optional, can be combined.

| Operation Parameter | Value | Details |
|---|---|---|
| Parameter1 | 0xFFFFFFFF | All the cards should be the targets. |
| | StorageID | The specified card should be the target. Refer to section 6.2.1.4. |
| Parameter2 (Option) | 0x00000000 | All the object formats should be the targets. |
| | No value | |
| | 0xFFFFFFFF | All the image formats in the card corresponding to the specified StorageID should be the targets. |
| | ObjectFormatCode | The specified object format should be the target. Refer to section 6.1. |
| Parameter3 (Option) | 0x00000000 | All the directories should be the targets. |
| | No value | |
| | 0xFFFFFFFF | The objects directly under the root should be the targets. |
| | ObjectHandle | The objects directly under the specified directory should be the targets. |

The format of the ObjectHandleArray that is sent by the camera is shown below.
Each field data is stored in the little endian format.

| Field | Size (Byte) | Data |
|---|---|---|
| NumElement | 4 | The element of the array is N (N indicates the number of objects). |
| ArrayEntry [0] | 4 | ObjectHandle [0] |
| ArrayEntry [1] | 4 | ObjectHandle [1] |
| ArrayEntry [2] | 4 | ObjectHandle [2] |
| --- | | |
| ArrayEntry [N - 1] | 4 | ObjectHandle [N - 1] |

Do Not Copy

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | StorageID |
| Parameter2 | [ObjectFormatCode] |
| Parameter3 | [ObjectHandle of the directory] |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_StorageID | The StorageID sent by the camera differs from the StorageID specified by the host. |
| Invalid_Object_Handle | For an invalid object handle |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Specification_By_Format_Unsupported | The specified ObjectFormatCode is not supported. |
| Invalid_Parent_Object | An ObjectHandle other than that indicating the directory in the card was specified for ObjectHandle (Parameter3) of the specified directory. |
| | The specified directory does not exist. |

Do Not Copy

### 6.2.1.8 GetObjectInfo

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x1008 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectInfo |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the information of the specified object (ObjectInfo). When a new object is added to the card and the host is informed of the addition of the object by the event, the host acquires the information of the object by this command.

#### ○ Command Details

The camera sends the information of the object corresponding to the ObjectHandle specified in Parameter1.

If the specified ObjectHandle is the data in the card, the camera returns the information of the object corresponding to the ObjectHandle.

The ObjectHandle passed by ObjectAddedInSdram should be specified in order to acquire the information of the object in the SDRAM. For the ObjectHandle passed by ObjectAddedInSdram, the image data information is sent to the host.

The ObjectInfo data set sent in the data phase differs depending on the directory and the file types.

The ObjectInfo data set of each object is shown in section 9.3.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |

Do Not Copy

### 6.2.1.9 GetObject

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1009 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DataObject |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode obtains the specified object (DataObject).

**○ Command Details**

The camera sends all the file data (DataObject) corresponding to the specified ObjectHandle to the host.

If the specified ObjectHandle is the data in the card, the camera returns the object corresponding to the ObjectHandle.

The ObjectHandle passed by ObjectAddedInSdram should be specified in order to acquire information of the object in the SDRAM. For the ObjectHandle passed by ObjectAddedInSdram, the image data is sent to the host.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | The ObjectHandle of the directory (Association) is specified. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |

Do Not Copy

### 6.2.1.10 GetThumb

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x100A |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ThumbnailObject |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the thumbnail data of the specified image/video object (ThumbnailObject).

#### ○ Command Details

The camera sends the thumbnail data (ThumbnailObject) corresponding to the specified ObjectHandle to the host.

If the specified ObjectHandle is the data in the card, the camera returns the thumbnail data corresponding to the ObjectHandle.

The ObjectHandle passed by ObjectAddedInSdram should be specified in order to acquire the information of the object in the SDRAM. For the ObjectHandle passed by ObjectAddedInSdram, the thumbnail data of the image or the video data is sent to the host.

ThumbnailObject is the thumbnail data of the object (file) specified in ObjectHandle. The size of the ThumbnailObject is 160 x 120 of the small thumbnail size.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| No_Thumbnail_Present | The object corresponding to the specified ObjectHandle does not have a thumbnail. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |

Do Not Copy

### 6.2.1.11 DeleteObject

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x100B |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode deletes a specific object saved in the card or all the objects saved in the card. The protected objects are not deleted.

**○ Command Details**

The protected objects are not deleted. An object with an image format that is not supported by the camera is not deleted.

The release is locked until the image deletion is completed.

| Operation Parameter | Value | Details |
|---|---|---|
| Parameter1 | 0xFFFFFFFF | All the objects in the card should be the targets. |
| | ObjectHandle | The specified object should be the target. |
| Parameter2 (Option) | 0x00000000 | All the object formats should be the targets. |
| | No value | |
| | ObjectFormatCode | The specified object format should be the target. When 0xFFFFFFFF is specified in Parameter1, all the objects that correspond to the specified object format are deleted. |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | [ObjectFormatCode] |
| Parameter3 | None |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| Object_Write_Protect | The object corresponding to the specified ObjectHandle is protected. |
| Partial_Deletion | When the deletion of two or more objects is specified, all the objects are not deleted. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Store_Read_Only | The card including the specified image is protected. |
| | All the inserted cards are protected. |
| Specification_By_Format_Unsupported | The specified ObjectFormatCode is not supported. |
| Device_Busy | The acquisition operation is being performed when the command processing is started. |
| Invalid_Parameter | The ObjectFormatCode is specified with the ObjectHandle set to a value other than 0xFFFFFFFF. |

Do Not Copy

### 6.2.1.12 SendObjectInfo

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x100C |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectInfo |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Refer to Response Parameter in this section. |

#### ○ Command Outline

The operation by this OperationCode sends the object information (ObjectInfo) from the host to the camera.

#### ○ Command Details

The operation by this OperationCode is effective when the StorageID is 0x00000000 (the storage destination is not specified) and the Parent ObjectHandle is 0x00000000 or 0xFFFFFFFF.

The camera retains the ObjectInfo received by this command until it receives the SendObject command to be sent from the host successively.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | [StorageID] |
| Parameter2 | [(Parent) ObjectHandle] |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block reception failed. |
| Invalid_StorageID | The StorageID is a value other than 0x00000000. |
| Invalid_ObjectFormatCode | The ObjectFormat field of the received ObjectInfo is set to a value other than 0x3002 (Script). |
| Store_Full | The object cannot be received with the size of the buffer prepared by the camera (32768Byte). |
| Access_Denied | A StorageID of the card is specified. |
| Specification_of_ Destination_Unsupported | The (Parent) ObjectHandle is a value other than 0x00000000 or 0xFFFFFFFF. |

#### ○ Response Parameter

| Response Parameter | Description |
|---|---|
| Parameter1 | StorageID [0x00000000] |
| Parameter2 | Parent ObjectHandle [0x00000000] |
| Parameter3 | Any ObjectHandle |

Do Not Copy

### 6.2.1.13 SendObject

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x100D |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DataObject |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode sends the object from the host to the camera.

**○ Command Details**

The camera records the ObjectData based on the information of the ObjectInfo received in advance. The recording destination is not a card but a virtual recording medium (SDRAM). The camera deletes the ObjectInfo data received in advance when receiving this command and completing the ObjectData reception.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block reception failed. |
| No_Valid_ObjectInfo | This command is received before the SendObjectInfo command is accepted. |

Do Not Copy

### 6.2.1.14 InitiateCapture

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x100E |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |
| Event Code | Refer to Event Code in this section. |

#### ○ Command Outline

The operation by this OperationCode starts the camera acquiring one or more new objects (release operation of the camera) according to the current setting. The acquired objects (image data) are always recorded in the card independently of the recording destination property.

#### ○ Command Details

The camera starts the release operation when receiving this command. The camera returns OK after starting the release operation. If the AF is required, the camera starts the release operation after finishing the AF. (This command is an activation command, so returns the response code before the release operation is completed.)

The camera starts acquiring one or more new objects (release operation of the camera) according to the current setting. The acquisition of the new object is performed asynchronously.

Whether the AF is operated when the release is started depends on the setting of the focus-mode selector.

Refer to Command Release Operation (section 11.7) for the details of the command release operation.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | [StorageID] | |
| | 0x00000000 | The recording destination specified in the camera. |
| | StorageID | The specified recording destination. |
| Parameter2 | [ObjectFormatCode] | |
| | 0x00000000 | The image quality mode set in the camera. |
| | ObjectFormatCode | The image quality mode according to the format. |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| General_Error | An error is generated in the camera body when the command processing is started. |
| | All of the following conditions are satisfied, and the AF operation is not focused.<br>- The AF operation mode is set to [Focus priority]. |
| | The aperture value is “F--”. |
| Invalid_StorageID | The StorageID sent by the camera differs from the StorageID specified by the host. |
| Invalid_ObjectFormatCode | The format of the ObjectFormatCode is not supported by the camera. |
| Store_Full | The free area for recording is not provided in the card. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |

Do Not Copy

| | |
|---|---|
| | The battery is exhausted. |
| Store_Read_Only | All the inserted cards are protected. |
| Device_Busy | The acquisition operation is being performed when the command processing is started. |
| | The shutter-release button is being fully pressed. |
| Access_Denied | When a retractable lens is mounted, the lens is retracting. |
| | STM lens is being initialized. |
| | Shooting cannot be performed depending on the state of the camera other than the above factors. |

### ○ Event Code

| Event Code | Details |
|---|---|
| ObjectAdded | A new object is recorded in the card. |
| StoreFull | The free area for recording is not provided in the card. |
| CaptureComplete | The acquisition operation of the new object is completed. |

Do Not Copy

### 6.2.1.15 FormatStore

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x100F |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode formats the card inserted in the camera.

#### ○ Command Details

The camera formats the card specified by the StorageID.
The StorageIDs supported by the camera are shown in section 6.2.1.4.
The FilesystemFormat parameter shown in Parameter2 is optional. However, setting of only 0x0003 indicating the DCF is permitted.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | StorageID |
| Parameter2 | [FilesystemFormat] |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_StorageID | The StorageID sent by the camera differs from the StorageID specified by the host. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Store_Read_Only | The specified card is protected. |
| Device_Busy | The acquisition operation is being performed when the command processing is started. |
| Invalid_Parameter | Parameter2 is neither 0x00000000 nor 0x00000003. |

Do Not Copy

### 6.2.1.16 GetDevicePropDesc

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x1014 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropDesc data set |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns the DevicePropDesc data set corresponding to the specified DevicePropCode.

**○ Command Details**

For the supported DevicePropCode, refer to section 6.5.

In this OperationCode, only the property of DevicePropCode of 2 bytes can be acquired.
To get the 4bytes property, use the GetDevicePropDescEx command (section 6.2.2.5).

For the DevicePropDesc, refer to section 9.4.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | DevicePropCode |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| DeviceProp_Not_Supported | The specified DevicePropCode is not supported. |
| | 4bytes DevicePropCode is specified. |

Do Not Copy

### 6.2.1.17 GetDevicePropValue

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x1015 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropValue |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode returns the current value corresponding to the specified DevicePropCode.

#### ○ Command Details

For the supported DevicePropCode and the details of the DevicePropValue, refer to section 6.5.

In this OperationCode, only the property of DevicePropCode of 2 bytes can be acquired.
To get the 4bytes property, use the GetDevicePropValueEx command (section 6.2.2.6).

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | DevicePropCode |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| DeviceProp_Not_Supported | The specified DevicePropCode is not supported. |
| | 4bytes DevicePropCode is specified. |

Do Not Copy

### 6.2.1.18 SetDevicePropValue

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x1016 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropValue |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |
| Event Code | Refer to Event Code in this section. |

#### ○ Command Outline

The operation by this OperationCode sets the DevicePropValue corresponding to the specified DevicePropCode to the camera.

#### ○ Command Details

For the supported DevicePropCode and the details of the DevicePropValue, refer to section 6.5.

In this OperationCode, only the property of DevicePropCode of 2 bytes can be acquired.
To get the 4bytes property, use the SetDevicePropValueEx command (section 6.2.2.7).

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | DevicePropCode |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block reception failed. |
| DeviceProp_Not_Supported | The specified DevicePropCode is not supported. |
| | 4bytes DevicePropCode is specified. |
| Access_Denied | An operation is denied depending on the status of the camera. |
| | The specified DevicePropCode is not permitted for setting. |
| Device_Busy | The acquisition operation is being performed when the command processing is started, or the AF is being operated. |
| Invalid_DeviceProp_Format | The size or the format of the DevicePropDesc data set is not appropriate. |
| Invalid_DeviceProp_Value | The specified DevicePropValue is not supported. |

Do Not Copy

### 6.2.1.19 GetPartialObject

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x101B |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DataObject |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | The number of bytes actually sent |

#### ○ Command Outline

The operation by this OperationCode is the same as that of GetObject.

However, the offset and the number of bytes to be acquired can be specified and the object (DataObject) can be acquired partially.

#### ○ Command Details

The camera sends the file data (DataObject) of the specified size corresponding to the specified ObjectHandle to the host. When the specified ObjectHandle is the data in the card, the camera returns the object corresponding to the ObjectHandle.

The ObjectHandle passed by ObjectAddedInSdram should be specified in order to acquire the information of the object in the SDRAM.

For the DataObject, which is the data to be sent, the file data corresponding to the specified ObjectHandle for MaxSize starting from the position set by the offset is sent to the host. In the case of “(File size - Offset) < MaxSize”, the data of “(File size - Offset)”, not MaxSize, is sent to the host. The sent number of bytes is stored in ResponseParameter and sent to the host.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | Offset (Byte) |
| Parameter3 | MaxSize (Byte) |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Incomplete_Transfer | The data block transmission failed. |
| | The size of the object corresponding to the specified ObjectHandle is 4 GB or more. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | The ObjectHandle of the directory (Association) is specified. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| Access_Denied | The video is playing. |

Do Not Copy

## 6.2.2 Vendor

### 6.2.2.1 GetVendorCodes

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9439 |
| Operation Parameter | None |
| Data | DevicePropCodeArray |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode acquires an array of vendor definition codes supported by the camera.

#### ○ Command Details

The format of the array sent by the camera is shown below.

| Field | Size (Byte) | Data |
|---|---|---|
| NumElement | 4 | The number of elements in the array. |
| ArrayEntry [0] | 4 | Vendor Code [0] |
| ArrayEntry [1] | 4 | Vendor Code [1] |
| ArrayEntry [2] | 4 | Vendor Code [2] |
| ・・・ | | |
| ArrayEntry [NumElement - 1] | 4 | Vendor Code [N - 1] |

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | Type of code | |
| | 0x09 | Vendor OperationCode |
| | 0x0D | Vendor DevicePropCode |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | The value of the specified parameter is out of range. |

Do Not Copy

### 6.2.2.2 OperationDataNone

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9440 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Depends on the command. |

**○ Command Outline**

This Operation Code executes the command corresponding to the 4-byte extension.

**○ Command Details**

In this Operation Code, it is possible to execute commands that do not have send / receive data.

The Operation Code command specified in Parameter1 is executed.
For Parameters 2 to 5, specify the Operation Parameter corresponding to the command specified in Parameter1.
Operation Parameters specified in Parameters 2 to 5 are shown in each Operation Code definition.
If the command specified in Parameter1 requires Operation Parameter5, it cannot be executed.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | Operation Code (4-byte extension) NOTE1 |
| Parameter2 | Operation Parameter1 of the command specified in Parameter1 |
| Parameter3 | Operation Parameter2 of the command specified in Parameter1 |
| Parameter4 | Operation Parameter3 of the command specified in Parameter1 |
| Parameter5 | Operation Parameter4 of the command specified in Parameter1 |

NOTE1) 2-byte Operation Code (upper 2 bytes are 0x0000) command can also be executed.

**○ Response Code**

For the command-specific Response Code, refer to the Response Code of each command.

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Operation_Not_Supported | An unsupported Operation Code is specified. |
| Invalid_Parameter | An Operation Code with data is specified. |
| | The Operation Code of this operation is specified. |

Do Not Copy

### 6.2.2.3 OperationReceiveData

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9441 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | Depends on the command. |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Depends on the command. |

**○ Command Outline**

This Operation Code executes the command corresponding to the 4-byte extension.

**○ Command Details**

In this Operation Code, it is possible to execute a command that has the data received from the camera.

The Operation Parameter is the same as the OperationDataNone command (section 6.2.2.2).

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | Operation Code (4-byte extension) NOTE1 |
| Parameter2 | Operation Parameter1 of the command specified in Parameter1 |
| Parameter3 | Operation Parameter2 of the command specified in Parameter1 |
| Parameter4 | Operation Parameter3 of the command specified in Parameter1 |
| Parameter5 | Operation Parameter4 of the command specified in Parameter1 |

NOTE1) 2-byte Operation Code (upper 2 bytes are 0x0000) command can also be executed.

**○ Response Code**

For the command-specific Response Code, refer to the Response Code of each command.

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Operation_Not_Supported | An unsupported Operation Code is specified. |
| Incomplete_Transfer | Data block transmission failed. |
| Invalid_Parameter | An Operation Code other than "Camera-> Host" is specified for the data direction. |
| | The Operation Code of this operation is specified. |

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### 6.2.2.4 OperationSendData

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9442 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | Depends on the command. |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Depends on the command. |

**○ Command Outline**

This Operation Code executes the command corresponding to the 4-byte extension.

**○ Command Details**

In this Operation Code, it is possible to execute a command that has the data sent to the camera.

The Operation Parameter is the same as the OperationDataNone command (section 6.2.2.2).

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | Operation Code (4-byte extension) NOTE1 |
| Parameter2 | Operation Parameter1 |
| Parameter3 | Operation Parameter2 |
| Parameter4 | Operation Parameter3 |
| Parameter5 | Operation Parameter4 |

NOTE1) 2-byte Operation Code (upper 2 bytes are 0x0000) command can also be executed.

**○ Response Code**

For the command-specific Response Code, refer to the Response Code of each command.

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Operation_Not_Supported | An unsupported Operation Code is specified. |
| Incomplete_Transfer | Data block transmission failed. |
| Invalid_Parameter | An Operation Code other than "Host -> Camera" is specified for the data direction. |
| | The Operation Code of this operation is specified. |

Do Not Copy

### 6.2.2.5 GetDevicePropDescEx

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x943A |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropDesc data set |
| Data direction | From camera to host |
| Response Code | Same as GetDevicePropDesc |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode returns the DevicePropDesc data set corresponding to the specified DevicePropCode.

For the DevicePropDesc, refer to section 9.4.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | DevicePropCode (4-byte extension) |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| DeviceProp_Not_Supported | The specified DevicePropCode is not supported. |

Do Not Copy

### 6.2.2.6 GetDevicePropValueEx

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x943B |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropValue |
| Data direction | From camera to host |
| Response Code | Same as GetDevicePropValue |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode returns the current value corresponding to the specified DevicePropCode.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | DevicePropCode (4-byte extension) |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| DeviceProp_Not_Supported | The specified DevicePropCode is not supported. |

Do Not Copy

### 6.2.2.7 SetDevicePropValueEx

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x943C |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropValue |
| Data direction | From host to camera |
| Response Code | Same as SetDevicePropValue. |
| Response Parameter | None |
| Event Code | Refer to Event Code in this section. |

#### ○ Command Outline

The operation by this OperationCode sets the DevicePropValue corresponding to the specified DevicePropCode to the camera.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | DevicePropCode (4-byte extension) |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block reception failed. |
| DeviceProp_Not_Supported | The specified DevicePropCode is not supported. |
| Access_Denied | An operation is denied depending on the status of the camera. |
| | The specified DevicePropCode is not permitted for setting. |
| Device_Busy | The acquisition operation is being performed when the command processing is started. |
| Invalid_DeviceProp_Format | The size or the format of the DevicePropDesc data set is not appropriate. |
| Invalid_DeviceProp_Value | The specified DevicePropValue is not supported. |

Do Not Copy

### 6.2.2.8 RegisterDevicePropGroup

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x943D |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropCode Array |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Number of properties registered in the property group |

**○ Command Outline**

The operation by this OperationCode registers property group.

**○ Command Details**

The registered property group information is retained until the session with the camera ends.
The property group numbers that can be registered are 1 to 32.
If the property group is registered again for the registered property group, it will be overwritten.

If you specify the registered property group when executing the GetDevicePropDescList command (section 6.2.2.9) and GetDevicePropValueList command (section 6.2.2.10), you can get DevicePropDesc DataSet and PropertyValue all at once.

If the property group to be registered contains a DevicePropertyCode that does not exist, that property will not be registered.

The format of the property group to be registered is shown below.

| Field | DataSize | Description |
|---|---|---|
| NumElement | 4 | Number of elements in the array. |
| ArrayEntry[0] | 4 | DevicePropCode 1 |
| ArrayEntry[1] | 4 | DevicePropCode 2 |
| … | | |
| ArrayEntry[NumElement-1] | 4 | DevicePropCode N |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | Property group number |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block reception failed. |
| Invalid_Parameter | The specified value of Parameter is out of the range. |
| Invalid_Data | The format of the property group to be registered is incorrect. |

Do Not Copy

### 6.2.2.9 GetDevicePropDescList

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x943E |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropDesc List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode is to get the list of the specified DevicePropDesc DataSet.

**○ Command Details**

If you specify the property group number registered with the RegisterDevicePropGroup command (section 6.2.2.8) in Parameter1, you will get a list of registered properties.

The format of the list of properties to be acquired is shown below.

| Field | DataSize | Description |
|---|---|---|
| NumberOfElements | 4 | Number of commands in the list. |
| Element1 | DTS | |
| Element Size | 2 | Size of the element. |
| DevicePropDesc DataSet | DTS | See section 9.4 |
| Element2 | DTS | |
| ～ | | |
| ElementN | DTS | |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | Property group number |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Access_Denied | A property group that is not registered is specified. |
| Invalid_Parameter | The specified value of Parameter is out of the range. |

Do Not Copy

### 6.2.2.10 GetDevicePropValueList

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x943F |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DevicePropValue List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode is to get the list of the current value of the specified DevicePropCode.

#### ○ Command Details

If you specify the property group number registered with the RegisterDevicePropGroup command (section 6.2.2.8) in Parameter1, you will get a list of registered properties.

The format of the list of properties to be acquired is shown below.

| Field | DataSize | Description |
|---|---|---|
| NumberOfElements | 4 | Number of properties in the list |
| Element1 | DTS | |
| DevicePropCode | 4 | DevicePropCode 1 |
| DataType | 2 | DataType of DevicePropCode 1 |
| DevicePropValue | DTS | Current value of DevicePropCode 1 |
| Element2 | DTS | |
| ～ | | |
| ElementN | DTS | |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | Property group number |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Access_Denied | A property group that is not registered is specified. |
| Invalid_Parameter | The specified value of Parameter is out of the range. |

Do Not Copy

### 6.2.2.11 DeviceReady

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90C8 |
| Operation Parameter | None |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode checks the action by the activation-type command.

#### ○ Command Details

This command is issued after issuing the following activation-type commands in order to check the operation.

| Activation-type commands | Reference |
|---|---|
| InitiateCapture | 6.2.1.14 |
| InitiateCaptureRecInMedia | 6.2.4.1 |
| InitiateCaptureRecInSdram | 6.2.4.2 |
| AfAndCaptureRecInSdram | 6.2.4.3 |
| AfDrive | 6.2.5.1 |
| MfDrive | 6.2.5.3 |
| StartLiveView | 6.2.6.1 |

This command should be issued repeatedly until the operation of the activation-type command is completed.

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Device_Busy | The operation of the activation-type command is not finished. |
| Out_of_Focus | The AF operation by the command is in the non-focused status. |
| Wb_Preset_Error | The preset measurement release failed. |
| MfDrive_Step_End | The MF driving reaches the end of steps. |
| MfDrive_Step_ Insufficiency | The driving amount is insufficient. |
| Bulb_Release_Busy | The bulb shooting is being performed. |
| Silent_Release_Busy | During the silent photography (Only models that support the silent photography). |
| MovieFrame_Release_Busy | The frames for the video recording are being saved. (Only for models that support video frame storage) |

Do Not Copy

### 6.2.2.12 GetEvent

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x90C7 |
| Operation Parameter | None |
| Data | Event array |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode sends the event retained in the camera to the host.

**○ Command Details**

This OperationCode is a command for compatibility with the GetEventEx command (section 6.2.2.13) and has the following restriction. The GetEventEx command (section 6.2.2.13) should be used for this camera.

This command cannot handle the events that have two or more parameters.

The format of the event array to be sent by the camera is shown below.

| Offset | Value | Name | Description |
|---|---|---|---|
| 0 | N | EventCount | The number of events |
| 2 | EventCode | EventCode [0] | The oldest event |
| 4 | | EventParameter [0] | Parameter attaching to the oldest event |
| 8 | EventCode | EventCode [1] | The second oldest event |
| 10 | | EventParameter [1] | Parameter attaching to the second oldest event |
| ・・・ | | | |
| 6 x (N – 1) + 2 | EventCode | EventCode [N - 1] | The newest event |
| 6 x (N – 1) + 4 | | EventParameter [N - 1] | Parameter attaching to the newest event |

If there is no event to be sent, the EventCount value is set to 0 and sent to the host.

When an event is sent by this command, the contents of the event queue buffer prepared for the GetEvent command are updated. The sent event is deleted.

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.2.13 GetEventEx

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x941C |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | Event array |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode sends the event retained in the camera to the host.

**○ Command Details**

The camera sends all the events saved in the event queue buffer prepared for the GetEventEx command.

When Parameter1 has no value or is 0, the event stored in the queue buffer is deleted after sending the event.

If 1 is specified for Parameter1, the event stored in the queue buffer is not deleted after sending the event.

The format of the event array to be sent by the camera is shown below.
Maximum element is 2048.

| Field | DataSize | DataType | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of events in the list. |
| Element1 Event Code | 2 | UINT16 | Event Code for element1. |
| Element1 NumParameters | 2 | UINT16 | Number of parameters for element1. |
| Element1 Parameter1 | 4 | UINT32 | Parameter1 for element1. |
| Element1 Parameter2 | 4 | UINT32 | Parameter2 for element1. |
| … | | | |
| Element1 ParameterN | 4 | UINT32 | Parameter N for element1. |
| … | | | |
| ElementN Event Code | 2 | UINT16 | Event Code for element N. |
| ElementN NumParameters | 2 | UINT16 | Number of parameters for element N. |
| ElementN Parameter1 | 4 | UINT32 | Parameter1 for element N. |
| ElementN Parameter2 | 4 | UINT32 | Parameter2 for element N. |
| … | | | |
| ElementN ParameterN | 4 | UINT32 | Parameter N for element N. |

If there is no event to be sent, the EventCount value is set to 0 and sent to the host.

**○ Operation Parameter**

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | [Whether to delete the buffer] | |
| | No value | The event stored in the queue buffer is deleted after sending the event. |
| | 0 | |
| | 1 | The event stored in the queue buffer is not deleted after sending the event |
| Parameter2 | None | |
| Parameter3 | None | |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Invalid_Parameter | The specified value is out of the range. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.2.14 ClearEvent

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9461 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode deletes the event held in the camera.

#### ○ Command Details

This command deletes the sent event when it is executed by specifying "1: The event stored in the queue buffer is not deleted after sending the event" in Parameter1 of the GetEventEx command.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |

Do Not Copy

### 6.2.2.15 ChangeCameraMode

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90C2 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode switches between the PC camera mode and the remote mode.

#### ○ Command Details

The camera is set to the mode specified by the ModeValue.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | ModeValue | |
| | 0 | Sets to the PC camera mode. |
| | 1 | Sets to the remote mode. |
| Parameter2 | None | |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value of ModeValue is out of the range. |
| Change_CameraMode_Failed | Mode cannot be changed due to shutter-release operation. |
| | Mode cannot be changed due to AF operation. |
| | Mode cannot be changed during cleaning image sensor. |

Do Not Copy

### 6.2.2.16 ChangeApplicationMode

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9435 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode switches the application mode.

#### ○ Command Details

Refer to section 2.4 for the application mode.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | Mode | |
| | 0 | Application mode OFF |
| | 1 | Application mode ON |
| Parameter2 | None | |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value of ModeValue is out of the range. |

Do Not Copy

### 6.2.2.17 ChangeMonitorOff

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9422 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode changes the monitor forced off state of the camera.

#### ○ Command Details

When the camera receives this command, it switches the monitor forced off state.
If “Forced light off” is specified for MonitorOffStatus of Parameter1 while the monitor is on, the monitor turns off.
If "Cancel forced light off" is specified while the monitor is off, the monitor turns on or off according to the camera status.

Monitors targeted by this command are as follows.
- Main monitor (Playback monitor)

The response of the normal end indicates that the request to switch the monitor off state has been received, completion of turning on / off the monitor is not guaranteed.

Refer to MonitorOffStatus property (section 6.5.6.16) for the monitor forced off state.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | MonitorOffStatus | |
| | 0 | Cancel forced light off. |
| | 1 | Forced light off. |
| Parameter2 | None | |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Device_Busy | The acquisition operation is being performed when the command processing is started. |
| Access_Denied | The battery is exhausted |
| Hardware_Error | When the command processing is started, some error is generated in the camera body. |

Do Not Copy

### 6.2.2.18 GetCameraSettingData

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x946D |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | CameraSettingData Dataset |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode gets the camera setting file saved on the memory card..

#### ○ Command Details

If you are using two memory cards, get the camera setting file from the card in the primary slot (*).
*The primary slot of this camera is slot 1.

CameraSettingData Dataset is as follows.

| Field | Size | Description |
|---|---|---|
| Camera setting file name | 64byte | ASCII：63 characters maximum +NULL |
| Camera setting data | - | Contents of the camera setting file name |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | Specified TransactionID is not a valid value. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Incomplete_Transfer | The data block transmission failed. |
| Access_Denied | The video is playing. |
| | Camera setting file does not exist on the card. |
| Store_Not_Available | The card does not exist. |
| | The card is being initialized. |
| | The battery is exhausted. |
| Store_Error | Card access error occurs. |
| Store_Unformatted | Card is not formatted. |

Do Not Copy

### 6.2.2.19 SendCameraSettingData

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x946E |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | Camera setting data |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode saves camera setting data to a memory card.

#### ○ Command Details

If you are using two memory cards, save the camera settings file to the card in the primary slot.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | Specified TransactionID is not a valid value. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Incomplete_Transfer | The data block transmission failed. |
| Object_Write_Protect | Camera settings file is protected. |
| Access_Denied | The video is playing. |
| Store_Full | There is no free space for saving on the card. |
| Store_Read_Only | The recording card is protected |
| Store_Not_Available | The card does not exist. |
| | The card is being initialized. |
| | The battery is exhausted. |
| Store_Error | Card access error occurs. |
| Store_Unformatted | Card is not formatted. |

Do Not Copy

## 6.2.2.20 SaveCameraSetting

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9420 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode saves the camera setting values. This is the same operation as that of saving the camera setting values when the power switch is turned off.

### ○ Command Details

The camera setting values are stored in the non-volatile area. If this operation is not completed in the case that the power supply is interrupted in saving the setting values, the operation of the camera is not guaranteed after that.

After responding to this command, do not execute any command (other than CloseSession) or do not operate the camera other than power OFF.

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Device_Busy | The shooting operation is being performed. |
| Access_Denied | During the PC camera mode. |
| | The camera cannot save the setting values. |

Do Not Copy

## 6.2.2.21 ResetMenu

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9436 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode resets the camera settings.

### ○ Command Details

The setting value that is specified in Parameter1 (Reset Type) is reset.

In Reset Type where you can specify the bank, specify Bank in Parameter2.
In Reset Type where bank cannot be specified, Bank specification is ignored.

White balance, white balance sub-item, fine-tuning values other than preset manual data, and preset manual data number are reset.

In the case of resetting file number sequence, the value of “Custom menu – Shooting/display – File number sequence” is reset.

### ○ Operation Parameter

| Operation Parameter | Detail | | |
|---|---|---|---|
| Parameter1 | Reset Type | | Bank can be specified |
| | 0x0001 | Photo shooting menu | yes |
| | 0x0002 | White balance (Photography) | no |
| | 0x0102 | White balance (Video) | no |
| | 0x0201 | Custom menu | yes |
| | 0x0202 | File number sequence | no |
| Parameter2 | [Bank] | | |
| | None | Bank in use (NOTE1) | |
| | 0 | | |
| | 1 | Bank A | |
| | 2 | Bank B | |
| | 3 | Bank C | |
| | 4 | Bank D | |
| Parameter3 | None | | |

NOTE1)
The shooting menu uses the bank set in the ShootingMenuBank property (section 6.5.2.1).
The custom menu uses the bank set in the CustomSettingBank property (section 6.5.4.1).

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Invalid_Parameter | The specified value is out of the range. |
| Access_Denied | In the case of the spot WB acquisition standby state. |
| | [White balance (Photography, Video)] is specified when the AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |
| | [File number sequence] is specified when the HDRMode property (section 6.5.2.72) is set to [On]. |
| Device_Busy | The shooting operation is being performed. |

Do Not Copy

## 6.2.2.22 GetPortraitImpressionBalanceDataList

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9454 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PortraitImpressionBalance List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode acquires the PortraitImpressionBalance List set in the adjustment value of “Portrait Impression Adjustment – Mode”.

### ○ Command Details

When the mode number is specified in Parameter1, the PortraitImpressionBalance DataSet (section 9.18) of the specified ModeNum is sent to the host.

If 0xFFFFFFFF is specified, all modes are sent to the host.

The data set of PortraitImpressionBalance List is shown below.

| Field | | DataSize | Description |
|---|---|---|---|
| NumberOfElements | | 2 | Number of modes in the list |
| Element1 | | 6 | |
| | Element Size | 2 | Size of the element |
| | ModeNum | 2 | Portrait impression adjustment mode number |
| | PortraitImpressionBalance DataSet | 2 | Adjustment value of Portrait impression adjustment |
| ~ | | | |
| ElementN | | 6 | |

### ○ Operation Parameter

| Operation Parameter | Description | |
|---|---|---|
| Parameter1 | ModeNum | Specified mode number<br>Same as PortraitImpressionBalance property (section 6.5.2.77), but "OFF" is an invalid value. |
| | 0xFFFFFFFF | All modes |
| Parameter2 | ShootingMode | |
| | 0 | Adjustment value of photo shooting menu |
| | 1 | Adjustment value of video recording menu |
| Parameter3 | DefaultFlag | |
| | 0 | Current setting |
| | 1 | Default value |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

## 6.2.2.23 SetPortraitImpressionBalanceData

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9453 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PortraitImpressionBalance DataSet |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode sets PortraitImpressionBalance DataSet to the adjustment value of “Portrait Impression Adjustment – Mode”.

### ○ Command Details

The contents of the PortraitImpressionBalance DataSet are shown in Section 9.18.

### ○ Operation Parameter

| Operation Parameter | Description | |
|---|---|---|
| Parameter1 | ModeNum | Specified mode number<br>Same as PortraitImpressionBalance property (section 6.5.2.77), but "OFF" is an invalid value. |
| Parameter2 | ShootingMode | |
| | 0 | Adjustment value of photo shooting menu |
| | 1 | Adjustment value of video recording menu |
| Parameter3 | None | |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |
| Device_Busy | During shooting operation. |
| Access_Denied | When ShootingMode is 1,<br>MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log]. |
| | When ShootingMode is 1,<br>MovieFileType property is set to [ProRes RAW HQ 10-bit(MOV)] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log]. |
| | When ShootingMode is 1,<br>MovieFileType property is set to [ProRes RAW HQ 12-bit(MOV)] and MovieProResRawHq12bitToneMode property (section 6.5.3.6) is set to [N-Log]. |
| | When ShootingMode is 1,<br>MovieFileType property is set to [N-RAW 12-bit(NEV)] and MovieNRaw12bitToneMode property (section 6.5.3.4) is set to [N-Log]. |
| Invalid_Data | Wrong PortraitImpressionBalance DataSet was specified. |

Do Not Copy

### 6.2.2.24 GetCommandFeature

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x944C |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | CommandFeature DataSet |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns the feature information about the specified operation code.

**○ Command Details**

The camera sends the CommandFeature DataSet about about the operation code specified by the parameter1 to the host.

Feature information can be acquired only for commands for which Feature information is defined.

**○ Operation Parameter**

| Operation Parameter | Description |
|---|---|
| Parameter1 | Operation Code |
| Parameter2 | Index (Fixed to 0 with this camera) |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | The specified operation code has no feature information. |

Do Not Copy

## 6.2.3 Vendor (Object)

### 6.2.3.1 GetVendorStorageIDs

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9209 |
| Operation Parameter | None |
| Data | StorageIDArray |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns a list of the currently valid StorageIDs.

In the application mode, the StorageID that can be obtained by GetStorageIDs is the one without the card installed, so it is possible to know the valid StorageID by using this Operation.

**○ Command Details**

The camera returns the StorageID of each slot.

The StorageID of each slot takes the following values.

| StorageID | Details |
|---|---|
| 0x00010001 | When the card is inserted in the slot1 |
| 0x00010000 | When the card is not inserted in the slot1 |
| | When the card in the slot1 is being formatted |
| | When the battery is exhausted |
| 0x00020001 | When the card is inserted in the slot2 |
| 0x00020000 | When the card is not inserted in the slot2 |
| | When the card in the slot2 is being formatted |
| | When the battery is exhausted |

The format of the StorageIDArray that is sent by the camera is shown below.
Each field data is stored in the little endian format.

| Field | Size (Byte) | Data |
|---|---|---|
| NumElement | 4 | 0x00000002 (Two elements for the array) |
| ArrayEntry1 | 4 | StorageID (slot1) |
| ArrayEntry2 | 4 | StorageID (slot2) |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.3.2 GetObjectSize

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9421 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectSize |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the size of the specified object.

#### ○ Command Details

When the value of the ObjectCompressedSize returned by GetObjectInfo command is 0xFFFFFFFF, the correct size of the object can be obtained by this operation code.

The ObjectSize transmitted by the camera is 8 bytes of data.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |

Do Not Copy

### 6.2.3.3 GetObjectsMetaData

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9434 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectMetaData Dataset |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode obtains the handle and the metadata of the object in the card. The ObjectHandle of the object recorded in the SDRAM is not included.

**○ Command Details**

If the new object is being recorded in the card when this command is received, the processing starts after the acquisition of all the data is completed.

The camera returns the ObjectHandle and the metadata in the card specified by the StorageID. In addition, Parameter2 and Parameter3, which are optional, can be combined.

| Operation Parameter | Value | Details |
|---|---|---|
| Parameter1 | 0xFFFFFFFF | All the cards should be the targets. |
| | StorageID | The specified object should be the target. |
| Parameter2 (Option) | 0x00000000 | All the object formats should be the targets. |
| | No value | |
| | 0xFFFFFFFF | All the image formats in the card corresponding to the specified StorageID should be the targets. |
| | ObjectFormatCode | The specified object format should be the target. |
| Parameter3 (Option) | 0x00000000 | All the directories should be the targets. (NOTE1) |
| | No value | |
| | ObjectHandle | The objects directly under the specified directory should be the targets. |

NOTE1: Since the processing time increases in proportion to the number of objects, if all directories are specified with a large number of objects, it may take time to execute.

The format of the ObjectMetaData Dataset sent by the camera is shown below.

ObjectMetaData Dataset

| Field | | Size(Byte) | DataType | Description |
|---|---|---|---|---|
| Version | | 4 | UINT32 | Dataset version<br>Version 1.00 |
| NumElement | | 4 | UINT32 | Number of metadata |
| ObjectMetaData[0] | | 16 | - | Metadata 1 |
| | ObjectHandle | 4 | UINT32 | Object handle |
| | ObjectAttribute | 4 | UINT32 | Bit2 and after: Reserved<br>Bit1: Protected<br>Bit0: Rating |
| | Reserved | 1 | UINT8 | - |
| | ModificationDate | 1 | UINT8 | Modification date: Second |
| | | 1 | UINT8 | Modification date: Minute |
| | | 1 | UINT8 | Modification date: Hour |
| | | 1 | UINT8 | Modification date: Day |

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| | | | |
|---|---|---|---|
| | 1 | UINT8 | Modification date: Month |
| | 2 | UINT16 | Modification date: Year |
| ObjectMetaData[1] | 16 | - | Metadata 2 |
| ... | | | |
| ObjectMetaData[N - 1] | 16 | - | Metadata N |

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | StorageID |
| Parameter2 | [ObjectFormatCode] |
| Parameter3 | [ObjectHandle of directory] |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_StorageID | The StorageID sent by the camera differs from the StorageID specified by the host. |
| Invalid_Object_Handle | For an invalid object handle |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Specification_By_Format_Unsupported | The specified ObjectFormatCode is not supported. |
| Invalid_Parent_Object | An ObjectHandle other than that indicating the directory in the card was specified for ObjectHandle (Parameter3) of the specified directory. |
| | The specified directory does not exist. |
| Device_Busy | The management file is not ready. |

Do Not Copy

### 6.2.3.4 GetPartialObjectEx

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9431 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DataObject |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Refer to Response Parameter in this section. |

#### ○ Command Outline

The operation by this OperationCode is an extension of GetPartialObject (section 6.2.1.19).

#### ○ Command Details

The Offset, MaxSize and actual number of bytes sent (Response Parameter) are expanded. Objects up to the size represented by 8 Bytes can be obtained.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | Offset (Lower 4Byte) |
| Parameter3 | Offset (Upper 4Byte) |
| Parameter4 | MaxSize (Lower 4Byte) |
| Parameter5 | MaxSize (Upper 4Byte) |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | The ObjectHandle of the directory (Association) is specified. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| Access_Denied | The video is playing. |

#### ○ Response Parameter

| Response Parameter | Description |
|---|---|
| Parameter1 | Actual number of bytes sent (Lower 4Byte) |
| Parameter2 | Actual number of bytes sent (Upper 4Byte) |

Do Not Copy

### 6.2.3.5 GetPartialObjectHighSpeed

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9400 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | DataObject |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Refer to Response Parameter in this section. |

#### ○ Command Outline

The operation by this OperationCode transfers the file data (DataObject) of the specified size corresponding to the specified ObjectHandle (parameter1) from the camera to the host.

This command is suited to transfer large-size file because the time for the camera to read the data from the card is shorter than that of the GetPartialObject command (section 6.2.1.19).

#### ○ Command Details

The maximum size of the data transferred by this command is the size of the file specified by the ObjectHandle (parameter1). The data size can be checked by the GetObjectInfo command (section 6.2.1.8).

The following must be satisfied to use this command.

- Turn on application mode by using ChangeApplicationMode command (section 6.2.2.16).

The command should be issued with Terminate (parameter3) set to “Terminate transferring” in order to cancel the transfer while all data of the specified file are not transferred.

When Terminate (parameter3) is set to “Terminate transferring”, data transfer with 0 byte is performed.

An example of using the virtual file data is shown below.

(ObjectHandle = 0x00000001, FileSize = 0x00100000)

| In the case that all data are transferred | In the case that data transfer is canceled |
|---|---|
| The first time:<br>ObjectHandle = 0x00000001<br>TransferMaxSize = 0x00020000<br>Terminate = 0x00000000<br>The second time:<br>ObjectHandle = 0x00000001<br>TransferMaxSize = 0x00020000<br>Terminate = 0x00000000<br>The eighth time (last data acquisition):<br>ObjectHandle = 0x00000001<br>TransferMaxSize = 0x00020000<br>Terminate = 0x00000000 | The first time:<br>ObjectHandle = 0x00000001<br>TransferMaxSize = 0x00020000<br>Terminate = 0x00000000<br>The second time:<br>ObjectHandle = 0x00000001<br>TransferMaxSize = 0x00020000<br>Terminate = 0x00000000<br>The third time (data transfer is canceled):<br>ObjectHandle = 0x00000001<br>TransferMaxSize = 0x00020000<br>Terminate = 0x00000001 |

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The information necessary to maintain this command is released in any of the following conditions.

After the information is released, the file data is acquired from the beginning if this command is issued again.

| Termination condition |
|---|
| The file data specified by the ObjectHandle (parameter1) has been transferred completely. |
| The parameter3 (Terminate) of this command is set to “Terminate transferring”. |
| The interval between the response of this command and the next issue of this command exceeds approx. 60 seconds.<br>* If the file data offset of the response parameter is 0 after the second transfer of divided data and before transfer of all data, it is regarded as a time-out. |
| A card has been inserted or ejected regardless of whether the card contains the file data specified by the ObjectHandle or not. |
| The file has been deleted. |
| Video recording or video playback has started. |
| The following command was issued.<br><br>CloseSession<br>GetObjectHandles<br>GetObjectInfo<br>GetObject<br>GetThumb<br>DeleteObject<br>SendObjectInfo<br>SendObject<br>InitiateCapture<br>FormatStore<br>GetPartialObject<br>ChangeApplicationMode<br>GetCameraSettingData<br>SendCameraSettingData<br>SaveCameraSetting<br>GetObjectSize<br>GetPartialObjectEx<br>GetLargeThumb<br>InitiateCaptureRecInMedia<br>StartMovieRecord<br>InitiatePixelMapping<br>GetObjectPropValue<br>GetObjectPropList |

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | ObjectHandle | |
| Parameter2 | TransferMaxSize (Byte) | |
| Parameter3 | Terminate | |
| | 0 | Continue transferring |
| | 1 | Terminate transferring |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | In the case of an ObjectHandle other than the image file |
| | In the case of an ObjectHandle of the SDRAM |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |
| Invalid_Parameter | “Terminate” is neither 0 nor 1. |
| Invalid_Status | The application mode is set to OFF. |

### ○ Response Parameter

| Response Parameter | Description |
|---|---|
| Parameter1 | The number of bytes actually sent |
| Parameter2 | File data offset before command execution and data transfer (Lower 4Byte) |
| Parameter3 | File data offset before command execution and data transfer (Upper 4Byte) |

Do Not Copy

### 6.2.3.6 GetLargeThumb

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x90C4 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | LargeThumbnail data |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode acquires the large thumbnail data of the specified object.

**○ Command Details**

The camera sends all the large thumbnail data corresponding to the specified ObjectHandle to the host.

When the specified ObjectHandle is the data in the card, the camera returns the large thumbnail data corresponding to the ObjectHandle.

The ObjectHandle passed by ObjectAddedInSdram should be specified in order to acquire the information of the object in the SDRAM.

The LargeThumbnail data is the large thumbnail data of the object (file) specified by the ObjectHandle. The size of the LargeThumbnail data shall be the reference JPEG image size acquired in order of priority shown in the table below.

| Order of priority | Reference (Still image object) | Reference (Video object) |
|---|---|---|
| 1 | MPF class 1 | MPF class 1 |
| 2 | VIEW | MPF class 2 |
| 3 | MPF class 2 | Small thumbnail |

The size of small thumbnail is the same as thumbnail that can be obtained with the GetThumb command (section 6.2.1.10).

| Aspect ratio | VIEW | MPF class 1 | MPF class 2 |
|---|---|---|---|
| 3:2 | 570x375 | 640x424 | 1620x1080 |
| 5:4 | | 600x480 | 1350x1080 |
| 4:3 | | 640x480 | 1440x1080 |
| 1:1 | | 480x480 | 1080x1080 |
| 16:9 | | 640x360 | 1920x1080 |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | An object in the SDRAM other than the ObjectHandle passed by |

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| | |
|---|---|
| | ObjectAddedInSdram is specified. |
| No_Thumbnail_Present | The object corresponding to the specified ObjectHandle does not include a thumbnail. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |

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### 6.2.3.7 GetFhdPicture

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x920F |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | FullHD image data |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode acquires the FullHD image data of the specified object in the card.

**○ Command Details**

The camera sends all the FullHD image data corresponding to the specified ObjectHandle in the card to the host.

The FullHD image data is a maximum of 1920x1080-size JPEG image.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist.<br>The object other than JPEG, HEIF, RAW or TIFF is specified. |
| No_FullHD_Present | An object that corresponds to the specified ObjectHandle is not the FullHD image. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |

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### 6.2.3.8 CancelImagesInSdram

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x940C |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode cancels the transfer of a specific object saved in the SDRAM.

#### ○ Command Details

This command cancels the transfer of the object corresponding to the specified ObjectHandle. For the ObjectHandle, the ObjectHandle passed by the ObjectAddedInSdram event should be specified. When the object corresponding to the specified ObjectHandle has been already sent to the host or deleted, an error response is made.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Device_Busy | The shooting operation is being performed. |
| | The ObjectAddedInSdram event is not ready for passing. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |

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### 6.2.3.9 DeleteImagesInSdram

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90C3 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode deletes a specific object saved in the SDRAM or all the objects saved in the SDRAM.

#### ○ Command Details

The object that corresponds to the specified ObjectHandle is deleted. All the objects in the SDRAM are deleted if the value of Parameter1 is 0. During the process of deleting all the objects, the camera does not issue a new ObjectAddedInSdram event.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Device_Busy | The shooting operation is being performed. |
| | The ObjectAddedInSdram event is not ready for passing. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |

Do Not Copy

# 6.2.4 Vendor (Shooting)

## 6.2.4.1 InitiateCaptureRecInMedia

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9207 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |
| Event Code | Refer to Event Code in this section. |

### ○ Command Outline

The operation by this OperationCode makes the camera start the acquisition of one or more new objects (release operation of the camera) according to the current setting. The acquired object (image data) is saved in the specified location.

### ○ Command Details

This command is an activation-type command. OK response is made when the start of the AF operation or the release operation is completed.

When the host gets the response of the normal termination, the host should issue the DeviceReady command (section 6.2.2.11) to confirm whether the release operation is terminated.

The following information can be returned for the DeviceReady command (section 6.2.2.11) when requesting the release operation.

| Response Code | Description |
|---|---|
| OK | The release operation is normally terminated. |
| Device_Busy | The AF, the shooting or the continuous shooting operation is being performed |
| Bulb_Release_Busy | The bulb shooting is being performed. |
| Silent_Release_Busy | The silent shooting is being performed. |
| Out_of_Focus | The AF operation is not focused before shooting or during continuous shooting. |
| Wb_Preset_Error | The preset measurement release failed. |
| Other Response Code | Some error has occurred when requesting the release. |

* For the details, refer to the DeviceReady command (section 6.2.2.11).

Whether the AF operation is performed or not depends on the value of CaptureSort (described later) in Parameter1 and the focus mode setting. When the live view is executed, it also depends on the live view mode.

For the details of the command release operation, refer to Command Release Operation (section 11.7).

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## ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | CaptureSort (The details are shown in the table below.) | |
| Parameter2 | SaveMedia | |
| | 0x0000 | Card |
| | 0x0001 | SDRAM |
| | 0x0002 | Card and SDRAM |
| Parameter3 | None | |

| CaptureSort | Operation | Description |
|---|---|---|
| 0xFFFFFFFE | Image acquisition release after AF driving | The AF driving is started and then the release operation of the camera is performed.<br><br>- Photo shooting<br>In the case of focus priority, the release operation is started only when the AF is focused after AF driving. Other than in the case of focus priority, the release operation is always started independent of the status after AF driving.<br><br>- Video recording<br>Because it is fixed to the release priority, the release operation is always started independent of the status after AF driving. |
| 0xFFFFFFFF | Image acquisition release | Normal release operation is performed. |
| 0x00000001 | Preset measurement release 1 | Stores the acquired preset gain in D1. |
| 0x00000002 | Preset measurement release 2 | Stores the acquired preset gain in D2. |
| 0x00000003 | Preset measurement release 3 | Stores the acquired preset gain in D3. |
| 0x00000004 | Preset measurement release 4 | Stores the acquired preset gain in D4. |
| 0x00000005 | Preset measurement release 5 | Stores the acquired preset gain in D5. |
| 0x00000006 | Preset measurement release 6 | Stores the acquired preset gain in D6. |
| 0x00000010 | Dust reference image release | Dust reference image release operation is performed.<br>When the shooting succeeds, the operation similar to the image acquisition release is performed hereafter. |

## ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified CaptureSort or SaveMedia is out of the setting range. |
| Device_Busy | When the command processing is started, the acquisition operation is being performed. |
| | The shutter-release button is being fully pressed. |
| | The STM lens is being initialized. |
| Hardware_Error | When the command processing is started, some error is generated in the camera body. |
| Out_of_Focus | All of the following conditions are satisfied and the AF operation is not focused.<br>- CaptureSort is the image acquisition release after AF driving.<br>- The AF operation mode is set to [Focus priority]. |
| Invalid_Status | The shutter speed is set to Time. |
| Dust_Reference_Error | The CPU internal lens is not mounted during the dust |

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| | |
|---|---|
| | reference image release. |
| | The dust reference image release failed. |
| | The dust reference image release is specified when the HDR mode is set to [On]. |
| | The dust reference image release is specified when the LiveViewSelector is set to [Video mode]. |
| Wb_Preset_Error | The preset manual WB data save destination that is specified in the preset measurement release is protected. |
| | The preset measurement release is specified when the HDR mode is set to ON. |
| | The preset measurement release is specified during PC camera mode. |
| CameraMode_Not_Adjust_Fnumber | An aperture value error occurs. |
| Store_Full | There is no free area for recording in the card. |
| | There is no free area for recording in the SDRAM. |
| Store_Not_Available | The card is being initialized |
| | The card does not exist when SaveMedia is set to [Card] or [Card and SDRAM]. |
| | The battery is exhausted. |
| Store_Error | SaveMedia is [Card] or [Card and SDRAM] and a card access error occurs in the camera. |
| Store_Unformatted | SaveMedia is [Card] or [Card and SDRAM] and the card is not formatted. |
| Store_Read_Only | The card that is ready for recording is protected. |
| Access_Denied | In the case of the spot WB acquisition standby state |
| | When a retractable lens is mounted, the lens is retracting. |
| | The WarningStatus property (section 6.5.6.22) is “Version upgrade warning when mount adapter is attached”. |
| | The preset measurement release is performed when StillCaptureMode property (section 6.5.1.15) is set to [High-speed frame capture]. |
| | SpecialShootingMode property (section 6.5.6.2) is set to ON (during interval timer shooting/focus shift shooting). |
| | Shooting cannot be performed depending on the camera status by a cause other than the above. |

### ○ Event Code

| Event Code | Description |
|---|---|
| ObjectAdded | A new object is recorded in the card. |
| StoreFull | There is no free area for recording in the card after shooting. |
| CaptureComplete | The acquisition operation of all the new objects is completed. |
| ObjectAddedInSdram | A new object is recorded in the SDRAM. |
| CaptureCompleteRecInSdram | All the images captured by this command are sent from the SDRAM to the host completely. |

Do Not Copy

### 6.2.4.2 InitiateCaptureRecInSdram

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90C0 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode makes the camera start the acquisition of one or more new objects (release operation of the camera) according to the current setting. The acquired object (image data) is saved in the SDRAM.

#### ○ Command Details

This command performs the same operation as that of the InitiateCaptureRecInMedia command with CaptureSort set to a value other than [Image acquisition release after AF driving] and SaveMedia set to [SDRAM].

The parameters to perform the same operation as that of InitiateCaptureRecInSdram by InitiateCaptureRecInMedia are shown below.

| Operation Parameter | |
|---|---|
| Parameter1(CaptureSort) | 0x00000010(DustOff)<br>0xFFFFFFFF(Image acquisition release) |
| Parameter2(SaveMedia) | 0x00000001(SDRAM) |

For the details, refer to the InitiateCaptureRecInMedia command (section 6.2.4.1).

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | CaptureSort<br>Refer to the InitiateCaptureRecInMedia command. |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

This section describes only ResponseCodes different from that of the InitiateCaptureRecInMedia command.

For the details of other response codes, refer to the InitiateCaptureRecInMedia command.

| Response Code | Description |
|---|---|
| Shutter_Speed_Bulb | The shutter speed is set to Bulb. |
| Shutter_Speed_Time | The shutter speed is set to Time. |

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### 6.2.4.3 AfAndCaptureRecInSdram

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90CB |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode starts the AF driving and makes the camera start the acquisition of one or more new objects (release operation of the camera). The acquired object (image data) is saved in the SDRAM.

#### ○ Command Details

This command performs the same operation as that of the InitiateCaptureRecInMedia command with CaptureSort set to [Image acquisition release after AF driving] and SaveMedia set to [SDRAM].

The parameters to perform the same operation as that of AfAndCaptureRecInSdram by InitiateCaptureRecInMedia are shown below.

| Operation Parameter | Description |
|---|---|
| Parameter1 (CaptureSort) | 0xFFFFFFFE (Image acquisition release after AF driving) |
| Parameter2 (SaveMedia) | 0x00000001(SDRAM) |

For the details, refer to the InitiateCaptureRecInMedia command (section 6.2.4.1).

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

This section describes only ResponseCodes different from that of the InitiateCaptureRecInMedia command.

For the details of other response codes, refer to the InitiateCaptureRecInMedia command.

| Response Code | Description |
|---|---|
| Shutter_Speed_Bulb | The shutter speed is set to Bulb. |
| Shutter_Speed_Time | The shutter speed is set to Time. |

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### 6.2.4.4 TerminateCapture

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x920C |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode stops the shooting operation executed by the host.

#### ○ Command Details

The camera stops the following shooting operation executed by the host.

| Shooting operation | Operation after stopping |
|---|---|
| Bulb shooting | The images captured until the operation is stopped are recorded in the card or the SDRAM. |
| Continuous shooting | If this command is executed during continuous shooting, the shooting operation is terminated. |

The transition to the response phase is performed when the start of the shooting stop processing is completed.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | Reserved (0) |
| Parameter2 | Reserved (0) |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |

Do Not Copy

### 6.2.4.5 ChangeAELock

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9426 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode changes the state of AE lock and AWB lock.

#### ○ Command Details

If the white balance is other than auto or natural auto, and LockValue is set to "Locked (hold)" and the lock type is set to "AE/AWB", only the AE lock is enabled.

Even if the ResponseCode of the command is OK, check the AELockStatus property (section 6.5.6.3) and AWBLockStatus property (section 6.5.6.6) to see if the AE / AWB lock status has actually changed.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | LockValue | |
| | 0 | Lock released |
| | 1 | Locked (hold) |
| Parameter2 | [LockType] | |
| | None | AE |
| | 0 | |
| | 1 | AWB |
| | 2 | AE/AWB |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Invalid_Parameter | The specified value is out of the range. |
| Access_Denied | The WarningStatus property (section 6.5.6.22) is [Sequence error], [Minimum aperture warning] or [Retractable zoom-ring warning]. |
| | Waiting for spot WB data acquisition |
| | During aperture preview. |
| | White balance is other than auto or natural light auto, and LockType is set to [AWB]. (NOTE1) |
| Device_Busy | When the command processing is started, the acquisition operation is being performed. |

NOTE1)
The white balance of the WhiteBalance property (section 6.5.1.4) is targeted in photo mode, and the white balance of the MovieWhiteBalance property (section 6.5.3.14) is targeted in video mode.

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## 6.2.4.6 InitiatePixelMapping

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9433 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode starts pixel mapping.

### ○ Command Details

This command is an activation command. The camera returns OK after starting pixel mapping operation.

When pixel mapping operation is completed, the camera sends PixelMappingComplete event (section 6.4.2.9).

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Device_Busy | The shooting operation is being performed. |
| | During recording image. |
| Access_Denied | Sequence error. |
| | The battery is exhausted. |
| | Temperature in the camera exceeds the specified value. |
| | Wireless remote control is connected. |

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### 6.2.4.7 InitiateOpenCaptureV

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9445 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Refer to Response Parameter in this section. |

#### ○ Command Outline

The operation by this Operation Code starts interval timer shooting and focus shift shooting.

#### ○ Command Details

If interval timer shooting and focus shift shooting have already started, an OK response is given.
If the start of the interval timer shooting is requested while the interval timer shooting is paused, the interval timer shooting is restarted.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | CaptureSort | |
| | 1 | Interval timer shooting |
| | 2 | Focus shift shooting |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Store_Full | There is no free space on the card. |
| | There is no free space on the SDRAM. |
| Invalid_Parameter | Invalid value is specified in CaptureSort. |
| Store_Read_Only | The card to be recorded is protected. |
| Access_Denied | Failed to start shooting. |
| | Focus shift shooting is specified in CaptureSort during interval timer shooting.<br>Interval timer shooting is specified in CaptureSort during focus shift shooting. |
| | Shutter-release mode is set to self-timer. |
| | Shutter-release mode is set to high-speed frame capture. |
| | Shutterspeed is set to bulb. |
| | Shutterspeed is set to time. |
| | HDRMode property (section 6.5.2.72) is set to ON. |
| | EnableBracketing property (section 6.5.8.1) is set to ON. |
| | When a retractable lens is mounted, the lens is retracting. |
| | The WarningStatus property (section 6.5.6.22) is “Version upgrade warning when mount adapter is attached”. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |
| Store_Not_Available | RecordingMedia property is set to [Card] or [Card and SDRAM] and the card is not inserted. |
| | All of the following conditions are met and there is no card.<br>- CaptureSort is interval timer shooting.<br>- Time-lapse video option is enabled for interval timer shooting. |
| | The battery is exhausted. |
| Device_Busy | The capturing operation is being performed at the start of command processing. |
| Store_Error | Card access error occurs. |
| Store_Unformatted | The card is not formatted. |

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### ○ Response Parameter

| Response Parameter | Description |
|---|---|
| Parameter1 | Number of shots |
| Parameter2 | Shooting time [second]<br>(Fixed to 0 for focus shift shooting) |

### 6.2.4.8 TerminateOpenCaptureV

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9446 |
| Operation Parameter | None |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this Operation Code stops interval timer shooting and focus shift shooting.

**○ Command Details**

This command is an activation command. The camera returns OK after accepting stop request.
When shooting operation is completed, the camera sends OpenCaptureComplete event.

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Capture_Already_Terminated | Shooting has already been completed. |

Do Not Copy

## 6.2.4.9 GetOpenCaptureInfo

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9447 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | OpenCaptureInfo Dataset |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this Operation Code acquires the information of interval timer shooting and focus shift shooting.

### ○ Command Details

You can cquire information on the number of shots and shooting time specified in Capture Sort.

Information can be acquired in the case of CaptureSort specified by the InitiateOpenCaptureV command during shooting, but the number of shootings and time are invalid values (fixed to 0) when the CaptureSort is different.
(Example: The number and time of focus shift shooting are invalid during interval timer shooting)

The format of OpenCaptureInfo Dataset is shown below.

| Field | Size(Byte) | Description |
|---|---|---|
| Version | 2 | Version of dataset<br>Version 1.00(0x0064) |
| Shooting condition | 2 | 0：Stop (If the interval timer shooting is paused, it will be stopped.)<br>1：During interval timer shooting<br>2：During focus shift shooting |
| Number of shots | 4 | - |
| Shooting time | 4 | The time it takes to finish shooting. Unit: seconds<br>Invalid value (fixed to 0) for focus shift shooting |

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | CaptureSort | |
| | 1 | Interval timer shooting |
| | 2 | Focus shift shooting |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_Parameter | The specified CaptureSort is out of the setting range. |

Do Not Copy

## 6.2.4.10 SetFlickerReductionShutterSpeed

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9463 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this Operation Code sets the shutter speed for high-frequency flicker reduction.

### ○ Command Details

Sets the shutter speed for high-frequency flicker reduction according to the current state (photo mode or video mode).

The change direction of the shutter speed specified by Parameter1 is as follows.
- On the high-speed side, it is equivalent to turning the main command dial counterclockwise.
- On the low-speed side, it is equivalent to turning the main command dial clockwise.

One step of the shutter speed change specified by Parameter2 corresponds to one click of the main command dial.

When the shutter speed for high-frequency flicker reduction is set with this command, the normal shutter speed is updated with an approximate value of the shutter speed for high-frequency flicker reduction.

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | The change direction of the shutter speed | |
| | 0 | High-speed side |
| | 1 | Low-speed side |
| Parameter2 | Shutter speed change steps | |
| | From1 to 10 | Number of steps |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Invalid_Parameter | The specified CaptureSort is out of the setting range. |
| Access_Denied | When the normal shutter speed is other than 1/30 second to 1/8000 second. |
| | When the following conditions are not met in photo mode.<br><br>HighFrequencyFlickerReduction property (section 6.5.2.65) is set to [On (high-resolution shutter setting)] and ExposureProgramMode property (section 6.5.1.11) is [M] or [S]. |
| | When the following conditions are not met in video mode.<br><br>MovieHighFrequencyFlickerReduction property (section 6.5.3.59) is set to [On (high-resolution shutter setting)] and ExposureProgramMode property is [M]. |

Do Not Copy

# 6.2.5 Vendor (Lens)

## 6.2.5.1 AfDrive

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90C1 |
| Operation Parameter | None |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode starts the AF driving and has the same function as that of pressing the shutter-release button of the camera body halfway.

### ○ Command Details

This command is an activation command. When the AF driving is started or the camera is in the AF non-driving condition described below, the OK response is made.

After the command is terminated normally, the host should issue the DeviceReady command to confirm the completion of the operation.

The following information can be returned for the DeviceReady command when the AF driving is started.

| Response Code | Description |
|---|---|
| OK | AF driving operation is terminated. |
| Device_Busy | AF is in operation. |
| Out_of_Focus | AF operation failed (out of focus). |
| Other Response Code | AF operation failed. |

For the details, refer to the DeviceReady command (section 6.2.2.11).

If the following conditions are true, the AF operation is not performed and the camera immediately returns OK.

| No | AF non-driving condition |
|---|---|
| 1 | The value of the FocusMode property (section 6.5.1.7) is 0x0001[MF]. |
| 2 | WarningStatus property (section 6.5.6.22) is [Retractable zoom-ring warning]. |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Access_Denied | WarningStatus property (section 6.5.6.22) is set to [A non-compatible lens is attached] or [A lens that failed to firmware upgrade is attached]. |
| Device_Busy | When the command processing is started, the acquisition operation or the AF operation is being performed. |
| | The STM lens is being initialized. |

Do Not Copy

### 6.2.5.2 AfDriveCancel

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9206 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode cancels the AF driving.

#### ○ Command Details

The camera cancels the AF driving in operation.

The transition to the response phase is performed when the cancellation of the AF driving is completed.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |

Do Not Copy

### 6.2.5.3 MfDrive

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9204 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode starts the MF driving.

**○ Command Details**

This command is an activation command. OK response is made when the MF driving is started. When the host gets the response of the normal termination, the host should issue the DeviceReady command to confirm whether the operation is terminated.

The following information can be returned for the DeviceReady command when the MF driving is started.

| Response Code | Description |
|---|---|
| OK | MF driving operation is terminated. |
| Device_Busy | MF is driving. |
| MfDrive_Step_End | The MF driving reaches the end of steps. |
| MfDrive_Step_Insufficiency | The driving amount is insufficient. |
| Other Response Code | The MF driving operation failed. |

* For the details, refer to the DeviceReady command (section 6.2.2.11).

The camera performs the MF driving based on the contents of DriveType specified by Parameter1. The MF driving amount is based on the contents of StepValue specified by Parameter2. The MF driving operates according to the current position.

**○ Operation Parameter**

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | DriveType | |
| | 0x00000001 | Infinite -> Nearest |
| | 0x00000002 | Nearest -> Infinite |
| Parameter2 | StepValue (driving amount (the number of pulses)) | |
| | From 1 to 32767 | |
| Parameter3 | None | |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Device_Busy | The AF operation is being performed in the camera. |
| | The AF-F is being set. |
| | The STM lens is being initialized. |
| | When the command processing is started, the acquisition operation is being performed. |
| Invalid_Parameter | The specified value of DriveType or StepValue is out of the range. |
| Invalid_Status | The MF driving cannot be performed due to a problem caused by the camera (an error caused by the camera, the CPU internal lens is not mounted, the lens cannot be used, a non-compatible lens is attached, a lens that failed to firmware upgrade is attached, etc.). |
| | The focus mode is MF. |

Do Not Copy

## 6.2.5.4 PowerZoomDrive

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9464 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | None |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode requests the start/stop of power zoom drive by a power zoom compatible lens.

### ○ Command Details

If a lens that supports power zoom is attached, a normal termination response is returned regardless of whether the power zoom drive started/stopped successfully.

Whether the power zoom is activated or stopped can be confirmed by changing the FocalLength property (section 6.5.1.6).

If this command is executed with "Start" while the power zoom is being driven, it will be re-driven with the specified parameters.

As for the correspondence between this command for the drive speed of the power zoom and the camera menu, the value obtained by subtracting 5 from the value of Parameter 3 of this command is same as the menu value.

| Speed | Low | | | | High |
|---|---|---|---|---|---|
| Parameter3 | 0 | … | 5 | … | 10 |
| Menu | -5 | | 0 | | +5 |

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | Power zoom driving | |
| | 0 | Stop |
| | 1 | Start |
| Parameter2 | Driving direction | |
| | 1 | Wide |
| | 2 | Tele |
| Parameter3 | Driving speed | |
| | From 0(Low speed) to 10(High speed) | |

If "Stop" is specified for Parameter1, Parameters2 and 3 are not used.

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_Parameter | The specified value is out of the range. |
| Access_Denied | When the power zoom compatible lens in the LensTypeML property (section 6.5.10.2) is set to zero. |

Do Not Copy

### 6.2.5.5 GetManualSettingLensData2

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9470 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ManualSettingLens DataSet |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode acquires the information about “Non-CPU lens data” in the setup menu.

**○ Command Details**

The camera sends the information (ManualSettingLens DataSet) about “Non-CPU lens data” of LensNo specified by the parameter1 to the host.

When 0xFFFFFFFF is set to LensNo, the camera sends all the information to the host.

The ManualSettingLens DataSet is shown below.

| Field | | Size | DataType | Description |
|---|---|---|---|---|
| Dataset version | | 2 | UINT16 | Version 1.00 : 0x0064 |
| Lens No. being set | | 4 | UINT32 | Refer to ManualSettingLensNo property (section 6.5.5.1). |
| Number of manual setting lens information (N) | | 4 | UINT32 | Acquire one data：1/ Acquire all data：20 |
| Manual setting lens information[0] (Information about the lens No. 1. See NOTE.) | Focal length | 4 | UINT32 | Refer to FmmManualSetting2 property (section 6.5.5.2). |
| | Maximum aperture | 2 | UINT16 | Refer to F0ManualSetting2 property (section 6.5.5.3). |
| | Lens name | DTS | String | Refer to ManualSettingLensName property (section 6.5.5.4). |
| Manual setting lens information [1] | Focal length | 4 | | Information about the lens No. 2. |
| | Maximum aperture | 2 | | |
| | Lens name | DTS | | |
| Manual setting lens information [N - 1] | Focal length | 4 | | Information about the lens No. N. |
| | Maximum aperture | 2 | | |
| | Lens name | DTS | | |

NOTE) When “Acquire one data” is selected, the information about the lens of LensNo specified by Parameter1.

**○ Operation Parameter**

| Operation Parameter | Description | |
|---|---|---|
| Parameter1 | LensNo | |
| | Refer to ManualSettingLensNo property. | Acquire one data |
| | 0xFFFFFFFF | Acquire all data |
| Parameter2 | None | |
| Parameter3 | None | |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

# 6.2.6 Vendor (Live view)

## 6.2.6.1 StartLiveView

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9201 |
| Operation Parameter | None |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode makes the camera enter the remote live view status.

### ○ Command Details

This command is an activation-type command. OK response is made when the entry into the remote live view status is started.

When the host gets the response code of normal termination, the host should issue the DeviceReady command to confirm whether the camera enters the remote live view status (and the acquisition of the live view image becomes enabled).

The following information can be returned for the DeviceReady command when entering the remote live view status.

| Response Code | Description |
|---|---|
| OK | The camera has entered the remote live view status. |
| Device_Busy | The camera is entering the remote live view status. |
| Other Response Code * | Some error has occurred when entering the remote live view status. |

* For the details, refer to the DeviceReady command (section 6.2.2.11).

When the live view is active on the camera but the camera does not enter the remote live view status because the StartLiveView command is not executed, commands such as GetLiveViewImageEx command cannot be executed.

The live view prohibition condition is shown in the LiveViewProhibitionCondition property (section 6.5.11.7). The StartLiveView command must be issued when there is no prohibition factor in the LiveViewProhibitionCondition property. If there is any prohibition factor in the LiveViewProhibitionCondition property, even if the StartLiveView command is issued and the live view is started, the operation is not guaranteed.

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Device_Busy | When the command processing is started, the acquisition operation is being performed. |
| Hardware_Error | When the command processing is started, some error is generated in the camera body. |
| Invalid_Status | An error caused by the camera (battery empty, warning information) occurs. |

Do Not Copy

### 6.2.6.2 EndLiveView

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9202 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode releases the live view status.

#### ○ Command Details

The remote live view status is released, and the camera becomes into the camera live view status.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Device_Busy | The live view cannot be terminated because of the camera factor. |

Do Not Copy

### 6.2.6.3 GetLiveViewImage

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9203 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | LiveViewObject (without version) |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode acquires the newest live view image.
The format of the live view image is JPEG.

**○ Command Details**

This OperationCode is a command for compatibility with the GetLiveViewImageEx command (section 6.2.6.4).

For this camera, GetLiveViewImageEx command should be used.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Access_Denied | The WarningStatus property (section 6.5.6.22) is “Version upgrade warning when mount adapter is attached”. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.6.4 GetLiveViewImageEx

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9428 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | LiveViewObject (with version) |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode acquires the newest live view image.
The format of the live view image is JPEG.

**○ Command Details**

This command is valid when the camera is in the remote live view condition after executing the StartLiveView command.

The camera sends the newest live view image (LiveViewObject) to the host. LiveViewObject is composed of the display information and the live view image (JPEG).
The data set of LiveViewObject is described in section 9.7.

Even if the through image display of the camera is being enlarged during video recording, the live view image transmitted by this command is the whole image.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Access_Denied | The WarningStatus property (section 6.5.6.22) is “Version upgrade warning when mount adapter is attached”. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.6.5 GetLiveViewCompressedSize

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9423 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | LiveViewObjectMaxSize Value |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode acquires the maximum size (Byte) of LiveViewObject returned by the GetLiveViewImageEx command.

#### ○ Command Details

The size of the live view image is calculated with the value set in the LiveViewImageCompression property and LiveViewImageSize property.

The format of LiveViewObjectMaxSize is UINT32.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.6.6 ChangeAfArea

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9205 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode changes the AF area in the live view condition.

**○ Command Details**

This command is valid when the camera is in the remote live view condition after getting the StartLiveView command.

The camera specifies the AF area with the coordinates specified by Parameter1 and Parameter2 set to the center. XValue of Parameter1 and YValue of Parameter2 are used to set the X-axis and the Y-axis, respectively.

The range of XValue and YValue should be the “Whole size” of the header information acquired by the GetLiveViewImageEx command. However, the range that can be specified should be smaller by the half size of the “AF frame size”. When a value that exceeds the setting permitted range is set, the maximum or the minimum value is reflected.

If the image is enlarged during video recording, only the enlarged position is changed without changing the AF area.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | XValue |
| Parameter2 | YValue |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Access_Denied | In the case of the spot WB acquisition standby state |
| Device_Busy | The AF operation is being performed in the camera. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.6.7 StartTracking

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9424 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode starts tracking action for auto-area AF and target tracking AF, or changes the target of subject recognition.

#### ○ Command Details

The specification method of OperationParameter is the same as the command details of ChangeAfArea (section 6.2.6.6).

After this command is executed, the changes in LiveViewObject (section 9.7) acquired by the GetLiveViewImageEx command are shown below.

| Condition | | Value of LiveViewObject that changes after this command is executed | | |
|---|---|---|---|---|
| AF area mode | Subject recognition target is detected or not | Tracking state | Center coordinates of AF frame | Index of selected subject |
| Auto-area AF | There is a detected subject recognition target at the specified coordinates. | Waiting | - | Index of the subject recognition target at the specified coordinates |
| | There is no detected subject recognition target at the specified coordinates. | Tracking | Specified coordinates | - |
| Target tracking AF | - | Tracking | Specified coordinates | - |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | XValue |
| Parameter2 | YValue |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Access_Denied | The StillFocusMeteringMode property (section 6.5.2.67) is set to the value other than [Auto-area AF] during photo mode. |
| | The MovieFocusMeteringMode property (section 6.5.3.62) is set to the value other than [Auto-area AF / Target tracking AF] during video mode. |
| | In the case of the spot WB acquisition standby state |
| Device_Busy | The image acquisition operation is being performed. |
| Invalid_Status | Failed to change the tracking state due to the camera factor. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.6.8 EndTracking

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9425 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode terminates the tracking operation in auto-area AF and target tracking AF.

#### ○ Command Details

After executing this command, the tracking state of LiveViewObject acquired by the GetLiveViewImageEx command (section 6.2.6.4) is set to [Waiting].

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Device_Busy | The image acquisition operation is being performed. |
| Invalid_Status | Failed to change the tracking state due to the camera factor. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

# 6.2.7 Vendor (Video)

## 6.2.7.1 StartMovieRecord

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x920A |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | Refer to Response Parameter in this section. |

### ○ Command Outline

The operation by this OperationCode starts video recording.

### ○ Command Details

This command is accepted only during the live view execution. It is recommended to check the video recording prohibition condition property (MovieRecProhibitionCondition (section 6.5.12.5)) before issuing this command.

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Access_Denied | In the case of the spot WB acquisition standby state |
| Invalid_Status | The video recording cannot be started due to an error caused by the camera. |
| Not_LiveView | The camera is not in the live view condition. |

### ○ Response Parameter

| Response Parameter | Description |
|---|---|
| Parameter1 | When the response code is OK, the recording media is specified in this field.<br>When the response code is not OK, this field is invalid.<br>0: Recording in the card<br>1: Recording in the external recorder<br>2: Recording in the card and the external recorder |
| Parameter2 | None |
| Parameter3 | None |

Do Not Copy

### 6.2.7.2 EndMovieRecord

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x920B |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode finishes video recording.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |

Do Not Copy

### 6.2.7.3 GetHLGQualityData

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x942E |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | HLGQuality DataSet |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode returns the HLGQuality DataSet set in "HLG Quality Adjustment" in the video recording menu.

#### ○ Command Details

The contents of the HLGQuality DataSet are shown in section 9.13.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.7.4 SetHLGQualityData

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x942F |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | HLGQuality DataSet |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode sets the HLGQuality DataSet in "HLG Quality Adjustment" in the video recording menu.

#### ○ Command Details

The contents of the HLGQuality DataSet are shown in section 9.13.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Device_Busy | During shooting operation. |
| Access_Denied | MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to other than [HLG]. |
| | MovieFileType property is set to [ProRes RAW HQ 12-bit(MOV)] or [N-RAW 12-bit(NEV)]. |
| Invalid_Data | Wrong HLGQuality DataSet was specified. |

Do Not Copy

# 6.2.8 Vendor (White balance)

## 6.2.8.1 SetPreWbData

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90C9 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PreWbThumImage |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode sets the data in the preset manual white balance data area of the camera.

### ○ Command Details

This command sets the data in the preset manual white balance data area of the camera.

The camera stores the preset white balance gain value specified by PreWbGainValue in the data area specified by PreWbDataIndex and stores the thumbnail image of PreWbThumImage.

The format of PreWbThumImage shall be the same as that of the thumbnail image (JPEG format) recorded in the JPEG file defined in the accompanying document “RAW Data Format for Digital Camera”.

When PreWbThumImageSize is 0, PreWbThumImage is not sent in the data phase and the thumbnail image is not recorded.

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | byte3: Reserved(0) | |
| | byte2: ShootingMode | |
| | 0 | Photography |
| | 1 | Video |
| | byte1: RotateThumb (*1) | |
| | 0 | Horizontal |
| | 1 | Grip side upward |
| | 2 | Grip side downward |
| | byte0: PreWbDataIndex (*2)<br>This is equivalent to the WbPresetDataNo property (section 6.5.2.32). | |
| Parameter2 | PreWbGainValue | |
| Parameter3 | PreWbThumImageSize | |

*1: RotateThumb is referred to by the camera when Parameter3 (PreWbThumImageSize) is a value other than 0.

*2: If the value of the WbPresetDataNo property is zero (in the case of the camera with acquired data), any data cannot be set in the preset manual white balance data area.

Do Not Copy

The contents of PreWbGainValue are shown below.

Upper 2 Bytes of PreWbGainValue are RGain and lower 2 Bytes are BGain.
The format of RGain and BGain is fixed point. RGain sets Bit 16 to 26 as decimal parts and BGain sets up to Bit 10 as decimal parts.
Since the number of significant digits of the decimal part is 9 bits, the lower 2 bits are ignored.

$$\mathrm{RGain} = \frac{\mathrm{G}}{\mathrm{R}} \times 2048$$

$$\mathrm{BGain} = \frac{\mathrm{G}}{\mathrm{B}} \times 2048$$

The range that can be set for RGain, BGain is as follows.

$$0 \leq \mathrm{RGain} < 8 \times 2048$$
$$0 \leq \mathrm{BGain} < 8 \times 2048$$

| PreWbGainValue | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RGain | Integer part | | | | | Decimal part | | | | | | | | | | |
| Bit | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| BGain | Integer part | | | | | Decimal part | | | | | | | | | | |

PreWbThumImageSize indicates the size of PreWbThumImage. In addition, PreWbThumImage should be the compression quality Fine (1/4 compression) and the size should be 13440 bytes or less.

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Incomplete_Transfer | The data block reception failed. |
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |
| Device_Busy | During shooting operation. |
| Invalid_Parameter | The specified value is out of the range. |
| Wb_Preset_Error | The specified index of the preset is protected. |

Do Not Copy

### 6.2.8.2 StartSpotWb

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9402 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode makes the camera enter into the spot WB acquisition standby state in the live view condition.

**○ Command Details**

This command is valid when the camera is in video mode and in the remote live view condition after executing the StartLiveView command.

The camera enters into the acquisition standby state for the storage destination specified by the MovieWbPresetDataNo property (section 6.5.3.26). When the MovieWhiteBalance (section 6.5.3.14) is set to [Same as photo settings], the storage destination specified by the WbPresetDataNo property is used.

A property other than the following cannot be set in the spot WB acquisition standby state. If it is set, the Access_Denied response is made.

| Property Code | Description | Reference item |
|---|---|---|
| 0xD1BD | LiveViewZoomArea | 6.5.11.6 |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Access_Denied | The white balance is set to anything other than [Preset manual]. |
| | During the PC camera mode. |
| Device_Busy | The shooting operation is being performed at the start of command processing. |
| Invalid_Status | The storage destination of the preset manual WB data is protected. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.8.3 EndSpotWb

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9403 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode cancels the spot WB acquisition standby state.

**○ Command Details**

This command cancels the spot WB acquisition standby state.
The OK response is made even when the camera is not in the spot WB acquisition standby state.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Device_Busy | During transition into the spot WB condition |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.8.4 ChangeSpotWbArea

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9404 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode changes the spot WB area in the spot WB acquisition standby state.

#### ○ Command Details

This command is valid when the camera is in the spot WB acquisition standby state after getting the StartSpotWb command.

For the details, refer to ChangeAfArea (section 6.2.6.6).

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | XValue |
| Parameter2 | YValue |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Invalid_Status | The camera is not in the spot WB acquisition standby state. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.8.5 MeasureSpotWb

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9405 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode acquires the spot WB in the spot WB acquisition standby state.

#### ○ Command Details

This command is valid when the camera is in the spot WB acquisition standby state after getting the StartSpotWb command (section 6.2.8.2).

This command is an activation command. When the spot WB acquisition is started, the transition to the response phase is performed.

When receiving this command, the camera acquires the spot WB with the coordinates set by ChangeSpotWbArea command (section 6.2.8.4).

The spot WB acquisition result is passed by the display information in LiveViewObject acquired by GetLiveViewImageEx command (section 6.2.6.4).

After releasing the acquisition result, the camera is in the spot WB acquisition standby state and the acquisition operation can be performed again by executing this command.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Device_Busy | During transition into the spot WB condition |
| Invalid_Status | The camera is not in the spot WB acquisition standby state. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

### 6.2.8.6 EndSpotWbResultDisp

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9406 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode cancels the spot WB acquisition result screen display.

**○ Command Details**

This command cancels the spot WB acquisition result screen display.
The OK response is made even while the spot WB acquisition result screen is not displayed.
This command is an activation command. When the cancellation of the spot WB acquisition result screen display is started, the transition to the response phase is performed.
The result of the spot WB acquisition result screen display is passed by the display information in LiveViewObject acquired by the GetLiveViewImageEx command (section 6.2.6.4).

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | None |
| Parameter2 | None |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Not_LiveView | The camera is not in the remote live view condition. |

Do Not Copy

# 6.2.9 Vendor (Picture Control)

## 6.2.9.1 GetPicCtrlData

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90CC |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControl DataSet |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode acquires PictureControl DataSet of the specified PicCtrlItem.

### ○ Command Details

The parameter3 is optional and the photo shooting menu or video recording menu can be specified.

For the format of the PictureControl DataSet, refer to section 9.9.

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlItem (equivalent to the ActivePicCtrlItem property (section 6.5.13.1) | |
| Parameter2 | DefaultFlag | |
| | 0 | The current setting value |
| | 1 | The default value |
| Parameter3 | [ShootingMode] | |
| | None | PictureControl data for the photo shooting menu |
| | 0 | |
| | 1 | PictureControl data for the video recording menu |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.9.2 GetPicCtrlDataList

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9443 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControl List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode acquires PictureControl List of the specified PicCtrlItem.

**○ Command Details**

When PicCtrlItem is specified in Parameter1, the current value and default value of PictureControl DataSet (section 9.9) of the specified PicCtrlItem are sent to the host. When 0xFFFFFFFF is specified, all PicCtrlItems are sent.

If the specified PicCtrlItem is an unregistered custom picture control, NumberOfElements is set to 0. If 0xFFFFFFFF is specified, it will not be added to the list.

The Dataset of Picture Control List is shown below.

| Field | | DataSize | Description |
|---|---|---|---|
| NumberOfElements | | 4 | Number of PictureControl in the list. |
| Element1 | | DTS | |
| | Element Size | 2 | Data size of an element |
| | PicCtrlItem | 2 | Type of PictureControl |
| | Default PictureControl DataSet | 36 or 614 (See NOTE) | Default value of PictureControl |
| | Current PictureControl DataSet | 36 or 614 (See NOTE) | Current value of PictureControl |
| … | | | |
| ElementN | | DTS | |

NOTE) The DataSize of the PictureControl DataSet differs depending on whether CustomCurveFlag is invalid (36 bytes) or valid (614 bytes).

**○ Operation Parameter**

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlItem | PicCtrlItem (equivalent to the ActivePicCtrlItem property (section 6.5.13.1) |
| | 0xFFFFFFFF | All PictureControl |
| Parameter2 | ShootingMode | |
| | 0 | Photo shooting menu |
| | 1 | Video recording menu |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

### 6.2.9.3 SetPicCtrlData

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x90CD |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControl DataSet |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode sets PictureControl DataSet to the specified PicCtrlItem.

**○ Command Details**

The parameter3 is optional and the photo shooting menu or video recording menu can be specified.

For the format of the PictureControl DataSet, refer to section 9.9.

**○ Operation Parameter**

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlItem (equivalent to the ActivePicCtrlItem property (section 6.5.13.1)) | |
| Parameter2 | ModifiedFlag | |
| | 0 | The contents of PictureControl DataSet are applied as a new picture control. |
| | 1 | The contents of PictureControl DataSet are applied to the current setting value of the existing picture control. |
| Parameter3 | [ShootingMode] | |
| | None | PictureControl data for the photo shooting menu |
| | 0 | |
| | 1 | PictureControl data for the video recording menu |

Do Not Copy

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |
| Device_Busy | During shooting by the camera |
| Access_Denied | When PicCtrlItem is a custom picture control and ModifiedFlag is set to 1, the registered base picture control and BasePictureControl in PictureControl DataSet do not match. |
| | When ShootingMode is set to 0, ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | When ShootingMode is set to 1, ToneMode is set to N-Log or HLG (If any of the following conditions are met).<br>- MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit(MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log] or [HLG].<br>- MovieFileType property is set to [ProRes 422 HQ 10-bit(MOV] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log].<br>- MovieFileType property is set to [N-RAW 12-bit(NEV] and MovieNRaw12bitToneMode property (section 6.5.3.4) is set to [N-Log].<br>- MovieFileType property is set to [ProRes RAW HQ 12-bit(MOV] and MovieProResRawHq12bitToneMode property (section 6.5.3.6) is set to [N-Log]. |
| | Invalid PictureControl DataSet (section 9.9) is specified. |

Do Not Copy

### 6.2.9.4 DeleteCustomPicCtrl

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90CE |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode deletes the specified custom picture control item.

#### ○ Command Details

The contents of CustomPicCtrlItem are the same as those of ActivePicCtrlItem property (section 6.5.13.1).

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | CustomPicCtrlItem | |
| Parameter2 | [ShootingMode] | |
| | None | PictureControl data for the photo shooting menu |
| | 0 | |
| | 1 | PictureControl data for the video recording menu |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Device_Busy | During shooting by the camera |

Do Not Copy

### 6.2.9.5 GetPicCtrlCapability

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x90CF |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControlCapabilityData |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode acquires PictureControlCapability DataSet of the picture control that is the basis of the specified PicCtrlItem.

#### ○ Command Details

The parameter2 is optional and the photo shooting menu or video recording menu can be specified.

For the format of the PictureControlCapability DataSet, refer to section 9.10.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlItem (equivalent to the ActivePicCtrlItem property (section 6.5.13.1)) | |
| Parameter2 | [ShootingMode] | |
| | None | PictureControl data for the photo shooting menu |
| | 0 | |
| | 1 | PictureControl data for the video recording menu |
| Parameter3 | None | |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

## 6.2.9.6 GetPicCtrlCapabilityList

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9444 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControlCapability List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode acquires PictureControlCapability List of the picture control that is the basis of the specified PicCtrlItem.

### ○ Command Details

When PicCtrlItem is specified in Parameter1, the PictureControlCapability DataSet (section 9.10) of the specified PicCtrlItem is sent to the host. When 0xFFFFFFFF is specified, all PicCtrlItems are sent.

If the specified PicCtrlItem is an unregistered custom picture control, NumberOfElements is set to 0. If 0xFFFFFFFF is specified, it will not be added to the list.

The data set of PictureContorl Capability List is shown below.

| Field | | DataSize | Description |
|---|---|---|---|
| NumberOfElements | | 4 | Number of PictureControl in the list. |
| Element1 | | 32 | |
| | Element Size | 2 | Data size of an element |
| | PicCtrlItem | 2 | Type of |
| | PictureContorlCapability DataSet | 28 | PictureControl function information |
| … | | | |
| ElementN | | 32 | |

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlItem | Specified PictureControl (equivalent to the ActivePicCtrlItem property (section 6.5.13.1)) |
| | 0xFFFFFFFF | All PictureControl |
| Parameter2 | ShootingMode | |
| | 0 | Photo shooting menu |
| | 1 | Video recording menu |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

## 6.2.9.7 GetPicCtrlHlgDataList

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x944D |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControlHlg List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this Operation Code acquires the PictureControlHlg DataSet of the PicCtrlHlgItem specified in "Picture Control (HLG)".

### ○ Command Details

When PicCtrlHlgItem is specified in Parameter1, the current value and default value of PictureControlHlg DataSet (Section 9.11) of the specified PicCtrlHlgItem are sent to the host.
When 0xFFFFFFFF is specified, all PicCtrlHlgItems are targeted.

The data set of PictureControlHlg List is shown below.

| Field | | DataSize | Description |
|---|---|---|---|
| NumberOfElements | | 4 | Number of Picture Controls (HLG) in the list |
| Element1 | | 76 | |
| | Element Size | 2 | Data size of element |
| | PicCtrlHlgItem | 2 | Types of Picture Control (HLG) |
| | Default PictureControlHlg DataSet | 36 | Default value of Picture Control (HLG) |
| | Current PictureControlHlg DataSet | 36 | Current value of Picture Control (HLG) |
| ～ | | | |
| ElementN | | 76 | |

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlHlgItem | Specified PictureControl (HLG)<br>Following the ActivePicCtrlHlgItem property (section 6.5.13.3). |
| | 0xFFFFFFFF | All PictureControl |
| Parameter2 | None | |
| Parameter3 | None | |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

### 6.2.9.8 SetPicCtrlHlgData

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x944E |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControlHlg DataSet |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode sets PictureControlHlg DataSet to PicCtrlHlgItem specified in "Picture Control (HLG)".

#### ○ Command Details

The contents of the PictureControlHlg DataSet are shown in Section 9.11.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | Following the ActivePicCtrlHlgItem property (section 6.5.13.3). |
| Parameter2 | None |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |
| Device_Busy | During shooting operation. |
| Access_Denied | Invalid PictureControlHlg DataSet (section 9.11) specified. |

Do Not Copy

## 6.2.9.9 GetPicCtrlHlgCapabilityList

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x944F |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | PictureControlHlgCapability List |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this Operation Code acquires the PictureContorlHlgCapability List of the Picture Control (HLG) that is the basis of the specified PicCtrlHlgItem.

### ○ Command Details

When PicCtrlHlgItem is specified in Parameter1, the PictureControlHlgCapability DataSet (Section 9.12) of the specified PicCtrlHlgItem is sent to the host. When 0xFFFFFFFF is specified, all PicCtrlHlgItems are targeted.

The data set of PictureControlHlgCapability List is shown below.

| Field | | DataSize | Description |
|---|---|---|---|
| NumberOfElements | | 4 | Number of Picture Controls (HLGs) in the list |
| Element1 | | 33 | |
| | Element Size | 2 | Data size of element |
| | PicCtrlHlgItem | 2 | Type of Picture Control (HLG) |
| | PictureContorlHlgCapability DataSet | 29 | Picture Control (HLG) function information |
| ～ | | | |
| ElementN | | 33 | |

### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PicCtrlHlgItem | Specified PictureControl (HLG)<br>Following the ActivePicCtrlHlgItem property (section 6.5.13.3). |
| | 0xFFFFFFFF | All PictureControl |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | The specified value is out of the range. |
| Incomplete_Transfer | The data block transmission failed. |

Do Not Copy

# 6.2.10 Vendor (IPTC)

## 6.2.10.1 GetIPTCInfo

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9401 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | Presence or absence of the IPTC information |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode obtains the presence or absence of the IPTC information of the specified object.

### ○ Command Details

The camera sends the presence or absence of the IPTC information corresponding to the ObjectHandle specified in Parameter1.

Information to be sent (the transfer data size is 4byte)

0: IPTC absent, 1: IPTC present

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | None |
| Parameter3 | None |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | Specified parameter is not supported. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object that corresponds to the specified ObjectHandle does not exist. |
| | An object in the SDRAM other than the ObjectHandle passed by ObjectAddedInSdram is specified. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |

Do Not Copy

### 6.2.10.2 GetIPTCPresetInfo

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9412 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | IPTCPreset list |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the specified IPTC preset information.

#### ○ Command Details

The camera sends IPTCPreset list with following format.

Each field data is stored in the little endian format.

| Field | Size(Byte) | Data |
|---|---|---|
| NumElement | 4 | The number of array elements (N is the number of object) |
| ArrayEntry [0] | DTS | IPTCPresetDataSet [0] |
| ArrayEntry [1] | DTS | IPTCPresetDataSet [1] |
| ArrayEntry [2] | DTS | IPTCPresetDataSet [2] |
| ... | | |
| ArrayEntry [N - 1] | DTS | IPTCPresetDataSet [N - 1] |

When PresetNo is set to 0, the number of entries in the list is 13.

IPTC preset information is stored in a list in the order of PresetNo.

When PresetNo is more than 1, the number of entry is 1 and the dataset specified by PresetNo is stored in IPTCPresetDataSet[0] field.

The dataset of IPTCPreset is described in section 9.8.

#### ○ Operation Parameter

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PresetNo | |
| | 0 | All preset |
| | 1 to 10 | Specified IPTC preset (NMS-IPTC) |
| | 11 to 13 | Specified IPTC preset (XMP-IPTC) |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | Specified parameter is not supported. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | The specified PresetNo is out of range. |

Do Not Copy

### 6.2.10.3 SetIPTCPresetInfo

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9413 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | IPTCPreset list |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode sets the specified IPTC preset information.

**○ Command Details**

The format of the preset list that is set to the camera is described in GetIPTCPresetInfo (section 6.2.10.2).

The IPTC format set in IPTCPresetDataSet differs depending on PresetNo.

When PresetNo is set to 0, 13 items of information should be set. Set IPTC preset information in ascending order of PresetNo in the storage order of the IPTCPreset list.

When PresetNo is more than one, the specified preset information should be set in IPTCPresetDataSet[0] field.

**○ Operation Parameter**

| Operation Parameter | Details | |
|---|---|---|
| Parameter1 | PresetNo | |
| | 0 | All preset |
| | 1 to 10 | Specified IPTC preset (NMS-IPTC) |
| | 11 to 13 | Specified IPTC preset (XMP-IPTC) |
| Parameter2 | None | |
| Parameter3 | None | |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | Specified parameter is not supported. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Device_Busy | During shooting by the camera |
| Invalid_Parameter | The specified PresetNo is out of range. |
| Invalid_Data | The format of the specified PrisetNo and IPTCPresetDataSet are different. |

Do Not Copy

# 6.2.11 Vendor (Speedlight)

## 6.2.11.1 GetSBHandles

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9414 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | SBHandleList |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode obtains the handle of the flash attached with the camera.

### ○ Command Details

When the parameter1 (SBGroupID, section 7.6) is specified, the handle of the flash in the specified group is transferred.

When the parameter1 (SBGroupID, section 7.6) is not specified or set to zero, the handles of all flash are transferred.

The data set of SBHandleList is shown below.

| Field | DataSize | DataType | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBHandle in the list. |
| Element1 SBHandle | 4 | SBHandle | The handle of the element1 flash. |
| Element2 SBHandle | 4 | SBHandle | The handle of the element2 flash. |
| … | | | |
| ElementN SBHandle | 4 | SBHandle | The handle of the element N flash. |

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | [SBGroupID] |
| Parameter2 | None |
| Parameter3 | None |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBGroupID is specified. |

Do Not Copy

### 6.2.11.2 GetSBAttrDesc

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9415 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | SBHandleAttributeDescList |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode obtains the descriptor of the attribute for the flash.

**○ Command Details**

When the parameter1 (SBHandle, section 7.5) is specified, the descriptor for the specified flash is transferred. When the parameter1 (SBHandle, section 7.5) is not specified or set to zero, the descriptors for all flash are transferred.

When the parameter2 (SBAttrID) is specified, the descriptor for the specified SBAttribute (section 6.7.1) is transferred. When the parameter2 (SBAttrID) is not specified, the descriptors for all SBAttribute are transferred.

The data set of SBHandleAttributeDescList is shown below.

| Field | DataSize | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBHandle in the list. |
| Element1 SBHandle | 4 | SBHandle | SBHandle for element1. |
| Element1 SBAttributeDescList | DTS | SBAttributeDescList (section 9.15) | Attribute descriptor of the element1 flash. |
| Element2 SBHandle | 4 | SBHandle | SBHandle for element2. |
| Element2 SBAttributeDescList | DTS | SBAttributeDescList | Attribute descriptor of the element2 flash. |
| … | | | |
| ElementN SBHandle | 4 | SBHandle | SBHandle for element N. |
| ElementN SBAttributeDescList | DTS | SBAttributeDescList | Attribute descriptor of the element N flash. |

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | [SBHandle] |
| Parameter2 | [SBAttrID] |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBHandle is specified. |
| | Invalid SBAttrID is specified. |
| | Valid SB information cannot be acquired. |

Do Not Copy

### 6.2.11.3 GetSBAttrValue

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9416 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | SBHandleAttributeList |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the current attribute value for the specified flash.

#### ○ Command Details

When the parameter1 (SBHandle, section 7.5) is specified, the value for the specified flash is transferred. When the parameter1 (SBHandle, section 7.5) is not specified or set to zero, the values for all flash are transferred.

When the parameter2 (SBAttrID) is specified, the value for the specified SBAttribute (section 6.7.1) is transferred. When the parameter2 (SBAttrID) is not specified, the values for all SBAttribute are transferred.

The data set of SBHandleAttributeList is shown below.

| Field | DataSize | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBHandle in the list. |
| Element1 SBHandle | 4 | SBHandle | SBHandle for element1. |
| Element1 SBAttributeList | DTS | SBAttributeList (section 9.17) | SBAttributeList for element1. |
| Element2 SBHandle | 4 | SBHandle | SBHandle for element2. |
| Element2 SBAttributeList | DTS | SBAttributeList | SBAttributeList for element2. |
| … | | | |
| ElementN SBHandle | 4 | SBHandle | SBHandle for element N. |
| ElementN SBAttributeList | DTS | SBAttributeList | SBAttributeList for element N. |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | [SBHandle] |
| Parameter2 | [SBAttrID] |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBHandle is specified. |
| | Invalid SBAttrID is specified. |
| | Valid SB information cannot be acquired. |

Do Not Copy

### 6.2.11.4 SetSBAttrValue

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9417 |
| Operation Parameter | None |
| Data | SBHandleAttributeList |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode updates the current attribute value for the specified flash.

#### ○ Command Details

The data set of SBHandleAttributeList is shown below.

| Field | DataSize | DataType or Data Set | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBHandle in the list. |
| Element1 SBHandle | 4 | SBHandle | SBHandle for element1. |
| Element1 SBAttributeList | DTS | SBAttributeList (section 9.17) | SBAttributeList for element1. |
| Element2 SBHandle | 4 | SBHandle | SBHandle for element2. |
| Element2 SBAttributeList | DTS | SBAttributeList | SBAttributeList for element2. |
| … | | | |
| ElementN SBHandle | 4 | SBHandle | SBHandle for element N. |
| ElementN SBAttributeList | DTS | SBAttributeList | SBAttributeList for element N. |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBHandle is specified. |
| | Invalid SBAttrID is specified. |
| Invalid_SBAttribute_Value | The specified attribute value is not supported. |
| Access_Denied | The specified flash is not in a settable state. |
| | An operation is denied depending on the status of the camera. |
| | The specified SBAttributeCode is not permitted for setting. |
| Device_Busy | The shooting operation or the AF operation is being performed when starting this command. |

Do Not Copy

### 6.2.11.5 GetSBGroupAttrDesc

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9418 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | SBGroupAttributeDescList |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the attribute descriptor for the flash group.

#### ○ Command Details

When the parameter1 (SBGroupID, section 7.6) is specified, the data for the specified flash is transferred. When the parameter1 (SBGroupID, section 7.6) is not specified or set to zero, the data for all flash are transferred.

When the parameter2 (SBGroupAttrID) is specified, the data for the specified SBGroupAttribute (section 6.7.2) is transferred. When the parameter2 (SBGroupAttrID) is not specified, the data for all SBGroupAttribute are transferred.

The data set of SBGroupAttributeList is shown below.

| Field | DataSize | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBGroupID in the list. |
| Element1 SBGroupID | 4 | SBGroupID | SBGroupID for element1. |
| Element1 SBAttributeDescList | DTS | SBAttributeDescList (section 9.15) | Attribute descriptor for the element1 flash. |
| Element2 SBGroupID | 4 | SBGroupID | SBGroupID for element2. |
| Element2 SBAttributeDescList | DTS | SBAttributeDescList | Attribute descriptor for the element2 flash. |
| … | | | |
| ElementN SBGroupID | 4 | SBGroupID | SBGroupID for element N. |
| ElementN SBAttributeDescList | DTS | SBAttributeDescList | Attribute descriptor for the element N flash. |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | [SBGroupID] |
| Parameter2 | [SBGroupAttrID] |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBGroupID is specified. |
| | Invalid SBGroupAttrID is specified. |

Do Not Copy

### 6.2.11.6 GetSBGroupAttrValue

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9419 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | SBGroupAttributeList |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode obtains the current attribute value for the flash group.

#### ○ Command Details

When the parameter1 (SBGroupID, section 7.6) is specified, the data for the specified flash group is transferred. When the parameter1 (SBGroupID, section 7.6) is not specified or set to zero, the data for all flash group are transferred.

When the parameter2 (SBGroupAttrID) is specified, the data for the specified SBGroupAttribute (section 6.7.2) is transferred. When the parameter2 (SBGroupAttrID) is not specified, the data for all SBGroupAttribute are transferred.

The data set of SBGroupAttributeList is shown below.

| Field | DataSize | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBGroupID in the list. |
| Element1 SBGroupID | 4 | SBGroupID | SBGroupID for element1. |
| Element1 SBAttributeList | DTS | SBAttributeList (section 9.17) | SBAttributeList for element1. |
| Element2 SBGroupID | 4 | SBGroupID | SBGroupID for element2. |
| Element2 SBAttributeList | DTS | SBAttributeList | SBAttributeList for element2. |
| … | | | |
| ElementN SBGroupID | 4 | SBGroupID | SBGroupID for element N. |
| ElementN SBAttributeList | DTS | SBAttributeList | SBAttributeList for element N. |

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | [SBGroupID] |
| Parameter2 | [SBGroupAttrID] |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBGroupID is specified. |
| | Invalid SBGroupAttrID is specified. |

Do Not Copy

### 6.2.11.7 SetSBGroupAttrValue

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x941A |
| Operation Parameter | None |
| Data | SBGroupAttributeList |
| Data direction | From host to camera |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode updates the current attribute value for the flash group.

#### ○ Command Details

The data set of SBGroupAttributeList is shown below.

| Field | DataSize | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of SBGroupID in the list. |
| Element1 SBGroupID | 4 | SBGroupID | SBGroupID for element1. |
| Element1 SBAttributeList | DTS | SBAttributeList (section 9.17) | SBAttributeList for element1. |
| Element2 SBGroupID | 4 | SBGroupID | SBGroupID for element2. |
| Element2 SBAttributeList | DTS | SBAttributeList | SBAttributeList for element2. |
| … | | | |
| ElementN SBGroupID | 4 | SBGroupID | SBGroupID for element N. |
| ElementN SBAttributeList | DTS | SBAttributeList | SBAttributeList for element N. |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Parameter | Invalid SBHandle is specified. |
| | Invalid SBAttrID is specified. |
| Invalid_SBAttribute_Value | The specified attribute value is not supported. |
| Access_Denied | An operation is denied depending on the status of the camera. |
| | The specified SBAttributeCode is not permitted for setting. |
| Device_Busy | The shooting operation or the AF operation is being performed when starting this command. |

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### 6.2.11.8 TestFlash

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x941B |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | None |
| Data direction | - |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode makes the flash fire for testing.

**○ Command Details**

The parameter1 (Flash mode) should be set to the flash testing.

When the parameter1 is set to the flash testing:
The flash specified by the parameter2 (SBHandle, section 7.5) fires. When the parameter2 is set to zero, all flashes fire.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | Flash mode (fixed to zero) |
| Parameter2 | SBHandle |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Parameter | Invalid flash mode is specified. |
| | Invalid SBHandle is specified. |
| Access_Denied | The specified flash cannot be operated. |

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# 6.2.12 Vendor (MTP)

## 6.2.12.1 GetObjectPropsSupported

### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9801 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectPropCodeArray |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

### ○ Command Outline

The operation by this OperationCode acquires an array of codes of the object property supported by the camera.

### ○ Command Details

The camera sends the Array of the property corresponding to ObjectFormatCode specified by Parameter1.

The format of ObjectPropCodeArray sent by the camera is shown below.
Each field data is stored in the little endian format.

For the supported ObjectPropCode, refer to section 6.6.

| Field | Size (Byte) | Data |
|---|---|---|
| NumElement | 4 | The element of the array is N (N means the number of objects). |
| ArrayEntry [0] | 2 | ObjectPropCode [0] |
| ArrayEntry [1] | 2 | ObjectPropCode [1] |
| ArrayEntry [2] | 2 | ObjectPropCode [2] |
| --- | | |
| ArrayEntry [N - 1] | 2 | ObjectPropCode [N - 1] |

The camera supports the following ObjectFormatCodes only.

| PropertyValue | ObjectFormat |
|---|---|
| 0x3000 | Undefined |
| 0x3001 | Association |
| 0x3008 | WAV |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x380D | TIFF (RGB) |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectFormatCode |
| Parameter2 | None |
| Parameter3 | None |

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### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| ObjectProp_Not_Supported | The specified ObjectFormatCode is not supported. |

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### 6.2.12.2 GetObjectPropDesc

#### ○ Command Specifications

| | |
|---|---|
| Operation Code | 0x9802 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectPropDesc data set |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

#### ○ Command Outline

The operation by this OperationCode returns the ObjectPropDesc data set corresponding to the specified ObjectPropCode and ObjectFormatCode.

#### ○ Command Details

For the supported ObjectPropCodes, refer to section 6.6.

#### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectPropCode |
| Parameter2 | ObjectFormatCode |
| Parameter3 | None |

#### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| ObjectProp_Not_Supported | The specified ObjectPropCode is not supported. |
| | The specified ObjectFormatCode is not supported. |

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### 6.2.12.3 GetObjectPropValue

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9803 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectPropValue |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns the current value corresponding to the specified ObjectPropCode.

**○ Command Details**

The camera sends the object property information specified by Parameter2 to the object corresponding to the ObjectHandle specified by Parameter1.

For the supported ObjectPropCode and the details of ObjectPropValue, refer to section 6.6.

**○ Operation Parameter**

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | ObjectPropCode |
| Parameter3 | None |

**○ Response Code**

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Invalid_Object_Handle | An object corresponding to the specified ObjectHandle does not exist, or it indicates an object in the SDRAM. |
| Incomplete_Transfer | The data block transmission failed. |
| Access_Denied | The video is playing. |
| ObjectProp_Not_Supported | The specified ObjectPropCode is not supported. |

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### 6.2.12.4 GetObjectPropList

**○ Command Specifications**

| | |
|---|---|
| Operation Code | 0x9805 |
| Operation Parameter | Refer to Operation Parameter in this section. |
| Data | ObjectPropList data set |
| Data direction | From camera to host |
| Response Code | Refer to Response Code in this section. |
| Response Parameter | None |

**○ Command Outline**

The operation by this OperationCode returns the ObjectPropList data set.

**○ Command Details**

The camera returns the ObjectPropList data set for the object corresponding to the ObjectHandle specified by Parameter1 and the ObjectPropCode (object property) specified by Parameter3. The target can be identified by specifying the optional Parameter2, Parameter4, and Parameter5.

For the supported ObjectPropCode and the details of ObjectPropValue, refer to section 6.6.

| Operation Parameter | Value | Details |
|---|---|---|
| Parameter1 | 0x00000000 | The objects directly under the root should be the targets. |
| | ObjectHandle | The object matching with the specified object handle should be the target. |
| | 0xFFFFFFFF | All the objects should be the targets. |
| Parameter2 (Optional) | 0x00000000 | Parameter2 is not specified.<br>The objects with all the object formats should be the targets. |
| | ObjectFormatCode | The object with the specified format should be the target. |
| Parameter3 | 0x00000000 | All the objects with the group code specified by Parameter4 should be the targets. |
| | ObjectPropCode | The object with the specified property should be the target. |
| | 0xFFFFFFFF | All the objects without the group code of 0xFFFFFFFF should be the targets. |
| Parameter4 (Optional) | ObjectPropGroupCode | The object with the specified group code should be the target.<br>*This is used only when Parameter3 is set to 0x00000000. |
| Parameter5 (Optional) | No value *1 | Only the object specified by Parameter1 should be the target.<br>If Parameter1 is not specified (0x00000000), the camera returns an empty ObjectPropList data set. |
| | 0x00000000 | |
| | Depth | All the objects from the objects specified by Parameter1 to the depth specified by Parameter5 should be the targets. |
| | 0xFFFFFFFF | All the objects included in the folder hierarchy of the object specified by Parameter1 should be the targets.<br>If an object of the file is specified in Parameter1, however, only the object specified by Parameter1 should be the target. |

*1 This is the case that ObjectHandle is specified in Parameter1. When Parameter1 is set to 0x00000000, all the objects should be the targets.

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The format of ObjectPropList sent by the camera is shown below.

| Field name | Field order | Size(Byte) | Datatype | Description |
|---|---|---|---|---|
| NumberOfElements | 1 | 4 | UINT32 | Number of properties (N)<br>The number of property fields is four times the number of properties. |
| Element1ObjectHandle | 2 | 4 | ObjectHandle | ObjectHandle of the object to which Property 1 is applied |
| Element1PropertyCode | 3 | 2 | Datacode | Datacode that specifies the ObjectPropDesc describing Property 1 |
| Element1Datatype | 4 | 2 | Datacode | Specifies DatatypeCode of Property 1. |
| Element1Value | 5 | DTS | DTS | Value of Property 1 |
| Element2ObjectHandle | 6 | 4 | ObjectHandle | ObjectHandle of the object to which Property 2 is applied |
| Element2PropertyCode | 7 | 2 | Datacode | Datacode that specifies the ObjectPropDesc describing Property 2 |
| Element2Datatype | 8 | 2 | Datacode | Specifies DatatypeCode of Property 2. |
| Element2Value | 9 | DTS | DTS | Value of Property 2 |
| ... | | | | |
| ElementNObjectHandle | N x 4 - 2 | 4 | ObjectHandle | ObjectHandle of the object to which Property N is applied |
| ElementNPropertyCode | N x 4 - 1 | 2 | Datacode | Datacode that specifies the ObjectPropDesc describing Property N |
| ElementNDatatype | N x 4 | 2 | Datacode | Specifies DatatypeCode of Property N. |
| ElementNValue | N x 4 + 1 | DTS | DTS | Value of Property N |

### ○ Operation Parameter

| Operation Parameter | Details |
|---|---|
| Parameter1 | ObjectHandle |
| Parameter2 | [ObjectFormatCode] |
| Parameter3 | ObjectPropCode |
| Parameter4 | [ObjectPropGroupCode] |
| Parameter5 | [Depth] |

### ○ Response Code

| Response Code | Description |
|---|---|
| OK | Normal termination |
| Parameter_Not_Supported | An unsupported parameter is specified. |
| Session_Not_Open | The session is not started. |
| Invalid_TransactionID | The TransactionID retained by the camera differs from the TransactionID specified by the host. |
| Incomplete_Transfer | The data block transmission failed. |
| Invalid_Object_Handle | An object corresponding to the specified ObjectHandle does not exist, or it indicates an object in the SDRAM. |
| Store_Not_Available | The card is being initialized. |
| | The card does not exist. |
| | The battery is exhausted. |
| Access_Denied | The video is playing. |
| Specification_By_Format_Unsupported | The specified ObjectFormatCode is not supported. |
| Invalid_ObjectPropCode | The specified ObjectPropCode is not supported. |

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# 6.3 Response Code

The camera returns the response for the processing in the response phase to the command sent from the host to the camera in the command phase. The ResponseCode shows the contents of the response. The relationship between the ResponseCode and the OperationCode sent in the command phase is as shown in section 6.2.

The ResponseCodes supported by the camera are shown below.

| Response Code | Description | Reference item |
|---|---|---|
| 0x2001 | OK | 6.3.1.1 |
| 0x2002 | General_Error | 6.3.1.2 |
| 0x2003 | Session_Not_Open | 6.3.1.3 |
| 0x2004 | Invalid_TransactionID | 6.3.1.4 |
| 0x2005 | Operation_Not_Supported | 6.3.1.5 |
| 0x2006 | Parameter_Not_Supported | 6.3.1.6 |
| 0x2007 | Incomplete_Transfer | 6.3.1.7 |
| 0x2008 | Invalid_StorageID | 6.3.1.8 |
| 0x2009 | Invalid_Object_Handle | 6.3.1.9 |
| 0x200A | DeviceProp_Not_Supported | 6.3.1.10 |
| 0x200B | Invalid_ObjectFormatCode | 6.3.1.11 |
| 0x200C | Store_Full | 6.3.1.12 |
| 0x200D | Object_Write_Protect | 6.3.1.13 |
| 0x200E | Store_Read_Only | 6.3.1.14 |
| 0x200F | Access_Denied | 6.3.1.15 |
| 0x2010 | No_Thumbnail_Present | 6.3.1.16 |
| 0x2012 | Partial_Deletion | 6.3.1.17 |
| 0x2013 | Store_Not_Available | 6.3.1.18 |
| 0x2014 | Specification_By_Format_Unsupported | 6.3.1.19 |
| 0x2015 | No_Valid_ObjectInfo | 6.3.1.20 |
| 0x2018 | Capture_Already_Terminated | 6.3.1.21 |
| 0x2019 | Device_Busy | 6.3.1.22 |
| 0x201A | Invalid_Parent_Object | 6.3.1.23 |
| 0x201B | Invalid_DeviceProp_Format | 6.3.1.24 |
| 0x201C | Invalid_DeviceProp_Value | 6.3.1.25 |
| 0x201D | Invalid_Parameter | 6.3.1.26 |
| 0x201E | Session_Already_Open | 6.3.1.27 |
| 0x201F | Transaction_Cancelled | 6.3.1.28 |
| 0x2020 | Specification_of_Destination_Unsupported | 6.3.1.29 |
| 0xA001 | Hardware_Error | 6.3.2.1 |
| 0xA002 | Out_of_Focus | 6.3.2.2 |
| 0xA003 | Change_CameraMode_Failed | 6.3.2.3 |
| 0xA004 | Invalid_Status | 6.3.2.4 |
| 0xA006 | Wb_Preset_Error | 6.3.2.5 |
| 0xA007 | Dust_Reference_Error | 6.3.2.6 |
| 0xA008 | Shutter_Speed_Bulb | 6.3.2.7 |
| 0xA00A | CameraMode_Not_Adjust_Fnumber | 6.3.2.8 |
| 0xA00B | Not_LiveView | 6.3.2.9 |
| 0xA00C | MfDrive_Step_End | 6.3.2.10 |

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| | | |
|---|---|---|
| 0xA00E | MfDrive_Step_Insufficiency | 6.3.2.11 |
| 0xA00F | No_FullHD_Present | 6.3.2.12 |
| 0xA021 | Store_Error | 6.3.2.13 |
| 0xA022 | Store_Unformatted | 6.3.2.14 |
| 0xA200 | Bulb_Release_Busy | 6.3.2.15 |
| 0xA201 | Silent_Release_Busy | 6.3.2.16 |
| 0xA202 | MovieFrame_Release_Busy | 6.3.2.17 |
| 0xA204 | Shutter_Speed_Time | 6.3.2.18 |
| 0xA209 | Invalid_SBAttribute_Value | 6.3.2.19 |
| 0xA210 | Invalid_Data | 6.3.2.20 |
| 0xA801 | Invalid_ObjectPropCode | 6.3.2.21 |
| 0xA802 | Invalid_ObjectProp_Format | 6.3.2.22 |
| 0xA80A | ObjectProp_Not_Supported | 6.3.2.23 |

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## 6.3.1 Standard

### 6.3.1.1 OK

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2001 |

- **Response Outline**

Indicates that the processing has been terminated normally.

### 6.3.1.2 General_Error

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2002 |

- **Response Outline**

Indicates that the processing cannot be terminated normally for some reason.

### 6.3.1.3 Session_Not_Open

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2003 |

- **Response Outline**

Indicates that the session is not started.

### 6.3.1.4 Invalid_TransactionID

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2004 |

- **Response Outline**

Indicates that the TransactionID retained by the camera differs from the TransactionID specified by the host.

### 6.3.1.5 Operation_Not_Supported

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2005 |

- **Response Outline**

Indicates that an OperationCode that is not passed by the DeviceInfo data set is specified.

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### 6.3.1.6 Parameter_Not_Supported

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2006 |

- **Response Outline**

Indicates that the specification of a parameter is inappropriate for the requested operation.

### 6.3.1.7 Incomplete_Transfer

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2007 |

- **Response Outline**

Indicates that the transmission/reception of the data block fails.
When the file access fails in the camera body, the camera may return this response.

### 6.3.1.8 Invalid_StorageID

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2008 |

- **Response Outline**

Indicates that a StorageID that differs from the StorageID sent by the camera is specified.

### 6.3.1.9 Invalid_Object_Handle

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2009 |

- **Response Outline**

Indicates that an invalid object handle is specified or the target object does not exist.

### 6.3.1.10 DeviceProp_Not_Supported

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x200A |

- **Response Outline**

Indicates that a DevicePropCode that is not passed by the DeviceInfo data set is specified.

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### 6.3.1.11 Invalid_ObjectFormatCode

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x200B |

- **Response Outline**

Indicates that the specified ObjectFormatCode is not supported.
It is also used to indicate that the contents specified in the ObjectCompressedSize field of ObjectInfo are not supported with the SendObjectInfo command.

### 6.3.1.12 Store_Full

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x200C |

- **Response Outline**

Indicates that the object cannot be received with the size of the buffer prepared by the camera with the SendObjectInfo command.

### 6.3.1.13 Object_Write_Protect

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x200D |

- **Response Outline**

Indicates that the target object is protected.

### 6.3.1.14 Store_Read_Only

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x200E |

- **Response Outline**

Indicates that a StorageID of the card is specified with the SendObjectInfo command.
The camera does not support the writing to the card from the host.

### 6.3.1.15 Access_Denied

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x200F |

- **Response Outline**

Indicates that the operation is denied because of the camera status.
This means that the operation will be denied unless the camera status is changed.
It is not an event that means the busy status.

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### 6.3.1.16 No_Thumbnail_Present

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2010 |

- **Response Outline**

Indicates that the target object does not have a thumbnail.

### 6.3.1.17 Partial_Deletion

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2012 |

- **Response Outline**

Indicates that although the deletion of two or more objects is commanded, only a part of those are deleted.

It may occur when a part of the target objects are protected.

### 6.3.1.18 Store_Not_Available

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2013 |

- **Response Outline**

Indicates that the card cannot be accessed because the card is being initialized, the card does not exist, or the battery is exhausted.

### 6.3.1.19 Specification_By_Format_Unsupported

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2014 |

- **Response Outline**

Indicates that the specified ObjectFormatCode is not supported.

### 6.3.1.20 No_Valid_ObjectInfo

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2015 |

- **Response Outline**

Indicates that the SendObject command is received before the SendObjectInfo command is accepted.

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### 6.3.1.21 Capture_Already_Terminated

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2018 |

- **Response Outline**

Indicates that shooting operation was already finished when trying to finish shooting.

### 6.3.1.22 Device_Busy

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x2019 |

- **Response Outline**

Indicates that the camera is in the busy status.

### 6.3.1.23 Invalid_Parent_Object

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x201A |

- **Response Outline**

Indicates that an ObjectHandle other than that indicating a directory is specified for a parameter with which an ObjectHandle of the directory should be specified.
It indicates that the specified directory does not exist.

### 6.3.1.24 Invalid_DeviceProp_Format

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x201B |

- **Response Outline**

Indicates that the size or the format of the DevicePropDesc data set is inappropriate.

### 6.3.1.25 Invalid_DeviceProp_Value

- **Response Specifications**

| | |
|---|---|
| Response Code | 0x201C |

- **Response Outline**

Indicates that the specified DevicePropValue is not supported.

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### 6.3.1.26 Invalid_Parameter

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0x201D |

○ **Response Outline**

Indicates that the specified parameter is out of the specifications.

### 6.3.1.27 Session_Already_Open

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0x201E |

○ **Response Outline**

Indicates that the OpenSession operation is specified with a session already started.
The camera supports only one session.

### 6.3.1.28 Transaction _Cancelled

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0x201F |

○ **Response Outline**

Indicates that the transfer was cancelled.

### 6.3.1.29 Specification_of_Destination_Unsupported

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0x2020 |

○ **Response Outline**

Indicates that the recording destination specified by the SendObjectInfo command is not supported.

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## 6.3.2 Vendor

### 6.3.2.1 Hardware_Error

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA001 |

○ **Response Outline**

Indicates that any error that prevents the camera from operating has occurred in the camera body.

### 6.3.2.2 Out_of_Focus

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA002 |

○ **Response Outline**

Indicates that the AF operation is terminated with the non-focused status.

### 6.3.2.3 Change_CameraMode_Failed

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA003 |

○ **Response Outline**

Indicates that the switching between the PC camera mode and the remote mode failed.

### 6.3.2.4 Invalid_Status

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA004 |

○ **Response Outline**

Indicates that the operation is invalid depending on the status of the camera.

### 6.3.2.5 Wb_Preset_Error

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA006 |

○ **Response Outline**

Indicates that the preset measurement release failed.

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### 6.3.2.6 Dust_Reference_Error

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA007 |

- **Response Outline**

Indicates that the dust reference image release failed.

### 6.3.2.7 Shutter_Speed_Bulb

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA008 |

- **Response Outline**

Indicates that the shutter speed is Bulb.

### 6.3.2.8 CameraMode_Not_Adjust_Fnumber

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA00A |

- **Response Outline**

Indicates that the aperture value is set to “F-".

### 6.3.2.9 Not_LiveView

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA00B |

- **Response Outline**

Indicates that the camera is not in the live view condition.

### 6.3.2.10 MfDrive_Step_End

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA00C |

- **Response Outline**

Indicates that the MF driving reaches the termination.

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### 6.3.2.11 MfDrive_Step_ Insufficiency

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA00E |

○ **Response Outline**

Indicates that the driving amount is insufficient.

### 6.3.2.12 No_FullHD_Present

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA00F |

○ **Response Outline**

Indicates that the target object does not include FullHD image data.

### 6.3.2.13 Store_Error

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA021 |

○ **Response Outline**

Indicates that a damaged card is included in the inserted cards.

### 6.3.2.14 Store_Unformatted

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA022 |

○ **Response Outline**

Indicates that an unformatted card is included in the inserted cards.

### 6.3.2.15 Bulb_Release_Busy

○ **Response Specifications**

| | |
|---|---|
| Response Code | 0xA200 |

○ **Response Outline**

Indicates that bulb shooting is being performed.

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### 6.3.2.16 Silent_Release_Busy

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA201 |

- **Response Outline**

Indicates that silent shooting is being performed.

### 6.3.2.17 MovieFrame_Release_Busy

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA202 |

- **Response Outline**

Indicates that the still image shooting is being performed during video recording.

### 6.3.2.18 Shutter_Speed_Time

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA204 |

- **Response Outline**

Indicates that the shutter speed is Time.

### 6.3.2.19 Invalid_SBAttribute_Value

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA209 |

- **Response Outline**

Indicates that the specified attribute value is not supported.

### 6.3.2.20 Invalid_Data

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA210 |

- **Response Outline**

Indicates that the data in the data phase is inappropriate.

Do Not Copy

### 6.3.2.21 Invalid_ObjectPropCode

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA801 |

- **Response Outline**

Indicates that the specified ObjectPropCode is not supported.

### 6.3.2.22 Invalid_ObjectProp_Format

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA802 |

- **Response Outline**

Indicates that the size or the type of the specified ObjectProp is not supported.

### 6.3.2.23 ObjectProp_Not_Supported

- **Response Specifications**

| | |
|---|---|
| Response Code | 0xA80A |

- **Response Outline**

Indicates that the specified ObjectPropCode is valid but the camera doesn’t support it.

Do Not Copy

# 6.4 Event Code

The EventCodes are used when an event is passed asynchronously from the camera to the host.

Following chart describes the EventCodes supported by the camera and whether event sending by command from host or by interrupt transfer is permitted.

| Event Code | Description | Reference | Interrupt transfer | GetEvent command | GetEventEx command |
|---|---|---|---|---|---|
| 0x4001 | CancelTransaction | 6.4.1.1 | ✓ | ✓ | ✓ |
| 0x4002 | ObjectAdded | 6.4.1.2 | ✓ | ✓ | ✓ |
| 0x4003 | ObjectRemoved | 6.4.1.3 | ✓ | ✓ | ✓ |
| 0x4004 | StoreAdded | 6.4.1.4 | ✓ | ✓ | ✓ |
| 0x4005 | StoreRemoved | 6.4.1.5 | ✓ | ✓ | ✓ |
| 0x4006 | DevicePropChanged | 6.4.1.6 | ✓ | ✓ | ✓ |
| 0x4007 | ObjectInfoChanged | 6.4.1.7 | ✓ | ✓ | ✓ |
| 0x4008 | DeviceInfoChanged | 6.4.1.8 | ✓ | ✓ | ✓ |
| 0x4009 | RequestObjectTransfer | 6.4.1.9 | ✓ | ✓ | ✓ |
| 0x400A | StoreFull | 6.4.1.10 | ✓ | ✓ | ✓ |
| 0x400C | StorageInfoChanged | 6.4.1.11 | ✓ | ✓ | ✓ |
| 0x400D | CaptureComplete | 6.4.1.12 | ✓ | ✓ | ✓ |
| 0xC101 | ObjectAddedInSdram | 6.4.2.1 | - | ✓ | ✓ |
| 0xC102 | CaptureCompleteRecInSdram | 6.4.2.2 | - | ✓ | ✓ |
| 0xC103 | CaptureSkip | 6.4.2.3 | - | ✓ | ✓ |
| 0xC105 | MovieRecordInterrupted | 6.4.2.6 | - | - | ✓ |
| 0xC108 | MovieRecordComplete | 6.4.2.5 | - | ✓ | ✓ |
| 0xC10A | MovieRecordStarted | 6.4.2.4 | - | ✓ | ✓ |
| 0xC10B | PictureControlAdjustChanged | 6.4.2.7 | - | - | ✓ |
| 0xC10C | LiveViewStateChanged | 6.4.2.8 | - | ✓ | ✓ |
| 0xC10D | PixelMappingComplete | 6.4.2.9 | - | ✓ | ✓ |
| 0xC10E | ManualSettingLensDataChanged | 6.4.2.10 | - | ✓ | ✓ |
| 0xC10F | OpenCaptureComplete | 6.4.2.11 | - | - | ✓ |
| 0xC120 | SBAdded | 6.4.2.12 | - | ✓ | ✓ |
| 0xC121 | SBRemoved | 6.4.2.13 | - | ✓ | ✓ |
| 0xC122 | SBAttrChanged | 6.4.2.14 | - | - | ✓ |
| 0xC123 | SBGroupAttrChanged | 6.4.2.15 | - | - | ✓ |
| 0xC125 | HlgQualityAdjustChanged | 6.4.2.16 | - | ✓ | ✓ |
| 0xC126 | PortraitImpressionBalanceAdjustChanged | 6.4.2.17 | - | - | ✓ |
| 0xC127 | PictureControlHlgAdjustChanged | 6.4.2.18 | - | ✓ | ✓ |

See Application Mode (section 2.4) for application mode event transmission.

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## 6.4.1 Standard

### 6.4.1.1 CancelTransaction

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4001 |
| Event Parameter1 | None |

○ **Event Outline**

It is used to inform the host that the processing is canceled. However, it is not used in the camera.

### 6.4.1.2 ObjectAdded

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4002 |
| Event Parameter1 | ObjectHandle |

○ **Event Outline**

It is used to inform the host that a new object is added to the card.

### 6.4.1.3 ObjectRemoved

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4003 |
| Event Parameter1 | ObjectHandle |

○ **Event Outline**

It is used to inform the host that a specific object in the card is deleted.

### 6.4.1.4 StoreAdded

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4004 |
| Event Parameter1 | StorageID |

○ **Event Outline**

It is used to inform the host that the card is inserted in the slot in which the card has not been inserted yet.

The StorageID corresponding to the slot in which the card is inserted is passed as an EventParameter.

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### 6.4.1.5 StoreRemoved

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4005 |
| Event Parameter1 | StorageID |

○ **Event Outline**

It is used to inform the host that the card is ejected from the slot in which the card has been inserted.

The StorageID corresponding to the slot in which the card has been inserted is passed as an EventParameter.

It is also used when the card is formatted to inform the host that the card information is invalid. When the formatting is completed, StoreAdded is used to inform the host that the card information is valid.

### 6.4.1.6 DevicePropChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4006 |
| Event Parameter1 | PropertyCode |

○ **Event Outline**

It is used to inform the host that the setting value of the camera is changed.

The setting value to be passed is that of DevicePropCode defined in section 6.5.

If the setting value of the camera is changed by the SetDevicePropValue command from the host, this event is not passed.

When the InterruptIN transfer is used, this event is passed only for the standard property.

When the GetEventEx command is used, this event is passed for all the properties.

### 6.4.1.7 ObjectInfoChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4007 |
| Event Parameter1 | ObjectHandle |

○ **Event Outline**

It is used to inform the host that the ObjectInfo data set corresponding to a specific object in the card has been changed.

### 6.4.1.8 DeviceInfoChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4008 |
| Event Parameter1 | None |

○ **Event Outline**

It is used to inform the host that the device function is changed.

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### 6.4.1.9 RequestObjectTransfer

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x4009 |
| Event Parameter1 | ObjectHandle |

○ **Event Outline**

It is used to request the GetObject operation for the ObjectHandle specified by the parameter.

### 6.4.1.10 StoreFull

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x400A |
| Event Parameter1 | StorageID |

○ **Event Outline**

It is used to inform the host that the card corresponding to the StorageID becomes full.

This event is passed when the card becomes full by operating the shutter-release button of the camera or by recording the video.

### 6.4.1.11 StorageInfoChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x400C |
| Event Parameter1 | StorageID |

○ **Event Outline**

It is used to inform the host that the free area in the card corresponding to the StorageID is changed.

This event is notified when the setting value of the image size and image quality mode applied to shooting is changed in [Photo shooting menu]-[Image size setting].

### 6.4.1.12 CaptureComplete

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0x400D |
| Event Parameter1 | TransactionID |

○ **Event Outline**

It is used to inform the host that the release operation started by the InitiateCapture command or the InitiateCaptureRecInMedia command is completed.

Do Not Copy

## 6.4.2 Vendor

### 6.4.2.1 ObjectAddedInSdram

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC101 |
| Event Parameter1 | ObjectHandle |

○ **Event Outline**

It is used to inform the host that a new object is added to the SDRAM (transmission of the image data to the host becomes enabled).

The ObjectHandle of the new object is passed as an EventParameter.

If the USB cutting occurs with the image data of the recording destination SDRAM saved in the SDRAM and then it is reconnected, the event is passed again.

### 6.4.2.2 CaptureCompleteRecInSdram

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC102 |
| Event Parameter1 | None |

○ **Event Outline**

It is used to inform the host that all the image data acquired by the release operation started by the InitiateCaptureRecInSdram, the AfAndCaptureRecInSdram, or the InitiateCaptureRecInMedia command is sent to the host completely.

### 6.4.2.3 CaptureSkip

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC103 |
| Event Parameter1 | SkipCount |

○ **Event Outline**

It is used to inform that shooting was skipped by interval timer shooting.

○ **Event Details**

If a skip occurs during shooting, this event will be notified.

In the case of "number of shots x number of frames per shot", one notification is given for each skip during shooting.

Even if multiple shots are skipped during one shooting, the notification is sent only once for each shooting.

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### 6.4.2.4 MovieRecordStarted

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC10A |
| Event Parameter1 | RecordKind |

**○ Event Outline**

It is used to inform the host that the video recording has started on the camera.
This event is not issued when the video recording has started by using StartMovieRecord command.

The recording method is sent as an EventParameter.

| RecordKind | Recording method |
|---|---|
| 0x00000000 | Card |
| 0x00000001 | External recorder |
| 0x00000002 | Card and external recorder |

### 6.4.2.5 MovieRecordComplete

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC108 |
| Event Parameter1 | RecordKind |
| Event Parameter2 | StopCode |

**○ Event Outline**

It is used to inform the host that the video recording is successfully completed.

The recording method and the couse of stop are sent as an EventParameter. For recording method, see the explanation of MovieRecordStarted (section 6.4.2.4).

| StopCode | Cause of stop |
|---|---|
| 0 | Stopping due to user operation |
| 3 | Stopping due to reaching recording time |

### 6.4.2.6 MovieRecordInterrupted

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC105 |
| Event Parameter1 | ErrorCode |
| Event Parameter2 | RecordKind |

**○ Event Outline**

It is used to inform the host that the video recording is interrupted.

The recording method and the interruption cause are sent as EventParameter. For recording method, see the explanation of MovieRecordStarted (section 6.4.2.4).

| ErrorCode | Interruption cause |
|---|---|
| 1 | A certain error |
| 2 | Low-speed card error |

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### 6.4.2.7 PictureControlAdjustChanged

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC10B |
| Event Parameter1 | PicCtrlItem (ActivePicCtrlItem property (section 6.5.13.1) |
| Event Parameter2 | ShootingMode |

**○ Event Outline**

It is used to inform the host that any setting value of the picture control was changed, or any setting value of the custom picture control was changed, registerd, renamed or deleted.

The changed menu (photo shooting menu or video recording menu) is set in Parameter 2.

| ShootingMode | Menu |
|---|---|
| 0x00000000 | Photo shooting menu |
| 0x00000001 | Video recording menu |

### 6.4.2.8 LiveViewStateChanged

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC10C |
| Event Parameter1 | CameraLiveViewStatus |

**○ Event Outline**

It is used to inform the host that the state of live view has changed on the camera.

| CameraLiveViewStatus | Description |
|---|---|
| 0 | OFF |
| 1 | ON |

### 6.4.2.9 PixelMappingComplete

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC10D |
| Event Parameter1 | Result |

**○ Event Outline**

It is used to inform the host that pixel mapping initiated by the InitiatePixelMapping command is complete.

| Code | Result |
|---|---|
| 0x00000000 | Failed |
| 0x00000001 | Completed |

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### 6.4.2.10 ManualSettingLensDataChanged

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC10E |
| Event Parameter1 | LensNo |

**○ Event Outline**

It is used to inform the host that the information of “Non-CPU lens data” in the setup menu has been changed.

The lens No. of the changed lens is passed as an EventParameter.

| LensNo | Description |
|---|---|
| Refer to ManualSettingLensNo property (section 6.5.5.1) | Lens No. of the changed lens |
| 0xFFFFFFFF | All lenses (Not supported on this camera) |

### 6.4.2.11 OpenCaptureComplete

**○ Event Specifications**

| | |
|---|---|
| Event Code | 0xC10F |
| Event Parameter1 | TransactionID |
| Event Parameter2 | EndFactor |

**○ Event Outline**

It is used to inform that interval timer shooting and focus shift shooting are completed.
The TransactionID of Parameter1 is the TransactionID of the InitiateOpenCaptureV command.
If shooting is started from the camera side, it will be 0.

The end factor is notified with EndFactor of Parameter2.

| EndFactor Code | End factor |
|---|---|
| 0x00000000 | Shooting completed |
| 0x00000001 | No free card space |
| 0x00000002 | No free SDRAM space |
| 0xFFFFFFFF | Other factors |

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### 6.4.2.12 SBAdded

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC120 |
| Event Parameter1 | SBHandle |

○ **Event Outline**

It is used to inform the host that the flash unit is newly attached to the camera.

### 6.4.2.13 SBRemoved

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC121 |
| Event Parameter1 | SBHandle |

○ **Event Outline**

It is used to inform the host that the flash unit is detached from the camera.

### 6.4.2.14 SBAttrChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC122 |
| Event Parameter1 | SBHandle |
| Event Parameter2 | SBAttrID |

○ **Event Outline**

It is used to inform the host that the attributes of the flash unit have changed.

This event is not issued when the attributes of the flash have changed by receiving SetSBAttrValue command from the host.

### 6.4.2.15 SBGroupAttrChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC123 |
| Event Parameter1 | SBGroupID |
| Event Parameter2 | SBGroupAttrID |

○ **Event Outline**

It is used to inform the host that the attribute descriptor for the flash group or the current value has changed.

This event is not issued when the attributes of the flash group have changed by receiving SetSBGroupAttrValue command from the host.

If the SBWirelessMode property (section 6.5.9.7) is set to other than “Stand-alone flash” and the SBWirelessMultipleFlashMode property (section 6.5.9.8) is set to “Repeating”, SBGroupID of the master flash is used to inform the changes for the following flash group regardless of the value of the SBUsableGroup property (section 6.5.9.9).

・SBGroupFlashLevel (section 6.7.2.4)
・SBGroupRepeatCount (section 6.7.2.7)
・SBGroupRepeatInterval (section 6.7.2.8)

Do Not Copy

### 6.4.2.16 HlgQualityAdjustChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC125 |
| Event Parameter1 | None |

○ **Event Outline**

It is used to inform that the HLG image quality adjustment settings have been changed.

### 6.4.2.17 PortraitImpressionBalanceAdjustChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC126 |
| Event Parameter1 | ModeNum<br>Following PortraitImpressionBalance property (section 6.5.2.77), but "OFF" is an invalid value. |
| Event Parameter2 | ShootingMode |

○ **Event Outline**

It is used to notify that the adjustment value of the Portrait impression adjustment has been changed and the changed mode number.

In Parameter2, specify whether the photo shooting menu or the video recording menu has changed.

| ShootingMode | Menu |
|---|---|
| 0x00000000 | Photo shooting menu |
| 0x00000001 | Video recording menu |

### 6.4.2.18 PictureControlHlgAdjustChanged

○ **Event Specifications**

| | |
|---|---|
| Event Code | 0xC127 |
| Event Parameter1 | PicCtrlHlgItem (ActivePicCtrlHlgItem property section 6.5.13.3) |

○ **Event Outline**

It is used to notify that each item of Picture Control (HLG) has been changed.

Do Not Copy

# 6.5 DevicePropCode

The camera has an attribute that can be changed as an option. The change is made by operating the device property. The property shows the device characteristics. Each property has a corresponding DevicePropCode.

When the setting value of each defined property is changed, the camera sends the DevicePropChanged event including the DevicePropCode in order to inform the host of the change. The camera sends the event as shown below.

1. The camera periodically (every 1 second) checks all the values of DeviceProperty. When a DeviceProperty whose value has changed since the last checking is detected, the camera sends the DevicePropChanged event in order to inform the host of the change. When detecting changes in two or more DeviceProperties, the camera sends the DevicePropChanged event for all the DeviceProperties in succession.

2. When the setting value of the DeviceProperty is changed by the SetDevicePropValue command, the camera does not send the DevicePropChanged event including the changed DevicePropCode. However, if any other DeviceProperty is changed under the influence of the change of the DeviceProperty, the camera sends the DevicePropChanged event for the property. For example, if the aperture value is changed by the SetDevicePropValue command when shooting is performed in the A mode (aperture priority), the camera changes the shutter speed automatically. In this case, the camera sends the DevicePropChanged event for the shutter speed.

However, depending on the type of PropertyCode, it is necessary to issue further events following the DevicePropChanged event. It is described in the explanation of each PropertyCode.

The error response is made to GetDevicePropDesc, GetDevicePropValue, SetDevicePropValue, and their extended commands as shown below.

1. When either or both of getting/setting are invalid depending on the setting status of the camera for each property, the response of the ResponseCode corresponding to the invalid status is made.

2. When setting is performed for the property that supports getting only, the Access_Denied error response is made.

The properties whose values cannot be set in photo mode, video mode or during video recording are described in the following table of DevicePropCodes instead of Response Code of each property. Access_Denied response is sent when the value is set in a state where setting is not possible.

The DevicePropCodes supported by the camera are shown below.

| Property Code | Description | Reference | Photo mode | Video mode | During video recording |
|---|---|---|---|---|---|
| 0x5001 | BatteryLevel | 6.5.1.1 | | | |
| 0x5003 | ImageSize | 6.5.1.2 | ✓ | ✓ | – |
| 0x5004 | CompressionSetting | 6.5.1.3 | ✓ | – | – |
| 0x5005 | WhiteBalance | 6.5.1.4 | ✓ | ✓ | ✓ |
| 0x5007 | Fnumber | 6.5.1.5 | ✓ | ✓ | ✓ |
| 0x5008 | FocalLength | 6.5.1.6 | | | |
| 0x500A | FocusMode | 6.5.1.7 | | | |
| 0x500B | ExposureMeteringMode | 6.5.1.8 | ✓ | ✓ | – |
| 0x500C | FlashMode | 6.5.1.9 | ✓ | – | – |

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| | | | | | |
|---|---|---|---|---|---|
| 0x500D | ExposureTime | 6.5.1.10 | ✓ | – | – |
| 0x500E | ExposureProgramMode | 6.5.1.11 | ✓ | ✓ | – |
| 0x500F | ExposureIndex | 6.5.1.12 | ✓ | ✓ | ✓ |
| 0x5010 | ExposureBiasCompensation | 6.5.1.13 | ✓ | ✓ | ✓ |
| 0x5011 | DateTime | 6.5.1.14 | ✓ | ✓ | – |
| 0x5013 | StillCaptureMode | 6.5.1.15 | ✓ | ✓ | ✓ |
| 0x5018 | BurstNumber | 6.5.1.16 | ✓ | ✓ | ✓ |
| 0x501E | Artist | 6.5.1.17 | | | |
| 0x501F | Copyright | 6.5.1.18 | | | |
| 0xD010 | ShootingMenuBank | 6.5.2.1 | ✓ | ✓ | – |
| 0xD011 | ShootingBankNameA | 6.5.2.2 | ✓ | ✓ | – |
| 0xD012 | ShootingBankNameB | 6.5.2.3 | ✓ | ✓ | – |
| 0xD013 | ShootingBankNameC | 6.5.2.4 | ✓ | ✓ | – |
| 0xD014 | ShootingBankNameD | 6.5.2.5 | ✓ | ✓ | – |
| 0xD016 | RawCompressionType | 6.5.2.13 | ✓ | ✓ | – |
| 0xD017 | WbTuneAuto | 6.5.2.21 | ✓ | ✓ | ✓ |
| 0xD018 | WbTuneIncandescent | 6.5.2.26 | ✓ | ✓ | ✓ |
| 0xD019 | WbTuneFluorescent | 6.5.2.28 | ✓ | ✓ | ✓ |
| 0xD01A | WbTuneSunny | 6.5.2.23 | ✓ | ✓ | ✓ |
| 0xD01B | WbTuneFlash | 6.5.2.29 | ✓ | ✓ | ✓ |
| 0xD01C | WbTuneCloudy | 6.5.2.24 | ✓ | ✓ | ✓ |
| 0xD01D | WbTuneShade | 6.5.2.25 | ✓ | ✓ | ✓ |
| 0xD01E | WbColorTemp | 6.5.2.30 | ✓ | ✓ | ✓ |
| 0xD01F | WbPresetDataNo | 6.5.2.32 | ✓ | ✓ | ✓ |
| 0xD021 | WbPresetDataComment1 | 6.5.2.33 | ✓ | ✓ | – |
| 0xD022 | WbPresetDataComment2 | 6.5.2.34 | ✓ | ✓ | – |
| 0xD023 | WbPresetDataComment3 | 6.5.2.35 | ✓ | ✓ | – |
| 0xD024 | WbPresetDataComment4 | 6.5.2.36 | ✓ | ✓ | – |
| 0xD026 | WbPresetDataValue1 | 6.5.2.39 | | | |
| 0xD027 | WbPresetDataValue2 | 6.5.2.40 | | | |
| 0xD028 | WbPresetDataValue3 | 6.5.2.41 | | | |
| 0xD029 | WbPresetDataValue4 | 6.5.2.42 | | | |
| 0xD030 | CaptureAreaCrop | 6.5.2.10 | ✓ | ✓ | – |
| 0xD032 | ColorSpace | 6.5.2.57 | ✓ | ✓ | – |
| 0xD034 | DecreaseFlicker | 6.5.3.56 | ✓ | ✓ | – |
| 0xD038 | WbPresetDataComment5 | 6.5.2.37 | ✓ | ✓ | – |
| 0xD039 | WbPresetDataComment6 | 6.5.2.38 | ✓ | ✓ | – |
| 0xD03A | WbTunePreset5 | 6.5.2.49 | ✓ | ✓ | ✓ |
| 0xD03B | WbTunePreset6 | 6.5.2.50 | ✓ | ✓ | ✓ |
| 0xD03C | WbPresetProtect5 | 6.5.2.55 | ✓ | ✓ | – |
| 0xD03D | WbPresetProtect6 | 6.5.2.56 | ✓ | ✓ | – |
| 0xD03E | WbPresetDataValue5 | 6.5.2.43 | | | |
| 0xD03F | WbPresetDataValue6 | 6.5.2.44 | | | |
| 0xD040 | CustomSettingBank | 6.5.4.1 | ✓ | ✓ | – |
| 0xD041 | CustomBankNameA | 6.5.4.2 | ✓ | ✓ | – |
| 0xD042 | CustomBankNameB | 6.5.4.3 | ✓ | ✓ | – |
| 0xD043 | CustomBankNameC | 6.5.4.4 | ✓ | ✓ | – |
| 0xD044 | CustomBankNameD | 6.5.4.5 | ✓ | ✓ | – |

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| | | | | | |
|---|---|---|---|---|---|
| 0xD046 | AFStillLockOnAcross | 6.5.4.8 | ✓ | ✓ | – |
| 0xD047 | AFStillLockOnMove | 6.5.4.9 | ✓ | ✓ | – |
| 0xD048 | DynamicAFonAFC | 6.5.4.6 | ✓ | ✓ | – |
| 0xD049 | DynamicAFonAFS | 6.5.4.7 | ✓ | ✓ | – |
| 0xD04F | FocusAreaSelect | 6.5.4.12 | ✓ | ✓ | – |
| 0xD053 | EnableCopyright | 6.5.5.11 | ✓ | ✓ | – |
| 0xD054 | ISOAutoControl | 6.5.2.15 | ✓ | ✓ | – |
| 0xD055 | IsoStep | 6.5.4.13 | ✓ | ✓ | – |
| 0xD056 | ExposureEVStep | 6.5.4.14 | ✓ | ✓ | – |
| 0xD059 | CenterWeightedExRange | 6.5.4.16 | ✓ | ✓ | – |
| 0xD05A | ExposureBaseCompMatrix | 6.5.4.17 | ✓ | ✓ | – |
| 0xD05B | ExposureBaseCompCenter | 6.5.4.18 | ✓ | ✓ | – |
| 0xD05C | ExposureBaseCompSpot | 6.5.4.19 | ✓ | ✓ | – |
| 0xD05D | StillFocusMeteringMode | 6.5.2.67 | ✓ | ✓ | – |
| 0xD060 | IPTCPresetSelect | 6.5.5.12 | ✓ | ✓ | – |
| 0xD061 | StillFocusMode | 6.5.2.66 | ✓ | ✓ | – |
| 0xD067 | AngleLevel | 6.5.6.19 | | | |
| 0xD068 | CspeedLow | 6.5.4.23 | ✓ | ✓ | – |
| 0xD069 | BurstMaxNumber | 6.5.4.24 | ✓ | ✓ | – |
| 0xD06A | ExposureDelay | 6.5.4.27 | ✓ | ✓ | – |
| 0xD06B | NoiseReduction | 6.5.2.59 | ✓ | ✓ | – |
| 0xD06C | NumberingMode | 6.5.4.29 | ✓ | ✓ | – |
| 0xD070 | NoiseReductionHiIso | 6.5.2.60 | ✓ | ✓ | – |
| 0xD072 | ArtistV | 6.5.5.9 | ✓ | ✓ | – |
| 0xD073 | CopyrightV | 6.5.5.10 | ✓ | ✓ | – |
| 0xD074 | FlashSyncSpeed | 6.5.4.39 | ✓ | ✓ | – |
| 0xD075 | FlashSlowSpeedLimit | 6.5.4.40 | ✓ | ✓ | – |
| 0xD078 | BracketingType | 6.5.8.2 | ✓ | ✓ | – |
| 0xD079 | SelectBracketChangeFactor | 6.5.4.43 | ✓ | ✓ | – |
| 0xD07A | BracketingOrder | 6.5.4.44 | ✓ | ✓ | – |
| 0xD07D | AngleLevelPitching | 6.5.6.20 | | | |
| 0xD07E | AngleLevelYawing | 6.5.6.21 | | | |
| 0xD07F | ExtendShootingMenu | 6.5.2.6 | ✓ | ✓ | – |
| 0xD08A | EnableShutter | 6.5.5.15 | ✓ | ✓ | – |
| 0xD08D | EnableAFAreaPoint | 6.5.4.10 | ✓ | ✓ | – |
| 0xD08F | ImageSensorCleaning | 6.5.5.6 | ✓ | ✓ | – |
| 0xD090 | CommentString | 6.5.5.7 | ✓ | ✓ | – |
| 0xD091 | EnableComment | 6.5.5.8 | ✓ | ✓ | – |
| 0xD092 | OrientationSensorMode | 6.5.6.17 | ✓ | ✓ | – |
| 0xD093 | ManualSettingLensNo | 6.5.5.1 | ✓ | ✓ | – |
| 0xD09C | RetractableLensWarning | 6.5.10.9 | | | |
| 0xD09F | MatrixMetering | 6.5.4.15 | ✓ | ✓ | – |
| 0xD0A0 | MovieRecordScreenSize | 6.5.3.7 | ✓ | ✓ | – |
| 0xD0A2 | MovieRecordMicrophoneLevel | 6.5.3.69 | ✓ | ✓ | ✓ |
| 0xD0A3 | MovieRecordDestination | 6.5.3.1 | ✓ | ✓ | – |
| 0xD0A4 | MovieRecProhibitionCondition | 6.5.12.5 | | | |
| 0xD0A8 | MovieRecordMicrophoneLevelValue | 6.5.3.70 | ✓ | ✓ | ✓ |
| 0xD0AA | MovieWindNoiseReduction | 6.5.3.73 | ✓ | ✓ | ✓ |

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| | | | | | |
|---|---|---|---|---|---|
| 0xD0AC | MovieRecordingZone | 6.5.3.72 | ✓ | ✓ | ✓ |
| 0xD0AD | MovieISOAutoControl | 6.5.3.12 | ✓ | ✓ | ✓ |
| 0xD0AE | MovieISOAutoHighLimit | 6.5.3.11 | ✓ | ✓ | ✓ |
| 0xD0AF | MovieFileType | 6.5.3.2 | ✓ | ✓ | – |
| 0xD0B4 | ExposureIndexEx | 6.5.2.14 | ✓ | ✓ | ✓ |
| 0xD0B5 | ISOControlSensitivity | 6.5.6.12 | | | |
| 0xD0B7 | FlickerReductionSetting | 6.5.2.64 | ✓ | ✓ | – |
| 0xD0BA | DiffractionCompensation | 6.5.2.62 | ✓ | ✓ | – |
| 0xD0BB | MovieLogOutput | 6.5.12.4 | | | |
| 0xD0BC | MovieAutoDistortion | 6.5.3.55 | ✓ | ✓ | – |
| 0xD0BE | RemainingExposureTime | 6.5.6.9 | | | |
| 0xD0C0 | EnableBracketing | 6.5.8.1 | ✓ | ✓ | – |
| 0xD0C1 | AEBracketingStep | 6.5.8.4 | ✓ | ✓ | – |
| 0xD0C2 | AEBracketingPattern | 6.5.8.3 | ✓ | ✓ | – |
| 0xD0C3 | AEBracketingCount | 6.5.8.9 | | | |
| 0xD0C4 | WBBracketingStep | 6.5.8.6 | ✓ | ✓ | – |
| 0xD0C5 | WBBracketingPattern | 6.5.8.5 | ✓ | ✓ | – |
| 0xD0C6 | ADLBracketingPattern | 6.5.8.7 | ✓ | ✓ | – |
| 0xD0C7 | ADLBracketingStep | 6.5.8.8 | ✓ | ✓ | – |
| 0xD0CD | SaveFocusPosition | 6.5.5.5 | ✓ | ✓ | – |
| 0xD0E0 | LensID | 6.5.10.4 | | | |
| 0xD0E1 | LensSort | 6.5.10.1 | | | |
| 0xD0E2 | LensTypeF | 6.5.10.3 | | | |
| 0xD0E3 | LensFocalMin | 6.5.10.5 | | | |
| 0xD0E4 | LensFocalMax | 6.5.10.6 | | | |
| 0xD0E5 | LensApatureMin | 6.5.10.7 | | | |
| 0xD0E6 | LensApatureMax | 6.5.10.8 | | | |
| 0xD0E7 | LensTypeML | 6.5.10.2 | | | |
| 0xD0F7 | VignetteControl | 6.5.2.61 | ✓ | ✓ | – |
| 0xD0F8 | AutoDistortion | 6.5.2.63 | ✓ | ✓ | – |
| 0xD100 | ShutterSpeed | 6.5.6.7 | ✓ | – | – |
| 0xD101 | ExternalDC-IN | 6.5.14.1 | | | |
| 0xD102 | WarningStatus | 6.5.6.22 | | | |
| 0xD104 | AFLockStatus | 6.5.6.4 | | | |
| 0xD105 | AELockStatus | 6.5.6.3 | | | |
| 0xD106 | FVLockStatus | 6.5.6.5 | | | |
| 0xD109 | FlexibleProgram | 6.5.6.10 | ✓ | – | – |
| 0xD10B | RecordingMedia | 6.5.7.1 | ✓ | ✓ | – |
| 0xD10C | USBSpeed | 6.5.15.1 | | | |
| 0xD10E | Orientation | 6.5.6.18 | | | |
| 0xD112 | TVLockSetting | 6.5.4.46 | ✓ | ✓ | – |
| 0xD113 | AVLockSetting | 6.5.4.47 | ✓ | ✓ | – |
| 0xD118 | ExposureCompFlashUsed | 6.5.4.41 | ✓ | ✓ | – |
| 0xD119 | Slot2JpegImageSize | 6.5.2.9 | ✓ | ✓ | – |
| 0xD120 | ExternalSpeedLightExist | 6.5.9.1 | | | |
| 0xD121 | ExternalSpeedLightStatus | 6.5.9.3 | | | |
| 0xD122 | ExternalSpeedLightSort | 6.5.9.2 | | | |
| 0xD124 | FlashCompensation | 6.5.9.5 | | | |

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| | | | | | |
|---|---|---|---|---|---|
| 0xD125 | NewExternalSpeedLightMode | 6.5.9.4 | | | |
| 0xD126 | InternalFlashCompensation | 6.5.6.11 | ✓ | – | – |
| 0xD12D | ExternalSpeedLightMultiFlashMode | 6.5.9.6 | | | |
| 0xD12E | ConnectionPath | 6.5.15.2 | | | |
| 0xD12F | AWBLockStatus | 6.5.6.6 | | | |
| 0xD130 | HDRMode | 6.5.2.72 | ✓ | ✓ | – |
| 0xD132 | HDRStrength | 6.5.2.73 | ✓ | ✓ | – |
| 0xD133 | HDRSaveIndividualImages | 6.5.2.74 | ✓ | ✓ | – |
| 0xD138 | VibrationReduction | 6.5.2.71 | ✓ | ✓ | – |
| 0xD141 | WbAutoType | 6.5.2.20 | ✓ | ✓ | ✓ |
| 0xD148 | Slot2ImageSaveMode | 6.5.2.8 | ✓ | ✓ | – |
| 0xD14E | Active-D-Lighting | 6.5.2.58 | ✓ | ✓ | – |
| 0xD14F | WbFluorescentType | 6.5.2.27 | ✓ | ✓ | ✓ |
| 0xD150 | WbTuneColorTemp | 6.5.2.31 | ✓ | ✓ | ✓ |
| 0xD152 | WbTunePreset1 | 6.5.2.45 | ✓ | ✓ | ✓ |
| 0xD153 | WbTunePreset2 | 6.5.2.46 | ✓ | ✓ | ✓ |
| 0xD154 | WbTunePreset3 | 6.5.2.47 | ✓ | ✓ | ✓ |
| 0xD155 | WbTunePreset4 | 6.5.2.48 | ✓ | ✓ | ✓ |
| 0xD156 | PrimarySlot | 6.5.2.7 | ✓ | ✓ | – |
| 0xD158 | WbPresetProtect1 | 6.5.2.51 | ✓ | ✓ | – |
| 0xD159 | WbPresetProtect2 | 6.5.2.52 | ✓ | ✓ | – |
| 0xD15A | WbPresetProtect3 | 6.5.2.53 | ✓ | ✓ | – |
| 0xD15B | ActiveFolder | 6.5.7.2 | | | |
| 0xD15C | WbPresetProtect4 | 6.5.2.54 | ✓ | ✓ | – |
| 0xD15E | WbTuneNatural | 6.5.2.22 | ✓ | ✓ | – |
| 0xD164 | ISOAutoShutterTime | 6.5.2.18 | ✓ | ✓ | – |
| 0xD16E | ExtendedShutterSpeedsManual | 6.5.4.28 | ✓ | ✓ | – |
| 0xD177 | FocusModeRestrictions | 6.5.4.11 | ✓ | ✓ | – |
| 0xD17B | ViewMode | 6.5.4.30 | ✓ | ✓ | – |
| 0xD17C | MovieAfSpeed | 6.5.4.52 | ✓ | ✓ | – |
| 0xD17D | MovieAfSpeedWhenToApply | 6.5.4.53 | ✓ | ✓ | – |
| 0xD17E | MovieAfTrackingSensitivity | 6.5.4.54 | ✓ | ✓ | – |
| 0xD180 | FlushBurstPrioritize | 6.5.4.45 | ✓ | ✓ | – |
| 0xD183 | ISOAutoHighLimit | 6.5.2.16 | ✓ | ✓ | – |
| 0xD186 | CSpeedHigh | 6.5.4.22 | ✓ | ✓ | – |
| 0xD199 | FlashISOAutoHighLimit | 6.5.2.17 | ✓ | ✓ | – |
| 0xD1A2 | RemoteLiveViewStatus | 6.5.11.2 | | | |
| 0xD1A4 | LiveViewProhibitionCondition | 6.5.11.7 | | | |
| 0xD1A6 | LiveViewSelector | 6.5.6.1 | ✓ | ✓ | – |
| 0xD1A8 | MovieShutterSpeed | 6.5.12.2 | – | ✓ | ✓ |
| 0xD1A9 | MovieFnumber | 6.5.12.1 | – | ✓ | ✓ |
| 0xD1AA | MovieExposureIndex | 6.5.3.13 | ✓ | ✓ | ✓ |
| 0xD1AB | MovieExposureBiasCompensation | 6.5.12.3 | – | ✓ | ✓ |
| 0xD1AC | LiveViewImageSize | 6.5.11.4 | ✓ | ✓ | – |
| 0xD1AF | MovieExposureMeteringMode | 6.5.3.60 | ✓ | ✓ | – |
| 0xD1B0 | ExposureDisplayStatus | 6.5.6.13 | | | |
| 0xD1B1 | ExposureIndicateStatus | 6.5.6.14 | | | |
| 0xD1B2 | InfoDisplayErrorStatus | 6.5.6.23 | | | |

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|---|---|---|---|---|---|
| 0xD1B3 | ExposureIndicateLightup | 6.5.6.15 | | | |
| 0xD1B4 | ContinuousShootingCount | 6.5.7.5 | | | |
| 0xD1B8 | CameraLiveViewStatus | 6.5.11.1 | | | |
| 0xD1BA | MonitorOffStatus | 6.5.6.16 | | | |
| 0xD1BB | LiveViewImageStatus | 6.5.11.3 | | | |
| 0xD1BC | LiveViewImageCompression | 6.5.11.5 | ✓ | ✓ | – |
| 0xD1BD | LiveViewZoomArea | 6.5.11.6 | ✓ | ✓ | ✓ |
| 0xD1C5 | AutoFlashIsoSensitivityControl | 6.5.4.42 | ✓ | ✓ | – |
| 0xD1C6 | KeepExposureWhenFNumberChanges | 6.5.4.21 | ✓ | ✓ | – |
| 0xD1DE | ExternalRecordingControl | 6.5.3.78 | ✓ | ✓ | – |
| 0xD1DF | HighlightBrightness | 6.5.4.55 | ✓ | ✓ | ✓ |
| 0xD1E2 | SBWirelessMode | 6.5.9.7 | ✓ | – | – |
| 0xD1E3 | SBWirelessMultipleFlashMode | 6.5.9.8 | ✓ | – | – |
| 0xD1E4 | SBUsableGroup | 6.5.9.9 | | | |
| 0xD1E5 | WirelessCLSEntryMode | 6.5.9.10 | ✓ | ✓ | – |
| 0xD1E6 | SBPINCode | 6.5.9.11 | ✓ | ✓ | – |
| 0xD1E7 | RadioMultipleFlashChannel | 6.5.9.12 | | | |
| 0xD1E8 | OpticalMultipleFlashChannel | 6.5.9.13 | ✓ | – | – |
| 0xD1E9 | FlashRangeDisplay | 6.5.9.14 | | | |
| 0xD1EA | AllTestFiringDisable | 6.5.9.15 | | | |
| 0xD1EC | SBSettingMemberLock | 6.5.9.16 | ✓ | – | – |
| 0xD1ED | SBIntegrationFlashReady | 6.5.9.17 | | | |
| 0xD1F1 | ExposureRemaining | 6.5.7.4 | | | |
| 0xD1F2 | ActiveSlot | 6.5.7.3 | | | |
| 0xD1F4 | ISOAutoShutterTimeCorrectionValue | 6.5.2.19 | ✓ | ✓ | – |
| 0xD1F8 | MovieFocusMeteringMode | 6.5.3.62 | ✓ | ✓ | ✓ |
| 0xD1F9 | MovieVibrationReduction | 6.5.3.67 | ✓ | ✓ | ✓ |
| 0xD1FA | MovieFocusMode | 6.5.3.61 | ✓ | ✓ | ✓ |
| 0xD1FB | RecordTimeCodes | 6.5.3.74 | ✓ | ✓ | – |
| 0xD1FC | CountUpMethod | 6.5.3.75 | ✓ | ✓ | – |
| 0xD1FD | TimeCodeOrigin | 6.5.3.76 | ✓ | ✓ | – |
| 0xD1FE | DropFrame | 6.5.3.77 | ✓ | ✓ | – |
| 0xD200 | ActivePicCtrlItem | 6.5.13.1 | ✓ | ✓ | – |
| 0xD20F | MovieCaptureAreaCrop | 6.5.3.9 | ✓ | ✓ | – |
| 0xD211 | MovieWbAutoType | 6.5.3.15 | ✓ | ✓ | ✓ |
| 0xD212 | MovieWbTuneAuto | 6.5.3.16 | ✓ | ✓ | ✓ |
| 0xD213 | MovieWbTuneIncandescent | 6.5.3.21 | ✓ | ✓ | ✓ |
| 0xD214 | MovieWbFluorescentType | 6.5.3.22 | ✓ | ✓ | ✓ |
| 0xD215 | MovieWbTuneFluorescent | 6.5.3.23 | ✓ | ✓ | ✓ |
| 0xD216 | MovieWbTuneSunny | 6.5.3.18 | ✓ | ✓ | ✓ |
| 0xD218 | MovieWbTuneCloudy | 6.5.3.19 | ✓ | ✓ | ✓ |
| 0xD219 | MovieWbTuneShade | 6.5.3.20 | ✓ | ✓ | ✓ |
| 0xD21A | MovieWbColorTemp | 6.5.3.24 | ✓ | ✓ | ✓ |
| 0xD21B | MovieWbTuneColorTemp | 6.5.3.25 | ✓ | ✓ | ✓ |
| 0xD21C | MovieWbPresetDataNo | 6.5.3.26 | ✓ | ✓ | ✓ |
| 0xD21D | MovieWbPresetDataComment1 | 6.5.3.27 | ✓ | ✓ | – |
| 0xD21E | MovieWbPresetDataComment2 | 6.5.3.28 | ✓ | ✓ | – |
| 0xD21F | MovieWbPresetDataComment3 | 6.5.3.29 | ✓ | ✓ | – |

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|---|---|---|---|---|---|
| 0xD220 | MovieWbPresetDataComment4 | 6.5.3.30 | ✓ | ✓ | – |
| 0xD221 | MovieWbPresetDataComment5 | 6.5.3.31 | ✓ | ✓ | – |
| 0xD222 | MovieWbPresetDataComment6 | 6.5.3.32 | ✓ | ✓ | – |
| 0xD223 | MovieWbPresetDataValue1 | 6.5.3.33 | | | |
| 0xD224 | MovieWbPresetDataValue2 | 6.5.3.34 | | | |
| 0xD225 | MovieWbPresetDataValue3 | 6.5.3.35 | | | |
| 0xD226 | MovieWbPresetDataValue4 | 6.5.3.36 | | | |
| 0xD227 | MovieWbPresetDataValue5 | 6.5.3.37 | | | |
| 0xD228 | MovieWbPresetDataValue6 | 6.5.3.38 | | | |
| 0xD229 | MovieWbTunePreset1 | 6.5.3.39 | ✓ | ✓ | ✓ |
| 0xD22A | MovieWbTunePreset2 | 6.5.3.40 | ✓ | ✓ | ✓ |
| 0xD22B | MovieWbTunePreset3 | 6.5.3.41 | ✓ | ✓ | ✓ |
| 0xD22C | MovieWbTunePreset4 | 6.5.3.42 | ✓ | ✓ | ✓ |
| 0xD22D | MovieWbTunePreset5 | 6.5.3.43 | ✓ | ✓ | ✓ |
| 0xD22E | MovieWbTunePreset6 | 6.5.3.44 | ✓ | ✓ | ✓ |
| 0xD22F | MovieWbPresetProtect1 | 6.5.3.45 | ✓ | ✓ | – |
| 0xD230 | MovieWbPresetProtect2 | 6.5.3.46 | ✓ | ✓ | – |
| 0xD231 | MovieWbPresetProtect3 | 6.5.3.47 | ✓ | ✓ | – |
| 0xD232 | MovieWbPresetProtect4 | 6.5.3.48 | ✓ | ✓ | – |
| 0xD233 | MovieWbPresetProtect5 | 6.5.3.49 | ✓ | ✓ | – |
| 0xD234 | MovieWbPresetProtect6 | 6.5.3.50 | ✓ | ✓ | – |
| 0xD236 | MovieNoiseReductionHiIso | 6.5.3.52 | ✓ | ✓ | – |
| 0xD237 | MovieActivePicCtrlItem | 6.5.13.2 | ✓ | ✓ | – |
| 0xD239 | ExposureBaseCompHighlight | 6.5.4.20 | ✓ | ✓ | – |
| 0xD23A | MovieWhiteBalance | 6.5.3.14 | ✓ | ✓ | ✓ |
| 0xD23B | MovieActive-D-Lighting | 6.5.3.51 | ✓ | ✓ | – |
| 0xD23C | MovieWbTuneNatural | 6.5.3.17 | ✓ | ✓ | ✓ |
| 0xD23D | MovieAttenuator | 6.5.3.71 | ✓ | ✓ | ✓ |
| 0xD23E | MovieVignetteControl | 6.5.3.53 | ✓ | ✓ | – |
| 0xD23F | MovieDiffractionCompensation | 6.5.3.54 | ✓ | ✓ | – |
| 0xD303 | UseDeviceStageFlag | 6.5.16.3 | | | |
| 0xD314 | ElectronicVR | 6.5.3.68 | ✓ | ✓ | – |
| 0xD406 | SessionInitiatorVersionInfo | 6.5.16.1 | ✓ | ✓ | – |
| 0xD407 | PerceivedDeviceType | 6.5.16.2 | | | |
| 0x0001D000 | MovieProResToneMode | 6.5.3.5 | ✓ | ✓ | – |
| 0x0001D001 | MovieH265ToneMode | 6.5.3.3 | ✓ | ✓ | – |
| 0x0001D002 | SpecialShootingMode | 6.5.6.2 | | | |
| 0x0001D005 | AFSubjectDetection | 6.5.2.70 | ✓ | ✓ | – |
| 0x0001D006 | MovieAFSubjectDetection | 6.5.3.65 | ✓ | ✓ | ✓ |
| 0x0001D007 | ViewModeEaseOfViewing | 6.5.4.32 | ✓ | ✓ | – |
| 0x0001D008 | ViewModeWhiteBalance | 6.5.4.33 | ✓ | ✓ | – |
| 0x0001D009 | ViewModeWbColorTemp | 6.5.4.34 | ✓ | ✓ | – |
| 0x0001D00A | ViewModeWbTuneColorTemp | 6.5.4.35 | ✓ | ✓ | – |
| 0x0001D00B | ViewModePictureControl | 6.5.4.36 | ✓ | ✓ | – |
| 0x0001D00C | ViewModeBrightenShadow | 6.5.4.37 | ✓ | ✓ | – |
| 0x0001D00E | StarlightView | 6.5.4.38 | ✓ | ✓ | – |
| 0x0001D00F | MovieTVLockSetting | 6.5.4.49 | ✓ | ✓ | – |
| 0x0001D010 | MovieAVLockSetting | 6.5.4.50 | ✓ | ✓ | – |

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| | | | | | |
|---|---|---|---|---|---|
| 0x0001D011 | SilentMode | 6.5.5.14 | ✓ | ✓ | ✓ |
| 0x0001D012 | ShutterSoundEffect | 6.5.5.13 | ✓ | ✓ | – |
| 0x0001D014 | AFWhenSubjectNotDetected | 6.5.3.66 | ✓ | ✓ | – |
| 0x0001D015 | FocusPointLock | 6.5.4.48 | ✓ | ✓ | – |
| 0x0001D016 | MovieFocusPointLock | 6.5.4.51 | ✓ | ✓ | – |
| 0x0001D017 | SkinSoftening | 6.5.2.76 | ✓ | ✓ | – |
| 0x0001D018 | MovieSkinSoftening | 6.5.3.57 | ✓ | ✓ | – |
| 0x0001D019 | PortraitImpressionBalance | 6.5.2.77 | ✓ | ✓ | – |
| 0x0001D01A | MoviePortraitImpressionBalance | 6.5.3.58 | ✓ | ✓ | – |
| 0x0001D01B | MovieExtendedShutterSpeedsManual | 6.5.4.57 | ✓ | ✓ | – |
| 0x0001D01C | ToneMode | 6.5.2.75 | ✓ | ✓ | – |
| 0x0001D01D | ActivePicCtrlHlgItem | 6.5.13.3 | ✓ | ✓ | – |
| 0x0001D01E | ExtendedOverSampling | 6.5.3.10 | ✓ | ✓ | – |
| 0x0001D01F | MovieExtendedIsoStepManual | 6.5.4.56 | ✓ | ✓ | – |
| 0x0001D025 | PreCapturePreReleaseBurst | 6.5.4.25 | ✓ | ✓ | – |
| 0x0001D026 | PreCapturePostReleaseBurst | 6.5.4.26 | ✓ | ✓ | – |
| 0x0001D028 | MovieNRaw12bitToneMode | 6.5.3.4 | ✓ | ✓ | – |
| 0x0001D029 | MovieProResRawHq12bitToneMode | 6.5.3.6 | ✓ | ✓ | – |
| 0x0001D02A | MovieQualityNRaw | 6.5.3.8 | ✓ | ✓ | – |
| 0x0001D031 | HighFrequencyFlickerReduction | 6.5.2.65 | ✓ | ✓ | – |
| 0x0001D032 | MovieHighFrequencyFlickerReduction | 6.5.3.59 | ✓ | ✓ | ✓ |
| 0x0001D033 | HiResZoom | 6.5.3.79 | ✓ | ✓ | – |
| 0x0001D036 | ViewModeShowEffectsOfSettings | 6.5.4.31 | ✓ | ✓ | – |
| 0x0001D03D | FlickerReductionShutterSpeed | 6.5.6.8 | | | |
| 0x0001D041 | EnableDXImageSize | 6.5.2.11 | ✓ | ✓ | – |
| 0x0001D042 | ImageSizeDX | 6.5.2.12 | ✓ | ✓ | – |
| 0x0001D046 | WideAreaAFC1 | 6.5.2.68 | ✓ | ✓ | – |
| 0x0001D045 | PixelShiftShootingMode | 6.5.2.78 | ✓ | ✓ | – |
| 0x0001D047 | WideAreaAFC2 | 6.5.2.69 | ✓ | ✓ | – |
| 0x0001D048 | MovieWideAreaAFC1 | 6.5.3.63 | ✓ | ✓ | ✓ |
| 0x0001D049 | MovieWideAreaAFC2 | 6.5.3.64 | ✓ | ✓ | ✓ |
| 0x0001D04A | FmmManualSetting2 | 6.5.5.2 | ✓ | ✓ | – |
| 0x0001D04B | F0ManualSetting2 | 6.5.5.3 | ✓ | ✓ | – |
| 0x0001D04C | ManualSettingLensName | 6.5.5.4 | ✓ | ✓ | – |
| 0x0001D050 | PixelShiftShootingInterval | 6.5.2.79 | ✓ | ✓ | – |

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# 6.5.1 Standard

## 6.5.1.1 BatteryLevel

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x5001 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 100 [100 %] |
| Property Value | From 1 [1 %] to 100 [100 %] |

### ○ Property Outline

Indicates “Remaining battery level” of the camera.

### ○ Property Details

When the battery level is reduced, the PropertyValues sent by the camera are: 1%, 20%, 40%, 60%, 80%, and 100% only.

The relationship between the PropertyValue and the remaining battery level display is shown below.

| Property Value | Remaining battery level display |
|---|---|
| 100 | 5 / 5 |
| 80 | 4 / 5 |
| 60 | 3 / 5 |
| 40 | 2 / 5 |
| 20 | 1 / 5 |
| 1 | 1 / 5 (blinking) |

When the remaining battery level is 1% and the shutter release is disabled, the following settings are made.

| No. | Description |
|---|---|
| 1 | The WarningStatus property (section 6.5.6.22) is set to “Battery exhausted”. |
| 2 | The LiveViewProhibitionCondition property (section 6.5.11.7) is set to “Battery exhausted”. |

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### 6.5.1.2 ImageSize

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5003 |
| DataType | String |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | Size L |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Image size” in the photo shooting menu.

**○ Property Value**

The valid PropertyValues are shown below.

| | PropertyValue | | | |
|---|---|---|---|---|
| CaptureAreaCrop | FX | DX | 1:1 | 16:9 |
| Size L | 8256x5504 | 5392x3592 | 5504x5504 | 8256x4640 |
| Size M | 6192x4128 | 4032x2688 | 4128x4128 | 6192x3480 |
| Size S | 4128x2752 | 2688x1792 | 2752x2752 | 4128x2320 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The CompressionSetting property (section 6.5.1.3) is set to [RAW]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |

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### 6.5.1.3 CompressionSetting

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5004 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 2 [JPEG (NORMAL)] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Image quality mode” that is set in the camera.

**○ Property Details**

This property indicates the control value in the camera, not a setting value in the menu. This is a value including RAW when the Plus RAW function is valid.

When StorageInfo DataSet (section 9.2) is changed according to the change of this property, the StorageInfoChanged event (section 6.4.1.11) is issued.

When the StillCaptureMode property (section 6.5.1.15) is "High-speed frame capture", it operates in JPEG (NORMAL).

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | JPEG (BASIC) |
| 1 | JPEG (BASIC★) |
| 2 | JPEG (NORMAL) |
| 3 | JPEG (NORMAL★) |
| 4 | JPEG (FINE) |
| 5 | JPEG (FINE★) |
| 7 | RAW |
| 8 | RAW+JPEG (BASIC) |
| 9 | RAW+JPEG (BASIC★) |
| 10 | RAW+JPEG (NORMAL) |
| 11 | RAW+JPEG (NORMAL★) |
| 12 | RAW+JPEG (FINE) |
| 13 | RAW+JPEG (FINE★) |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |

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### 6.5.1.4 WhiteBalance

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5005 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0002 [Auto] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “White Balance” that is set in the camera.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0002 | Auto |
| 0x0004 | Sunny |
| 0x0005 | Fluorescent |
| 0x0006 | Incandescent |
| 0x0007 | Flash |
| 0x8010 | Cloudy |
| 0x8011 | Shade |
| 0x8012 | Choose color temp. |
| 0x8013 | Preset manual |
| 0x8016 | Natural light auto |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | During video recording, the MovieWhiteBalance (section 6.5.3.14) is set to anything other than [Same as photo settings]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.1.5 Fnumber

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5007 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | (Minimum value in the setting range) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Aperture value” with the CPU internal lens mounted.

**○ Property Details**

When the LensSort property (section 6.5.10.1) is [Not mounted], the enumeration cannot be created. Therefore, the number of enumeration values shall be 1 and the enumeration value, the DefaultValue, and the PropertyValue shall be the same value.

For models that support the exposure meter interlocking lever, it changes depending on the F0ManualSetting2 property (section 6.5.5.3) and the state of the exposure meter interlocking lever. The value shall be 1 EV except the maximum aperture value.

It is set to 0 if the model does not support the exposure meter interlocking lever or if the aperture value cannot be acquired.

If an aperture value error occurs, the number of enumeration values shall be 1 and the enumeration value, the DefaultValue, and the PropertyValue shall be 0xFFFF.

**○ Property Value**

The PropertyValue should be a hundred times the aperture value.

The PropertyValue changes depending on the value of the ExposureEVStep property (section 6.5.4.14).

The setting range of PropertyValue changes depending on the lens and the magnification setting.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExposureProgramMode property (section 6.5.1.11) is anything other than [A] or [M]. |
| | The LensSort property (section 6.5.10.1) is [Not mounted]. |
| | If the WarningStatus property (section 6.5.6.22) is as follows:<br>[Sequence error]<br>[Bulb warning / Time warning]<br>[Retractable zoom-ring warning]<br>[Shutter speed incompatibility, HDR shooting not possible] |
| | The AVLockSetting property (section 6.5.4.47) is set to [On]. |
| | The aperture value error. |

Do Not Copy

### 6.5.1.6 FocalLength

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5008 |
| DataType | UINT32 |
| Description form | Range |
| Get / Set | Get |
| Default Value | (Minimum value in the setting range) |
| Property Value | The value of a hundred times the focal length |

**○ Property Outline**

Indicates “Focal length” with the CPU internal lens mounted.

**○ Property Details**

The setting range of PropertyValue changes depending on the lens and the magnification setting.
When the LensSort property (section 6.5.10.1) is [Not mounted], the PropertyValue is not fixed.

Do Not Copy

### 6.5.1.7 FocusMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x500A |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0x8010 [Single AF] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Focus mode” that is set in the camera.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description | Photo mode | Video mode |
|---|---|---|---|
| 0x0001 | Manual focus (MF) | ✓ | ✓ |
| 0x8010 | Single AF(AF-S) | ✓ | ✓ |
| 0x8011 | Continuous AF(AF-C) | ✓ | ✓ |
| 0x8013 | Full time AF(AF-F) | – | ✓ |

Do Not Copy

### 6.5.1.8 ExposureMeteringMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x500B |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0003 [Matrix metering] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Metering mode” that is set in the camera.
Use the MovieExposureMeteringMode property (section 6.5.3.60) for the exposure metering mode in video mode.

**○ Property Details**

In the following cases, this property operates with center-weighted metering.

- The LensSort property (section 6.5.10.1) is set to [Not mounted] and highlight-weighted metering is selected.

- When the lens information is not registered in “Non-CPU lens data” menu, the LensSort property is set to [Not mounted] and matrix metering is selected.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0002 | Center-weighted metering |
| 0x0003 | Matrix metering |
| 0x0004 | Spot metering |
| 0x8010 | Highlight-weighted metering |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The AELockStatus property (section 6.5.6.3) is [Locked]. |

Do Not Copy

### 6.5.1.9 FlashMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x500C |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x8010 [Normal] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Flash mode” that is set in the camera.

**○ Property Details**

If Rear curtain is set when the NewExternalSpeedLightMode property (section 6.5.9.4) is set to Multi-flash, normal flash is set.

The camera display status depends on the value of the ExposureProgramMode property (section 6.5.1.11). In addition, it also differs according to the speedlight condition.

When the camera is in the Access_Denied condition of this property or in video mode, this property is set to [Flash off].

The camera display status with the speedlight firing is shown below.

| Shooting mode or selected scene | Normal flash | Normal flash + slow shutter | Rear curtain flash | Red-eye reduction flash | Red-eye reduction flash + slow shutter | Flash off |
|---|---|---|---|---|---|---|
| P / A | Normal flash | Normal flash + slow shutter | Rear curtain flash | Red-eye reduction flash | Red-eye reduction flash + slow shutter | Flash off |
| S / M | Normal flash | - | Rear curtain flash | Red-eye reduction flash | - | Flash off |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0002 | Flash off |
| 0x0004 | Red-eye reduction flash |
| 0x8010 | Normal flash |
| 0x8011 | Normal flash + slow shutter |
| 0x8012 | Rear curtain flash |
| 0x8013 | Red-eye reduction + slow shutter |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |
| | The SilentMode property (section 6.5.5.14) is set to [On]. |
| | During SB setting warning. |
| | During image dust off data acquiring mode |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON" and the PixelShiftShootingInterval property (section 6.5.2.79) is set to "0" |

Do Not Copy

### 6.5.1.10 ExposureTime

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x500D |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | (Minimum value in the setting range) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Shutter speed” that is set in the camera.

Since there are shutter speeds that cannot be expressed with this property, use the ShutterSpeed (section 6.5.6.7) property to obtain an accurate shutter speed.

For the shutter speed in video mode, the MovieShutterSpeed property (section 6.5.12.2) should be used.

**○ Property Details**

Setting prohibition conditions, contents of enumeration values, and factors for changing enumeration values are the same as for the ShutterSpeed property, but since the unit of PropertyValue is 1/10000, there is a range that is grouped together.

The handling within the range summarized below is described.

| PropertyValue | Shutter speed | |
|---|---|---|
| | Get | Set (NOTE1) |
| 1 | Shutter speed faster than 1/5000 | 1/10000<br>1/8000 |
| 2 | 1/5000<br>1/4000 | 1/4000 |
| 3 | 1/3200<br>1/3000 | 1/3200<br>1/3000 |
| 6 | 1/1600<br>1/1500 | 1/1600<br>1/1500 |

NOTE1) Where there are multiple descriptions, the shutter speed that can be set depending on the state of the camera is set.

**○ Property Value**

Shutter speed:

Shutter speed x 10000 [unit: 1/10000 sec.]

Special values other than shutter speed:

| PropertyValue | Description |
|---|---|
| 0xFFFFFFFF | Bulb shooting |
| 0xFFFFFFFD | Time shooting |

(Example) PropertyValue of shutter speed.

Shutter speed 1/250 sec.: PropertyValue = 40

Shutter speed 25 sec.: PropertyValue = 250000

Do Not Copy

### 6.5.1.11 ExposureProgramMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x500E |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0002 [Program auto] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Shooting mode” that is set in the camera.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description | Used setting |
|---|---|---|
| 0x0001 | [M] Manual | - |
| 0x0002 | [P] Program auto | - |
| 0x0003 | [A] Aperture priority auto | - |
| 0x0004 | [S] Shutter priority auto | - |

Do Not Copy

### 6.5.1.12 ExposureIndex

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x500F |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 100 [ISO100] |
| Property Value | Refer to the ExposureIndexEx property. |

**○ Property Outline**

Indicates “ISO sensitivity” that is set in the camera.
Refer to the MovieExposureIndex property (section 6.5.3.13) for the ISO sensitivity in video mode.

**○ Property Details**

The maximum value of ISO sensitivity that can be expressed by this property is up to the ISO sensitivity that can be handled with 16Bit.

Therefore, if the camera is set to an ISO sensitivity that exceeds 16Bit, the value of this property will be the maximum ISO sensitivity that can be expressed with 16Bit.

Use the ExposureIndexEx property (section 6.5.2.14) when dealing with ISO sensitivities over 16Bit.

Do Not Copy

## 6.5.1.13 ExposureBiasCompensation

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x5010 |
| DataType | INT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the compensation value of “Exposure compensation” that is set in the camera.
For the exposure compensation in video mode, the MovieExposureBiasCompensation property (section 6.5.12.3) should be used.

### ○ Property Value

The valid PropertyValues are shown below.
The value of PropertyValue changes depending on the value of the ExposureEVStep property (section 6.5.4.14).

| Property Value | Description | ExposureEVStep | | |
|---|---|---|---|---|
| | | 1/3 step | 1/2 step | 1 step |
| +5000 | +5.0 EV | ✓ | ✓ | ✓ |
| +4666 | +4.7 EV | ✓ | – | ✓ |
| +4500 | +4.5 EV | – | ✓ | – |
| +4333 | +4.3 EV | ✓ | – | ✓ |
| +4000 | +4.0 EV | ✓ | ✓ | ✓ |
| +3666 | +3.7 EV | ✓ | – | ✓ |
| +3500 | +3.5 EV | – | ✓ | – |
| +3333 | +3.3 EV | ✓ | – | ✓ |
| +3000 | +3.0 EV | ✓ | ✓ | ✓ |
| +2666 | +2.7 EV | ✓ | – | ✓ |
| +2500 | +2.5 EV | – | ✓ | – |
| +2333 | +2.3 EV | ✓ | – | ✓ |
| +2000 | +2.0 EV | ✓ | ✓ | ✓ |
| +1666 | +1.7 EV | ✓ | – | ✓ |
| +1500 | +1.5 EV | – | ✓ | – |
| +1333 | +1.3 EV | ✓ | – | ✓ |
| +1000 | +1.0 EV | ✓ | ✓ | ✓ |
| +666 | +0.7 EV | ✓ | – | ✓ |
| +500 | +0.5 EV | – | ✓ | – |
| +333 | +0.3 EV | ✓ | – | ✓ |
| 0 | 0.0 EV | ✓ | ✓ | ✓ |
| -333 | -0.3 EV | ✓ | – | ✓ |
| -500 | -0.5 EV | – | ✓ | – |
| -666 | -0.7 EV | ✓ | – | ✓ |
| -1000 | -1.0 EV | ✓ | ✓ | ✓ |
| -1333 | -1.3 EV | ✓ | – | ✓ |
| -1500 | -1.5 EV | – | ✓ | – |
| -1666 | -1.7 EV | ✓ | – | ✓ |
| -2000 | -2.0 EV | ✓ | ✓ | ✓ |
| -2333 | -2.3 EV | ✓ | – | ✓ |

Do Not Copy

| | | | | |
|---|---|---|---|---|
| -2500 | -2.5 EV | – | ✓ | – |
| -2666 | -2.7 EV | ✓ | – | ✓ |
| -3000 | -3.0 EV | ✓ | ✓ | ✓ |
| -3333 | -3.3 EV | ✓ | – | ✓ |
| -3500 | -3.5 EV | – | ✓ | – |
| -3666 | -3.7 EV | ✓ | – | ✓ |
| -4000 | -4.0 EV | ✓ | ✓ | ✓ |
| -4333 | -4.3 EV | ✓ | – | ✓ |
| -4500 | -4.5 EV | – | ✓ | – |
| -4666 | -4.7 EV | ✓ | – | ✓ |
| -5000 | -5.0 EV | ✓ | ✓ | ✓ |

Do Not Copy

### 6.5.1.14 DateTime

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5011 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 20210101T000000 [00:00:00, Jan. 1, 2021] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates “Date and time” that is set in the camera.

**○ Property Details**

The format of PropertyValue is a Unicode string of “YYYYMMDDThhmmss” where YYYY is the year, MM is the month, DD is the day of the month, T is a constant character, hh is the hours, mm is the minutes, and ss is the seconds past the minute, in accordance with the ISO8601 standards.

The PropertyValue is the date and time obtained by “UTC + Time zone difference + Daylight saving time”.

The setting range of PropertyValue is from 20000101T000000 to 20991231T235959.

The format of PropertyValue is shown below.

| Field | Size (Byte) | Data | Description |
|---|---|---|---|
| NumChar | 1 | 0x10 | The number of characters in the string.<br>It is sixteen (including the null character). |
| StringChars | 32 | | Unicode string<br>“YYYYMMDDThhmmss” |

The DevicePropChanged event (section 6.4.1.6) is not issued even if this PropertyValue has changed.

Do Not Copy

### 6.5.1.15 StillCaptureMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5013 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0001 [Single-frame shooting] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Release mode” that is set in the camera.

**○ Property Details**

If the release mode of the camera is "High-speed frame capture (C120)" when the DX lens is attached, the Property Value of this property is fixed to "High-speed frame capture (C60)".

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0001 | Single-frame shooting |
| 0x0002 | Continuous high-speed shooting |
| 0x8010 | Continuous low-speed shooting |
| 0x8011 | Self-timer |
| 0x811E | High-speed frame capture (C30) |
| 0x813C | High-speed frame capture (C60) |
| 0x8178 | High-speed frame capture (C120) |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | When the ToneMode property (section 6.5.2.75) is set to [HLG], [High-speed frame capture] is set. |
| | When the PixelShiftShootingMode property (section 6.5.2.78) is set to "ON", "Self-timer" or "High-speed frame capture" is set. |

Do Not Copy

### 6.5.1.16 BurstNumber

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x5018 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [One frame] |
| Property Value | From 1 [One frame] to 65535 [65535 frames] |

**○ Property Outline**

Indicates “The number of continuous shooting frames” captured by the command.

**○ Property Details**

For shooting by the command processing, the continuous shooting is performed until the number of frames set by this property is reached.

Refer to Continuous Shooting (section 11.7.1) about the maximum number of continuous shooting.

Do Not Copy

### 6.5.1.17 Artist

○ **Property Specifications**

| | |
|---|---|
| Property Code | 0x501E |
| DataType | String |
| Description form | None |
| Get / Set | Get |
| Default Value | NULL (0x00) 0 character |
| Property Value | (Depends on the user setting) |

○ **Property Outline**

Indicates “Artist” that is set in the camera.

○ **Property Details**

For setting “Artist”, the ArtistV property (section 6.5.5.9) should be used.

The PropertyValue is an optional string of 36 characters or shorter (not including the null character).

If the artist is not set in the camera, the value is zero length string (not including the null character).

### 6.5.1.18 Copyright

○ **Property Specifications**

| | |
|---|---|
| Property Code | 0x501F |
| DataType | String |
| Description form | None |
| Get / Set | Get |
| Default Value | NULL (0x00) 0 character |
| Property Value | (Depends on the user setting) |

○ **Property Outline**

Indicates “Copyright” that is set in the camera.

○ **Property Details**

For setting “Copyright”, the CopyrightV property (section 6.5.5.10) should be used.

The PropertyValue is an optional string of 54 characters or shorter (not including the null character).

If the copyright is not set in the camera, the value is zero length string (not including the null character).

Do Not Copy

# 6.5.2 Vendor (Photo Shooting Menu)

## 6.5.2.1 ShootingMenuBank

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD010 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 1[A] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting menu bank” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Do Not Copy

### 6.5.2.2 ShootingBankNameA

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD011 |
| DataType | String |
| Description form | None |
| Get / Set | Get, Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “A” in the photo shooting menu.

**○ Property Details**

The PropertyValue is an optional string of 20 characters or shorter (not including the null character).

When the string is shorter than 20 characters, the shortage is padded with spaces (0x20).

For the characters that can be input (ASCII code), refer to section 10.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

### 6.5.2.3 ShootingBankNameB

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD012 |
| DataType | String |
| Description form | None |
| Get / Set | Get, Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “B” in the photo shooting menu.

The method of using this property is the same as that of the ShootingBankNameA property (section 6.5.2.2).

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

Do Not Copy

### 6.5.2.4 ShootingBankNameC

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD013 |
| DataType | String |
| Description form | None |
| Get / Set | Get, Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “C” in the shooting menu.

The method of using this property is the same as that of the ShootingBankNameA property (section 6.5.2.2).

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

### 6.5.2.5 ShootingBankNameD

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD014 |
| DataType | String |
| Description form | None |
| Get / Set | Get, Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “D” in the shooting menu.

The method of using this property is the same as that of the ShootingBankNameA property (section 6.5.2.2).

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

Do Not Copy

### 6.5.2.6 ExtendShootingMenu

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD07F |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Extended menu banks” in the photo shooting menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

### 6.5.2.7 PrimarySlot

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD156 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Slot 1] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Primary slot” in the photo shooting menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | Slot 1 |
| 2 | Slot 2 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |

Do Not Copy

## 6.5.2.8 Slot2ImageSaveMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD148 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Overflow] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Slot2 function” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Overflow |
| 1 | Backup |
| 2 | RAW primary - JPEG secondary |
| 3 | JPEG primary - JPEG secondary |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |

Do Not Copy

## 6.5.2.9 Slot2JpegImageSize

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD119 |
| DataType | String |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | Size M |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Secondary slot function – JPEG+JPEG division recording” in the photo shooting menu. Specify the image size to be saved in JPEG + JPEG division recording.

### ○ Property Value

The valid PropertyValues are shown below.

PropertyValue changes depending on the value of the CaptureAreaCrop property (section 6.5.2.10). The size is the same as the image size applied to shooting in [Photo shooting menu]-[Image size setting].

| Property Value | Description |
|---|---|
| Same as size L of ImageSize property (section 6.5.1.2) or ImageSizeDX property (section 6.5.2.12). | Size L |
| Same as size M of ImageSize property or ImageSizeDX property. | Size M |
| Same as size S of ImageSize property or ImageSizeDX property. | Size S |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |

Do Not Copy

### 6.5.2.10 CaptureAreaCrop

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD030 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [FX] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Image area - Image area setting” in the photo shooting menu.

**○ Property Details**

If the StillCaptureMode property (section 6.5.1.15) is set to high-speed frame capture, this property will be as follows.

| StillCaptureMode property | Property Value |
|---|---|
| High speed frame capture (C30) | If [1:1] or [16:9], FX. |
| High speed frame capture (C60) | DX |
| High speed frame capture (C120) | FX |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | FX |
| 2 | DX |
| 4 | 1:1 |
| 5 | 16:9 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | A DX lens is mounted on the camera. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High-speed frame capture (C60/C120)]. |
| | [1:1] or [16:9] is set when the StillCaptureMode property is set to [High-speed frame capture (C30)]. |

Do Not Copy

### 6.5.2.11 EnableDXImageSize

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D041 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [OFF] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Image size – Enable image size (DX)” in the photo shooting menu.

**○ Property Details**

If this property is ON, CaptureAreaCrop (section 6.5.2.10) is "DX", and the setting prohibition conditions are not met, the setting value of ImageSizeDX (section 6.5.2.12) will be the image size that is applied when shooting.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | OFF |
| 1 | ON |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The CompressionSetting property (section 6.5.1.3) is set to [RAW]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |

Do Not Copy

### 6.5.2.12 ImageSizeDX

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D042 |
| DataType | String |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | Size L |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Image size – Image size (DX)” in the photo shooting menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| Size L | 5392x3592 |
| Size M | 4032x2688 |
| Size S | 2688x1792 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The CompressionSetting property (section 6.5.1.3) is set to [RAW]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |

Do Not Copy

### 6.5.2.13 RawCompressionType

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD016 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 3 [High efficiency★] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “NEF (RAW) recording” in the photo shooting menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Lossless compressed |
| 3 | High efficiency★ |
| 4 | High efficiency |

Do Not Copy

## 6.5.2.14 ExposureIndexEx

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0B4 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 100 [ISO100] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “ISO sensitivity” that is set in the camera.

This property is 32bit-expanded version of ExposureIndex property (section 6.5.1.12) and allows you to handle ISO sensitivities over 16Bit.

Refer to the MovieExposureIndex property (section 6.5.3.13) for the ISO sensitivity in video mode.

### ○ Property Details

If the following conditions are met, the lower and upper limits of ISO sensitivity will be limited. The setting valid range can be confirmed in the descriptor of this property.

| No | Conditions |
|---|---|
| 1 | ToneMode property (section 6.5.2.75) is set to [HLG]. |

### ○ Property Value

The valid PropertyValues are shown below.

The value of PropertyValue changes depending on the value of IsoStep property (section 6.5.4.13).

| Property Value | Description | IsoStep | |
|---|---|---|---|
| | | 1/3 step | 1 step |
| 32 | Lo 1.0 | ✓ | ✓ |
| 40 | Lo 0.7 | ✓ | – |
| 50 | Lo 0.3 | ✓ | – |
| 64 | 64 | ✓ | ✓ |
| 80 | 80 | ✓ | – |
| 100 | 100 | ✓ | ✓ |
| 125 | 125 | ✓ | – |
| 160 | 160 | ✓ | – |
| 200 | 200 | ✓ | ✓ |
| 250 | 250 | ✓ | – |
| 320 | 320 | ✓ | – |
| 400 | 400 | ✓ | ✓ |
| 500 | 500 | ✓ | – |
| 640 | 640 | ✓ | – |
| 800 | 800 | ✓ | ✓ |
| 1000 | 1000 | ✓ | – |
| 1250 | 1250 | ✓ | – |
| 1600 | 1600 | ✓ | ✓ |
| 2000 | 2000 | ✓ | – |
| 2500 | 2500 | ✓ | – |
| 3200 | 3200 | ✓ | ✓ |
| 4000 | 4000 | ✓ | – |

Do Not Copy

| | | | |
|---|---|---|---|
| 5000 | 5000 | ✓ | – |
| 6400 | 6400 | ✓ | ✓ |
| 8000 | 8000 | ✓ | – |
| 10000 | 10000 | ✓ | – |
| 12800 | 12800 | ✓ | ✓ |
| 16000 | 16000 | ✓ | – |
| 20000 | 20000 | ✓ | – |
| 25600 | 25600 | ✓ | ✓ |
| 32000 | Hi 0.3 | ✓ | – |
| 40000 | Hi 0.7 | ✓ | – |
| 51200 | Hi 1.0 | ✓ | ✓ |
| 102400 | Hi 2.0 | ✓ | ✓ |

Do Not Copy

## 6.5.2.15 ISOAutoControl

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD054 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “ISO sensitivity settings – Auto ISO sensitivity control” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.2.16 ISOAutoHighLimit

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD183 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 25600 [ISO25600] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “ISO sensitivity settings – Auto ISO sensitivity control – Maximum sensitivity” in the photo shooting menu.

### ○ Property Details

If the following conditions are met, the lower and upper limits of ISO sensitivity will be limited. The setting valid range can be confirmed in the descriptor of this property.

| No | Conditions |
|---|---|
| 1 | ToneMode property (section 6.5.2.75) is set to [HLG]. |

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| From 100 to 102400 | See the ExposureIndexEx property (section 6.5.2.14). |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ISOAutoControl property (section 6.5.2.15) is set to [Off]. |

Do Not Copy

## 6.5.2.17 FlashISOAutoHighLimit

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD199 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0xFFFFFFFF [The setting in the case that the flash is not attached] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “ISO sensitivity settings – Auto ISO sensitivity control – Maximum sensitivity with flash” in the photo shooting menu.

### ○ Property Details

If the following conditions are met, the lower and upper limits of ISO sensitivity will be limited. The setting valid range can be confirmed in the descriptor of this property.

| No | Conditions |
|---|---|
| 1 | ToneMode property (section 6.5.2.75) is set to [HLG]. |

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| From 100 to 102400 | Sane as the property value of ISOAutoHighLimit (section 6.5.2.16). |
| 0xFFFFFFFF | The setting in the case that the flash is not attached. |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ISOAutoControl property (section 6.5.2.15) is set to [Off]. |

Do Not Copy

## 6.5.2.18 ISOAutoShutterTime

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD164 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0xFFFFFFFF [Auto] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “ISO sensitivity settings – Auto ISO sensitivity control – Minimum shutter speed” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00013E80 | 1/16000 sec. |
| 0x000132C8 | 1/13000 sec. |
| 0x00012EE0 | 1/12000 sec. |
| 0x00012710 | 1/10000 sec. |
| 0x00011F40 | 1/8000 sec. |
| 0x00011900 | 1/6400 sec. |
| 0x00011770 | 1/6000 sec. |
| 0x00011388 | 1/5000 sec. |
| 0x00010FA0 | 1/4000 sec. |
| 0x00010C80 | 1/3200 sec. |
| 0x000109C4 | 1/2500 sec. |
| 0x000107D0 | 1/2000 sec. |
| 0x00010640 | 1/1600 sec. |
| 0x000104E2 | 1/1250 sec. |
| 0x000103E8 | 1/1000 sec. |
| 0x00010320 | 1/800 sec. |
| 0x00010280 | 1/640 sec. |
| 0x000101F4 | 1/500 sec. |
| 0x00010190 | 1/400 sec. |
| 0x00010140 | 1/320 sec. |
| 0x000100FA | 1/250 sec. |
| 0x000100C8 | 1/200 sec. |
| 0x000100A0 | 1/160 sec. |
| 0x0001007D | 1/125 sec. |
| 0x00010064 | 1/100 sec. |
| 0x00010050 | 1/80 sec. |
| 0x0001003C | 1/60 sec. |
| 0x00010032 | 1/50 sec. |
| 0x00010028 | 1/40 sec. |
| 0x0001001E | 1/30 sec. |
| 0x00010019 | 1/25 sec. |
| 0x00010014 | 1/20 sec. |
| 0x0001000F | 1/15 sec. |

Do Not Copy

| | |
|---|---|
| 0x0001000D | 1/13 sec. |
| 0x0001000A | 1/10 sec. |
| 0x00010008 | 1/8 sec. |
| 0x00010006 | 1/6 sec. |
| 0x00010005 | 1/5 sec. |
| 0x00010004 | 1/4 sec. |
| 0x00010003 | 1/3 sec. |
| 0x000A0019 | 1/2.5 sec. |
| 0x00010002 | 1/2 sec. |
| 0x000A0010 | 1/1.6 sec. |
| 0x000A000D | 1/1.3 sec. |
| 0x00010001 | 1 sec. |
| 0x00020001 | 2 sec. |
| 0x00040001 | 4 sec. |
| 0x00080001 | 8 sec. |
| 0x000F0001 | 15 sec. |
| 0x001E0001 | 30 sec. |
| 0xFFFFFFFF | Auto |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ISOAutoControl property (section 6.5.2.15) is set to [Off]. |

Do Not Copy

### 6.5.2.19 ISOAutoShutterTimeCorrectionValue

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1F4 |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 |
| Property Value | From -2 to +2 |

**○ Property Outline**

Indicates the correction value when “ISO sensitivity settings – Auto ISO sensitivity control – Minimum shutter speed” in the photo shooting menu is [Auto].

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ISOAutoControl property (section 6.5.2.15) is set to [Off]. |
| | The ISOAutoShutterTime property (section 6.5.2.18) is not set to [Auto]. |

Do Not Copy

## 6.5.2.20 WbAutoType

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD141 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 2 [Keep white] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the type of “White balance - Auto” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Keep overall atmosphere |
| 1 | Keep warm lighting colors |
| 2 | Keep white |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.21 WbTuneAuto

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD017 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance - Auto” in the photo shooting menu.
For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.22 WbTuneNatural

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD15E |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance – Natural light auto” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

## 6.5.2.23 WbTuneSunny

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD01A |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance – Direct sunlight” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.24 WbTuneCloudy

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD01C |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance - Cloudy” in the photo shooting menu.
For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

### 6.5.2.25 WbTuneShade

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD01D |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance – Shade” in the photo shooting menu.
For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.26 WbTuneIncandescent

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD018 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance - Incandescent” in the photo shooting menu. For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.27 WbFluorescentType

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD14F |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 3 [White fluorescent lamp] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the light source of “White balance - Fluorescent” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 3 | White fluorescent lamp |
| 4 | Day white fluorescent lamp |
| 5 | Daylight fluorescent lamp |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.28 WbTuneFluorescent

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD019 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance - Fluorescent” in the photo shooting menu. For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.29 WbTuneFlash

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD01B |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance - Flash” in the photo shooting menu.
For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.30 WbColorTemp

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD01E |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 5000 [K] |
| Property Value | From 2500 K to 10000 K (increase or decrease in units of 10 K) |

**○ Property Outline**

Indicates the color temperature (K) of “White balance – Choose color temp.” in the photo shooting menu.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

### 6.5.2.31 WbTuneColorTemp

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD150 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance – Choose color temp.” in the photo shooting menu.

See Section 11.5 for details on fine-tuning value.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.32 WbPresetDataNo

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD01F |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [d-1] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “White balance – Preset manual” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | d-1 |
| 2 | d-2 |
| 3 | d-3 |
| 4 | d-4 |
| 5 | d-5 |
| 6 | d-6 |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |
| | The MovieWhiteBalance property (section 6.5.3.14) is set to anything other than [Same as photo settings] during video recording. |

Do Not Copy

### 6.5.2.33 WbPresetDataComment1

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD021 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the comment of “White balance – Preset manual – d-1” in the photo shooting menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: MovieWbPresetDataComment1 property (section 6.5.3.27)

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The WbPresetProtect1 property (section 6.5.2.51) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.34 WbPresetDataComment2

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD022 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the comment of “White balance – Preset manual – d-2” in the photo shooting menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: MovieWbPresetDataComment2 property (section 6.5.3.28)

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The WbPresetProtect2 property (section 6.5.2.52) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.35 WbPresetDataComment3

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD023 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the comment of “White balance – Preset manual – d-3” in the photo shooting menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: MovieWbPresetDataComment3 property (section6.5.3.29)

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The WbPresetProtect3 property (section 6.5.2.53) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.36 WbPresetDataComment4

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD024 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the comment of “White balance – Preset manual – d-4” in the photo shooting menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: MovieWbPresetDataComment4 property (section 6.5.3.30)

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The WbPresetProtect4 property (section 6.5.2.54) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.37 WbPresetDataComment5

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD038 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the comment of “White balance – Preset manual – d-5” in the photo shooting menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: MovieWbPresetDataComment5 property (section 6.5.3.31)

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The WbPresetProtect5 property (section 6.5.2.55) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.38 WbPresetDataComment6

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD039 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the comment of “White balance – Preset manual – d-6” in the photo shooting menu.

### ○ Property Details

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: MovieWbPresetDataComment6 property (section 6.5.3.32)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The WbPresetProtect6 property (section 6.5.2.56) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.2.39 WbPresetDataValue1

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD026 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the white balance data of “White balance – Preset manual – d-1” in the photo shooting menu.

**○ Property Details**

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetDataValue1 property (section 6.5.3.33)

### 6.5.2.40 WbPresetDataValue2

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD027 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the white balance data of “White balance – Preset manual – d-2” in the photo shooting menu.

**○ Property Details**

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetDataValue2 property (section 6.5.3.34)

Do Not Copy

## 6.5.2.41 WbPresetDataValue3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD028 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the white balance data of “White balance – Preset manual – d-3” in the photo shooting menu.

### ○ Property Details

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetDataValue3 property (section 6.5.3.35)

## 6.5.2.42 WbPresetDataValue4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD029 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the white balance data of “White balance – Preset manual – d-4” in the photo shooting menu.

### ○ Property Details

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetDataValue4 property (section 6.5.3.36)

Do Not Copy

## 6.5.2.43 WbPresetDataValue5

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD03E |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the white balance data of “White balance – Preset manual – d-5” in the photo shooting menu.

### ○ Property Details

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetDataValue5 property (section 6.5.3.37)

## 6.5.2.44 WbPresetDataValue6

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD03F |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the white balance data of “White balance – Preset manual – d-6” in the photo shooting menu.

### ○ Property Details

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetDataValue6 property (section 6.5.3.38)

Do Not Copy

## 6.5.2.45 WbTunePreset1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD152 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-1” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbTunePreset1 property (section 6.5.3.39)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The WbPresetProtect1 property (section 6.5.2.51) is [On]. |
| | The WbPresetDataNo property (section 6.5.2.32) is set to anything other than [d-1]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.46 WbTunePreset2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD153 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-2” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbTunePreset2 property (section 6.5.3.40)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The WbPresetProtect2 property (section 6.5.2.52) is [On]. |
| | The WbPresetDataNo property (section 6.5.2.32) is set to anything other than [d-2]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.47 WbTunePreset3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD154 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-3” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbTunePreset3 property (section 6.5.3.41)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The WbPresetProtect3 property (section 6.5.2.53) is [On]. |
| | The WbPresetDataNo property (section 6.5.2.32) is set to anything other than [d-3]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.48 WbTunePreset4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD155 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-4” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbTunePreset4 property (section 6.5.3.42)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The WbPresetProtect4 property (section 6.5.2.54) is [On]. |
| | The WbPresetDataNo property (section 6.5.2.32) is set to anything other than [d-4]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.49 WbTunePreset5

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD03A |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-5” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbTunePreset5 property (section 6.5.3.43)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The WbPresetProtect5 property (section 6.5.2.55) is [On]. |
| | The WbPresetDataNo property (section 6.5.2.32) is set to anything other than [d-5]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

## 6.5.2.50 WbTunePreset6

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD03B |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-6” in the photo shooting menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbTunePreset6 property (section 6.5.3.44)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The WbPresetProtect6 property (section 6.5.2.56) is [On]. |
| | The WbPresetDataNo property (section 6.5.2.32) is set to anything other than [d-6]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.51 WbPresetProtect1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD158 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-1” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetProtect1 property (section 6.5.3.45)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.52 WbPresetProtect2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD159 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-2” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetProtect2 property (section 6.5.3.46)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.53 WbPresetProtect3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD15A |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-3” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetProtect3 property (section 6.5.3.47)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.54 WbPresetProtect4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD15C |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-4” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetProtect4 property (section 6.5.3.48)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.55 WbPresetProtect5

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD03C |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-5” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetProtect5 property (section 6.5.3.49)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.56 WbPresetProtect6

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD03D |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-6” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieWbPresetProtect6 property (section 6.5.3.50)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.2.57 ColorSpace

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD032 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [sRGB] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Color space” in the photo shooting menu.

### ○ Property Details

If the ToneMode property (section 6.5.2.75) is "HLG", "BT.2100" is added to PropertyValue to this property.

This property is "BT.2100" when the ToneMode property is "HLG".

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | sRGB |
| 1 | Adobe RGB |
| 100 | BT.2100<br>(Only when the ToneMode property is [HLG]) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [HLG]. |

Do Not Copy

## 6.5.2.58 Active-D-Lighting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD14E |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0000 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Active D-Lighting” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |
| 0x0105 | Extra high 1 |
| 0x0205 | Extra high 2 |
| 0xFFFF | Auto |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [HLG]. |

Do Not Copy

## 6.5.2.59 NoiseReduction

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD06B |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Long exposure NR” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The SilentMode property (section 6.5.5.14) is set to [On]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.2.60 NoiseReductionHilso

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD070 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0003 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “High ISO NR” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |

Do Not Copy

## 6.5.2.61 VignetteControl

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0F7 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0003 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Vignette control” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieVignetteControl property (section 6.5.3.53)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |

Do Not Copy

## 6.5.2.62 DiffractionCompensation

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0BA |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Diffraction compensation” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieDiffractionCompensation property (section 6.5.3.54)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.2.63 AutoDistortion

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0F8 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Auto distortion control” in the photo shooting menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: MovieAutoDistortion property (section 6.5.3.55)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The LensTypeML property (section 6.5.10.2) is set to [Lens whose automatic distortion correction is always effective]. |

Do Not Copy

## 6.5.2.64 FlickerReductionSetting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0B7 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Flicker reduction shooting” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |

Do Not Copy

## 6.5.2.65 HighFrequencyFlickerReduction

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D031 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “High-frequency flicker reduction shooting” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On (High resolution shutter setting) |

Do Not Copy

## 6.5.2.66 StillFocusMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD061 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [AF-S] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Focus mode” in photo shooting.

### ○ Property Details

Even if this property is set to AF, this property is set to MF when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | AF-S |
| 1 | AF-C |
| 4 | MF |

Do Not Copy

## 6.5.2.67 StillFocusMeteringMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD05D |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x8010 [Single-point AF] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “AF-area mode” in photo shooting.

### ○ Property Details

In the following cases, this property is set to the single-point AF mode. When the following state is canceled without changing the AF area mode, it returns to the AF area mode which was set immediately before.

- In the case that the StillFocusMode property (section 6.5.2.66) is set to [MF].

- In the case that the AF area mode that cannot be selected is set when the StillFocusMode property (section 6.5.2.66) is changed.

### ○ Property Value

The valid PropertyValues are shown below.
The value of PropertyValue changes depending on the still image focus mode setting.

| Property Value | Description | Still image focus mode setting | | |
|---|---|---|---|---|
| | | MF | AF-S | AF-C |
| 0x0002 | Dynamic AF (S) | - | - | ✓ |
| 0x8010 | Single-point AF | ✓ | ✓ | ✓ |
| 0x8011 | Auto-area AF | - | ✓ | ✓ |
| 0x8012 | 3D-Tracking | - | - | ✓ |
| 0x8013 | Dynamic AF(M) | - | - | ✓ |
| 0x8014 | Dynamic AF(L) | - | - | ✓ |
| 0x8017 | Pinpoint AF | - | ✓ | - |
| 0x8018 | Wide-area AF (S) | - | ✓ | ✓ |
| 0x8019 | Wide-area AF (L) | - | ✓ | ✓ |
| 0x801E | Wide-area AF (C1) | - | ✓ | ✓ |
| 0x801F | Wide-area AF (C2) | - | ✓ | ✓ |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The photo focus mode is set to the manual focus.<br>NOTE) Except when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch. |

Do Not Copy

## 6.5.2.68 WideAreaAFC1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D046 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0D03 [13×3] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the focus area size of “AF-area mode – Wide area AF(C1)” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0101 | 1×1 |
| 0x0103 | 1×3 |
| 0x0107 | 1×7 |
| 0x010B | 1×11 |
| 0x0501 | 5×1 |
| 0x0503 | 5×3 |
| 0x0507 | 5×7 |
| 0x050B | 5×11 |
| 0x0901 | 9×1 |
| 0x0903 | 9×3 |
| 0x0907 | 9×7 |
| 0x090B | 9×11 |
| 0x0D01 | 13×1 |
| 0x0D03 | 13×3 |
| 0x0D07 | 13×7 |
| 0x0D0B | 13×11 |
| 0x1301 | 19×1 |
| 0x1303 | 19×3 |
| 0x1307 | 19×7 |
| 0x130B | 19×11 |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The photo focus mode is set to the manual focus.<br>NOTE) Except when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch. |

Do Not Copy

## 6.5.2.69 WideAreaAFC2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D047 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0507 [5x7] |
| Property Value | Same as the WideAreaAFC1 property (section 6.5.2.68). |

### ○ Property Outline

Indicates the focus area size of “AF-area mode – Wide area AF(C2)” in the photo shooting menu.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The photo focus mode is set to the manual focus.<br>NOTE) Except when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch. |

Do Not Copy

## 6.5.2.70 AFSubjectDetection

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D005 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Auto] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Subject detection setting in AF” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Auto |
| 2 | Person |
| 3 | Animal |
| 4 | Vehicle |
| 5 | Birds |
| 6 | Airplane |

Do Not Copy

## 6.5.2.71 VibrationReduction

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD138 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Vibration reduction” in the photo setting menu.

### ○ Property Details

The supported Property Value differs depending on the type of attached lens.

### ○ Property Value

The valid PropertyValues are shown below.

When the lens anti-vibration mode is normal or sports:

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Normal |
| 2 | Sports |

When the lens anti-vibration mode is normal only:

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The mounted lens has VR switch. |
| | The LensSort property (section 6.5.10.1) is set to [Not mounted] and the FmmManualSetting2 property (section 6.5.5.2) is set to [0.0mm]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.2.72 HDRMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD130 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x000 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “HDR (high dynamic range) – HDR mode” in the photo shooting menu.

### ○ Property Details

When the StillCaptureMode property (section 6.5.1.15) is [Continuous low-speed shooting] or [Continuous high-speed shooting], HDR is applied to only the first image with the continuous shot operation.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0001 | On (single photo) |
| 0x0101 | On (series) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The EnableBracketing property (section 6.5.8.1) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.2.73 HDRStrength

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD132 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0xFFFF [Auto] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “HDR (high dynamic range) – HDR strength” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |
| 0x0005 | Extra high |
| 0xFFFF | Auto |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The EnableBracketing property (section 6.5.8.1) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.2.74 HDRSaveIndividualImages

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD133 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “HDR (high dynamic range) – Save individual images (RAW)” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The EnableBracketing property (section 6.5.8.1) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.2.75 ToneMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D01C |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [SDR] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Tone Mode” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | SDR |
| 2 | HLG |

Do Not Copy

## 6.5.2.76 SkinSoftening

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D017 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0000 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Skin Softening” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The HDRMode property (section 6.5.2.72) is set to [On]. |

Do Not Copy

## 6.5.2.77 PortraitImpressionBalance

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D019 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Portrait Impression Adjustment” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Mode 1 |
| 2 | Mode 2 |
| 3 | Mode 3 |

Do Not Copy

## 6.5.2.78 PixelShiftShootingMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D045 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0000 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Pixel shift shooting – Pixel shift shooting mode” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0001 | On (Single) |
| 0x0101 | On (Continuous) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |
| | The EnableBracketing property (section 6.5.8.1) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [Self-timer] or [High speed frame capture]. |

Do Not Copy

## 6.5.2.79 PixelShiftShootingInterval

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D050 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [0 sec.] |
| Property Value | From 0 [0 sec.] to 30 [30 sec.] |

### ○ Property Outline

Indicates “Pixel shift shooting – Interval time” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0001 | On (Single) |
| 0x0101 | On (Continuous) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |
| | The EnableBracketing property (section 6.5.8.1) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [Self-timer] or [High speed frame capture]. |

Do Not Copy

# 6.5.3 Vendor (Video Recording Menu)

## 6.5.3.1 MovieRecordDestination

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0A3 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Slot 1] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Destination” in the video recording menu.

If a card is not inserted in the slot that is set in this property, the video is recorded on the card that is inserted in the other slot.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | Slot 1 |
| 2 | Slot 2 |

Do Not Copy

### 6.5.3.2 MovieFileType

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0AF |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x00010800 [H.265 8-bit(MOV)] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Video file type” in the video recording menu.

**○ Property Value**

The meaning of each Bit of Property Value is as follows.

| Bit | Bit size | Description |
|---|---|---|
| 24-31 | 8 | Reserved(0) |
| 16-23 | 8 | Video codec |
| 8-15 | 8 | Bit depth |
| 0-7 | 8 | File container |

Video codec:

| Value | Description |
|---|---|
| 0 | H.264/AVC |
| 1 | H.265/HEVC |
| 2 | N-RAW |
| 16 | ProRes 422 HQ |
| 17 | ProRes RAW HQ |

Bit depth:

Assign the bit depth value as it is.

File container:

| Property Value | Description |
|---|---|
| 0 | MOV |
| 1 | MP4 |
| 2 | NEV |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00000801 | H.264 8-bit(MP4) |
| 0x00010800 | H.265 8-bit(MOV) |
| 0x00010A00 | H.265 10-bit(MOV) |
| 0x00020C02 | N-RAW 12-bit(NEV) |
| 0x00100A00 | ProRes 422 HQ 10-bit(MOV) |
| 0x00110C00 | ProRes RAW HQ 12-bit(MOV) |

Do Not Copy

### 6.5.3.3 MovieH265ToneMode

**Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D001 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [SDR] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates “Video file type - H.265 10-bit (MOV) tone mode” in the video recording menu.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | SDR |
| 1 | N-Log |
| 2 | HLG |

### 6.5.3.4 MovieNRaw12bitToneMode

**Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D028 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [SDR] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates “Video file type – N-RAW 12-bit (NEV) tone mode” in the video recording menu.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | SDR |
| 1 | N-Log |

Do Not Copy

### 6.5.3.5 MovieProResToneMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D000 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [SDR] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Video file type - ProRes 422 HQ 10-bit(MOV) tone mode” in the video recording menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | SDR |
| 1 | N-Log |

### 6.5.3.6 MovieProResRawHq12bitToneMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D029 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [SDR] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Video file type - ProRes RAW HQ 12-bit(MOV) tone mode” in the video recording menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | SDR |
| 1 | N-Log |

Do Not Copy

## 6.5.3.7 MovieRecordScreenSize

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0A0 |
| DataType | UINT64 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0F000870001E0000 [3840×2160 30p] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Image size/frame rate” in the video recording menu.

### ○ Property Details

This property represents the setting value of the menu, but if the setting prohibition condition is met, it represents the control value of the camera.

The following values are assigned to each bit of Property Value.

| Bit | Bit number | Description |
|---|---|---|
| 48-63 | 16 | Horizontal resolution |
| 32-47 | 16 | Vertical resolution |
| 24-31 | 8 | Reserved (0) |
| 16-23 | 8 | Frame rate |
| 14-15 | 2 | Scan type |
| 8-13 | 6 | Special shooting |
| 0-7 | 8 | Special shooting supplementary information |

Horizontal resolution / Vertical resolution / Frame rate

The resolution or frame rate values are assigned.

Scan type

| Value | Description |
|---|---|
| 0 | p (Progressive) |
| 1 | i (Interlace) |
| From 2 to 3 | Reserved |

Special shooting

| Value | Description |
|---|---|
| 0 | Normal |
| 1 | Slow-motion |
| 2 | Crop |
| From 3 to 63 | Reserved |

Special shooting supplementary information

If there is no supplementary information, it is fixed to 0.
In the case of slow motion, the value is 10 times the slow magnification.

| Value | Description |
|---|---|
| 40 | x4 slow-motion |
| 50 | x5 slow-motion |

Do Not Copy

### ○ Property Value

The valid PropertyValues are shown below.
PropertyValue changes depending on the setting of MovieFileType property (section 6.5.3.2).

| Property Value | Description | MovieFileType | | | | |
|---|---|---|---|---|---|---|
| | | H.264 | H.265 | N-RAW | ProRes 422 HQ | ProRes RAW HQ |
| 0x20401224003C0000 | 8256×4644 60p | - | - | ✓ | - | - |
| 0x2040122400320000 | 8256×4644 50p | - | - | ✓ | - | - |
| 0x20401224001E0000 | 8256×4644 30p | - | - | ✓ | - | - |
| 0x2040122400190000 | 8256×4644 25p | - | - | ✓ | - | - |
| 0x2040122400180000 | 8256×4644 24p | - | - | ✓ | - | - |
| 0x1020091200780000 | 4128×2322 120p | - | - | ✓ | - | - |
| 0x1020091200640000 | 4128×2322 100p | - | - | ✓ | - | - |
| 0x10200912003C0000 | 4128×2322 60p | - | - | ✓ | - | ✓ |
| 0x1020091200320000 | 4128×2322 50p | - | - | ✓ | - | ✓ |
| 0x10200912001E0000 | 4128×2322 30p | - | - | ✓ | - | ✓ |
| 0x1020091200190000 | 4128×2322 25p | - | - | ✓ | - | ✓ |
| 0x1020091200180000 | 4128×2322 24p | - | - | ✓ | - | ✓ |
| 0x15100BD8003C0000 | 5392×3032 60p | - | - | ✓ | - | - |
| 0x15100BD800320000 | 5392×3032 50p | - | - | ✓ | - | - |
| 0x15100BD8001E0000 | 5392×3032 30p | - | - | ✓ | - | ✓ |
| 0x15100BD800190000 | 5392×3032 25p | - | - | ✓ | - | ✓ |
| 0x15100BD800180000 | 5392×3032 24p | - | - | ✓ | - | ✓ |
| 0x1E0010E0001E0000 | 7680×4320 30p | - | ✓ | - | - | - |
| 0x1E0010E000190000 | 7680×4320 25p | - | ✓ | - | - | - |
| 0x1E0010E000180000 | 7680×4320 24p | - | ✓ | - | - | - |
| 0x0F00087000780000 | 3840×2160 120p | - | ✓ | ✓ | - | - |
| 0x0F00087000640000 | 3840×2160 100p | - | ✓ | ✓ | - | - |
| 0x0F000870003C0000 | 3840×2160 60p | - | ✓ | - | ✓ | ✓ |
| 0x0F00087000320000 | 3840×2160 50p | - | ✓ | - | ✓ | ✓ |
| 0x0F000870001E0000 | 3840×2160 30p | - | ✓ | - | ✓ | - |
| 0x0F00087000190000 | 3840×2160 25p | - | ✓ | - | ✓ | - |
| 0x0F00087000180000 | 3840×2160 24p | - | ✓ | - | ✓ | - |
| 0x0780043800780000 | 1920×1080 120p | - | ✓ | - | ✓ | - |
| 0x0780043800640000 | 1920×1080 100p | - | ✓ | - | ✓ | - |
| 0x07800438003C0000 | 1920×1080 60p | ✓ | ✓ | - | ✓ | - |
| 0x0780043800320000 | 1920×1080 50p | ✓ | ✓ | - | ✓ | - |
| 0x07800438001E0000 | 1920×1080 30p | ✓ | ✓ | - | - | - |
| 0x0780043800190000 | 1920×1080 25p | ✓ | ✓ | - | - | - |
| 0x0780043800180000 | 1920×1080 24p | ✓ | ✓ | - | - | - |
| 0x07800438001E0128 | 1920×1080 30p x4 slow | ✓ | – | | | |
| 0x0780043800190128 | 1920×1080 25p x4 slow | ✓ | – | | | |
| 0x0780043800180132 | 1920×1080 24p x5 slow | ✓ | – | | | |

Do Not Copy

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | [8256×4644], [7680×4320] or [4128×2322] is set when a DX lens is mounted. |

### 6.5.3.8 MovieQualityNRaw

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D02A |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [High Quality] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Video Quality” in the video recording menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Normal Quality |
| 1 | High Quality |

Do Not Copy

## 6.5.3.9 MovieCaptureAreaCrop

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD20F |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [FX] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Image area - Choose image area” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | FX |
| 2 | DX |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieRecordScreenSize property (section 6.5.3.7) is set to [7680×4320]. |
| | DX lens is mounted on the camera. |
| | MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |

Do Not Copy

### 6.5.3.10 ExtendedOverSampling

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D01E |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Extended Over Sampling” in the video recording menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.3.11 MovieISOAutoHighLimit

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0AE |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 25600 [ISO 25600] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video ISO sensitivity settings – Maximum sensitivity” in the video recording menu.

### ○ Property Details

When the following conditions are satisfied, the minimum value and maximum value for Video ISO sensitivity are limited. The setting effective range can be confirmed in the descriptor of this property.

| No | Condition |
|---|---|
| 1 | The MovieLogOutput property (section 6.5.12.4) is [On]. |
| 2 | MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |

### ○ Property Value

The valid PropertyValues are shown below.

The value of PropertyValue changes depending on the value of IsoStep property (section 6.5.4.13).

| Property Value | Description | IsoStep | |
|---|---|---|---|
| | | 1/3 step | 1 step |
| 200 | 200 | ✓ | ✓ |
| 250 | 250 | ✓ | – |
| 320 | 320 | ✓ | – |
| 400 | 400 | ✓ | ✓ |
| 500 | 500 | ✓ | – |
| 640 | 640 | ✓ | – |
| 800 | 800 | ✓ | ✓ |
| 1000 | 1000 | ✓ | – |
| 1250 | 1250 | ✓ | – |
| 1600 | 1600 | ✓ | ✓ |
| 2000 | 2000 | ✓ | – |
| 2500 | 2500 | ✓ | – |
| 3200 | 3200 | ✓ | ✓ |
| 4000 | 4000 | ✓ | – |
| 5000 | 5000 | ✓ | – |
| 6400 | 6400 | ✓ | ✓ |
| 8000 | 8000 | ✓ | – |
| 10000 | 10000 | ✓ | – |
| 12800 | 12800 | ✓ | ✓ |
| 16000 | 16000 | ✓ | – |
| 20000 | 20000 | ✓ | – |
| 25600 | 25600 | ✓ | ✓ |
| 32000 | Hi 0.3 | ✓ | – |

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| | | | |
|---|---|---|---|
| 40000 | Hi 0.7 | ✓ | - |
| 51200 | Hi 1.0 | ✓ | ✓ |
| 102400 | Hi 2.0 | ✓ | ✓ |

Do Not Copy

### 6.5.3.12 MovieISOAutoControl

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0AD |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Video ISO sensitivity settings – Auto ISO control (mode M)” in the video recording menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than M during video recording. |

Do Not Copy

## 6.5.3.13 MovieExposureIndex

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1AA |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 100 [ISO100] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video ISO sensitivity settings – ISO sensitivity (mode M)” in the video recording menu.

### ○ Property Details

When the following conditions are satisfied, the minimum value and maximum value for video ISO sensitivity are limited. The setting effective range can be confirmed in the descriptor of this property.

| No | Condition |
|---|---|
| 1 | The MovieLogOutput property (section 6.5.12.4) is [On]. |
| 2 | MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |
| 3 | MovieFileType property is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log] or [HLG]. |
| 4 | MovieFileType property (section 6.5.3.2) is set to [ProRes 422 HQ 10-bit (MOV)] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log]. |

### ○ Property Value

The valid PropertyValues are shown below.

The value of PropertyValue changes depending on the value of IsoStep property (section 6.5.4.13).

When the MovieExtendedIsoStepManual property (section 6.5.4.56) is [On (1/6 steps)], the values of 1/6 steps are listed regardless of the value of the IsoStep property. However, for Lo sensitivity, if the IsoStep property is "1 step", the value of "1 step" is listed.

If the tone mode of the video file format is N-Log, values below ISO800 are Lo sensitivity.
* PropertyValue with Lo notation in the content is targeted.

| Property Value | Description | Extension (1/6 step) | IsoStep | |
|---|---|---|---|---|
| | | | 1/3 step | 1 step |
| 64 | 64 | ✓ | ✓ | ✓ |
| 72 | 72 | ✓ | – | – |
| 80 | 80 | ✓ | ✓ | – |
| 90 | 90 | ✓ | – | – |
| 100 | 100 | ✓ | ✓ | ✓ |
| 110 | 110 | ✓ | – | – |
| 125 | 125 | ✓ | ✓ | – |
| 140 | 140 | ✓ | – | – |
| 160 | 160 | ✓ | ✓ | – |
| 180 | 180 | ✓ | – | – |
| 200 | 200 | ✓ | ✓ | ✓ |
| 220 | 220 | ✓ | – | – |
| 250 | 250 | ✓ | ✓ | – |
| 280 | 280 | ✓ | – | – |

Do Not Copy

| | | | | |
|---|---|---|---|---|
| 320 | 320 | ✓ | ✓ | – |
| 360 | 360 | ✓ | – | – |
| 400 | 400 | ✓ | ✓ | ✓ |
| 450 | 450 | ✓ | – | – |
| 500 | 500 | ✓ | ✓ | – |
| 560 | 560 | ✓ | – | – |
| 640 | 640 | ✓ | ✓ | – |
| 720 | 720 | ✓ | – | – |
| 800 | 800 | ✓ | ✓ | ✓ |
| 900 | 900 | ✓ | – | – |
| 1000 | 1000 | ✓ | ✓ | – |
| 1100 | 1100 | ✓ | – | – |
| 1250 | 1250 | ✓ | ✓ | – |
| 1400 | 1400 | ✓ | – | – |
| 1600 | 1600 | ✓ | ✓ | ✓ |
| 1800 | 1800 | ✓ | – | – |
| 2000 | 2000 | ✓ | ✓ | – |
| 2200 | 2200 | ✓ | – | – |
| 2500 | 2500 | ✓ | ✓ | – |
| 2800 | 2800 | ✓ | – | – |
| 3200 | 3200 | ✓ | ✓ | ✓ |
| 3600 | 3600 | ✓ | – | – |
| 4000 | 4000 | ✓ | ✓ | – |
| 4500 | 4500 | ✓ | – | – |
| 5000 | 5000 | ✓ | ✓ | – |
| 5600 | 5600 | ✓ | – | – |
| 6400 | 6400 | ✓ | ✓ | ✓ |
| 7200 | 7200 | ✓ | – | – |
| 8000 | 8000 | ✓ | ✓ | – |
| 9000 | 9000 | ✓ | – | – |
| 10000 | 10000 | ✓ | ✓ | – |
| 11400 | 11400 | ✓ | – | – |
| 12800 | 12800 | ✓ | ✓ | ✓ |
| 14400 | 14400 | ✓ | – | – |
| 16000 | 16000 | ✓ | ✓ | – |
| 18000 | 18000 | ✓ | – | – |
| 20000 | 20000 | ✓ | ✓ | – |
| 22800 | 22800 | ✓ | – | – |
| 25600 | 25600 | ✓ | ✓ | ✓ |
| 32000 | Hi 0.3 | ✓ | ✓ | – |
| 40000 | Hi 0.7 | ✓ | ✓ | – |
| 51200 | Hi 1.0 | ✓ | ✓ | ✓ |
| 102400 | Hi 2.0 | ✓ | ✓ | ✓ |

Do Not Copy

○ **Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than M during video recording. |

## 6.5.3.14 MovieWhiteBalance

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD23A |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x8015 [Same as photo settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “White Balance” in the video recording menu.

### ○ Property Details

When the property value is set to [Same as photo settings], the white balance value that is set in the WhiteBalance (section 6.5.1.4) is applied to the video white balance. In this case, the photo white balance setting is used for video setting as in the following table.

| Item | Property |
|---|---|
| Auto – Type | WbAutoType(6.5.2.20) |
| Auto - FineTune | WbTuneAuto(6.5.2.21) |
| Incandescent - FineTune | WbTuneIncandescent(6.5.2.26) |
| Fluorescent - Type | WbFluorescentType(6.5.2.27) |
| Fluorescent - FineTune | WbTuneFluorescent(0) |
| Sunny - FineTune | WbTuneSunny(6.5.2.23) |
| Flash - FineTune | WbTuneFlash(6.5.2.29) |
| Cloudy - FineTune | WbTuneCloudy(6.5.2.24) |
| Shade - FineTune | WbTuneShade(6.5.2.25) |
| Choose color temp - Value | WbColorTemp(6.5.2.30) |
| Choose color temp - FineTune | WbTuneColorTemp(6.5.2.31) |
| Natural light auto - FineTune | WbTuneNatural(6.5.2.22) |
| Preset manual | WbPresetDataNo(6.5.2.32) |

Regardless of whether the PropertyValue is [Same as photo settings], each of the following white balance settings is linked with the still image setting.

| Item | Property |
|---|---|
| Preset manual-D1- Comment | WbPresetDataComment1(6.5.2.33) |
| Preset manual-D2- Comment | WbPresetDataComment2(6.5.2.34) |
| Preset manual-D3- Comment | WbPresetDataComment3(6.5.2.35) |
| Preset manual-D4- Comment | WbPresetDataComment4(6.5.2.36) |
| Preset manual-D5- Comment | WbPresetDataComment5(6.5.2.37) |
| Preset manual-D6- Comment | WbPresetDataComment6(6.5.2.38) |
| Preset manual-D1- Data | WbPresetDataValue1(6.5.2.39) |
| Preset manual-D2- Data | WbPresetDataValue2(6.5.2.40) |
| Preset manual-D3- Data | WbPresetDataValue3(6.5.2.41) |
| Preset manual-D4- Data | WbPresetDataValue4(6.5.2.42) |
| Preset manual-D5- Data | WbPresetDataValue5(6.5.2.43) |
| Preset manual-D6- Data | WbPresetDataValue6(6.5.2.44) |
| Preset manual-D1- FineTune | WbTunePreset1(6.5.2.45) |
| Preset manual-D2- FineTune | WbTunePreset2(6.5.2.46) |
| Preset manual-D3- FineTune | WbTunePreset3(6.5.2.47) |

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| | |
|---|---|
| Preset manual-D4- FineTune | WbTunePreset4(6.5.2.48) |
| Preset manual-D5- FineTune | WbTunePreset5(6.5.2.49) |
| Preset manual-D6- FineTune | WbTunePreset6(6.5.2.50) |
| Preset manual-D1- Protect | WbPresetProtect1(6.5.2.51) |
| Preset manual-D2- Protect | WbPresetProtect2(6.5.2.52) |
| Preset manual-D3- Protect | WbPresetProtect3(6.5.2.53) |
| Preset manual-D4- Protect | WbPresetProtect4(6.5.2.54) |
| Preset manual-D5- Protect | WbPresetProtect5(6.5.2.55) |
| Preset manual-D6- Protect | WbPresetProtect6(6.5.2.56) |

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0002 | Auto |
| 0x0004 | Sunny |
| 0x0005 | Fluorescent |
| 0x0006 | Incandescent |
| 0x8010 | Cloudy |
| 0x8011 | Shade |
| 0x8012 | Choose color temp. |
| 0x8013 | Preset manual |
| 0x8015 | Same as photo settings |
| 0x8016 | Natural light auto |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | [Same as photo settings] was set during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.15 MovieWbAutoType

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD211 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 2 [Keep white] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the type of “White balance - Auto” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Keep overall atmosphere |
| 1 | Keep warm lighting colors |
| 2 | Keep white |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.3.16 MovieWbTuneAuto

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD212 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance - Auto” in the video recording menu.
For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.3.17 MovieWbTuneNatural

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD23C |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance – Natural light auto” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

### 6.5.3.18 MovieWbTuneSunny

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD216 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance – Direct sunlight” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.3.19 MovieWbTuneCloudy

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD218 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance - Cloudy” in the video recording menu.
For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

### 6.5.3.20 MovieWbTuneShade

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD219 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine-tuning volume of “White balance – Shade” in the video recording menu.
For the details of the fine-tuning volume, refer to section 11.5.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.21 MovieWbTuneIncandescent

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD213 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance - Incandescent” in the video recording menu. For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.22 MovieWbFluorescentType

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD214 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 3 [White fluorescent lamp] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the light source of “White balance - Fluorescent” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 3 | White fluorescent lamp |
| 4 | Day white fluorescent lamp |
| 5 | Daylight fluorescent lamp |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.23 MovieWbTuneFluorescent

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD215 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance - Fluorescent” in the video recording menu. For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.24 MovieWbColorTemp

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD21A |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 5000 [K] |
| Property Value | From 2500 K to 10000 K (increase or decrease in units of 10 K) |

### ○ Property Outline

Indicates the color temperature (K) of “White balance – Choose color temp.” in the video recording menu.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

## 6.5.3.25 MovieWbTuneColorTemp

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD21B |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the fine-tuning volume of “White balance – Choose color temp.” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.26 MovieWbPresetDataNo

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD21C |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [d-1] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “White balance – Preset manual” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | d-1 |
| 2 | d-2 |
| 3 | d-3 |
| 4 | d-4 |
| 5 | d-5 |
| 6 | d-6 |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |
| | The MovieWhiteBalance property (section 6.5.3.14) is set to [Same as photo settings] during video recording. |

Do Not Copy

## 6.5.3.27 MovieWbPresetDataComment1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD21D |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the comment of “White balance – Preset manual – d-1” in the video recording menu.

### ○ Property Details

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: WbPresetDataComment1 property (section 6.5.2.33)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The MovieWbPresetProtect1 property (section 6.5.3.45) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.28 MovieWbPresetDataComment2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD21E |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the comment of “White balance – Preset manual – d-2” in the video recording menu.

### ○ Property Details

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: WbPresetDataComment2 property (section 6.5.2.34)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The MovieWbPresetProtect2 property (section 6.5.3.46) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.29 MovieWbPresetDataComment3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD21F |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the comment of “White balance – Preset manual – d-3” in the video recording menu.

### ○ Property Details

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: WbPresetDataComment3 property (section 6.5.2.35)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The MovieWbPresetProtect3 property (section 6.5.3.47) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.30 MovieWbPresetDataComment4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD220 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the comment of “White balance – Preset manual – d-4” in the video recording menu.

### ○ Property Details

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: WbPresetDataComment4 property (section 6.5.2.36)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The MovieWbPresetProtect4 property (section 6.5.3.48) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.31 MovieWbPresetDataComment5

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD221 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the comment of “White balance – Preset manual – d-5” in the video recording menu.

### ○ Property Details

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: WbPresetDataComment5 property (section 6.5.2.37)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The MovieWbPresetProtect5 property (section 6.5.3.49) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.3.32 MovieWbPresetDataComment6

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD222 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 36 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the comment of “White balance – Preset manual – d-6” in the video recording menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).
When the string is shorter than 36 characters, the shortage is padded with spaces (0x20).
For the characters that can be input (ASCII code), refer to section 10.1.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.
Linked properties: WbPresetDataComment6 property (section 6.5.2.38)

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |
| Access_Denied | The MovieWbPresetProtect6 property (section 6.5.3.50) is [On]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

### 6.5.3.33 MovieWbPresetDataValue1

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD223 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the white balance data of “White balance – Preset manual – d-1” in the video recording menu.

**○ Property Details**

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetDataValue1 property (section 6.5.2.39)

### 6.5.3.34 MovieWbPresetDataValue2

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD224 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the white balance data of “White balance – Preset manual – d-2” in the video recording menu.

**○ Property Details**

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetDataValue2 property (section 6.5.2.40)

Do Not Copy

## 6.5.3.35 MovieWbPresetDataValue3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD225 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the white balance data of “White balance – Preset manual – d-3” in the video recording menu.

### ○ Property Details

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetDataValue3 property (section 6.5.2.41)

## 6.5.3.36 MovieWbPresetDataValue4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD226 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

### ○ Property Outline

Indicates the white balance data of “White balance – Preset manual – d-4” in the video recording menu.

### ○ Property Details

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetDataValue4 property (section 6.5.2.42)

Do Not Copy

### 6.5.3.37 MovieWbPresetDataValue5

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD227 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the white balance data of “White balance – Preset manual – d-5” in the video recording menu.

**○ Property Details**

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetDataValue5 property (section 6.5.2.43)

### 6.5.3.38 MovieWbPresetDataValue6

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD228 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x08000800 [Rgain: 1.0, Bgain: 1.0] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the white balance data of “White balance – Preset manual – d-6” in the video recording menu.

**○ Property Details**

Refer to PreWbGainValue in SetPreWbData (section 6.2.8.1) about the format of the PropertyValue.

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetDataValue6 property (section 6.5.2.44)

Do Not Copy

## 6.5.3.39 MovieWbTunePreset1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD229 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-1” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbTunePreset1 property (section 6.5.2.45)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWbPresetProtect1 property (section 6.5.3.45) is [On]. |
| | The MovieWbPresetDataNo property (section 6.5.3.26) is set to anything other than [d-1]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.40 MovieWbTunePreset2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD22A |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-2” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbTunePreset2 property (section 6.5.2.46)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWbPresetProtect2 property (section 6.5.3.46) is [On]. |
| | The MovieWbPresetDataNo property (section 6.5.3.26) is set to anything other than [d-2]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.41 MovieWbTunePreset3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD22B |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-3” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbTunePreset3 property (section 6.5.2.47)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWbPresetProtect3 property (section 6.5.3.47) is [On]. |
| | The MovieWbPresetDataNo property (section 6.5.3.26) is set to anything other than [d-3]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.42 MovieWbTunePreset4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD22C |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-4” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbTunePreset4 property (section 6.5.2.48)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWbPresetProtect4 property (section 6.5.3.48) is [On]. |
| | The MovieWbPresetDataNo property (section 6.5.3.26) is set to anything other than [d-4]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.43 MovieWbTunePreset5

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD22D |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-5” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbTunePreset5 property (section 6.5.2.49)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWbPresetProtect5 property (section 6.5.3.49) is [On]. |
| | The MovieWbPresetDataNo property (section 6.5.3.26) is set to anything other than [d-5]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.44 MovieWbTunePreset6

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD22E |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

### ○ Property Outline

Indicates the white balance fine-tuning volume of “White balance – Preset manual – d-6” in the video recording menu.

For the details of the fine-tuning volume, refer to section 11.5.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbTunePreset6 property (section 6.5.2.50)

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieWbPresetProtect6 property (section 6.5.3.50) is [On]. |
| | The MovieWbPresetDataNo property (section 6.5.3.26) is set to anything other than [d-6]. |
| | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.45 MovieWbPresetProtect1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD22F |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-1” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetProtect1 property (section 6.5.2.51)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.46 MovieWbPresetProtect2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD230 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-2” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetProtect2 property (section 6.5.2.52)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.47 MovieWbPresetProtect3

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD231 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-3” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetProtect3 property (section 6.5.2.53)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.48 MovieWbPresetProtect4

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD232 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-4” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetProtect4 property (section 6.5.2.54)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.49 MovieWbPresetProtect5

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD233 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-5” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetProtect5 property (section 6.5.2.55)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.50 MovieWbPresetProtect6

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD234 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the protection of “White balance – Preset manual – d-6” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: WbPresetProtect6 property (section 6.5.2.56)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | AWBLockStatus property (section 6.5.6.6) is set to [Locked]. |

Do Not Copy

## 6.5.3.51 MovieActive-D-Lighting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD23B |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0000 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Active D-Lighting” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |
| 0x0005 | Extra high |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log] or [HLG]. |
| | MovieFileType property (section 6.5.3.2) is set to [ProRes 422 HQ 10-bit (MOV)] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log]. |
| | MovieFileType property is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |

Do Not Copy

## 6.5.3.52 MovieNoiseReductionHiIso

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD236 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0003 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “High ISO NR” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log]. |
| | MovieFileType property (section 6.5.3.2) is set to [ProRes 422 HQ 10-bit (MOV)] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log]. |
| | MovieFileType property is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |

Do Not Copy

## 6.5.3.53 MovieVignetteControl

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD23E |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x003 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Vignette control” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: VignetteControl property (section 6.5.2.61)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |

Do Not Copy

## 6.5.3.54 MovieDiffractionCompensation

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD23F |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Diffraction compensation” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: DiffractionCompensation property (section 6.5.2.62)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |

Do Not Copy

## 6.5.3.55 MovieAutoDistortion

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0BC |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Auto distortion control” in the video recording menu.

### ○ Property Details

The properties of this setting are in both the photo shooting menu and the video recording menu, and if you change one of them, the settings will be the same.

Linked properties: AutoDistortion property (section 6.5.2.63)

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The LensTypeML property (section 6.5.10.2) is set to [Lens whose automatic distortion correction is always effective]. |

Do Not Copy

### 6.5.3.56 DecreaseFlicker

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD034 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0xFF [Auto] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the setting of “Flicker reduction” in the video recording menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00 | 50 Hz |
| 0x01 | 60 Hz |
| 0xFF | Auto |

Do Not Copy

## 6.5.3.57 MovieSkinSoftening

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D018 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0064 [Same as photo settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “Skin Softening” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Off |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |
| 0x0064 | Same as photo settings |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log]. |
| | MovieFileType property (section 6.5.3.2) is set to [ProRes 422 HQ 10-bit (MOV)] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log]. |
| | MovieFileType property is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |
| | The MovieRecordScreenSize property (section 6.5.3.7) is set to [3840 x 2160 120p], [3840 x 2160 100p], [1920 x 1080 120p] or [1920 x 1080 100p]. |

Do Not Copy

## 6.5.3.58 MoviePortraitImpressionBalance

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D01A |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “Portrait Impression Adjustment” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Mode 1 |
| 2 | Mode 2 |
| 3 | Mode 3 |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit (MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log]. |
| | MovieFileType property is set to [ProRes 422 HQ 10-bit (MOV)] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log]. |
| | MovieFileType property is set to [ProRes RAW HQ 12-bit(MOV)] and MovieProResRawHq12bitToneMode property (section 6.5.3.6) is set to [N-Log]. |
| | MovieFileType property is set to [N-RAW 12-bit(NEV)] and MovieNRaw12bitToneMode property (section 6.5.3.4) is set to [N-Log]. |

Do Not Copy

## 6.5.3.59 MovieHighFrequencyFlickerReduction

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D032 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [OFF] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “High frequency flicker reduction shooting” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | OFF |
| 1 | ON (High resolution shutter setting) |

Do Not Copy

## 6.5.3.60 MovieExposureMeteringMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1AF |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0003 [Matrix metering] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “Exposure metering mode” in the video Lv.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0002 | Center-weighted metering |
| 0x0003 | Matrix metering |
| 0x8010 | Highlight-weighted metering |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The AELockStatus property (section 6.5.6.3) is [Locked]. |

Do Not Copy

## 6.5.3.61 MovieFocusMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1FA |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 2 [AF-F] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Focus mode” in video recording.

### ○ Property Details

If the following conditions are satisfied, it operates with MF mode even when this property is set to AF.

- CPU lens is not mounted.
- The switches on the camera or lens are set to MF.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | AF-S |
| 1 | AF-C |
| 2 | AF-F |
| 4 | MF |

Do Not Copy

## 6.5.3.62 MovieFocusMeteringMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1F8 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x8010 [Single-point AF] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “AF-area mode” in video recording.

### ○ Property Details

In the following cases, this property is set to the single-point AF mode. When the following state is canceled without changing the AF area mode, it returns to the AF area mode which was set immediately before.

- In the case that the MovieFocusMode property (section 6.5.3.61) is [MF].

### ○ Property Value

The valid PropertyValues are shown below.
The value of PropertyValue changes depending on the value of the video focus mode.

| Property Value | Description | Video focus mode setting | | | |
|---|---|---|---|---|---|
| | | MF | AF-S | AF-C | AF-F |
| 0x8010 | Single-point AF mode | ✓ | ✓ | ✓ | ✓ |
| 0x8011 | Auto-area AF mode | - | ✓ | ✓ | ✓ |
| 0x8018 | Wide-area AF (S) | - | ✓ | ✓ | ✓ |
| 0x8019 | Wide-area AF (L) | - | ✓ | ✓ | ✓ |
| 0x801E | Wide-area AF (C1) | - | ✓ | ✓ | ✓ |
| 0x801F | Wide-area AF (C2) | - | ✓ | ✓ | ✓ |
| 0x8033 | Target tracking AF | - | ✓ | ✓ | ✓ |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The video focus mode is set to the manual focus.<br>NOTE) Except when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch. |
| | When the HiResZoom property (section 6.5.3.79) is ON and HiResZoom is controlled ON.<br><br>HiResZoom is ON control when all the following conditions are satisfied.<br>- MovieCaptureAreaCrop property (section 6.5.3.9) is [FX].<br>- MovieFileType property (section 6.5.3.2) is other than [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)].<br>- MovieRecordScreenSize property (section 6.5.3.7) is other than [7680×4320] and [3840×2160 120p/100p/60p/50p]. |

Do Not Copy

## 6.5.3.63 MovieWideAreaAFC1

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D048 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0D03 [13×3] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “AF-area mode – Wide area AF(C1)” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0101 | 1×1 |
| 0x0103 | 1×3 |
| 0x0107 | 1×7 |
| 0x0501 | 5×1 |
| 0x0503 | 5×3 |
| 0x0507 | 5×7 |
| 0x0901 | 9×1 |
| 0x0903 | 9×3 |
| 0x0907 | 9×7 |
| 0x0D01 | 13×1 |
| 0x0D03 | 13×3 |
| 0x0D07 | 13×7 |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The video focus mode is set to the manual focus.<br>NOTE) Except when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch. |
| | When the HiResZoom property (section 6.5.3.79) is ON and HiResZoom is controlled ON.<br><br>Refer to the MovieFocusMeteringMode property (section 6.5.3.62) for conditions for HiResZoom ON control. |

Do Not Copy

## 6.5.3.64 MovieWideAreaAFC2

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D049 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0507 [5×7] |
| Property Value | Same as the MovieWideAreaAFC1 property (section 6.5.3.63) |

### ○ Property Outline

Indicates “AF-area mode – Wide area AF(C2)” in the video recording menu.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The video focus mode is set to the manual focus.<br>NOTE) Except when the focus mode is set to MF due to the CPU lens not attached or the operation with the camera dial/switch. |
| | When the HiResZoom property (section 6.5.3.79) is ON and HiResZoom is controlled ON.<br><br>Refer to the MovieFocusMeteringMode property (section 6.5.3.62) for conditions for HiResZoom ON control. |

Do Not Copy

## 6.5.3.65 MovieAFSubjectDetection

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D006 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Auto] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Subject detection setting in AF” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Auto |
| 2 | Person |
| 3 | Animal |
| 4 | Vehicle |
| 5 | Birds |
| 6 | Airplane |

Do Not Copy

### 6.5.3.66 AFWhenSubjectNotDetected

**Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D014 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates “AF drive when subject is not detected” in the video recording menu.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.3.67 MovieVibrationReduction

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1F9 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 100 [Same as photo settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “Vibration reduction” in the video recording menu.

### ○ Property Details

The supported value changes depending on the type of mounted lens.

### ○ Property Value

The valid PropertyValues are shown below.

When the lens anti- vibration mode is [Normal] or [Sports]:

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Normal |
| 2 | Sports |
| 100 | Same as photo settings |

When the lens anti- vibration mode is only [Normal]:

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |
| 100 | Same as photo settings |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The mounted lens has VR switch. |
| | The LensSort property (section 6.5.10.1) is set to [Not mounted] and the FmmManualSetting2 property (section 6.5.5.2) is set to [0.0mm]. |

Do Not Copy

## 6.5.3.68 ElectronicVR

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD314 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “Electronic VR” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieRecordScreenSize property (section 6.5.3.7) is set to [7680 x 4320], [3840 x 2160 120p], [3840 x 2160 100p], [1920 x 1080 120p], [1920 x 1080 100p] or [Slow-motion]. |
| | MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |
| | When the HiResZoom property (section 6.5.3.79) is ON and HiResZoom is controlled ON.<br><br>HiResZoom is ON control when all the following conditions are satisfied.<br>- MovieCaptureAreaCrop property (section 6.5.3.9) is [FX].<br>- MovieFileType property (section 6.5.3.2) is other than [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)].<br>- MovieRecordScreenSize property (section 6.5.3.7) is other than [7680×4320] and [3840×2160 120p/100p/60p/50p]. |

Do Not Copy

## 6.5.3.69 MovieRecordMicrophoneLevel

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0A2 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Auto sensitivity] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Microphone sensitivity” in the video recording menu.

### ○ Property Details

Switching between [Auto sensitivity/Manual sensitivity] and [Microphone off] cannot be performed during video recording.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Auto sensitivity |
| 1 | Manual sensitivity |
| 2 | Microphone off |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | [Microphone off] is set during video recording. |
| | [Microphone off] is set while [Auto sensitivity] or [Manual sensitivity] is specified during video recording. |
| | The MovieRecordScreenSize property (section 6.5.3.7) is set to [Slow-motion]. |

Do Not Copy

## 6.5.3.70 MovieRecordMicrophoneLevelValue

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0A8 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 15 |
| Property Value | From 1 to 20 |

### ○ Property Outline

Indicates the microphone sensitivity of “Microphone sensitivity – Manual sensitivity” in the video recording menu.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieRecordMicrophoneLevel property (section 6.5.3.69) is set to anything other than [Manual sensitivity]. |
| | The MovieRecordScreenSize property (section 6.5.3.7) is set to [Slow-motion]. |

Do Not Copy

## 6.5.3.71 MovieAttenuator

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD23D |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Attenuator” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieRecordMicrophoneLevel property (section 6.5.3.69) is set to [Microphone off] during video recording. |
| | The MovieRecordScreenSize property (section 6.5.3.7) is set to [Slow-motion]. |

Do Not Copy

## 6.5.3.72 MovieRecordingZone

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0AC |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Wide range] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Frequency response” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Wide range |
| 1 | Vocal range |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The MovieRecordMicrophoneLevel property (section 6.5.3.69) is set to [Microphone off] during video recording. |
| | The MovieRecordScreenSize property (section 6.5.3.7) is set to [Slow-motion]. |

Do Not Copy

## 6.5.3.73 MovieWindNoiseReduction

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0AA |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Wind noise reduction” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | An external microphone is connected. |
| | The MovieRecordMicrophoneLevel property (section 6.5.3.69) is set to [Microphone off] during video recording. |
| | The MovieRecordScreenSize property (section 6.5.3.7) is set to [Slow-motion]. |

Do Not Copy

## 6.5.3.74 RecordTimeCodes

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1FB |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Time codes – Recording time codes” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |
| 2 | On (with HDMI external output) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | During time not set warning. |

Do Not Copy

## 6.5.3.75 CountUpMethod

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1FC |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Rec Run] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Time codes – Count up method” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Rec Run |
| 1 | Free Run |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The RecordTimeCodes property (section 6.5.3.74) is set to [Off]. |
| | During time not set warning. |
| | When connection with ATOMOS AirGlu BT is ON. |

Do Not Copy

## 6.5.3.76 TimeCodeOrigin

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1FD |
| DataType | AUINT8 |
| Description form | None |
| Get / Set | Get, Get / Set |
| Default Value | Array[0]：2 [Current time]<br>From Array[1] to [4]：0 |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Time codes – Time code origin” in the video recording menu.

### ○ Property Details

If Array[0] (Time code origin nformation) is set to the value other than 1 (Manual input), it is as follows.

| Get/Set | Description |
|---|---|
| Get operation | The values from Array[1] to Array[4] are not fixed. |
| Set operation | Any values set from Array[1] to Array[4] are ignored. |

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description | |
|---|---|---|
| NumElement | Time code origin nformation, Number of elements of time | |
| Array[0] | Time code origin nformation | |
| | 0 | Reset |
| | 1 | Manual input |
| | 2 | Current time |
| Array[1] | Hour (From 00 to 23) | |
| Array[2] | Minute (From 00 to 59) | |
| Array[3] | Second (From 00 to 59) | |
| Array[4] | Frame *1<br>24fps (From 00 to 23)<br>25/50/100fps (From 00 to 24)<br>30/60/120fps (From 00 to 29) | |

*1 The range of the settable value changes depending on the frame rate of the MovieRecordScreenSize property (section 6.5.3.7).

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The RecordTimeCodes property (section 6.5.3.74) is set to [Off]. |
| | During time not set warning. |
| | When connection with ATOMOS AirGlu BT is ON. |

Do Not Copy

## 6.5.3.77 DropFrame

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1FE |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Time codes – Drop frame” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The RecordTimeCodes property (section 6.5.3.74) is set to [Off]. |
| | During time not set warning. |
| | When connection with ATOMOS AirGlu BT is ON. |

Do Not Copy

## 6.5.3.78 ExternalRecordingControl

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1DE |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “External recording control (HDMI)” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.3.79 HiResZoom

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D033 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [OFF] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Hi-Res Zoom” in the video recording menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | OFF |
| 1 | ON |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | MovieCaptureAreaCrop property (section 6.5.3.9) is other than [FX]. |
| | MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |
| | MovieRecordScreenSize property (section 6.5.3.7) is set to [7680×4320] or [3840×2160 120p/100p/60p/50p]. |

Do Not Copy

# 6.5.4 Vendor (Custom Settings Menu)

## 6.5.4.1 CustomSettingBank

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD040 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [A] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Custom settings bank” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Do Not Copy

### 6.5.4.2 CustomBankNameA

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD041 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “A” in the custom settings menu.

**○ Property Details**

The PropertyValue is an optional string of 20 characters or shorter (not including the null character).

When the string is shorter than 20 characters, the shortage is padded with spaces (0x20).

For the characters that can be input (ASCII code), refer to section 10.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

### 6.5.4.3 CustomBankNameB

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD042 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “B” in the custom settings menu.

The method of using this property is the same as that of the CustomBankNameA property (section 6.5.4.2).

**○ Response Code**

When the value is set under the following conditions, an error response is made as shown below and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

Do Not Copy

### 6.5.4.4 CustomBankNameC

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD043 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “C” in the custom settings menu.

The method of using this property is the same as that of the CustomBankNameA property (section 6.5.4.2).

**○ Response Code**

When the value is set under the following conditions, an error response is made as shown below and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

### 6.5.4.5 CustomBankNameD

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD044 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | 20 characters of spaces (0x20) |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates the name of “D” in the custom settings menu.

The method of using this property is the same as that of the CustomBankNameA property (section 6.5.4.2).

**○ Response Code**

When the value is set under the following conditions, an error response is made as shown below and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 20 characters is set. |

Do Not Copy

### 6.5.4.6 DynamicAFonAFC

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD048 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Release] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Focus – AF-C priority selection” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Release |
| 2 | Focus/ Release |
| 3 | Focus |

### 6.5.4.7 DynamicAFonAFS

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD049 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 3 [Focus] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Focus – AF-S priority selection” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Release |
| 3 | Focus |

Do Not Copy

### 6.5.4.8 AFStillLockOnAcross

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD046 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 3 |
| Property Value | From 1 [High sensitivity] to 5 [Low sensitivity] |

**Property Outline**

Indicates “Focus – Focus tracking with lock-on – Across” in the custom settings menu.

### 6.5.4.9 AFStillLockOnMove

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD047 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [Smooth] |
| Property Value | From 1 [Smooth] to 2 [Random] |

**Property Outline**

Indicates “Focus – Focus tracking with lock-on – Move” in the custom settings menu.

Do Not Copy

## 6.5.4.10 EnableAFAreaPoint

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD08D |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [All points] |
| Property Value | Refer to Property Value in this section. |

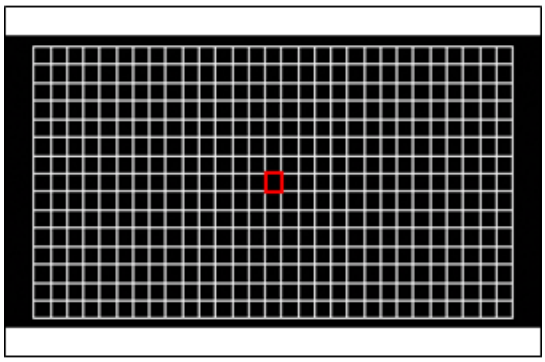
### ○ Property Outline

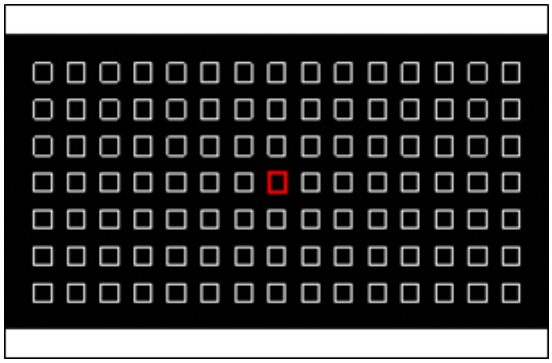
Indicates “Focus – Number of focus points” in the custom settings menu.

### ○ Property Details

If the selected focus point is not included in the skip when the focus points are switched from all points to skip, the selected focus point is changed automatically.

A change pattern of focus points is shown below.
The focus points on the left figure (All points) are changed to the focus points on the right figure (Skip).





### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | All points |
| 1 | Skip |

Do Not Copy

### 6.5.4.11 FocusModeRestrictions

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD177 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0xFF [No restrictions] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Focus – Focus mode restrictions” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | AF-S |
| 1 | AF-C |
| 4 | Manual focus |
| 0xFF | No restrictions |

Do Not Copy

### 6.5.4.12 FocusAreaSelect

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD04F |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [No wrap] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Focus – Focus point wrap-around” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | No wrap |
| 1 | Wrap |

### 6.5.4.13 IsoStep

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD055 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 2 [1/3 step (correction 1/3 step)] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Metering/exposure – ISO sensitivity step value” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 2 | 1/3 step |
| 6 | 1 step |

### 6.5.4.14 ExposureEVStep

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD056 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 2 [1/3 step] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Metering/exposure – EV steps for exposure cntrl” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 2 | 1/3 step (correction 1/3 step) |
| 3 | 1/2 step (correction 1/2 step) |
| 6 | 1 step (correction 1/3 step) |

Do Not Copy

## 6.5.4.15 MatrixMetering

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD09F |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Metering/exposure – Face detection on matrix metering” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

## 6.5.4.16 CenterWeightedExRange

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD059 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Metering/exposure – Center-weighted area” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Small |
| 1 | Normal |
| 0xFF | Average |

### 6.5.4.17 ExposureBaseCompMatrix

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD05A |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | From -6 [-1.0 EV] to +6 [+1.0 EV] |

**Property Outline**

Indicates “Metering/exposure – Fine-tune optimal exposure - Matrix metering” in the custom settings menu.

### 6.5.4.18 ExposureBaseCompCenter

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD05B |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | From -6 [-1.0 EV] to +6 [+1.0 EV] |

**Property Outline**

Indicates “Metering/exposure – Fine-tune optimal exposure - Center-weighted metering” in the custom settings menu.

### 6.5.4.19 ExposureBaseCompSpot

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD05C |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | From -6 [-1.0 EV] to +6 [+1.0 EV] |

**Property Outline**

Indicates “Metering/exposure – Fine-tune optimal exposure - Spot metering” in the custom settings menu.

### 6.5.4.20 ExposureBaseCompHighlight

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD239 |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | From -6 [-1.0 EV] to +6 [+1.0 EV] |

**Property Outline**

Indicates “Metering/exposure – Fine-tune optimal exposure - Highlight-weighted metering” in the custom settings menu.

Do Not Copy

## 6.5.4.21 KeepExposureWhenFNumberChanges

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1C6 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Metering/exposure – Maintaining exposure when the aperture value changes” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | Shutter speed |
| 2 | ISO sensitivity |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | [Shutter speed] is set when the HighFrequencyFlickerReduction property (section 6.5.2.65) is [ON (high-resolution shutter setting)]. |

Do Not Copy

### 6.5.4.22 CSpeedHigh

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD186 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 20 [20 fps] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Shooting/display – Continuous shooting speed – Continuous high-speed” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 10 | 10 fps |
| 12 | 12 fps |
| 15 | 15 fps |
| 20 | 20 fps |

### 6.5.4.23 CspeedLow

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD068 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 5 [5 fps] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Shooting/display – Continuous shooting speed – Continuous low-speed” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | 1 fps |
| 2 | 2 fps |
| 3 | 3 fps |
| 4 | 4 fps |
| 5 | 5 fps |
| 6 | 6 fps |
| 8 | 8 fps |
| 10 | 10 fps |

Do Not Copy

## 6.5.4.24 BurstMaxNumber

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD069 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [No limit] |
| Property Value | 0 [No limit], From 1 [One frame] to 200 [200 frames] |

### ○ Property Outline

Indicates “Shooting/display – Max. continuous release” in the custom settings menu.

### ○ Property Details

1 to 50 frames can be set in 1 frame units, 50 to 100 frames can be set in 10 frame units, and 100 to 200 frames can be set in 20 frame units.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Value | A value that does not satisfy the frame unit is set. |

Do Not Copy

## 6.5.4.25 PreCapturePreReleaseBurst

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D025 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – PreCapture setting – Pre-capture time” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 3 | 0.3 sec. |
| 5 | 0.5 sec. |
| 10 | 1.0 sec. |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [HLG]. |

Do Not Copy

### 6.5.4.26 PreCapturePostReleaseBurst

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D026 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0xFF [Maximum] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Shooting/display – PreCapture setting – Post-release time” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 10 | 1.0 sec. |
| 20 | 2.0 sec. |
| 30 | 3.0 sec. |
| 0xFF | Maximum |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [HLG]. |

Do Not Copy

## 6.5.4.27 ExposureDelay

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD06A |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [OFF] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – Exposure delay mode” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | OFF |
| 2 | 0.2 second |
| 5 | 0.5 second |
| 10 | 1 second |
| 20 | 2 second |
| 30 | 3 second |

Do Not Copy

## 6.5.4.28 ExtendedShutterSpeedsManual

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD16E |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – Extended Shutter Speed (M)” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.4.29 NumberingMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD06C |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – File number sequence” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |

Do Not Copy

## 6.5.4.30 ViewMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD17B |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Priority to shooting settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – View mode settings (photo Lv)” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Priority to shooting settings |
| 1 | Ease of viewing |

Do Not Copy

## 6.5.4.31 ViewModeShowEffectsOfSettings

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D036 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Only when flash is not used] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – View mode (photo Lv) – Show effects of settings” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Only when flash is not used |
| 1 | Always |

## 6.5.4.32 ViewModeEaseOfViewing

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D007 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Auto] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – View mode settings (photo Lv) – Ease of viewing” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Auto |
| 1 | Custom |

Do Not Copy

## 6.5.4.33 ViewModeWhiteBalance

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D008 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Use shooting settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shooting/display – View mode settings (photo Lv) – Ease of viewing – Custom – White balance” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Use shooting settings |
| 1 | Auto |
| 2 | Color temperature setting |

Do Not Copy

### 6.5.4.34 ViewModeWbColorTemp

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D009 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 5000 [K] |
| Property Value | From 2500[K] to 10000[K] (Increase / decrease in 10K increments) |

**○ Property Outline**

Indicates the color temperature value (K) of “Shooting/display – View mode settings (photo Lv) – Ease of viewing – Custom – White balance – Color temperature settings” in the custom settings menu.

### 6.5.4.35 ViewModeWbTuneColorTemp

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D00A |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 612 |
| Property Value | From 0 to 1224 |

**○ Property Outline**

Indicates the fine adjustment value of “Shooting/display – View mode settings (photo Lv) – Ease of viewing – Custom – White balance – Color temperature settings” in the custom settings menu.

For the details of the fine-tuning volume, refer to section 11.5.

Do Not Copy

## 6.5.4.36 ViewModePictureControl

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D00B |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Use shooting settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the settings of “Shooting/display – View mode settings (photo Lv) – Ease of viewing – Custom – Picture control” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Use shooting settings |
| 1 | Ease of viewing |

## 6.5.4.37 ViewModeBrightenShadow

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D00C |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the settings of “Shooting/display – View mode settings (photo Lv) – Ease of viewing – Custom – Shadow correction” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | +1 |
| 2 | +2 |
| 3 | +3 |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [HLG]. |

Do Not Copy

## 6.5.4.38 StarlightView

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D00E |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the settings of “Shooting/display – Starlight view (photo Lv)” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.4.39 FlashSyncSpeed

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD074 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x000100C8 [1/200 s] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Bracketing/flash – Flash sync speed” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x800100FA | 1/250 s (auto FP) |
| 0x800100C8 | 1/200 s (auto FP) |
| 0x000100C8 | 1/200 s |
| 0x000100A0 | 1/160 s |
| 0x0001007D | 1/125 s |
| 0x00010064 | 1/100 s |
| 0x00010050 | 1/80 s |
| 0x0001003C | 1/60 s |

Do Not Copy

## 6.5.4.40 FlashSlowSpeedLimit

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD075 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0x0001003C [1/60 s] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Bracketing/flash – Flash shutter speed” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0001003C | 1/60 s |
| 0x0001001E | 1/30 s |
| 0x0001000F | 1/15 s |
| 0x00010008 | 1/8 s |
| 0x00010004 | 1/4 s |
| 0x00010002 | 1/2 s |
| 0x00010001 | 1 s |
| 0x00020001 | 2 s |
| 0x00040001 | 4 s |
| 0x00080001 | 8 s |
| 0x000F0001 | 15 s |
| 0x001E0001 | 30 s |

Do Not Copy

### 6.5.4.41 ExposureCompFlashUsed

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD118 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Entire frame] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Bracketing/flash – Exposure comp. for flash” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Entire frame |
| 1 | Background only |

### 6.5.4.42 AutoFlashIsoSensitivityControl

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1C5 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Subject and background] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Bracketing/flash – Auto Flash ISO Sensitivity Control” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Subject and background |
| 1 | Subject only |

Do Not Copy

### 6.5.4.43 SelectBracketChangeFactor

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD079 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Flash/speed] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Bracketing/flash – Auto bracketing (mode M)” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Flash/speed |
| 1 | Flash/speed/aperture |
| 2 | Flash/aperture |
| 3 | Flash only |
| 4 | Flash/ISO sensitivity |

### 6.5.4.44 BracketingOrder

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD07A |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [[0] -> [-] -> [+]] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Bracketing/flash – Bracketing order” in the custom settings menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | [0] -> [-] -> [+] |
| 1 | [-] -> [0] -> [+] |

Do Not Copy

## 6.5.4.45 FlushBurstPrioritize

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD180 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Dimming accuracy priority] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Bracketing/flash – Continuous shooting when using flash” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Speed priority |
| 1 | Dimming accuracy priority |

## 6.5.4.46 TVLockSetting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD112 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Controls – Control lock – Shutter speed lock” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than [S] or [M]. |
| | Error / warning displayed on shutter speed or aperture value display. (Either the shutter speed or aperture value turns off) |
| | On the card unformatted error. |
| | Attached a lens that failed firmware upgrade. |

Do Not Copy

## 6.5.4.47 AVLockSetting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD113 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Controls – Control lock – Aperture lock” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| AAccess_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than [A] or [M]. |
| | Error / warning displayed on shutter speed or aperture value display. (Either the shutter speed or aperture value turns off) |
| | On the card unformatted error. |
| | Attached a lens that failed firmware upgrade. |

Do Not Copy

### 6.5.4.48 FocusPointLock

**Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D015 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates “Controls – Control lock – Focus point lock” in the custom settings menu.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.4.49 MovieTVLockSetting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D00F |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video – Control lock – Shutter speed lock” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| AAccess_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than [M]. |
| | Error / warning displayed on shutter speed or aperture value display. (Either the shutter speed or aperture value turns off) |
| | On the card unformatted error. |
| | Attached a lens that failed firmware upgrade. |

Do Not Copy

## 6.5.4.50 MovieAVLockSetting

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D010 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video – Control lock – Aperture lock” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| AAccess_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than [A] or [M]. |
| | Error / warning displayed on shutter speed or aperture value display. (Either the shutter speed or aperture value turns off) |
| | On the card unformatted error. |
| | Attached a lens that failed firmware upgrade. |

Do Not Copy

### 6.5.4.51 MovieFocusPointLock

**Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D016 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates “Video – Control lock – Focus point lock” in the custom settings menu.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

## 6.5.4.52 MovieAfSpeed

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD17C |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 |
| Property Value | From -5[Low speed] to 5[High speed] |

### ○ Property Outline

Indicates “Video – AF speed” in the custom settings menu.

## 6.5.4.53 MovieAfSpeedWhenToApply

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD17D |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Always] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video – AF speed – When to apply” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Always |
| 1 | Only while recording |

Do Not Copy

## 6.5.4.54 MovieAfTrackingSensitivity

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD17E |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 4 |
| Property Value | From 1 [High sensitivity] to 7 [Low sensitivity] |

### ○ Property Outline

Indicates “Video – AF tracking sensitivity” in the custom settings menu.

Do Not Copy

### 6.5.4.55 HighlightBrightness

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD1DF |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 250 |
| Property Value | From 120 to 255 (step size is 5) |

**Property Outline**

Indicates “Video – Zebra Pattern - Brightness detection range” in the custom settings menu.

Do Not Copy

## 6.5.4.56 MovieExtendedIsoStepManual

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D01F |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video – Extended ISO Step (M mode)” in the custom settings menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On (1/6 step) |

Do Not Copy

## 6.5.4.57 MovieExtendedShutterSpeedsManual

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D01B |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Video – Extended Shutter Speed (M mode)” in the custom settings menu.

### ○ Property Details

If the MovieRecordScreenSize property (section 6.5.3.7) is set to slow-motion video, this property setting is disabled.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

# 6.5.5 Vendor (Setup Menu)

## 6.5.5.1 ManualSettingLensNo

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD093 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [No.1] |
| Property Value | From 0 [No. 1] to 19 [No. 20] |

### ○ Property Outline

Indicates “Non-CPU lens data – Lens number” in the setup menu.

Do Not Copy

### 6.5.5.2 FmmManualSetting2

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D04A |
| DataType | UINT32 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [0.0mm] |
| Property Value | From 0 [0.0mm] to 999900 [9999.0mm] (StepSize 10) |

**○ Property Outline**

Indicates “Non-CPU lens data – Focal length (mm)” in the setup menu.

**○ Property Details**

The PropertyValue is 100 times the focal length.

Values less than 100.0 mm can be set in units of 0.1 mm, and values over 100.0 mm can be set in units of 1.0 mm.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Value | When the focal length is 100.0mm or more, decimal places are set. |

### 6.5.5.3 F0ManualSetting2

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D04B |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [F0.00] |
| Property Value | From 0 [F0.00] to 9990 [F99.90] |

**○ Property Outline**

Indicates “Non-CPU lens data – Maximum aperture” in the setup menu.

**○ Property Details**

The PropertyValue is 100 times the maximum aperture value.

Values less than F10.00 can be set in units of F0.01, and values over F10.00 can be set in units of F0.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Value | When the maximum aperture value is F10.00 or more, the second decimal place is set. |

Do Not Copy

### 6.5.5.4 ManualSettingLensName

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D04C |
| DataType | String |
| Description form | None |
| Get / Set | Get, Get / Set |
| Default Value | NULL (0x00) 0 characters |
| Property Value | (depends on user settings) |

**○ Property Outline**

Indicates “Non-CPU lens data – Lens name” in the setup menu.

**○ Property Details**

PropertyValue is an arbitrary string of 36 characters (not including NULL characters).

If the "lens name" is not set in the camera, it will be a string of 0 characters (not including NULL characters).

For characters that can be entered (ASCII code), refer to section 10.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A character string exceeding 36 characters was set. |

Do Not Copy

### 6.5.5.5 SaveFocusPosition

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0CD |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the setting of “Save focus position” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

### 6.5.5.6 ImageSensorCleaning

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD08F |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 2 [Clean at shutdown] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the setting of “Clean image sensor – Clean at startup/shutdown” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Cleaning off |
| 2 | Clean at shutdown |

Do Not Copy

### 6.5.5.7 CommentString

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD090 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | NULL (0x00) 0 character |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates “Image comment” in the setup menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters (not including the null character).

When the comment is not set in the camera, the length of the string is 0 (not including the null character).

The camera does not send the DevicePropChanged event (section 6.4.1.6) even if the PropertyValue is changed.

For the characters that can be input (ASCII code), refer to section 10.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |

### 6.5.5.8 EnableComment

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD091 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Not attached] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Image comment – Attach comment” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not attached |
| 1 | Attached |

Do Not Copy

### 6.5.5.9 ArtistV

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD072 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | NULL (0x00) 0 character |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates “Artist” in the setup menu.

**○ Property Details**

The PropertyValue is an optional string of 36 characters or shorter (not including the null character).

If the artist is not set in the camera, the value is zero length string (not including the null character).

For the characters that can be input (ASCII code), refer to section 10.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made as shown below and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 36 characters is set. |

### 6.5.5.10 CopyrightV

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD073 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | NULL (0x00) 0 character |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates “Copyright” in the setup menu.

**○ Property Details**

The PropertyValue is an optional string of 54 characters or shorter (not including the null character).

If the copyright is not set in the camera, the value is zero length string (not including the null character).

For the characters that can be input (ASCII code), refer to section 10.1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_DeviceProp_Format | A string exceeding 54 characters is set. |

Do Not Copy

### 6.5.5.11 EnableCopyright

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD053 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Not attached] |
| Property Value | (Depends on the user setting) |

**○ Property Outline**

Indicates “Copyright information– Attach copyright information” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not attached |
| 1 | Attached |

Do Not Copy

### 6.5.5.12 IPTCPresetSelect

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD060 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get / Set |
| Default Value | 0 [Not attached] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “IPTC – Attach IPTC when shooting” in the setup menu.

**○ Property Details**

Preset Nos. 1 to 10 are NMS-IPTC, and Preset Nos. 11 to 13 are XMP-IPTC presets.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not attached |
| From 1 to 13 | Preset No. (from 1 to 13) |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | A number with unregistered preset data is specified. |

Do Not Copy

### 6.5.5.13 ShutterSoundEffect

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D012 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [On] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Shutter sound effect” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The SilentMode property (section 6.5.5.14) is set to [On] and headphones are not connected. |

Do Not Copy

### 6.5.5.14 SilentMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D011 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Silent mode” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

### 6.5.5.15 EnableShutter

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD08A |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 1 [Enable release] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Slot empty release lock” in the setup menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Release locked |
| 1 | Enable release |

Do Not Copy

# 6.5.6 Vendor (Camera Information)

## 6.5.6.1 LiveViewSelector

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1A6 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Photo mode] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the state of live view mode.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Photo mode |
| 1 | Video mode |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The camera is in the PC camera mode. |

Do Not Copy

## 6.5.6.2 SpecialShootingMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0x0001D002 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the shooting status of interval timer shooting and focus shift shooting.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On (interval timer shooting) |
| 2 | On (focus shift shooting) |

When PropertyValue is the interval timer shooting, the command processing and property setting shown below are prohibited, and the Access_Denied response is performed.

| Operation Code | Description | Conditions |
|---|---|---|
| 0x100F | FormatStore | - |
| 0x90C0 | InitiateCaptureRecInSdram | - |
| 0x9207 | InitiateCaptureRecInMedia | - |
| 0x9402 | StartSpotWb | - |
| 0x9420 | SaveCameraSetting | - |
| 0x9433 | InitiatePixelMapping | - |
| 0x9436 | ResetMenu | - |

| Property Code | Description | Conditions |
|---|---|---|
| 0x500C | FlashMode | When shooting interval priority is enabled |
| 0x5011 | DateTime | - |
| 0xD010 | ShootingMenuBank | - |
| 0xD011 | ShootingBankNameA | - |
| 0xD012 | ShootingBankNameB | - |
| 0xD013 | ShootingBankNameC | - |
| 0xD014 | ShootingBankNameD | - |
| 0xD040 | CustomSettingBank | - |
| 0xD041 | CustomBankNameA | - |
| 0xD042 | CustomBankNameB | - |
| 0xD043 | CustomBankNameC | - |
| 0xD044 | CustomBankNameD | - |
| 0xD078 | BracketingType | - |
| 0xD07F | ExtendShootingMenu | - |
| 0xD08F | ImageSensorCleaning | - |
| 0xD0C0 | EnableBracketing | - |
| 0xD0C1 | AEBracketingStep | - |
| 0xD0C2 | AEBracketingPattern | - |

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| | | |
|---|---|---|
| 0xD0C4 | WBBracketingStep | - |
| 0xD0C5 | WBBracketingPattern | - |
| 0xD0C6 | ADLBracketingPattern | - |
| 0xD0C7 | ADLBracketingStep | - |
| 0xD126 | InternalFlashCompensation | When shooting interval priority is enabled |
| 0xD130 | HDRMode | - |
| 0xD132 | HDRStrength | - |
| 0xD133 | HDRSaveIndividualImages | - |
| 0x0001D01C | ToneMode | - |
| 0x0001D045 | PixelShiftShootingMode | - |
| 0x0001D050 | PixelShiftShootingInterval | - |

When PropertyValue is the focus shift shooting, the command processing and property setting shown below are prohibited, and the Access_Denied response is performed.

| Operation Code | Description | Conditions |
|---|---|---|
| 0x100B | DeleteObject | - |
| 0x100F | FormatStore | - |
| 0x90C0 | InitiateCaptureRecInSdram | - |
| 0x90CD | SetPicCtrlData | When registering a custom picture control |
| 0x90CE | DeleteCustomPicCtrl | - |
| 0x9207 | InitiateCaptureRecInMedia | - |
| 0x9402 | StartSpotWb | - |
| 0x9420 | SaveCameraSetting | - |
| 0x9433 | InitiatePixelMapping | - |
| 0x9436 | ResetMenu | - |

| Property Code | Description | Conditions |
|---|---|---|
| 0x5003 | ImageSize | - |
| 0x5004 | CompressionSetting | - |
| 0x5011 | DateTime | - |
| 0xD010 | ShootingMenuBank | - |
| 0xD011 | ShootingBankNameA | - |
| 0xD012 | ShootingBankNameB | - |
| 0xD013 | ShootingBankNameC | - |
| 0xD014 | ShootingBankNameD | - |
| 0xD016 | RawCompressionType | - |
| 0xD030 | CaptureAreaCrop | - |
| 0xD040 | CustomSettingBank | - |
| 0xD041 | CustomBankNameA | - |
| 0xD042 | CustomBankNameB | - |
| 0xD043 | CustomBankNameC | - |
| 0xD044 | CustomBankNameD | - |
| 0xD05D | StillFocusMeteringMode | - |
| 0xD061 | StillFocusMode | - |
| 0xD078 | BracketingType | - |
| 0xD07F | ExtendShootingMenu | - |
| 0xD08F | ImageSensorCleaning | - |
| 0xD0C0 | EnableBracketing | - |

Do Not Copy

| | | |
|---|---|---|
| 0xD0C1 | AEBracketingStep | - |
| 0xD0C2 | AEBracketingPattern | - |
| 0xD0C4 | WBBracketingStep | - |
| 0xD0C5 | WBBracketingPattern | - |
| 0xD0C6 | ADLBracketingPattern | - |
| 0xD0C7 | ADLBracketingStep | - |
| 0xD130 | HDRMode | - |
| 0xD132 | HDRStrength | - |
| 0xD133 | HDRSaveIndividualImages | - |
| 0xD148 | Slot2ImageSaveMode | - |
| 0xD156 | PrimarySlot | - |
| 0x0001D01C | ToneMode | - |
| 0x0001D041 | EnableDXImageSize | - |
| 0x0001D042 | ImageSizeDX | - |
| 0x0001D045 | PixelShiftShootingMode | - |
| 0x0001D050 | PixelShiftShootingInterval | - |

Do Not Copy

### 6.5.6.3 AELockStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD105 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Lock released] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the AE lock status.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Lock released |
| 1 | Locked |

### 6.5.6.4 AFLockStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD104 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Lock released] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the AF lock status.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Lock released |
| 1 | Locked |

Do Not Copy

### 6.5.6.5 FVLockStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD106 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Lock released] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the FV lock status.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Lock released |
| 1 | Locked |

### 6.5.6.6 AWBLockStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD12F |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Lock released] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the AWB lock status.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Lock released |
| 1 | Locked |

Do Not Copy

## 6.5.6.7 ShutterSpeed

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD100 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | (Minimum value in the setting range) |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Shutter speed” that is set in the camera.

For the shutter speed in video mode, the MovieShutterSpeed property (section 6.5.12.2) should be used.

### ○ Property Details

This property can handle the 1/1000 or low shutter speed in addition to the ExposureTime property (section 6.5.1.10). In this property, the upper 2 bytes of PropertyValue are treated as the numerator of the shutter speed, and the lower 2 bytes are treated as the denominator.

If the enum value changes, update the enum value and DefaultValue. Enumerated values vary depending on the following factors.

**Table 6.5.6-1 Associated properties**

| Property | Ref. | |
|---|---|---|
| ExposureProgramMode | 6.5.1.11 | Shooting mode |
| ExtendedShutterSpeedsManual | 6.5.4.28 | Extended shutter speeds (M) |
| StillCaptureMode | 6.5.1.15 | Release mode |
| ExposureEVStep | 6.5.4.14 | Step value for exposure setting |
| ExternalSpeedLightExist | 6.5.9.1 | Speedlight attached |
| NewExternalSpeedLightMode | 6.5.9.4 | Speedlight flash mode |
| FlashSyncSpeed | 6.5.4.39 | Flash sync speed |
| FlashMode | 6.5.1.9 | Flash mode |
| FlashSlowSpeedLimit | 6.5.4.40 | Flash shutter speed limit |

When the shooting mode is M, values for bulb shooting and time shooting are added.

When the shooting mode is S, a value corresponding to the current value of bulb shooting/time shooting are added to the enumerated value.

If the shooting mode is M and the shutter speed extension in M mode is set to "On", a value exceeding 30 seconds is added.

Do Not Copy

When the StillCaptureMode property (section 6.5.1.15) is set to "High-speed frame capture (C15/C30/C60)", the slow shutter speed is limited to 1/60 second. When the StillCaptureMode property (section 6.5.1.15) is set to "High-speed frame capture (C120)", the slow shutter speed is limited to 1/125 second.

If the camera is equipped with a speedlight, it depends on the following factors:

Shutter speed on the high-speed side:
The value is limited to the slower value between flash shooting synchronization speed and the limit of the speedlight.
The value is not limited when the auto FP is supported by the flash mode of the speedlight and the flash shooting synchronization speed is set to the auto FP.

Shutter speed on the low-speed side:
If both the flash mode and shooting mode are set to the values shown below, the shutter speed is limited to the value set in flash shutter speed limit.

| Flash mode | Shooting mode |
|---|---|
| Red-eye reduction flash | P |
| Normal flash | A |

### ○ Property Value

The valid PropertyValues are shown below.

Shutter speed:

| Property Value | Description |
|---|---|
| Upper 2 bytes | Numerator of the shutter speed |
| Lower 2 bytes | Denominator of the shutter speed |

Special values other than shutter speed:

| Property Value | Description |
|---|---|
| 0xFFFFFFFF | Bulb shooting |
| 0xFFFFFFFD | Time shooting |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExposureProgramMode property (section 6.5.1.11) is not set to [S] or [M]. |
| | The WarningStatus property (section 6.5.6.22) is as follows:<br>[Sequence error]<br>[Retractable zoom-ring warning]. |
| | Bulb shooting or time shooting is set when the WarningStatus property is [Bulb warning/Time warning]. |
| | The TVLockSetting property (section 6.5.4.46) is set to [On]. |

Do Not Copy

### 6.5.6.8 FlickerReductionShutterSpeed

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D03D |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | (Minimum shutter speed for high frequency flicker reduction) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates shutter speed for high frequency flicker reduction set in the camera.

**○ Property Details**

You can get the shutter speed according to the current state (photo mode or video mode).

This property expresses the denominator value of the shutter speed with PropertyValue to the first decimal place.

If the Access_Denied condition of the SetFlickerReductionShutterSpeed command (section 6.2.4.10) is satisfied, the high-frequency flicker reduction control will be turned OFF, and the property value will be invalid and set to "0".

**○ Property Value**

Valid PropertyValues are listed below.

The denominator value of the shutter speed for high frequency flicker reduction x 10.

(Example) Shutter speed 1/8192.0 seconds: PropertyValue = 81920
(Example) Shutter speed 1/32.2 sec: PropertyValue = 322

Do Not Copy

### 6.5.6.9 RemainingExposureTime

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0BE |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 second |
| Property Value | From 0 to 900 second |

**○ Property Outline**

Indicates the remaining exposure time for photo shooting.

**○ Property Details**

The DevicePropChanged event is transmitted when the shooting mode is set to M and shooting is performed at the extended shutter speed.

Do Not Copy

### 6.5.6.10 FlexibleProgram

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD109 |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | From -30 [-5 EV] to +30 [+5 EV] |

**○ Property Outline**

Indicates the program shift value in units of 1/6 EV.

**○ Property Details**

When the value of the ExposureProgramMode property (section 6.5.1.11) is a value other than [P], the value of PropertyValue is not valid but set to 0.

The StepSize of the property changes depending on the value of the ExposureEVStep property (section 6.5.4.14).

| ExposureEVStep | StepSize |
|---|---|
| 0 (1/3 EV) | 2 |
| 1 (1/2 EV) | 3 |
| 2 (1 EV) | 6 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExposureProgramMode property (section 6.5.1.11) is set to anything other than [P]. |
| | Displaying an error / warning that prohibits program shift setting. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |

Do Not Copy

## 6.5.6.11 InternalFlashCompensation

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD126 |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 |
| Property Value | From -18 to +6 |

### ○ Property Outline

Indicates “Flash compensation value” of the camera in units of 1/6 EV.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description | ExposureEVStep | | |
|---|---|---|---|---|
| | | 1/3 step | 1/2 step | 1 step |
| -18 | -3.0EV | ✓ | ✓ | ✓ |
| -16 | -2.7EV | ✓ | - | ✓ |
| -15 | -2.5EV | - | ✓ | - |
| -14 | -2.3EV | ✓ | - | ✓ |
| -12 | -2.0EV | ✓ | ✓ | ✓ |
| -10 | -1.7EV | ✓ | - | ✓ |
| -9 | -1.5EV | - | ✓ | - |
| -8 | -1.3EV | ✓ | - | ✓ |
| -6 | -1.0EV | ✓ | ✓ | ✓ |
| -4 | -0.7EV | ✓ | - | ✓ |
| -3 | -0.5EV | - | ✓ | - |
| -2 | -0.3EV | ✓ | - | ✓ |
| 0 | 0.0EV | ✓ | ✓ | ✓ |
| 2 | 0.3EV | ✓ | - | ✓ |
| 3 | 0.5EV | - | ✓ | - |
| 4 | 0.7EV | ✓ | - | ✓ |
| 6 | 1.0EV | ✓ | ✓ | ✓ |

The StepSize of this property changes depending on the value of the ExposureEVStep property (section 6.5.4.14).

| ExposureCompEVStep | StepSize |
|---|---|
| 0 (1/3 EV) | 2 |
| 1 (1/2 EV) | 3 |
| 2 (1 EV) | 2 |

Do Not Copy

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is set to [On]. |
| | The SilentMode property (section 6.5.5.14) is set to [On]. |
| | During SB setting warning. |
| | During image dust off data acqusition mode. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON" and the PixelShiftShootingInterval property (section 6.5.2.79) is set to "0" |

Do Not Copy

## 6.5.6.12 ISOControlSensitivity

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0B5 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 100 [ISO 100] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates ISO sensitivity that is controlled by the camera.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description | Property Value | Description |
|---|---|---|---|
| 32 | Lo1.0 | 6400 | ISO6400 |
| 40 | Lo0.7 | 7200 | ISO7200 |
| 45 | Lo0.5 | 8000 | ISO8000 |
| 50 | Lo0.3 | 9000 | ISO9000 |
| 56 | ISO56 | 10000 | ISO10000 |
| 64 | ISO64 | 11000 | ISO11000 |
| 72 | ISO72 | 12800 | ISO12800 |
| 80 | ISO80 | 14400 | ISO14400 |
| 90 | ISO90 | 16000 | ISO16000 |
| 100 | ISO100 | 18000 | ISO18000 |
| 110 | ISO110 | 20000 | ISO20000 |
| 125 | ISO125 | 22000 | ISO22000 |
| 140 | ISO140 | 25600 | ISO25600 |
| 160 | ISO160 | 28800 | Hi0.2 |
| 180 | ISO180 | 32000 | Hi0.3 |
| 200 | ISO200 | 36000 | Hi0.5 |
| 220 | ISO220 | 40000 | Hi0.7 |
| 250 | ISO250 | 45600 | Hi0.8 |
| 280 | ISO280 | 51200 | Hi1.0 |
| 320 | ISO320 | 57600 | Hi1.2 |
| 360 | ISO360 | 64000 | Hi1.3 |
| 400 | ISO400 | 72000 | Hi1.5 |
| 450 | ISO450 | 80000 | Hi1.7 |
| 500 | ISO500 | 91200 | Hi1.8 |
| 560 | ISO560 | 102400 | Hi2.0 |
| 640 | ISO640 | | |
| 720 | ISO720 | | |
| 800 | ISO800 | | |
| 900 | ISO900 | | |
| 1000 | ISO1000 | | |
| 1100 | ISO1100 | | |
| 1250 | ISO1250 | | |
| 1400 | ISO1400 | | |
| 1600 | ISO1600 | | |
| 1800 | ISO1800 | | |

Do Not Copy

| | | | |
|---|---|---|---|
| 2000 | ISO2000 | | |
| 2200 | ISO2200 | | |
| 2500 | ISO2500 | | |
| 2800 | ISO2800 | | |
| 3200 | ISO3200 | | |
| 3600 | ISO3600 | | |
| 4000 | ISO4000 | | |
| 4500 | ISO4500 | | |
| 5000 | ISO5000 | | |
| 5600 | ISO5600 | | |

Do Not Copy

## 6.5.6.13 ExposureDisplayStatus

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1B0 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Normal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the display status of the shutter speed and the aperture value in the camera.

### ○ Property Details

It is 1 (only shutter speed is blinking) during bulb/time warning.

### ○ Property Value

The valid PropertyValues are shown below.

| | | Shutter speed | |
|---|---|---|---|
| | | Normal | Blinking |
| Aperture value | Normal | 0 | 1 |
| | Blinking | 2 | 3 |

### 6.5.6.14 ExposureIndicateStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1B1 |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [0.0 EV] |
| Property Value | From -60 [-10 EV] to +60 [+10 EV] |

**○ Property Outline**

Indicates the display value of the indicator in units of 1/6 EV.

When the ExposureIndicateLightup property (section 6.5.6.15) is [Off], the value of PropertyValue is not fixed.

### 6.5.6.15 ExposureIndicateLightup

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1B3 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [On] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the indicator display On/Off.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | On |
| 1 | Off |

Do Not Copy

### 6.5.6.16 MonitorOffStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1BA |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Not monitor-off] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the camera monitor off state. The actual monitor control status is not shown.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not monitor-off |
| 1 | Monitor-off |

Do Not Copy

### 6.5.6.17 OrientationSensorMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD092 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [ON] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates “Recording rotation information” in the playback menu

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | OFF |
| 1 | ON |

### 6.5.6.18 Orientation

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD10E |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Landscape or not fixed] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the orientation information.

**○ Property Details**

If the OrientationSensorMode property (section 6.5.6.17) is OFF, the PropertyValue will be “Landscape or not fixed”.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Landscape or not fixed |
| 1 | Portrait (grip side upward) |
| 2 | Portrait (grip side downward) |
| 3 | Landscape (upside down) |

Do Not Copy

## 6.5.6.19 AngleLevel

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD067 |
| DataType | INT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 [Horizontal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the level angle information of the camera.

### ○ Property Details

The valid angle information is shown below.

From 0.0 degrees to 359.9999847412109375 degrees

The angle information is 0.0 degrees when the camera is in the horizontal state. The angle increases as the camera is rotated counterclockwise viewing from the person who operates the camera.

When the camera is rotated clockwise starting from 0.0 degrees, the angle becomes 359.9999847412109375 degrees or smaller.
When the camera is rotated counterclockwise starting from 359.9999847412109375 degrees, the angle becomes 0.0 degrees or larger.

The angle information is displayed in green when it is horizontal or vertical in the virtual horizon of the setup menu.
Add +0.5 degrees to the angle information and omit the value below the decimal section. When the result is 0 or a multiple of 90 degrees, it is judged to be horizontal or vertical.

Even if the PropertyValue is changed, the camera does not send a DevicePropChanged event (section 6.4.1.6).

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| Upper 16 bits | Integer section of the angle information |
| Lower 16 bits | Decimal section of the angle information |

*When the angle information cannot be acquired or it is not reliable, the PropertyValue is set to 0xFFFFFFFF.

Do Not Copy

## 6.5.6.20 AngleLevelPitching

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD07D |
| DataType | INT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 [Horizontal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the level angle information (pitching) of the camera.

### ○ Property Details

The valid angle information is shown below.

For the data type, the signed 32-bit fixed-point system is used, and the integer section and the decimal section use the upper 16 bits and the lower 16 bits, respectively.

The angle information is 0.0 or 180.0 degrees when the camera is in the horizontal state. The angle range is ±15 degrees from 0.0 or 180.0 degrees.

| Camera orientation | Lens orientation | Angle range | Horizontal |
|---|---|---|---|
| Landscape | Up | From 0.0 to 60.0 | 0.0 |
| | Down | 0.0, from 359.9999847412109375 to 300.0 | |
| Landscape (upside down) | Up | From 180.0 to 120.0 | 180.0 |
| | Down | From 180.0 to 240.0 | |

The angle information is displayed in green when it is horizontal in the virtual horizon of the setup menu. Add +0.5 degrees to the angle information and omit the value below the decimal section. When the result is 0.0 or 180.0 degrees, it is judged to be horizontal.

Even if the PropertyValue is changed, the camera does not send a DevicePropChanged event (section 6.4.1.6).

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| Upper 16 bits | Integer section of the angle information |
| Lower 16 bits | Decimal section of the angle information |

*When the angle information cannot be acquired or it is not reliable, the PropertyValue is set to 0xFFFFFFFF.

Do Not Copy

## 6.5.6.21 AngleLevelYawing

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD07E |
| DataType | INT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 [Horizontal] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the level angle information (yawing) of the camera.

### ○ Property Details

The valid angle information is shown below.

For the data type, the signed 32-bit fixed-point system is used, and the integer section and the decimal section use the upper 16 bits and the lower 16 bits, respectively.

The angle information is 0.0 or 180.0 degrees when the camera is in the horizontal state. The angle range is ±15 degrees from 0.0 or 180.0 degrees.

| Camera orientation | Lens orientation | Angle range | Horizontal |
|---|---|---|---|
| Portrait (grip side upward) | Up | From 0.0 to 60.0 | 0.0 |
| | Down | 0.0, from 359.9999847412109375 to 300.0 | |
| Portrait (grip side downward) | Up | From 180.0 to 120.0 | 180.0 |
| | Down | From 180.0 to 240.0 | |

The angle information is displayed in green when it is horizontal in the virtual horizon of the setup menu. Add +0.5 degrees to the angle information and omit the value below the decimal section. When the result is 0.0 or 180.0 degrees, it is judged to be horizontal.

Even if the PropertyValue is changed, the camera does not send a DevicePropChanged event (section 6.4.1.6).

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| Upper 16 bits | Integer section of the angle information |
| Lower 16 bits | Decimal section of the angle information |

*When the angle information cannot be acquired or it is not reliable, the PropertyValue is set to 0xFFFFFFFF.

Do Not Copy

## 6.5.6.22 WarningStatus

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD102 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 [No warning] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the camera warning information.

### ○ Property Value

The valid PropertyValues are shown below.

When the PropertyValue is set to any value other than 0, the release is locked.
The PropertyValue takes the following values in the bit assignment. (1: Valid, 0: Invalid)

| Bit | Description |
|---|---|
| From Bit15 to Bit31 | Reserved |
| Bit14 | A non-compatible lens is attached. |
| Bit13 | A lens that failed to firmware upgrade is attached. |
| Bit12 | Shutter speed incompatibility HDR shooting not possible Warning |
| Bit11 | Version upgrade warning when mount adapter is attached |
| Bit10 | Retractable zoom-ring warning |
| Bit9 | Reserved |
| Bit8 | Shooting error |
| Bit7 | Reserved |
| Bit6 | Bulb warning/Time warning |
| Bit5 | Minimum aperture warning |
| Bit4 | Reserved |
| Bit3 | Reserved |
| Bit2 | Reserved |
| Bit1 | Battery exhausted |
| Bit0 | Sequence error |

Do Not Copy

## 6.5.6.23 InfoDisplayErrorStatus

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1B2 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the error display status of the INFO display on the monitor.

### ○ Property Details

The error display conditions indicated by this property are as shown in the table below.

| Name | Timing for starting message of errors and warning | Finish timing |
|---|---|---|
| Release sequence error | After the release sequence is completed | Generation cause released |
| Card access error | When the card is inserted or the card is accessed | Generation cause released |
| Battery ID unauthentication error | When a battery other than the exclusive ID battery is mounted | Generation cause released |
| Minimum aperture warning | When the Fmin detection switch is turned Off | Generation cause released |
| Card write-protected & not-formatted warning | When a card is inserted with the recording destination set to “Card (Card and SDRAM)” or when the recording destination is set to “Card (Card and SDRAM)” while a card is inserted | Generation cause released |
| Card not-formatted warning | Same as the above | Generation cause released |
| Card write-protected warning | Same as the above | Generation cause released |
| Battery exhausted (shutter release disabled) warning | After the shutter-release button is pressed fully | Generation cause released |

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

# 6.5.7 Vendor (Recording Destination)

## 6.5.7.1 RecordingMedia

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD10B |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Card] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the recording destination of the images.

### ○ Property Details

For more information, see Recording Destination (section 2.6).

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Card |
| 1 | SDRAM |
| 2 | Card and SDRAM |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The SpecialShootingMode property (section 6.5.6.2) is set to [On (interval timer shooting/focus shift shooting)]. |

Do Not Copy

### 6.5.7.2 ActiveFolder

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD15B |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 100 |
| Property Value | From 100 to 999 |

**○ Property Outline**

Indicates the folder number of the recording location.

Do Not Copy

### 6.5.7.3 ActiveSlot

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1F2 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Card not inserted] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the recording destination slot when the recording destination is a card.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Card not inserted |
| 1 | Slot 1 |
| 2 | Slot 2 |
| 3 | Slot 1 & slot 2 |

Do Not Copy

### 6.5.7.4 ExposureRemaining

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1F1 |
| DataType | UINT32 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [0 frames] |
| Property Value | 0 [0 frames] to 4294967295 [4294967295 frames] |

**○ Property Outline**

Indicates the number of frames that can be recorded on the card.

**○ Property Details**

The value changes depending on the setting of the camera. When a card is not inserted in the camera, it should be 0 frames.

Do Not Copy

## 6.5.7.5 ContinuousShootingCount

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1B4 |
| DataType | UINT16 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 450 [450 frames] |
| Property Value | From 0 [0 frames] to 450 [450 frames] |

### ○ Property Outline

Indicates the number of frames that can be recorded in continuous shooting by the command.

The number of frames that can be recorded in continuous shooting can be acquired with any setting value of the RecordingMedia property (section 6.5.7.1).

### ○ Property Details

The number of continuous shooting frames changes depending on the following setting values on the camera.

| Set contents | Property |
|---|---|
| Image quality mode | CompressionSetting property (section 6.5.1.3) |
| Image size | ImageSize property (section 6.5.1.2)<br>EnableDXImageSize property (section 6.5.2.11)<br>ImageSizeDX property (section 6.5.2.12) |
| Long exposure noise reduction | NoiseReduction property (section 6.5.2.59) |
| High ISO noise reduction | NoiseReductionHiIso property (section 6.5.2.60) |
| HDR | HDRMode property (section 6.5.2.72) |
| Max. continuous release | BurstMaxNumber property (section 6.5.4.24) |
| Choose image area | CaptureAreaCrop property (section 6.5.2.10) |
| NEF (RAW) recording – Compression Type | RawCompressionType property (section 6.5.2.13) |
| Pre-release time | PreCapturePreReleaseBurst property (section 6.5.4.25) |
| Post-release time | PreCapturePostReleaseBurst property (section 6.5.4.26) |
| Photo/Video switching lever | LiveViewSelector property (section 6.5.6.1) |

Do Not Copy

# 6.5.8 Vendor (Bracketing)

## 6.5.8.1 EnableBracketing

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0C0 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the status of bracketing.

### ○ Property Details

If the property of the number of shots corresponding to the BracketingType property setting is "0", the PropertyValue will be "Off".

| BracketingType property | Property of the number of shots | Number of shots | |
|---|---|---|---|
| | | 0 | Other than 0 |
| AE・Flash/AE/Flash | AEBracketingPattern | Off | On |
| WB | WBBracketingPattern | Off | On |
| ADL | ADLBracketingPattern | Off | On |

PropertyValue becomes [Off] due to the combination of the setting of BracketingType property (section 6.5.8.2) and other settings.

The conditions for canceling the bracketing are described below.

| Conditions | | BracketingType property | | | | |
|---|---|---|---|---|---|---|
| | | AE Flash | AE | Flash | WB | ADL |
| Change BracketingType property | WB⇔Changes between other types | On | On | On | On | On |
| | ADL⇔Changes between other types | On | On | On | On | On |
| Change ExposureEVStep property (section 6.5.4.14) | | On | On | On | Off | Off |
| Change CompressionSetting property to RAW or RAW + JPEG | | Off | Off | Off | On | Off |
| Change SilentMode property | | Off | Off | On* | Off | Off |
| Change the StillCaptureMode property (section 6.5.1.15) to "High-speed frame capture". | | On* | On* | On* | On* | On* |
| Change the value of the FlashMode property (section 6.5.1.9) to “Flash-off”. | | Off | Off | On* | Off | Off |

NOTE) PropertyValue temporarily becomes "Off".

The property of the number of shots does not become "0", and when the condition is canceled, the PropertyValue returns to "On".

Do Not Copy

When this property is changed, a value is also set for the number of shots property.
The values set in the number of shots property by changing this property are described below.

| BracketingType property | Property of the number of shots | EnableBracketing | |
|---|---|---|---|
| | | Off | On |
| AE・Flash/AE/Flash | AEBracketingPattern | 0 | 3 images (normal, under and over) |
| WB | WBBracketingPattern | 0 | 3 images (normal, under and over) |
| ADL | ADLBracketingPattern | 0 | 2 images |

When video mode, bracketing shooting is not performed even if this property is set to [On].

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The HDRMode property (section 6.5.2.72) is [On]. |
| | The BracketingType property (section 6.5.8.2) is set to [WB bracketing] and the CompressionSetting property (section 6.5.1.3) is set to [RAW] or [RAW + JPEG]. |
| | The BracketingType property (section 6.5.8.2) is set to [Flash only] and the SilentMode property (section 6.5.5.14) is set to [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to "High-speed frame capture". |
| | The BracketingType property (section 6.5.8.2) is set to [Flash only] and the FlashMode property (section 6.5.1.9) is set to [Flash-off]. |
| | The BracketingType property is set to [ADL Bracketing] and the ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.8.2 BracketingType

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD078 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [AE & flash] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates “Auto bracketing - Auto bracketing set” in the photo shooting menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | AE & flash |
| 1 | AE only |
| 2 | Flash only |
| 3 | WB bracketing |
| 4 | ADL bracketing |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

### 6.5.8.3 AEBracketingPattern

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0C2 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0000 [0 image] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the compensation direction and the number of images to be captured for AE bracketing.

**○ Property Details**

The conditions for PropertyValue to be 0 are described below.

| No. | Description |
|---|---|
| 1 | When the value of the ExposureEVStep property (section 6.5.4.14) is changed. |

When the BracketingType property (section 6.5.8.2) is set to [WB bracketing] or [ADL bracketing], the value of PropertyValue is not fixed.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | 0 image |
| 0x0403 | 3 images (normal, under and over) |
| 0x0405 | 5 images (normal, 2 unders and 2 overs) |
| 0x0407 | 7 images (normal, 3 unders and 3 overs) |
| 0x0409 | 9 images (normal, 4 unders and 4 overs) |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The BracketingType property (section 6.5.8.2) is set to [WB bracketing] or [ADL bracketing]. |
| | When the AEBracketingStep property (section 6.5.8.4) is set to [2.0], [2.3], [2.5], [2.7] or [3.0] is set. |
| | The SilentMode property (section 6.5.5.14) is set to [On] and the BracketingType property (section 6.5.8.2) is set to [Flash Bracketing]. |
| | The HDRMode property (section 6.5.2.72) is [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.8.4 AEBracketingStep

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0C1 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 6 [1 EV] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the step range of AE bracketing.

### ○ Property Details

When the value of the ExposureEVStep property (section 6.5.4.14) is changed, the PropertyValue is set to 1.0.

When the BracketingType property (section 6.5.8.2) is set to [WB bracketing] or [ADL bracketing], the value of PropertyValue is not fixed.

The value of PropertyValue changes depending on the value of the ExposureEVStep property.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description | ExposureEVStep | | |
|---|---|---|---|---|
| | | 1/3 step | 1/2 step | 1 step |
| 2 | 0.3 | ✓ | - | - |
| 3 | 0.5 | - | ✓ | - |
| 4 | 0.7 | ✓ | - | - |
| 6 | 1.0 | ✓ | ✓ | ✓ |
| 8 | 1.3 | ✓ | - | - |
| 9 | 1.5 | - | ✓ | - |
| 10 | 1.7 | ✓ | - | - |
| 12 | 2.0 | ✓ | ✓ | ✓ |
| 14 | 2.3 | ✓ | - | - |
| 15 | 2.5 | - | ✓ | - |
| 16 | 2.7 | ✓ | - | - |
| 18 | 3.0 | ✓ | ✓ | ✓ |

Do Not Copy

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The BracketingType property (section 6.5.8.2) is set to [WB bracketing] or [ADL bracketing]. |
| | The SilentMode property (section 6.5.5.14) is set to [On] and the BracketingType property (section 6.5.8.20) is set to [Flash Bracketing]. |
| | The HDRMode property (section 6.5.2.72) is [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.8.5 WBBracketingPattern

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0C5 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0000 [0 image] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the compensation direction and the number of images to be captured for WB bracketing.

### ○ Property Details

When the BracketingType property (section 6.5.8.2) is set to anything other than [WB bracketing], the value of PropertyValue is not fixed.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | 0 image |
| 0x0403 | 3 images (normal, under and over) |
| 0x0405 | 5 images (normal, 2 unders and 2 overs) |
| 0x0407 | 7 images (normal, 3 unders and 3 overs) |
| 0x0409 | 9 images (normal, 4 unders and 4 overs) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The BracketingType property (section 6.5.8.2) is set to anything other than [WB bracketing]. |
| | The CompressionSetting property (section 6.5.1.3) is set to [RAW] or [RAW + JPEG]. |
| | The HDRMode property (section 6.5.2.72) is [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

### 6.5.8.6 WBBracketingStep

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0C4 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [1 EV] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the step range for WB bracketing.

**○ Property Details**

When the BracketingType property (section 6.5.8.2) is set to anything other than [WB bracketing], the value of PropertyValue is not fixed.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | 1 EV |
| 2 | 2 EV |
| 3 | 3 EV |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The BracketingType property (section 6.5.8.2) is set to anything other than [WB bracketing]. |
| | The CompressionSetting property (section 6.5.1.3) is set to [RAW] or [RAW + JPEG]. |
| | The HDRMode property (section 6.5.2.72) is [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.8.7 ADLBracketingPattern

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0C6 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0x0000 [0 image] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the number of images to be captured for ADL bracketing.

### ○ Property Details

When the BracketingType property (section 6.5.8.2) is set to anything other than [ADL bracketing], the value of PropertyValue is not fixed.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | 0 shot |
| 0x0002 | 2 shots (Off -> User setting) |
| 0x0003 | 3 shots (Off -> Low -> Normal) |
| 0x0004 | 4 shots (Off -> Low -> Normal -> High) |
| 0x0105 | 5 shots (Off -> Low -> Normal -> High -> Extra high1) |
| 0x0205 | 5 shots (Low -> Normal -> High -> Extra high1 -> Extra high2) |

* For the user setting, the setting value of the ADLBracketingStep property (section 6.5.8.8) should be used.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The BracketingType property (section 6.5.8.2) is set to anything other than [ADL bracketing]. |
| | The HDRMode property (section 6.5.2.72) is [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

### 6.5.8.8 ADLBracketingStep

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0C7 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0xFFFF [Auto] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the selected ADL with ADL bracketing set to two shots.

**○ Property Details**

When the BracketingType property (section 6.5.8.2) is set to anything other than [ADL bracketing], the value of PropertyValue is not fixed.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0xFFFF | Auto |
| 0x0002 | Low |
| 0x0003 | Normal |
| 0x0004 | High |
| 0x0105 | Extra high 1 |
| 0x0205 | Extra high 2 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The BracketingType property (section 6.5.8.2) is set to anything other than [ADL bracketing]. |
| | The HDRMode property (section 6.5.2.72) is [On]. |
| | The StillCaptureMode property (section 6.5.1.15) is set to [High speed frame capture]. |
| | The ToneMode property (section 6.5.2.75) is set to [HLG]. |
| | The PixelShiftShootingMode property (section 6.5.2.78) is set to "ON". |

Do Not Copy

## 6.5.8.9 AEBracketingCount

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0C3 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 1 |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the number of the image (nth image) to be captured next for AE bracketing and ADL bracketing.

### ○ Property Details

When the EnableBracketing property (section 6.5.8.1) is set to [Off] and the BracketingType property (section 6.5.8.2) is set to [WB bracketing], the value of PropertyValue is not fixed.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| From 1 to 9 | AE bracketing |
| From 1 to 5 | ADL bracketing |

Do Not Copy

# 6.5.9 Vendor (Speedlight)

## 6.5.9.1 ExternalSpeedLightExist

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD120 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Not mounted] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the mounting status of the speedlight.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not mounted |
| 1 | Mounted |

Do Not Copy

## 6.5.9.2 ExternalSpeedLightSort

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD122 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Noncommunication] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the communication status of the speedlight.

### ○ Property Details

When the ExternalSpeedLightExist property (section 6.5.9.1) is set to [Not mounted], the value of PropertyValue is not fixed.

If the speedlight for old-type communication is mounted, the PropertyValue becomes [Noncommunication].

For the communication status types of the speedlight, refer to “Speedlight Types” (section 11.6).

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Noncommunication |
| 2 | CLS (New-type communication) with the operating and setting section |
| 3 | CLS (New-type communication) without the operating and setting section |

Do Not Copy

### 6.5.9.3 ExternalSpeedLightStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD121 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Not charged] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the charge status of the speedlight.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not charged |
| 1 | Ready |

*When the ExternalSpeedLightExist property (section 6.5.9.1) is set to [Not mounted], the value of PropertyValue is not fixed.

Do Not Copy

## 6.5.9.4 NewExternalSpeedLightMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD125 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the flash mode of the speedlight (CLS).

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | i-TTL balanced fill flash |
| 2 | i-TTL fill flash |
| 3 | Auto aperture |
| 4 | Non-TTL auto |
| 5 | Distance-priority manual |
| 6 | Manual |
| 7 | Repeating-flash |

*When the ExternalSpeedLightExist property (section 6.5.9.1) is set to [Not mounted], the value of PropertyValue is not fixed.
*When the ExternalSpeedLightSort property (section 6.5.9.2) is set to anything other than [CLS], the value of PropertyValue is not fixed.

Do Not Copy

## 6.5.9.5 FlashCompensation

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD124 |
| DataType | INT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [0.0 EV] |
| Property Value | From -18 [-3.0 EV] to +18 [+3.0 EV] |

### ○ Property Outline

Indicates the flash compensation value in units of 1/6 EV.

### ○ Property Details

When the ExternalSpeedLightExist property (section 6.5.9.1) is set to [Not mounted], the value of PropertyValue is not fixed.

When the ExternalSpeedLightSort property (section 6.5.9.2) is set to anything other than [CLS], the value of PropertyValue is not fixed.

The value of PropertyValue is valid when the values of the ExternalSpeedLightSort property (section 6.5.9.2) and the NewExternalSpeedLightMode property (section 6.5.9.4) are as shown in the table below. In the cases other than those in the table below, the value of PropertyValue is 0.

| ExternalSpeedLightSort | NewExternalSpeedLightMode |
|---|---|
| Noncommunication | (Invalid) |
| CLS | i-TTL balanced fill flash<br>i-TTL fill flash<br>Auto aperture<br>Distance-priority manual |

Do Not Copy

### 6.5.9.6 ExternalSpeedLightMultiFlashMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD12D |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Stand-alone flash] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the commander mode of the speedlight.

**○ Property Details**

When the ExternalSpeedLightExist property (section 6.5.9.1) is set to [Not mounted], the value of PropertyValue is not fixed.

When the ExternalSpeedLightSort property (section 6.5.9.2) is set to anything other than [CLS], the value of PropertyValue is not fixed.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Stand-alone flash (Commander mode is invalid or not supported) |
| 1 | Other flash units (Commander mode is valid) |

Do Not Copy

### 6.5.9.7 SBWirelessMode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1E2 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Stand-alone flash] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the wireless setting of the speedlight.

**○ Property Details**

The supported property values change depending on the type of the flash attached to the camera or whether the wireless communication accessory is equipped or not. If there's no value supported, the property value is not fixed.

If the supported property values change, it is informed by the DevicePropChanged event (section 6.4.1.6).

If the camera's wireless settings work with optical + radio, "optical + radio" is added to the enumerated values.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Stand-alone flash |
| 1 | Other flash units (Radio) |
| 2 | Other flash units (Optical) |
| 3 | Other flash units (Optical + Radio) |

Do Not Copy

## 6.5.9.8 SBWirelessMultipleFlashMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1E3 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Group flash units] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the flash mode of the multiple speedlight.

### ○ Property Details

The supported property values change depending on the type of the flash attached to the camera or whether the wireless communication accessory is equipped or not. If there's no value supported, the property value is not fixed.

When the SBWirelessMode property (section 6.5.9.7) is [Other flash units (Optical + Radio)], this property is fixed to [Group flash units].

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Group flash units |
| 1 | Quick wireless lighting |
| 2 | Repeating |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash]. |

Do Not Copy

### 6.5.9.9 SBUsableGroup

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1E4 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the available group of the speedlight.

**○ Property Details**

The property value is the logical OR of the SBGroupIDs for available groups.
If the property value is zero, any group cannot be used.
For the details of SBGroupID, refer to section 7.6

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00000000 | Cannot be used |
| Other than 0x00000000 | Logical OR of the available SBGroupID |

Do Not Copy

## 6.5.9.10 WirelessCLSEntryMode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1E5 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [Pairing mode] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the setting of “Remote (WR) setting – Link mode” in the setup menu.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Pairing mode |
| 1 | PIN code mode |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The wireless remote controller is not installed. |
| | The installed wireless remote controller doesn’t support the radio-controlled AWL. |

Do Not Copy

### 6.5.9.11 SBPINCode

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1E6 |
| DataType | UINT32 |
| Description form | Range |
| Get / Set | Get, Get / Set |
| Default Value | 0 |
| Property Value | From 0 to 9999 |

**○ Property Outline**

Indicates the PIN code of “Remote (WR) setting – Link mode” in the setup menu.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The wireless remote controller is not installed. |
| | The installed wireless remote controller doesn’t support the radio-controlled AWL. |

Do Not Copy

## 6.5.9.12 RadioMultipleFlashChannel

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1E7 |
| DataType | UINT8 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the communication channel of the wireless remote controller.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 5 | 5CH |
| 10 | 10CH |
| 15 | 15CH |

Do Not Copy

## 6.5.9.13 OpticalMultipleFlashChannel

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1E8 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [CH] |
| Property Value | From 1 to 4 [CH] |

### ○ Property Outline

Indicates the channel of the optical communication to the speedlight.

If the channel of the optical communication is not fixed, this property returns 0xFF as an invalid value.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash] or [Other flash units (Radio)]. |
| | The flash supporting the SB control is not attached. |

Do Not Copy

## 6.5.9.14 FlashRangeDisplay

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1E9 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [meter] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the unit of distance for the setting of the speedlight.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Meter (m) |
| 1 | Feet (ft) |

Do Not Copy

### 6.5.9.15 AllTestFiringDisable

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1EA |
| DataType | UINT8 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 [Enable firing] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the disabled state for the all test firing of the speedlight.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Enable firing |
| Other than 0 | Disable firing |

Do Not Copy

### 6.5.9.16 SBSettingMemberLock

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1EC |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Unlock] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the lock state of the dials and the buttons on the speedlight.

**○ Property Details**

The camera sets the default value when the speedlight is disconnected from the camera or the session to the camera terminates.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Unlock |
| 1 | Lock |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ExternalSpeedLightExist property (section 6.5.9.1) is set to [Not mounted]. |

Do Not Copy

### 6.5.9.17 SBIntegrationFlashReady

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1ED |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Uncharged] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the information about the ready state of the speedlight displayed in the flash info of the INFO display.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Uncharged |
| 1 | Ready |

Do Not Copy

# 6.5.10 Vendor (Lens)

## 6.5.10.1 LensSort

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0E1 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 1 [CPU lens mounted] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the mounting status of the CPU lens.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not mounted (lens not mounted or non-CPU lens mounted) |
| 1 | CPU lens mounted |

Do Not Copy

## 6.5.10.2 LensTypeML

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0E7 |
| DataType | UINT64 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates information on the ML lens.

The LensTypeF property (section 6.5.10.3) becomes effective only when the mount adapter of this property is valid.

### ○ Property Value

The valid PropertyValues are shown below. It is invalid when bit is 0, and valid when it is 1.

| Bit | Description |
|---|---|
| From Bit9 to Bit63 | Reserved |
| Bit8 | Power zoom lens |
| Bit7 | Lens whose automatic distortion correction is always effective. |
| Bit6 | Lens supporting automatic distortion correction |
| Bit5 | DX lens (Lens for Nikon digital cameras) |
| Bit4 | VR lens (with anti-vibration mechanism) |
| From Bit1 to Bit3 | Reserved |
| Bit0 | Mount adapter |

Do Not Copy

### 6.5.10.3 LensTypeF

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0E2 |
| DataType | UINT64 |
| Description form | None |
| Get / Set | Get |
| Default Value | 1 [D-type lens] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates information on the F-CPU lens.

When the mount adapter is set to invalid in the LensTypeML property (section 6.5.10.2), the value of this property is fixed to zero.

**○ Property Value**

The valid PropertyValues are shown below. It is invalid when bit is 0, and valid when it is 1.

| Bit | Description |
|---|---|
| From Bit9 to Bit63 | Reserved |
| Bit8 | STM lens |
| Bit7 | E-type lens (electromagnetic diaphragm) |
| Bit6 | Retractable lens |
| Bit5 | Lens supporting automatic distortion correction |
| Bit4 | AF-S lens |
| Bit3 | DX lens (Lens for Nikon digital cameras) |
| Bit2 | VR lens (with anti-vibration mechanism) |
| Bit1 | G-type lens (without aperture dial) |
| Bit0 | D-type lens (with distance encoder) |

Do Not Copy

### 6.5.10.4 LensID

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0E0 |
| DataType | UINT16 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0 |
| Property Value | (Depends on the lens type) |

**○ Property Outline**

Indicates the ID of the CPU lens.

**○ Property Details**

The value of PropertyValue indicates an ID (2Byte).

When the value of the LensSort property (section 6.5.10.1) is set to [Not mounted], the value of PropertyValue is fixed to zero.

Do Not Copy

### 6.5.10.5 LensFocalMin

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0E3 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 5000 [50 mm] |
| Property Value | (Depends on the lens type) |

**○ Property Outline**

Indicates the focal length at the Wide-end with the CPU lens mounted.

**○ Property Details**

The value of PropertyValue should be a hundred times the focal length (mm).

When the value of the LensSort property (section 6.5.10.1) is set to [Not mounted], the value of PropertyValue is not fixed.

### 6.5.10.6 LensFocalMax

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0E4 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 5000 [50 mm] |
| Property Value | (Depends on the lens type) |

**○ Property Outline**

Indicates the focal length at the Tele-end with the CPU lens mounted.

**○ Property Details**

The value of PropertyValue should be a hundred times the focal length (mm).

When the value of the LensSort property (section 6.5.10.1) is set to [Not mounted], the value of PropertyValue is not fixed.

Do Not Copy

### 6.5.10.7 LensApatureMin

○ **Property Specifications**

| | |
|---|---|
| Property Code | 0xD0E5 |
| DataType | UINT16 |
| Description form | None |
| Get / Set | Get |
| Default Value | 140 [F 1.4] |
| Property Value | (Depends on the lens type) |

○ **Property Outline**

Indicates the maximum aperture value at the Wide-end with the CPU lens mounted.

○ **Property Details**

The value of PropertyValue should be a hundred times the maximum aperture value.
When the value of the LensSort property (section 6.5.10.1) is set to [Not mounted], the value of PropertyValue is not fixed.

### 6.5.10.8 LensApatureMax

○ **Property Specifications**

| | |
|---|---|
| Property Code | 0xD0E6 |
| DataType | UINT16 |
| Description form | None |
| Get / Set | Get |
| Default Value | 1600 [F16] |
| Property Value | (Depends on the lens type) |

○ **Property Outline**

Indicates the maximum aperture value at the Tele-end with the CPU lens mounted.

○ **Property Details**

The value of PropertyValue should be a hundred times the maximum aperture value.
When the value of the LensSort property (section 6.5.10.1) is set to [Not mounted], the value of PropertyValue is not fixed.

Do Not Copy

## 6.5.10.9 RetractableLensWarning

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD09C |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the retractable lens warning condition.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | (Retractable lens warning) Off |
| 1 | (Retractable lens warning) On |

Do Not Copy

# 6.5.11 Vendor (Live View)

## 6.5.11.1 CameraLiveViewStatus

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1B8 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the status of the live view in the camera.

### ○ Property Details

This property is set to “On” when live view is active.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

## 6.5.11.2 RemoteLiveViewStatus

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1A2 |
| DataType | UINT8 |
| Description form | Range |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the status of the remote live view.

### ○ Property Details

This property is set to “On” when the StartLiveView command is executed.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

Do Not Copy

### 6.5.11.3 LiveViewImageStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1BB |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Live view image is not available] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the status of the live view image acquired by the GetLiveViewImage command.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Live view image is not available. |
| 1 | Live view image is available. |

Do Not Copy

### 6.5.11.4 LiveViewImageSize

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD1AC |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 2 [Equivalent to VGA] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates the size of the live view image acquired by the GetLiveViewImage command.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | Equivalent to QVGA (up to 320x240) |
| 2 | Equivalent to VGA (up to 640x480) |
| 3 | Equivalent to XGA (up to 1024x768) |

### 6.5.11.5 LiveViewImageCompression

**Property Specifications**

| | |
|---|---|
| Property Code | 0xD1BC |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Basic – Size priority] |
| Property Value | Refer to Property Value in this section. |

**Property Outline**

Indicates the quality of the live view image acquired by the GetLiveViewImage command.

**Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Basic – Size priority |
| 1 | Basic – Quality priority |
| 2 | Normal – Size priority |
| 3 | Normal – Quality priority |
| 4 | Fine – Size priority |
| 5 | Fine – Quality priority |

Do Not Copy

### 6.5.11.6 LiveViewZoomArea

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1BD |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 0 [Entire display] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the zoom-area of the live view image.

**○ Property Details**

Available property value changes depending on the display device on the camera. If you set a value that is not supported on the display device, it is changed to another area that has the same magnification.

When switching the image monitor and the electronic viewfinder during zoom preview, the area before switching is maintained as the current value. NOTE) During video recording, returns to full display without maintaining zooming.

If recording video or being connected to HDMI in video mode, the live view image of the GetLiveViewImageEx command (section 6.2.6.4) will not be enlarged even if this property is set to anything other than entire display. (The live view image on the camera side is enlarged)

With this property, only the same size (100%) can be used for enlargement during video recording.

In the following cases, the values that can be specified are limited, and the enumerated values vary.

- When the spot WB acquisition standby state.
  Magnification: 200% or more cannot be used.

- When in video mode, MovieRecordScreenSize property (section 6.5.3.7) is "1920x1080", and video recording or HDMI connection.
  Magnification: 50% cannot be used.

Do Not Copy

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Display device | | Description |
|---|---|---|---|
| | Image monitor | Electronic viewfinder | |
| 0 | ✓ | ✓ | Entire display |
| 256 | ✓ | - | Zoom area: 256 x 170<br>Magnification: High magnification (400%) |
| 320 | - | ✓ | Zoom area: 320 x 214<br>Magnification: High magnification (400%) |
| 512 | ✓ | - | Zoom area: 512x 342<br>Magnification: Zoom-in (200%) |
| 640 | - | ✓ | Zoom area: 640x 426<br>Magnification: Zoom-in (200%) |
| 1024 | ✓ | - | Zoom area: 1024x 682<br>Magnification: Equal (100%) |
| 1280 | - | ✓ | Zoom area: 1280x 854<br>Magnification: Equal (100%) |
| 2048 | ✓ | - | Zoom area: 2048x1366<br>Magnification: Zoom-out (50%) |
| 2560 | - | ✓ | Zoom area: 2560x1706<br>Magnification: Zoom-out (50%) |

The magnification is the magnification relative to the size of the display device.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Not_LiveView | The property was set in a state other than during live view. |
| Access_Denied | The zoom value was changed to other than the same size (100%) or the entire display while recording a video. |
| | The MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)]. |

Do Not Copy

### 6.5.11.7 LiveViewProhibitionCondition

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1A4 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the live view prohibition condition. If there is any prohibition factor in this property, the StartLiveView command must not be issued. If the StartLiveView command is issued and the live view is started when there is any prohibition factor in this property, the operation is not guaranteed.

**○ Property Value**

The valid PropertyValues are shown below.
The PropertyValue takes the following values in the bit assignment. (1: Valid, 0: Invalid)

| Bit | Description |
|---|---|
| Bit29 to Bit31 | Reserved |
| Bit28 | Lens error<br>Version upgrade warning when mount adapter is attached |
| Bit18 to Bit27 | Reserved |
| Bit17 | The live view cannot be started when the temperature rises. |
| Bit16 | Reserved |
| Bit15 | Processing by shooting<br>Until the shooting operation ends |
| Bit9 to Bit14 | Reserved |
| Bit8 | Battery exhausted |
| Bit3 to Bit7 | Reserved |
| Bit2 | Sequence error |
| Bit0 to Bit1 | Reserved |

*When the PropertyValue is a value other than 0, the live view cannot be started.

Do Not Copy

# 6.5.12 Vendor (Video)

## 6.5.12.1 MovieFnumber

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1A9 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | The minimum value of the enumeration |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the aperture value that is used in video mode.

### ○ Property Details

When the LensSort property (section 6.5.10.1) is [Not mounted], the enumeration cannot be created. Therefore, the number of enumeration values shall be 1 and the enumeration value, the DefaultValue, and the PropertyValue shall be the same value.

For models that support the exposure meter interlocking lever, it changes depending on the F0ManualSetting2 property (section 6.5.5.3) and the state of the exposure meter interlocking lever. The value shall be 1 EV except the maximum aperture value.

It is set to 0 if the model does not support the exposure meter interlocking lever or if the aperture value cannot be acquired.

If an aperture value error occurs, the number of enumeration values shall be 1 and the enumeration value, the DefaultValue, and the PropertyValue shall be 0xFFFF.

If not in video mode, the number of enumeration values shall be 1 and the enumeration value, the DefaultValue, and the PropertyValue shall be 0 because the value is not fixed.

### ○ Property Value

When the ExposureProgramMode property (section 6.5.1.11) is M or A, the PropertyValue is set to the value of the Fnumber property (section 6.5.1.5) just before the start of video mode. When the ExposureProgramMode property (section 6.5.1.11) is other than M or A, the PropertyValue is set to a value depending on the camera.

The PropertyValue should be a hundred times the aperture value.

The PropertyValue changes depending on the value of the ExposureEVStep property (section 6.5.4.14).

The setting range of PropertyValue changes depending on the lens and the magnification setting.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The value of the ExposureProgramMode property (section 6.5.1.11) is not [A] or [M]. |
| | The value of the LensSort property (section 6.5.10.1) is [Not mounted]. |
| | The WarningStatus property (section 6.5.6.22) is [Sequence error] or [Retractable zoom-ring warning]. |
| | The MovieAVLockSetting property (section 6.5.4.50) is [On]. |
| | An aperture value error. |

Do Not Copy

## 6.5.12.2 MovieShutterSpeed

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD1A8 |
| DataType | UINT32 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | The minimum value of the enumeration |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the shutter speed that is used in video mode.

### ○ Property Details

In this property, the upper 2 bytes of PropertyValue are treated as the numerator of the shutter speed, and the lower 2 bytes are treated as the denominator.

If the enum value changes, update the enum value and DefaultValue. Enumerated values vary depending on the following factors.

**Table 6.5.12-1 Associated properties**

| Property | Ref. | |
|---|---|---|
| ExposureEVStep | 6.5.4.14 | Exposure step value |
| ExternalSpeedLightExist | 6.5.9.1 | Speedlight attached |
| MovieRecordScreenSize | 6.5.3.7 | Image size / Frame rate |
| MovieExtendedShutterSpeedsManual | 6.5.4.57 | Extended shutter speed (M mode) |

The shutter speed on the low-speed side changes depending on the setting of the video frame rate. The relationship between each frame rate and the shutter speed is shown below.

In addition, if the extended shutter speed is valid and the video frame rate setting is not 100 fps or the 120fps, shutter speed becomes 1/4 seconds regardless of the setting of the video frame rate.

| Video frame rate | ExposureEVStep | | |
|---|---|---|---|
| | 1/3 step | 1/2 step | 1 step |
| 120fps | 1/125 | 1/125 | 1/125 |
| 100fps | 1/100 | 1/90 | 1/125 |
| 60fps | 1/60 | 1/60 | 1/60 |
| 50fps | 1/50 | 1/60 | 1/60 |
| 30fps | 1/30 | 1/30 | 1/30 |
| 25fps | 1/25 | 1/30 | 1/30 |
| 24fps | 1/25 | 1/30 | 1/30 |

Do Not Copy

### ○ Property Value

The valid PropertyValues are shown below.

Shutter speed:

| Property Value | Description |
|---|---|
| Upper 2 bytes | Numerator of the shutter speed |
| Lower 2 bytes | Denominator of the shutter speed |

Example of the shutter speed:

For 1/250 sec. PropertyValue = 0x000100FA
For 25 sec. PropertyValue = 0x00190001

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| | The value of the ExposureProgramMode property (section 6.5.1.11) is anything other than M. |
| | The WarningStatus property (section 6.5.6.22) is [Sequence error], [Minimum aperture warning] or [Retractable zoom-ring warning]. |
| | The MovieTVLockSetting property (section 6.5.4.49) is [On]. |

Do Not Copy

### 6.5.12.3 MovieExposureBiasCompensation

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD1AB |
| DataType | INT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 0 [0.0 EV] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the compensation value of the exposure compensation that is used in video mode.

**○ Property Value**

The valid PropertyValues are shown below.

The value of PropertyValue changes depending on the value of the ExposureEVStep property (section 6.5.4.14).

| Property Value | Description | ExposureEVStep | | |
|---|---|---|---|---|
| | | 1/3 step | 1/2 step | 1 step |
| +3000 | +3.0 EV | ✓ | ✓ | ✓ |
| +2666 | +2.7 EV | ✓ | - | ✓ |
| +2500 | +2.5 EV | - | ✓ | - |
| +2333 | +2.3 EV | ✓ | - | ✓ |
| +2000 | +2.0 EV | ✓ | ✓ | ✓ |
| +1666 | +1.7 EV | ✓ | - | ✓ |
| +1500 | +1.5 EV | - | ✓ | - |
| +1333 | +1.3 EV | ✓ | - | ✓ |
| +1000 | +1.0 EV | ✓ | ✓ | ✓ |
| +666 | +0.7 EV | ✓ | - | ✓ |
| +500 | +0.5 EV | - | ✓ | - |
| +333 | +0.3 EV | ✓ | - | ✓ |
| 0 | 0.0 EV | ✓ | ✓ | ✓ |
| -333 | -0.3 EV | ✓ | - | ✓ |
| -500 | -0.5 EV | - | ✓ | - |
| -666 | -0.7 EV | ✓ | - | ✓ |
| -1000 | -1.0 EV | ✓ | ✓ | ✓ |
| -1333 | -1.3 EV | ✓ | - | ✓ |
| -1500 | -1.5 EV | - | ✓ | - |
| -1666 | -1.7 EV | ✓ | - | ✓ |
| -2000 | -2.0 EV | ✓ | ✓ | ✓ |
| -2333 | -2.3 EV | ✓ | - | ✓ |
| -2500 | -2.5 EV | - | ✓ | - |
| -2666 | -2.7 EV | ✓ | - | ✓ |
| -3000 | -3.0 EV | ✓ | ✓ | ✓ |

Do Not Copy

### 6.5.12.4 MovieLogOutput

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD0BB |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Off] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the state of N-log/HDR(HLG) control in the camera.

**○ Property Details**

The output destination of the control state changes depending on the value of the MovieFileType property (Section 6.5.3.2).

If the MovieFileType property is "H.265 10-bit (MOV)", refer to the MovieH265ToneMode property (Section 6.5.3.3).

For "ProRes 422 HQ 10-bit (MOV)", see MovieProResToneMode property (Section 6.5.3.5).

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Off (Not outputting) |
| 1 | On (Outputting) |

Do Not Copy

## 6.5.12.5 MovieRecProhibitionCondition

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD0A4 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the video recording prohibition condition.

### ○ Property Details

When the PropertyValue is a value other than 0, the video recording cannot be started.
When the live view is not started, the value of PropertyValue is not fixed. Even if a value has been entered, it is not guaranteed.

### ○ Property Value

The valid PropertyValues are shown below.
The PropertyValue takes the following values in the bit assignment. (1: Valid, 0: Invalid)

| Bit | Description |
|---|---|
| From Bit21 to Bit31 | Reserved |
| Bit20 | If any of the following conditions are met when MovieFileType property (section 6.5.3.2) is set to [ProRes RAW HQ 12-bit (MOV)] or [N-RAW 12-bit (NEV)].<br>- Video Record Destination is set to SD card.<br>- The format of the card that is set as Video Record Destination is FAT32. |
| Bit19 | During warning / error of lens factor |
| Bit18 | During sequence error |
| Bit15 to Bit17 | Reserved |
| Bit14 | The camera is not in the application mode. |
| Bit13 | During photo mode |
| Bit12 | During enlarged display of live view |
| Bit11 | Card is protected. |
| Bit10 | During video recording |
| Bit9 | There are images or videos not recorded in the buffer. |
| Bit4 to Bit8 | Reserved |
| Bit3 | No free area in the card |
| Bit2 | Card not formatted |
| Bit1 | Card access error |
| Bit0 | No card inserted.<br>(When waiting for synchronous recording by an external recording device, 0: Invalid even if there is no card.) |

Do Not Copy

# 6.5.13 Vendor (Picture Control)

## 6.5.13.1 ActivePicCtrlItem

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD200 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 8 [Auto] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the picture control item whose setting is currently valid.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | Standard |
| 2 | Neutral |
| 3 | Vivid |
| 4 | Monochrome |
| 5 | Portrait |
| 6 | Landscape |
| 7 | Flat |
| 8 | Auto |
| 9 | Flat Monochrome |
| 10 | Deep Tone Monochrome |
| 11 | Rich Tone Portrait |
| 101 | Dream |
| 102 | Morning |
| 103 | Pop |
| 104 | Sunday |
| 105 | Somber |
| 106 | Drama |
| 107 | Silence |
| 108 | Bleach |
| 109 | Melancholic |
| 110 | Pure |
| 111 | Denim |
| 112 | Toy |
| 113 | Sepia |
| 114 | Blue |
| 115 | Red |
| 116 | Pink |
| 117 | Charcoal |
| 118 | Graphite |
| 119 | Binary |
| 120 | Carbon |
| From 201 to 209 | Custom picture control (from 1 to 9) |

Do Not Copy

○ **Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_Status | Setting to an unregistered area was performed for the custom picture control. |
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [HLG]. |

## 6.5.13.2 MovieActivePicCtrlItem

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD237 |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get / Set |
| Default Value | 100 [Same as photo settings] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the picture control item whose setting is currently valid in the video recording menu.

### ○ Property Details

When the property value is set to [Same as photography setting], the picture control specified by ActivePicCtrlItem (section 6.5.13.1) is used.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | Standard |
| 2 | Neutral |
| 3 | Vivid |
| 4 | Monochrome |
| 5 | Portrait |
| 6 | Landscape |
| 7 | Flat |
| 8 | Auto |
| 9 | Flat Monochrome |
| 10 | Deep Tone Monochrome |
| 11 | Rich Tone Portrait |
| 100 | Same as photography setting |
| 101 | Dream |
| 102 | Morning |
| 103 | Pop |
| 104 | Sunday |
| 105 | Somber |
| 106 | Drama |
| 107 | Silence |
| 108 | Bleach |
| 109 | Melancholic |
| 110 | Pure |
| 111 | Denim |
| 112 | Toy |
| 113 | Sepia |
| 114 | Blue |
| 115 | Red |
| 116 | Pink |
| 117 | Charcoal |
| 118 | Graphite |
| 119 | Binary |

Do Not Copy

| | |
|---|---|
| 120 | Carbon |
| From 201 to 209 | Custom picture control (from 1 to 9) |

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Invalid_Status | Setting to an unregistered area was performed for the custom picture control. |
| Access_Denied | If the ToneMode is set to N-Log or HLG (If any of the following conditions are met).<br>- MovieFileType property (section 6.5.3.2) is set to [H.265 10-bit(MOV)] and MovieH265ToneMode property (section 6.5.3.3) is set to [N-Log] or [HLG].<br>- MovieFileType property is set to [ProRes 422 HQ 10-bit(MOV] and MovieProResToneMode property (section 6.5.3.5) is set to [N-Log].<br>- MovieFileType property is set to [N-RAW 12-bit(NEV] and MovieNRaw12bitToneMode property (section 6.5.3.4) is set to [N-Log].<br>- MovieFileType property is set to [ProRes RAW HQ 12-bit(MOV] and MovieProResRawHq12bitToneMode property (section 6.5.3.6) is set to [N-Log]. |

Do Not Copy

### 6.5.13.3 ActivePicCtrlHlgItem

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0x0001D01D |
| DataType | UINT16 |
| Description form | Enumeration |
| Get / Set | Get, Get / Set |
| Default Value | 1 [Standard] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the setting of “Picture Control (HLG)” in the photo shooting menu.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | Standard |
| 4 | Monochrome |
| 7 | Flat |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The ToneMode property (section 6.5.2.75) is set to [SDR]. |

Do Not Copy

# 6.5.14 Vendor (Power Supply)

## 6.5.14.1 ExternalDC-IN

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD101 |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [Not connected] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the AC adapter connection status.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | Not connected |
| 1 | Connected |

Do Not Copy

# 6.5.15 Vendor (USB)

## 6.5.15.1 USBSpeed

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD10C |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 2 [Super-Speed] |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

Indicates the maximum data transfer speed of USB.
When the Super-Speed is not selected, the property value is High-Speed.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 1 | High-Speed |
| 2 | Super-Speed |

Do Not Copy

### 6.5.15.2 ConnectionPath

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD12E |
| DataType | UINT8 |
| Description form | Enumeration |
| Get / Set | Get |
| Default Value | 0 [USB] |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

Indicates the connection path for PTP communication.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0 | USB |
| 2 | Built-in Wi-Fi |
| 3 | Wired LAN |

Do Not Copy

# 6.5.16 Vendor (MTP)

## 6.5.16.1 SessionInitiatorVersionInfo

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD406 |
| DataType | String |
| Description form | None |
| Get / Set | Get / Set |
| Default Value | “Windows/6.0.5330.0 MTPClassDriver/6.0.5330.0”<br>[Session initiator version information character string] |
| Property Value | (Depends on the camera setting) |

### ○ Property Outline

Indicates the version information of the host in open session.

### ○ Property Details

The PropertyValue should be a PTP string of 255 characters or shorter.
(HTTP 1.1 spec (RFC 2068) User Agent string format)

## 6.5.16.2 PerceivedDeviceType

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xD407 |
| DataType | UINT32 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x00000001 [Digital still camera] |
| Property Value | 0x00000001 [Digital still camera] |

### ○ Property Outline

Indicates the type of the device.

Do Not Copy

### 6.5.16.3 UseDeviceStageFlag

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xD303 |
| DataType | UINT8 |
| Description form | None |
| Get / Set | Get |
| Default Value | 0x01 [On] |
| Property Value | 0x01 [On] |

**○ Property Outline**

Indicates that the device can use Device Stage when the PropertyValue is a value other than 0. Windows searches the metadata of Device Stage in the metadata service until the device installation is completed. If the metadata of Device Stage in the server of Microsoft cannot be acquired, Windows displays Baseline Experience instead of Device Stage.

Do Not Copy

# 6.6 ObjectPropCode

Each of the objects in the camera has various sorts of specific information. As a method of transmission/reception of object information, an operation already exists in the PTP for operating the ObjectInfo data set. However, this is a static data set that cannot be expanded and includes basic information on the object. Various metadata concerning objects can be handled by operating the object property. Each object property has a corresponding ObjectPropCode.

The ObjectPropCodes supported by the camera are shown below.

| Object Code | Description | Reference item |
|---|---|---|
| 0xDC01 | StorageID | 6.6.1.1 |
| 0xDC02 | ObjectFormat | 6.6.1.2 |
| 0xDC03 | ProtectionStatus | 6.6.1.3 |
| 0xDC04 | ObjectSize | 6.6.1.4 |
| 0xDC07 | ObjectFilename | 6.6.1.5 |
| 0xDC08 | DateCreated | 6.6.1.6 |
| 0xDC09 | DateModified | 6.6.1.7 |
| 0xDC0B | ParentObject | 6.6.1.8 |
| 0xDC0D | Hidden | 6.6.1.9 |
| 0xDC41 | PersistentUniqueObjectIdentifier | 6.6.1.10 |
| 0xDC44 | Name | 6.6.1.11 |
| 0xDC81 | RepresentativeSampleFormat | 6.6.2.1 |
| 0xDC82 | RepresentativeSampleSize | 6.6.2.2 |
| 0xDC83 | RepresentativeSampleHeight | 6.6.2.3 |
| 0xDC84 | RepresentativeSampleWidth | 6.6.2.4 |
| 0xDC86 | RepresentativeSampleData | 6.6.2.5 |
| 0xDC87 | Width | 6.6.3.1 |
| 0xDC88 | Height | 6.6.3.2 |
| 0xDC8A | Rating | 6.6.3.3 |
| 0xDCD3 | ImageBitDepth | 6.6.3.4 |
| 0xDC89 | Duration | 6.6.4.1 |
| 0xDE93 | SampleRate | 6.6.4.2 |
| 0xDE94 | NumberOfChannels | 6.6.4.3 |
| 0xDE97 | ScanType | 6.6.4.4 |
| 0xDE9A | AudioBitRate | 6.6.4.5 |
| 0xDE9B | VideoFourCCCode | 6.6.4.6 |
| 0xDE9C | VideoBitRate | 6.6.4.7 |
| 0xDE9D | FramesPerThousandSeconds | 6.6.4.8 |

Do Not Copy

# 6.6.1 Object Information

## 6.6.1.1 StorageID

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xDC01 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00010001 |
| Group Code | 0x00000001 |
| Form Flag | 0x00 (None) |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

This indicates the StorageID of the object. (It is the same value as that of the first field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00010001 | Slot 1 |
| 0x00020001 | Slot 2 |

Do Not Copy

### 6.6.1.2 ObjectFormat

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC02 |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x3000 |
| Group Code | 0x00000001 |
| Form Flag | 0x00 (None) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the ObjectFormatCode of the object. (It is the same value as that of the second field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x3000 | Undefined |
| 0x3001 | Association |
| 0x3002 | Script |
| 0x3008 | WAV |
| 0x300D | MOV<br>Apple QuickTime Video Format（H.264/AVC） |
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.1.3 ProtectionStatus

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC03 |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x0000 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the protection status of the object. (It is the same value as that of the third field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Protection is not set. |
| 0x0001 | Protection is set. |

Do Not Copy

### 6.6.1.4 ObjectSize

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC04 |
| DataType | UINT64 |
| Get / Set | Get |
| Default Value | 0x0000000000000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x00 (None) |
| Property Value | (Object size) |

**○ Property Outline**

This indicates the size of the object in units of bytes. (It is the same value as that of the fourth field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

Do Not Copy

### 6.6.1.5 ObjectFilename

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC07 |
| DataType | String |
| Get / Set | Get |
| Default Value | 0x00 (Null) |
| Group Code | 0x00000001 |
| Form Flag | 0x05 (RegEx) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates an optional string that shows a file name of an object. (It is the same value as that of the sixteenth field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

However, the ObjectPropDesc returned in the Image format differs from that returned in the Association format.

**○ Property Value**

The valid PropertyValues are shown below.

| ObjectFormat | RegEx |
|---|---|
| Association | [0-9]{3}[_a-zA-Z0-9]{5} |
| Other than Association | [_a-zA-Z]{4}[0-9]{4}¥.[a-zA-Z]{3} |

Do Not Copy

### 6.6.1.6 DateCreated

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC08 |
| DataType | String |
| Get / Set | Get |
| Default Value | 0x00 (Null) |
| Group Code | 0x00000001 |
| Form Flag | 0x03 (DateTime) |
| Property Value | (Date and time of object creation) |

**○ Property Outline**

This indicates a string that shows the date and time of object creation. (It is the same value as that of the seventeenth field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

Do Not Copy

### 6.6.1.7 DateModified

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC09 |
| DataType | String |
| Get / Set | Get |
| Default Value | 0x00 (Null) |
| Group Code | 0x00000001 |
| Form Flag | 0x03 (DateTime) |
| Property Value | (Date and time of object update) |

**○ Property Outline**

This indicates a string that shows the date and time of object update. (It is the same value as that of the eighteenth field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

Do Not Copy

### 6.6.1.8 ParentObject

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC0B |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x00 (None) |
| Property Value | (ObjectHandle of the parent object of the specified object) |

**○ Property Outline**

This indicates the ObjectHandle of the parent object of the object. (It is the same value as that of the twelfth field of the ObjectInfo data set.)

It is applied to the objects of all the formats supported by the camera.

Do Not Copy

### 6.6.1.9 Hidden

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC0D |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x0000 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | (Hidden attribute of the object) |

**○ Property Outline**

This indicates a hidden attribute of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3000 | Undefined |
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Displayed object |
| 0x0001 | Hidden object |

Do Not Copy

### 6.6.1.10 PersistentUniqueObjectIdentifier

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC41 |
| DataType | UINT128 |
| Get / Set | Get |
| Default Value | 0x00^16 |
| Group Code | 0x00000001 |
| Form Flag | 0x00 (None) |
| Property Value | (Object-specific identifier) |

**○ Property Outline**

This indicates an object-specific identifier (PUOID).
It is applied to the objects of all the formats supported by the camera.

The generation rule of the PUOID is shown below.

| 0 through 3 | 4 through 15 |
|---|---|
| ObjectHandle (4Byte) | Reserved (0) |

Do Not Copy

### 6.6.1.11 Name

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC44 |
| DataType | String |
| Get / Set | Get |
| Default Value | 0x00 (Null) |
| Group Code | 0x00000001 |
| Form Flag | 0x00 (None) |
| Property Value | (File name of an object) |

**○ Property Outline**

This indicates an optional string that shows a file name of an object. (It should be the same value as that of ObjectFilename.)

It is applied to the objects of all the formats supported by the camera.

Do Not Copy

## 6.6.2 Thumbnail

### 6.6.2.1 RepresentativeSampleFormat

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC81 |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x3000 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the ObjectFormatCode of the thumbnail image.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

**○ Property Value**

The valid PropertyValues are shown below.

| PropertyValue | Setting |
|---|---|
| 0x3000 | Undefined |
| 0x3808 | JFIF |
| 0x380D | TIFF |

Do Not Copy

### 6.6.2.2 RepresentativeSampleSize

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC82 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 0x00010000 |

**○ Property Outline**

This indicates the size of the thumbnail in bytes.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.2.3 RepresentativeSampleHeight

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC83 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 120 |

**○ Property Outline**

This indicates the height of the thumbnail in pixels.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.2.4 RepresentativeSampleWidth

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC84 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 160 |

**○ Property Outline**

This indicates the width of the thumbnail in pixels.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.2.5 RepresentativeSampleData

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC86 |
| DataType | AUINT8 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0xFFFFFFFF |
| Form Flag | 0x06 (ByteArray) |
| Property Value | From 0 to 0x00010000 |

**○ Property Outline**

This indicates the thumbnail data.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

# 6.6.3 Image Information

## 6.6.3.1 Width

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xDC87 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 10000 |

### ○ Property Outline

This indicates the width of the object in pixels.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.3.2 Height

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDC88 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 10000 |

**○ Property Outline**

This indicates the height of the object in pixels.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.3.3 Rating

#### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xDC8A |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x0000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 100 |

#### ○ Property Outline

This indicates the rating of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0x380D | TIFF |
| 0xB200 | HEIF |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

#### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Rating setting |
|---|---|
| 0x0000 | None |
| 0x0001 | ★ |
| 0x0019 | ★★ |
| 0x0032 | ★★★ |
| 0x004B | ★★★★ |
| 0x0063 | ★★★★★ |

Do Not Copy

### 6.6.3.4 ImageBitDepth

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDCD3 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x0000000C |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the bit depth of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3800 | Unknown Image Object |
| 0x3801 | EXIF/JPEG |
| 0x380D | TIFF |
| 0xB200 | HEIF |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000000C | 12bit |
| 0x0000000E | 14bit |
| 0x00000018 | 24bit |

Do Not Copy

# 6.6.4 Video Information

## 6.6.4.1 Duration

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xDC89 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0 to 0x001B773F (1799999msec) |

### ○ Property Outline

This indicates the length of the object in msec.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3008 | WAV |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.4.2 SampleRate

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDE93 |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the sample rate of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3008 | WAV |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00000000 | 0 Hz (no sound)/Unknown |
| 0x00002EE0 | 12.000 Hz |
| 0x00005DC0 | 24.000 Hz |
| 0x0000BB80 | 48.000 Hz |

Do Not Copy

### 6.6.4.3 NumberOfChannels

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDE94 |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x0000 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the number of channels of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3008 | WAV |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB982 | MP4 Container |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Unused (no sound)/Unknown |
| 0x0001 | Monaural (1ch) |
| 0x0002 | Stereo (2ch) |

Do Not Copy

### 6.6.4.4 ScanType

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDE97 |
| DataType | UINT16 |
| Get / Set | Get |
| Default Value | 0x0000 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

**○ Property Outline**

This indicates the scan type of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

**○ Property Value**

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x0000 | Unused |

Do Not Copy

### 6.6.4.5 AudioBitRate

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDE9A |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0x00000000 (No sound/Unknown) to 0x00177000 |

**○ Property Outline**

This indicates the audio bit rate of the object.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x3008 | WAV |
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

## 6.6.4.6 VideoFourCCCode

### ○ Property Specifications

| | |
|---|---|
| Property Code | 0xDE9B |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x31637661 |
| Group Code | 0x00000001 |
| Form Flag | 0x02 (Enumeration) |
| Property Value | Refer to Property Value in this section. |

### ○ Property Outline

This indicates the FourCC code for the video codec.
It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

### ○ Property Value

The valid PropertyValues are shown below.

| Property Value | Description |
|---|---|
| 0x00000000 | Indefinite (ProRes 422 HQ/ProRes RAW/TICO RAW) |
| 0x31637661 | “avc1” |
| 0x31637668 | “hvc1” |

Do Not Copy

### 6.6.4.7 VideoBitRate

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDE9C |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | From 0x00000000 (Unknown) to 0xFFFFFFFF |

**○ Property Outline**

This indicates the number of bits of the object processed per sec.

The PropertyValue should be a value obtained by multiplying the maximum number of bytes per sec. of the object by 8 (bit).

It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

### 6.6.4.8 FramesPerThousandSeconds

**○ Property Specifications**

| | |
|---|---|
| Property Code | 0xDE9D |
| DataType | UINT32 |
| Get / Set | Get |
| Default Value | 0x00000000 |
| Group Code | 0x00000001 |
| Form Flag | 0x01 (Range) |
| Property Value | (Frame rate of the object) |

**○ Property Outline**

This indicates the frame rate of the object.
Since the value of the property is represented by 1/1000 frame, 29.97fps becomes 29970.

It is applied to the objects of the following formats.

| ObjectFormatCode | Description |
|---|---|
| 0x300D | MOV<br>Apple QuickTime Video Format (H.264/AVC) |
| 0xB201 | NEV |
| 0xB982 | MP4 Container |

Do Not Copy

# 6.7 SBAttributeCode

The flash has various peculiar information. The various metadata for the flash can be handled by operating the flash attributes. Each flash attribute has corresponding SBAttrID and SBGroupAttrID.

The camera supports the following SBAttrID and SBGroupAttrID.

| SBAttrID | Description | Reference |
|---|---|---|
| 0x1001 | SBName | 6.7.1.1 |
| 0x1002 | SBGroupID | 6.7.1.2 |
| 0x1003 | SBStatus | 6.7.1.3 |
| 0x1005 | SBTestFlashDisable | 6.7.1.4 |

| SBGroupAttrID | Description | Reference |
|---|---|---|
| 0x4001 | SBGroupFlashMode | 6.7.2.1 |
| 0x4002 | SBGroupFlashCompensation | 6.7.2.2 |
| 0x4003 | SBGroupFlashRatio | 6.7.2.3 |
| 0x4004 | SBGroupFlashLevel | 6.7.2.4 |
| 0x4005 | SBGroupFlashRange | 6.7.2.5 |
| 0x4006 | SBGroupRepeat | 6.7.2.6 |
| 0x4007 | SBGroupRepeatCount | 6.7.2.7 |
| 0x4008 | SBGroupRepeatInterval | 6.7.2.8 |
| 0x4009 | SBGroupValid | 6.7.2.9 |

Do Not Copy

# 6.7.1 SBAttribute

## 6.7.1.1 SBName

### ○ Attribute Specifications

| | |
|---|---|
| SBAttrID | 0x1001 |
| DataType | String |
| Get / Set | Get / Set |
| Default Value | “Noused” |
| FormFlag | None |
| Attribute Value | (SB model name / nickname) |

### ○ Attribute Outline

This indicates the string to express the name or the nickname of the remote flash.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | When in video mode. |

## 6.7.1.2 SBGroupID

### ○ Attribute Specifications

| | |
|---|---|
| SBAttrID | 0x1002 |
| DataType | SBGroupID |
| Get / Set | Get |
| Default Value | 0x00000000 |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

### ○ Attribute Outline

This indicates the SBGroupID of the flash.

### ○ Attribute Value

For the AttributeValue, refer to the SBGroupID (section 7.6).

Do Not Copy

### 6.7.1.3 SBStatus

**○ Attribute Specifications**

| | |
|---|---|
| SBAttrID | 0x1003 |
| DataType | UINT8 |
| Get / Set | Get |
| Default Value | 0x01 |
| FormFlag | None |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the ready state of the remote flash.

**○ Attribute Details**

When getting this attribute with specifying the master flash, the GetSBAttrDesc and the GetSBAttrValue commands respond Invalid_Parameter.

Use the ExternalSpeedLightStatus property (section 6.5.9.3) to get the ready state of the master flash.

**○ Attribute Value**

The valid AttributeValues are shown below.

| Attribute Value | Description |
|---|---|
| From Bit3 to Bit7 | (Reserved) |
| Bit2 | High temperature error |
| Bit1 | Battery warning |
| Bit0 | Ready |

### 6.7.1.4 SBTestFlashDisable

**○ Attribute Specifications**

| | |
|---|---|
| SBAttrID | 0x1005 |
| DataType | UINT8 |
| Get / Set | Get |
| Default Value | 0 |
| FormFlag | None |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the disabled state for the test firing of the flash.

**○ Attribute Value**

The valid AttributeValues are shown below.

| Attribute Value | Description |
|---|---|
| 0 | Enable firing |
| Other than 0 | Disable firing |

Do Not Copy

# 6.7.2 SBGroupAttribute

## 6.7.2.1 SBGroupFlashMode

### ○ Attribute Specifications

| | |
|---|---|
| SBGroupAttrID | 0x4001 |
| DataType | UINT8 |
| Get / Set | Get / Set |
| Default Value | The default value differs depending on the flash group. |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

### ○ Attribute Outline

This indicates the flash mode of the flash group.

### ○ Attribute Value

The valid AttributeValues are shown below.

| Attribute Value | Description |
|---|---|
| 0 | Off |
| 1 | [TTL-BL] i-TTL balanced fill flash * |
| 2 | [TTL] i-TTL fill flash |
| 3 | [AA] Auto aperture |
| 4 | [A] Non-TTL auto * |
| 5 | [GN] Distance-priority manual * |
| 6 | [M] Manual |
| 7 | [RPT] Repeating flash * |

* This can be used on the case of the stand-alone flash or the main flash.

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The flash group that cannot be used was specified.<br>(The specified group is not in the values of the SBUsableGroup (section 6.5.9.9).) |
| | The value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is [Repeating]. |

* Except that the value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is [Quick wireless lighting].

Do Not Copy

### 6.7.2.2 SBGroupFlashCompensation

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4002 |
| DataType | INT8 |
| Get / Set | Get, Get / Set |
| Default Value | 0 [0.0EV] |
| FormFlag | Range |
| Attribute Value | From -18[-3.0EV] to +18[+3.0EV] |

**○ Attribute Outline**

This indicates the flash group compensation value in units of 1/6 EV.
When the attached flash does not support this function, the value of this attribute is not fixed.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The flash group that cannot be used was specified.<br>(The specified group is not in the values of the SBUsableGroup (section 6.5.9.9).) |
| | The value of the SBGroupFlashMode property (section 6.7.2.1) for the specified flash group is [Off], [Manual] or [Repeating flash]. |
| | The value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is [Repeating]. |

* Except that the value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is [Quick wireless lighting].

Do Not Copy

### 6.7.2.3 SBGroupFlashRatio

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4003 |
| DataType | UINT8 |
| Get / Set | Get, Get / Set |
| Default Value | 7 [1:1] |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the ratio of the flash group.

**○ Attribute Value**

The valid AttributeValues are shown below.
When the attached flash does not support this function, the value of this attribute is not fixed.

| Attribute Value | Description |
|---|---|
| 0 | 1:-- |
| 1 | 8:1 |
| 2 | 6:1 |
| 3 | 4:1 |
| 4 | 3:1 |
| 5 | 2:1 |
| 6 | 1.5:1 |
| 7 | 1:1 |
| 8 | 1:1.5 |
| 9 | 1:2 |
| 10 | 1:3 |
| 11 | 1:4 |
| 12 | 1:6 |
| 13 | 1:8 |
| 14 | --:1 |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash]. |
| | The value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is other than [Quick wireless lighting]. |

Do Not Copy

### 6.7.2.4 SBGroupFlashLevel

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4004 |
| DataType | UINT8 |
| Get / Set | Get, Get / Set |
| Default Value | The default value differs depending on the flash mode. |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the flash level of the flash group.

**○ Attribute Value**

The valid AttributeValues are shown below.

| Attribute Value | Description |
|---|---|
| 1 | 1/1(Full) |
| 2 | 1/1 – 0.3(Full - 1/3 step) |
| 3 | 1/1 – 0.7(Full - 2/3 step) |
| 4 | 1/2 |
| 5 | 1/2 – 0.3 |
| 6 | 1/2 – 0.7 |
| 7 | 1/4 |
| 8 | 1/4 – 0.3 |
| 9 | 1/4 – 0.7 |
| 10 | 1/8 |
| 11 | 1/8 – 0.3 |
| 12 | 1/8 – 0.7 |
| 13 | 1/16 |
| 14 | 1/16 – 0.3 |
| 15 | 1/16 – 0.7 |
| 16 | 1/32 |
| 17 | 1/32 – 0.3 |
| 18 | 1/32 – 0.7 |
| 19 | 1/64 |
| 20 | 1/64 – 0.3 |
| 21 | 1/64 – 0.7 |
| 22 | 1/128 |
| 23 | 1/128 – 0.3 |
| 24 | 1/128 – 0.7 |
| 25 | 1/256 |

Do Not Copy

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The flash group that cannot be used was specified.<br>(The specified group is not in the values of the SBUsableGroup (section 6.5.9.9).) |
| | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash] and the value of the SBGroupFlashMode property (section 6.7.2.1) for the master flash is not [Off], [Manual] or [Repeating flash]. |
| | The value of the SBGroupFlashMode property (section 6.7.2.1) is other than [Manual]. |

* Except that the value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is [Quick wireless lighting] or [Repeating].

Do Not Copy

### 6.7.2.5 SBGroupFlashRange

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4005 |
| DataType | UINT32 |
| Get / Set | Get, Get / Set |
| Default Value | 13 [2.0[m]/6.6[ft]] |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the distance of the flash group.

**○ Attribute Value**

The valid AttributeValues are shown below.
When the attached flash does not support this function, the value of this attribute is not fixed.

| Attribute Value | Description | Attribute Value | Description |
|---|---|---|---|
| 0 | 0.3[m]/1.0[ft] | 17 | 3.2[m]/10[ft] |
| 1 | 0.4[m]/1.3[ft] | 18 | 3.6[m]/12[ft] |
| 2 | 0.5[m]/1.7[ft] | 19 | 4.0[m]/13[ft] |
| 3 | 0.6[m]/2.0[ft] | 20 | 4.5[m]/15[ft] |
| 4 | 0.7[m]/2.3[ft] | 21 | 5.0[m]/17[ft] |
| 5 | 0.8[m]/2.6[ft] | 22 | 5.6[m]/19[ft] |
| 6 | 0.9[m]/2.9[ft] | 23 | 6.3[m]/21[ft] |
| 7 | 1.0[m]/3.3[ft] | 24 | 7.1[m]/23[ft] |
| 8 | 1.1[m]/3.7[ft] | 25 | 8.0[m]/26[ft] |
| 9 | 1.3[m]/4.1[ft] | 26 | 9.0[m]/29[ft] |
| 10 | 1.4[m]/4.6[ft] | 27 | 10[m]/33[ft] |
| 11 | 1.6[m]/5.2[ft] | 28 | 11[m]/37[ft] |
| 12 | 1.8[m]/5.8[ft] | 29 | 13[m]/42[ft] |
| 13 | 2.0[m]/6.6[ft] | 30 | 14[m]/47[ft] |
| 14 | 2.2[m]/7.4[ft] | 31 | 16[m]/52[ft] |
| 15 | 2.5[m]/8.3[ft] | 32 | 18[m]/59[ft] |
| 16 | 2.8[m]/9.3[ft] | 33 | 20[m]/66[ft] |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The group of the master flash is invalid in the SBUsableGroup property (section 6.5.9.9). |
| | The value of the SBWirelessMode property (section 6.5.9.7) is other than [Stand-alone flash]. |
| | The value of the SBGroupFlashMode property (section 6.7.2.1) for the master flash is other than [Distance-priority manual]. |

Do Not Copy

### 6.7.2.6 SBGroupRepeat

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4006 |
| DataType | UINT8 |
| Get / Set | Get, Get / Set |
| Default Value | The default value differs depending on the flash group. |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the repeat setting of the flash group.

**○ Attribute Value**

The valid AttributeValues are shown below.
When the attached flash does not support this function, the value of this attribute is not fixed.

| Attribute Value | Description |
|---|---|
| 0 | Off |
| 1 | On |

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The flash group that cannot be used was specified.<br>(The specified group is not in the values of the SBUsableGroup (section 6.5.9.9).) |
| | The value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is not [Repeating]. |
| | The value of the SBUsableGroup property (section 6.5.9.9) is zero (Cannot be used). |

Do Not Copy

### 6.7.2.7 SBGroupRepeatCount

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4007 |
| DataType | UINT8 |
| Get / Set | Get, Get / Set |
| Default Value | 10 |
| FormFlag | Range |
| Attribute Value | From 2 to 90 |

**○ Attribute Outline**

This indicates the firing count of the flash group.
When the attached flash does not support this function, the value of this attribute is not fixed.

**○ Attribute Details**

The upper limit of the attribute value varies depending on the setting of the flash level and frequency. If the upper limit value varies, it is informed by the SBGroupAttrChanged event (section 6.4.2.15).

The number from 2 to 10 and the even number over 10 can be set.
When the odd number from 10 to 90 was set, it is converted to the even number by adding 1.

**○ Response Code**

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash] and the group of the master flash is invalid in the SBUsableGroup property (section 6.5.9.9). |
| | The value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is not [Repeating]. |
| | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash] and the value of the SBGroupFlashMode property (section 6.7.2.1) for the master flash is not [Repeating flash]. |
| | The value of the SBUsableGroup property (section 6.5.9.9) is zero (Cannot be used). |

Do Not Copy

### 6.7.2.8 SBGroupRepeatInterval

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4008 |
| DataType | UINT16 |
| Get / Set | Get, Get / Set |
| Default Value | 9[10Hz] |
| FormFlag | Enumeration |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the repeat interval of the flash group.

**○ Attribute Value**

The valid AttributeValues are shown below.
When the attached flash does not support this function, the value of this attribute is not fixed.

| AttributeValue | Description |
|---|---|
| 0 | 1Hz |
| 1 | 2Hz |
| 2 | 3Hz |
| 3 | 4Hz |
| 4 | 5Hz |
| 5 | 6Hz |
| 6 | 7Hz |
| 7 | 8Hz |
| 8 | 9Hz |
| 9 | 10Hz |
| 10 | 20Hz |
| 11 | 30Hz |
| 12 | 40Hz |
| 13 | 50Hz |
| 14 | 60Hz |
| 15 | 70Hz |
| 16 | 80Hz |
| 17 | 90Hz |
| 18 | 100Hz |

Do Not Copy

### ○ Response Code

When the value is set under the following conditions, an error response is made and the value cannot be set.

| Response Code | Description |
|---|---|
| Access_Denied | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash] and the group of the master flash is invalid in the SBUsableGroup property (section 6.5.9.9). |
| | The value of the SBWirelessMode property (section 6.5.9.7) is anything other than [Stand-alone flash] and the value of the SBWirelessMultipleFlashMode property (section 6.5.9.8) is not [Repeating]. |
| | The value of the SBWirelessMode property (section 6.5.9.7) is [Stand-alone flash] and the value of the SBGroupFlashMode property (section 6.7.2.1) for the master flash is not [Repeating flash]. |
| | The value of the SBUsableGroup property (section 6.5.9.9) is zero (Cannot be used). |

### 6.7.2.9 SBGroupValid

**○ Attribute Specifications**

| | |
|---|---|
| SBGroupAttrID | 0x4009 |
| DataType | UINT8 |
| Get / Set | Get |
| Default Value | 0 |
| FormFlag | None |
| Attribute Value | Refer to Attribute Value in this section. |

**○ Attribute Outline**

This indicates the valid state of the flash group.

**○ Attribute Value**

The valid AttributeValues are shown below.

| AttributeValue | Description |
|---|---|
| 0 | Invalid group |
| Other than 0 | Valid group |

Do Not Copy

# 7 Data Type

## 7.1 DataTypeCode

The standard data types used by the camera are shown below.

| DataTypeCode | Type | Description |
|---|---|---|
| 0x0001 | INT8 | Signed 8-bit integer |
| 0x0002 | UINT8 | Unsigned 8-bit integer |
| 0x0003 | INT16 | Signed 16-bit integer |
| 0x0004 | UINT16 | Unsigned 16-bit integer |
| 0x0005 | INT32 | Signed 32-bit integer |
| 0x0006 | UINT32 | Unsigned 32-bit integer |
| 0x0007 | INT64 | Signed 64-bit integer |
| 0x0008 | UINT64 | Unsigned 64-bit integer |
| 0x0009 | INT128 | Signed 128-bit integer |
| 0x000A | UINT128 | Unsigned 128-bit integer |
| 0x4001 | AINT8 | Signed 8-bit integer array |
| 0x4002 | AUINT8 | Unsigned 8-bit integer array |
| 0x4003 | AINT16 | Signed 16-bit integer array |
| 0x4004 | AUINT16 | Unsigned 16-bit integer array |
| 0x4005 | AINT32 | Signed 32-bit integer array |
| 0x4006 | AUINT32 | Unsigned 32-bit integer array |
| 0x4007 | AINT64 | Signed 64-bit integer array |
| 0x4008 | AUINT64 | Unsigned 64-bit integer array |
| 0x4009 | AINT128 | Signed 128-bit integer array |
| 0x400A | AUINT128 | Unsigned 128-bit integer array |
| 0xFFFF | STR | Variable length Unicode character string |

## 7.2 Character String Format

The field representing the character string complies with the following format.
Each field data is stored in the little endian format.

| Field | Size (Byte) | Data | Description |
|---|---|---|---|
| NumChar | 1 | N | Represents the number of characters in the string.<br>The terminating null character is included.<br>The maximum number of characters is 255. |
| StringChars [0] | 2 | | Unicode character |
| StringChars [1] | 2 | | Unicode character |
| --- | | | |
| StringChars [N - 1] | 2 | 0x0000 | Unicode character (null) |

Do Not Copy

## 7.3 Date Format

The character string representing the date complies with the following format.
The date and time is shown in the form of the most significant value through the least significant value according to the format of ISO8601 standard. This is a Unicode string format of “YYYYMMDDThhmmss” where YYYY is the year, MM is the month, DD is the day of the month, T is a constant character, hh is the hours, mm is the minutes, and ss is the seconds past the minute. The data is stored in the following array for the transmission/reception between the camera and the host.

| Field | Size (Byte) | Data | Description |
|---|---|---|---|
| NumChar | 1 | 0x10 | Represents the number of characters in the string.<br>The terminating null character is included.<br>The number of characters in the string representing the time is sixteen. |
| StringChars | 32 | | Unicode string<br>“YYYYMMDDThhmmss” |

When the format setting is “YYYYMMDDThhmmss.xx”, the data following “YYYYMMDDThhmmss” should be ignored for use.

## 7.4 Array Format

The array type complies with the following format. Each field data is stored in the little endian format.

| Field | Size (Byte) | Data |
|---|---|---|
| NumElement | 4 | The number of array elements is N (N is the number of objects). |
| ArrayEntry [0] | ElementSize | ArrayData [0] |
| ArrayEntry [1] | ElementSize | ArrayData [1] |
| ArrayEntry [2] | ElementSize | ArrayData [2] |
| --- | | |
| ArrayEntry [N - 1] | ElementSize | ArrayData [N - 1] |

ElementSize: Data size of ArrayData

Do Not Copy

## 7.5 SBHandle

The SBHandle is used for identifying the flashes (the internal flash, speedlight or other flash units) connected to the camera.

The SBHandle takes an unsigned 32-bit value, and the unique number is assigned to each flash.

## 7.6 SBGroupID

The SBGroupID is the ID number for identifying the flash group.

| SBGroupID | Description |
|---|---|
| 0x00000001 | Master flash |
| 0x00000002 | Group A |
| 0x00000004 | Group B |
| 0x00000008 | Group C |
| 0x00000010 | Group D |
| 0x00000020 | Group E |
| 0x00000040 | Group F |

Do Not Copy

# 8 ObjectHandle

The ObjectHandle is used to represent the individual objects in the camera (image file, non-image file, directories, and the virtual association representing the relations of the images and the DCF objects conforming to the DCF standards).

The ObjectHandle is unsigned 32 bits. A unique value is set for the ObjectHandle indicating each object. The specified ObjectHandle is fixed in one session.

## 8.1 ObjectHandle of the Object Recorded in the Card

The camera sets a 4-byte unique value freely for the ObjectHandle created for the data in the card by the camera. The host application does not care the contents.

## 8.2 ObjectHandle of the Object Recorded in the SDRAM

The object recorded in the SDRAM is the image file only.

The host can access the image data in the SDRAM by specifying the ObjectHandle passed by ObjectAddedInSdram.

## 8.3 Addition of the ObjectHandle

The camera acts as shown below when an object is newly added to the card during one session.

| No. | Description |
|---|---|
| 1 | In accordance with the ObjectHandle format defined in this section, the camera specifies a unique ObjectHandle that is not coordinated with other ObjectHandles already specified for the newly added object. |
| 2 | The camera sends the ObjectAdded event including the specified ObjectHandle as a parameter to the host. At this time, FreeSpaceInBytes and FreeSpaceInImages indicated in the StorageInfo data set are updated immediately. |

Do Not Copy

# 9 DataSet

The camera transmits the information about the camera to the host by using some data sets.
The data sets supported by the camera and their contents are shown below.

## 9.1 DeviceInfo DataSet

The DeviceInfo data set is sent by the operation of the GetDeviceInfo command.
Each field data is stored in the little endian format.
The information sent by the DeviceInfo data set is shown below.

| Field | Description |
|---|---|
| StandardVersion | The highest version of the standard that can support the device. |
| VendorExtensionID | The vendor extension ID used by the device. |
| VendorExtensionVersion | The vendor-specific version number of extensions that are supported. |
| VendorExtensionDesc | An optional string used to hold a human-readable description of the VendorExtensionID. |
| FunctionalMode | An optional field used to hold the functional mode. |
| OperationsSupported | An array of OperationCodes supported by the camera. |
| EventsSupported | An array of EventCodes supported by the camera. |
| DevicePropertiesSupported | An array of DevicePropCodes supported by the camera. |
| CaptureFormats | An array of ObjectFormatCodes that can be created by the camera with InitiateCapture. |
| ImageFormats | An array of ObjectFormatCodes supported by the camera. |
| Manufacture | An optional human-readable string used to indicate the device manufacturer. |
| Model | An optional human-readable string used to indicate the device name. |
| DeviceVersion | An optional human-readable string used to indicate the version of the device. |
| SerialNumber | An optional human-readable string used to indicate the serial number of the camera. |

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StandardVersion | 2 | 0x0064 | | Version 1.00 |
| VendorExtensionID | 4 | 0xFFFFFFFF | | - |
| VendorExtensionVersion | 2 | 0x006E | | Version 1.10 |
| VendorExtensionDesc | 69 | 0x22<br>0x4D00<br>0x6900<br>0x6300<br>0x7200<br>0x6F00<br>0x7300<br>0x6F00<br>0x6600<br>0x7400<br>0x2E00<br>0x6300<br>0x6F00<br>0x6D00<br>0x2F00<br>0x4400<br>0x6500<br>0x7600 | String | UNICODE character string<br>“Microsoft.com/DeviceServices: 1.0” |

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| | | | | |
|---|---|---|---|---|
| | | 0x6900<br>0x6300<br>0x6500<br>0x5300<br>0x6500<br>0x7200<br>0x7600<br>0x6900<br>0x6300<br>0x6500<br>0x7300<br>0x3A00<br>0x2000<br>0x3100<br>0x2E00<br>0x3000<br>0x0000 | | |
| FunctionalMode | 2 | 0x0000 | | Normal mode |
| OperationsSupported | 248 | 0x0000007A<br>0x1001<br>0x1002<br>0x1003<br>0x1004<br>0x1005<br>0x1006<br>0x1007<br>0x1008<br>0x1009<br>0x100A<br>0x100B<br>0x100E<br>0x100F<br>0x1014<br>0x1015<br>0x1016<br>0x101B<br>0x90C0<br>0x90C1<br>0x90C2<br>0x90C3<br>0x90C4<br>0x90C7<br>0x90C8<br>0x90C9<br>0x90CB<br>0x90CC<br>0x90CD<br>0x90CE<br>0x90CF<br>0x9201<br>0x9202<br>0x9203<br>0x9204<br>0x9205<br>0x9206<br>0x9207<br>0x9209<br>0x920A<br>0x920B<br>0x920C<br>0x920F<br>0x9400<br>0x9401<br>0x9402<br>0x9403<br>0x9404<br>0x9405<br>0x9406 | Array | OperationCode supported by the camera<br>This varies depending on whether the connection destination is MTP or not.<br>MTP:<br>0x100C Not supported.<br>0x100D Not supported.<br>Other than MTP:<br>0x9801 Not supported.<br>0x9802 Not supported.<br>0x9803 Not supported.<br>0x9805 Not supported. |

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| | | | | |
|---|---|---|---|---|
| | | 0x940C<br>0x9412<br>0x9413<br>0x9414<br>0x9415<br>0x9416<br>0x9417<br>0x9418<br>0x9419<br>0x941A<br>0x941B<br>0x941C<br>0x9420<br>0x9421<br>0x9422<br>0x9423<br>0x9424<br>0x9425<br>0x9426<br>0x9428<br>0x942E<br>0x942F<br>0x9431<br>0x9433<br>0x9434<br>0x9435<br>0x9436<br>0x9439<br>0x943A<br>0x943B<br>0x943C<br>0x943D<br>0x943E<br>0x943F<br>0x9440<br>0x9441<br>0x9442<br>0x9443<br>0x9444<br>0x9445<br>0x9446<br>0x9447<br>0x944C<br>0x944D<br>0x944E<br>0x944F<br>0x9453<br>0x9454<br>0x9455<br>0x9456<br>0x9457<br>0x9458<br>0x9459<br>0x945A<br>0x9461<br>0x9463<br>0x9464<br>0x946D<br>0x946E<br>0x946F<br>0x9470<br>0x9520<br>0x9521<br>0x9522<br>0x9523<br>0x9527<br>0x9528<br>0x9529 | | |

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| | | | | |
|---|---|---|---|---|
| | | 0x9530<br>0x9801<br>0x9802<br>0x9803<br>0x9805 | | |
| EventsSupported | 58 | 0x0000001B<br>0x4001<br>0x4002<br>0x4003<br>0x4004<br>0x4005<br>0x4006<br>0x4007<br>0x4008<br>0x4009<br>0x400A<br>0x400C<br>0x400D<br>0xC101<br>0xC102<br>0xC105<br>0xC108<br>0xC10A<br>0xC10B<br>0xC10C<br>0xC10D<br>0xC10E<br>0xC120<br>0xC121<br>0xC122<br>0xC123<br>0xC128<br>0xC700 | Array | EventCode supported by the camera |
| DevicePropertiesSupported | 46 | 0x00000015<br>0x5001<br>0x5003<br>0x5004<br>0x5005<br>0x5007<br>0x5008<br>0x500A<br>0x500B<br>0x500C<br>0x500D<br>0x500E<br>0x500F<br>0x5010<br>0x5011<br>0x5013<br>0x5018<br>0x501E<br>0x501F<br>0xD303<br>0xD406<br>0xD407 | Array | DevicePropertyCode supported by the camera<br><br>The vendor codes are not enumerated in this field.<br>The vendor codes can be acquired by the GetVendorCodes command (section 6.2.2.1). |
| CaptureFormats | 8 | 0x00000002<br>0x3801<br>0x3800 | Array | ObjectFormatCode that can be created by the camera with InitiateCapture.<br>This varies depending on whether the connection destination is MTP or not.<br>MTP:<br>0x3000 Not supported.<br>Other than MTP:<br>0x3800 Not supported |
| ImageFormats | 24 | 0x0000000A<br>0x3801<br>0x3800<br>0x3001<br>0x3008 | Array | ObjectFormatCode supported by the camera.<br>This varies depending on whether the connection destination is MTP or not.<br>MTP: |

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| | | | | |
|---|---|---|---|---|
| | | 0x380D<br>0x3000<br>0x300D<br>0xB200<br>0xB201<br>0xB982 | | 0x3002 Not supported.<br>Other than MTP:<br>0x3800 Not supported. |
| Manufacture | 37 | 0x12<br>0x4E00<br>0x6900<br>0x6B00<br>0x6F00<br>0x6E00<br>0x2000<br>0x4300<br>0x6F00<br>0x7200<br>0x7000<br>0x6F00<br>0x7200<br>0x6100<br>0x7400<br>0x6900<br>0x6F00<br>0x6E00<br>0x0000 | String | UNICODE character string<br>“Nikon Corporation” |
| Model | 9 | 0x04<br>0x5A00<br>0x2000<br>0x3800<br>0x0000 | String | UNICODE character string<br>“Z 8” |
| DeviceVersion | 13 | 0x06<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0x0000 | String | UNICODE character string<br><br>The device version is stored in XX.<br>In the case of V1.00, it is as follows.<br>0x5600<br>0x3100<br>0x2E00<br>0x3000<br>0x3000 |
| SerialNumber | 67 | 0x21<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0xXX00 | String | UNICODE character string<br>“XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX” |

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| | | | | |
|---|---|---|---|---|
| | | 0xXX00<br>0xXX00<br>0xXX00<br>0xXX00<br>0x0000 | | |

# 9.2 StorageInfo DataSet

The StorageInfo data set is sent by the operation of the GetStorageInfo command.
This data set indicates information about the storage medium (card).
Each field data is stored in the little-endian format.
Information sent by the StorageInfo data set is shown below.

| Field | Description |
|---|---|
| StorageType | Indicates the type of the card. Removable Ram is set. |
| FilesystemType | Indicates the filesystem of the card. It conforms to the DCF. |
| AccessCapability | Indicates the access right for the card.<br>When the card is not locked, it is read-only and the image deletion is allowed.<br>When the card is locked, it is read-only and the image deletion is not allowed. |
| MaxCapacity | Indicates the capacity of the card. It depends on the card used. |
| FreeSpaceInBytes | Indicates the free space in the card. It depends on the card used and the space that is currently used. |
| FreeSpaceInImages | Indicates the number of images that can be recorded in the free space of the card.<br>It is the number of images captured in the mode that is currently set in the camera.<br>It depends on the card used and the space that is currently used. |
| StorageDescription | Indicates a human-readable text description of the card. This field is not used for the camera. |
| VolumeLabel | Indicates the volume label of the card.<br>It is described in a human-readable character string (Unicode character string). |

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StorageType | 2 | 0x0004 | | Removable Ram |
| FilesystemType | 2 | 0x0003 | | Based on the DCF |
| AccessCapability | 2 | 0x0002<br>0x0001 (Card lock) | | Read-Only with Object Deletion<br>Read-Only without Object Deletion (Card lock) |
| MaxCapacity | 8 | | | Depends on the card. |
| FreeSpaceInBytes | 8 | | | Depends on the card and the space used. |
| FreeSpaceInImages | 4 | | | Depends on the card and the space used. |
| StorageDescription | 1 | 0x00 | | - |
| VolumeLabel | | | String | Unicode character string |

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# 9.3 ObjectInfo DataSet

The ObjectInfo data set is sent by the operation of the GetObjectInfo command.
This data set indicates information about the objects in the card.
Each field data in which the data type is not specified is stored in the little endian format.

| Field | Description |
|---|---|
| StorageID | Indicates the StorageID of the card. |
| ObjectFormat | Indicates the ObjectFormatCode of the object. |
| ProtectionStatus | Indicates the protection status of the object. |
| ObjectCompressedSize | Indicates the size of the object in bytes.<br>0xFFFFFFFF for over 4GB. |
| ThumbFormat | Indicates the ObjectFormat of the thumbnail. |
| ThumbCompressedSize | Indicates the size of the thumbnail in bytes. |
| ThumbPixWidth | Indicates the thumbnail width in pixels. |
| ThumbPixHeight | Indicates the thumbnail height in pixels. |
| ImagePixWidth | Indicates the image width in pixels. |
| ImagePixHeight | Indicates the image height in pixels. |
| ImageBitDepth | Indicates the bit depth of the image. |
| ParentObject | Indicates the ObjectHandle of the parent object of this object. |
| AssociationType | Indicates the association type.<br>It is used for the object of the association type. |
| AssociationDesc | Indicates the descriptor parameter of the association.<br>It is not used in the camera. |
| SequenceNumber | Indicates the component of the association.<br>It is not used in the camera. |
| Filename | Indicates an optional character string showing the file name of the object. |
| CaptureDate | Indicates the character string showing the date and time of object creation. |
| ModificationDate | Indicates the character string showing the date and time of object modification.<br>When the object does not have a date and time of modification, the same date and time as that of the object creation is stored. |
| Keywords | Indicates the character string showing the image-related keyword.<br>It is not used in the camera. |

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### 9.3.1 Directory/Virtual Association DataSet

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StorageID | 4 | StorageID | | StorageID |
| ObjectFormat | 2 | 0x3001 | | Association (Refer to ObjectFormatCode.) |
| ProtectionStatus | 2 | 0x0000 | | - |
| ObjectCompressedSize | 4 | 0x00000000 | | - |
| ThumbFormat | 2 | 0x0000 | | - |
| ThumbCompressedSize | 4 | 0x00000000 | | - |
| ThumbPixWidth | 4 | 0x00000000 | | - |
| ThumbPixHeight | 4 | 0x00000000 | | - |
| ImagePixWidth | 4 | 0x00000000 | | - |
| ImagePixHeight | 4 | 0x00000000 | | - |
| ImageBitDepth | 4 | 0x00000000 | | - |
| ParentObject | 4 | ObjectHandle | | ObjectHandle of the parent directory<br>DCIM folder: 0x00000000 |
| AssociationType | 2 | 0x0001 | | GenericFolder |
| AssociationDesc | 4 | 0x00000000 | | - |
| SequenceNumber | 4 | 0x00000000 | | - |
| Filename | | | String | Unicode character string |
| CaptureDate | | | String | Date and time of capture (Unicode character string)<br>(Not used for the virtual association) |
| ModificationDate | | | String | Date and time of modification (Unicode character string)<br>(Not used for the virtual association) |
| Keywords | 1 | 0x00 | | - |

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## 9.3.2 Image File DataSet

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StorageID | 4 | StorageID | | StorageID<br>SDRAM image: 0x00000000 |
| ObjectFormat | 2 | | | 0x3000 (Undefined)<br>0x3801 (EXIF)<br>0x380D (TIFF) |
| ProtectionStatus | 2 | | | 0x0001 (with protection setting),<br>0x0000 (without protection setting) |
| ObjectCompressedSize | 4 | | | File size |
| ThumbFormat | 2 | 0x3808 | | JFIF (Refer to ObjectFormatCode.) |
| ThumbCompressedSize | 4 | | | Thumbnail size |
| ThumbPixWidth | 4 | 0x000000A0 | | Horizontal size of the thumbnail (160) |
| ThumbPixHeight | 4 | 0x00000078 | | Vertical size of the thumbnail (120) |
| ImagePixWidth | 4 | | | Horizontal size of the main image |
| ImagePixHeight | 4 | | | Vertical size of the main image |
| ImageBitDepth | 4 | 0x00000000 | | - |
| ParentObject | 4 | ObjectHandle | | ObjectHandle of the parent directory |
| AssociationType | 2 | 0x0000 | | - |
| AssociationDesc | 4 | 0x00000000 | | - |
| SequenceNumber | 4 | 0x00000000 | | - |
| Filename | | | String | File name character string (Unicode character string)<br>“File name.extension” is set for the images recorded in the card with the recording destination set to “Card” or “Card and SDRAM”.<br>“DSC_0000.extension” is set for the images whose recording destination is the SDRAM.<br>For the images recorded in the SDRAM with the recording destination set to “Card and SDRAM”, the name including the folder name and the file name of the image recorded in the card simultaneously is set. “Folder name¥(backslash)file name.extension”. If the image deletion is performed by operating the camera during the card recording while “Card and SDRAM” recording is set in the application mode, the file name of the image with the recording destination SDRAM may be “DSC_0000.extension” in some cases.<br>When the object format is “Undefined”, the extension is NEF (RAW) or NDF (dust reference image). |
| CaptureDate | | | String | Date and time of capture (Unicode character string) |
| ModificationDate | | | String | Date and time of modification (Unicode character string) |
| Keywords | 1 | 0x00 | | - |

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## 9.3.3 Script File DataSet

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StorageID | 4 | 0x00000000 | | |
| ObjectFormat | 2 | 0x3002 | | Script (Refer to ObjectFormatCode.) |
| ProtectionStatus | 2 | 0x0000 | | No protection setting |
| ObjectCompressedSize | 4 | | | File size |
| ThumbFormat | 2 | 0x0000 | | - |
| ThumbCompressedSize | 4 | 0x00000000 | | - |
| ThumbPixWidth | 4 | 0x00000000 | | - |
| ThumbPixHeight | 4 | 0x00000000 | | - |
| ImagePixWidth | 4 | 0x00000000 | | - |
| ImagePixHeight | 4 | 0x00000000 | | - |
| ImageBitDepth | 4 | 0x00000000 | | - |
| ParentObject | 4 | 0x00000000 | | - |
| AssociationType | 2 | 0x0000 | | - |
| AssociationDesc | 4 | 0x00000000 | | - |
| SequenceNumber | 4 | 0x00000000 | | - |
| Filename | | | String | File name (Unicode character string) (“DDISCVRY.DPS” or “DREQUEST.DPS”) |
| CaptureDate | | | String | Date and time of creation (Unicode character string) |
| ModificationDate | | | String | Date and time of modification (Unicode character string) |
| Keywords | 1 | 0x00 | | - |

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## 9.3.4 Video File DataSet

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StorageID | 4 | StorageID | | StorageID<br>Image: 0x00000000 |
| ObjectFormat | 2 | | | 0x300D (MOV)<br>0xB201 (NEV)<br>0xB982 (MP4) |
| ProtectionStatus | 2 | | | 0x0001 (with protection setting)<br>0x0000 (without protection setting) |
| ObjectCompressedSize | 4 | | | File size |
| ThumbFormat | 2 | 0x3808 | | JFIF (Refer to ObjectFormatCode.) |
| ThumbCompressedSize | 4 | | | Thumbnail size |
| ThumbPixWidth | 4 | 0x000000A0 | | Horizontal size of the thumbnail (160) |
| ThumbPixHeight | 4 | 0x00000078 | | Vertical size of the thumbnail (120) |
| ImagePixWidth | 4 | | | Horizontal size of the main video |
| ImagePixHeight | 4 | | | Vertical size of the main video |
| ImageBitDepth | 4 | 0x00000000 | | - |
| ParentObject | 4 | ObjectHandle | | ObjectHandle of the parent directory |
| AssociationType | 2 | 0x0000 | | - |
| AssociationDesc | 4 | 0x00000000 | | - |
| SequenceNumber | 4 | 0x00000000 | | - |
| Filename | | | String | File name character string (Unicode character string)<br>“File name.MOV” |
| CaptureDate | | | String | Date and time of capture (Unicode character string) |
| ModificationDate | | | String | Date and time of modification (Unicode character string) |
| Keywords | 1 | 0x00 | | - |

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## 9.3.5 WAV File Data Set

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| StorageID | 4 | StorageID | | StorageID |
| ObjectFormat | 2 | 0x3008 | | WAV |
| ProtectionStatus | 2 | | | 0x0001 (with protection setting)<br>0x0000 (without protection setting) |
| ObjectCompressedSize | 4 | | | File size |
| ThumbFormat | 2 | 0x0000 | | - |
| ThumbCompressedSize | 4 | 0x00000000 | | - |
| ThumbPixWidth | 4 | 0x00000000 | | - |
| ThumbPixHeight | 4 | 0x00000000 | | - |
| ImagePixWidth | 4 | 0x00000000 | | - |
| ImagePixHeight | 4 | 0x00000000 | | - |
| ImageBitDepth | 4 | 0x00000000 | | - |
| ParentObject | 4 | ObjectHandle | | ObjectHandle of the parent directory |
| AssociationType | 2 | 0x0000 | | - |
| AssociationDesc | 4 | 0x00000000 | | - |
| SequenceNumber | 4 | 0x00000000 | | - |
| Filename | | | String | File name character string (Unicode character string) |
| CaptureDate | | | String | Date and time of capture (Unicode character string) |
| ModificationDate | | | String | Date and time of modification (Unicode character string) |
| Keywords | 1 | 0x00 | | - |

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# 9.4 DevicePropDesc DataSet

The DevicePropDesc data set is sent by the operation of the GetDevicePropDesc command.
This data set indicates information about the settings and the attribute of the device.
Each field data in which the data type is not specified is stored in the little endian format.

| Field | Description |
|---|---|
| DevicePropCode | Indicates DevicePropCode of the property. |
| DataType | Indicates the data type of the property. |
| GetSet | Indicates the access attribute of the property. |
| Factory Default Value | Indicates the default value of the property. |
| Current Value | Indicates the current value of the property. |
| FormFlag | Indicates the property description data set. |

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| DevicePropertyCode | 2 or 4 | DevicePropCode | | DevicePropCode supported by the camera.<br>When acquiring with the GetDevicePropDescEx command and GetDevicePropDescList command, the Field Size of DevicePropCode is 4Byte. |
| DataType | 2 | | | Indicates the data type of the property.<br>It differs depending on each property.<br>Refer to DataTypeCode (section 7.1). |
| GetSet | 1 | | | Indicates whether the property is for reading only or for both reading and writing.<br>0x00: Reading only (Get)<br>0x01: Reading/writing (Get/Set) |
| FactoryDefaultValue | DTS | | | Default value.<br>It differs depending on each property. |
| CurrentValue | DTS | | | Current value.<br>It differs depending on each property. |
| FormFlag | 1 | | | Indicates the property description data set.<br>0x00: None<br>0x01: Range<br>0x02: Enumeration |
| FORM | DTS | | | The contents of the field depend on the FormFlag field.<br>It does not exist when the FormFlag field is set to 0. |

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# 9.5 ObjectPropDesc DataSet

The ObjectPropDesc data set is sent by the operation of the GetObjectPropDesc command.
This data set indicates information about the settings and the attribute of the object.
Each field data in which the data type is not specified is stored in the little endian format.

| Field | Description |
|---|---|
| ObjectPropertyCode | Indicates ObjectPropCode of the property. |
| DataType | Indicates the data type of the property. |
| GetSet | Indicates the access attribute of the property. |
| Default Value | Indicates the default value of the property. |
| Group Code | Indicates the search group of the property. |
| FormFlag | Indicates the property description data set. |

| Field | Size (Byte) | Data | DataType | Description |
|---|---|---|---|---|
| ObjectPropertyCode | 2 | ObjectPropCode | UINT16 | ObjectPropCode supported by the camera |
| DataType | 2 | DataTypeCode | UINT16 | Indicates the data type of the property.<br>It differs depending on each property.<br>Refer to DataTypeCode (section 7.1). |
| GetSet | 1 | | UINT8 | Indicates whether the property is for reading only or for both reading and writing.<br>0x00: Reading only (Get)<br>0x01: Reading/writing (Get/Set) |
| DefaultValue | DTS | | | The default value in the camera.<br>It differs depending on each property. |
| Group Code | 4 | | UINT32 | Search group |
| FormFlag | 1 | | UINT8 | Indicates the property description data set.<br>0x00: None<br>0x01: Range<br>0x02: Enumeration<br>0x03: Time<br>0x04: Fixed-length array<br>0x05: Regular expression<br>0x06: Byte string<br>0xFF: LongString |
| FORM | DTS | | | The contents of the field depend on the FormFlag field.<br>It does not exist when the FormFlag field is set to 0. |

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# 9.6 Property Description DataSet

The property description data set is set in the FORM field of the DevicePropDesc data set and the ObjectPropDesc data set.

## 9.6.1 Range Form

| Field | Size (Byte) | Description |
|---|---|---|
| MinimumValue | DTS | The minimum value supported by the PropertyValue |
| MaximumValue | DTS | The maximum value supported by the PropertyValue |
| StepSize | DTS | The property supports the value indicated as shown below.<br>MinimumValue + N x StepSize<br>* N: From 0 to the maximum value<br>* PropertyValue: Smaller than the Maximum Value |

## 9.6.2 Enumeration Form

| Field | Size (Byte) | Description |
|---|---|---|
| NumberOfValue | 2 | Indicates the number of values of the PropertyValue supported by the property. |
| SupportedValue1 | DTS | The property supports this PropertyValue. |
| SupportedValue2 | DTS | The property supports this PropertyValue. |
| SupportedValue3 | DTS | The property supports this PropertyValue. |
| --- | | |
| SupportedValueM | DTS | The property supports this PropertyValue. |

## 9.6.3 Time Form

For the time form, the FORM field does not exist.

The time form is represented by a Unicode character string in the ISO standard format. (Refer to ISO8601.)

| YYYYMMDDThhmmss.s | |
|---|---|
| YYYY | Year |
| MM | Month (from 01 to 12) |
| DD | Date (from 01 to 31) |
| T | Fixed character |
| hh | Hour starting from 0 a.m. (from 00 to 23) |
| mm | Minutes (from 00 to 59) |
| ss.s | Seconds |

## 9.6.4 Fixed-Length Array Form

| Field | Size (Byte) | Description |
|---|---|---|
| Length | 2 | It is an unsigned 16-bit integer and indicates the number of array elements. |

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### 9.6.5 Regular Expression Form

| Field | Size (Byte) | Description |
|---|---|---|
| RegEx | DTS | It indicates the regular expression for creating the PropertyValue correctly. |

### 9.6.6 Byte String Form

| Field | Description |
|---|---|
| MaxLength | It indicates the maximum length of the byte string. |

### 9.6.7 LongString Form

| Field | Description |
|---|---|
| MaxLength | It indicates the maximum length of the LongString. The property includes the data type of AUINT16.<br>(Characters coded by 2-byte Unicode characters as defined in ISO10646.) |

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# 9.7 LiveViewObject DataSet

## 9.7.1 DataSet with Version

| | Field | | Offset | Size | Description |
|---|---|---|---|---|---|
| Display information | Version | | 0 | 2Byte | Major version (0x02) |
| | | | 2 | 2Byte | Minor version (0x01) |
| | Reserved | | 4 | 4Byte | |
| | Display information area size | | 8 | 4Byte | |
| | Live view image area size | | 12 | 4Byte | |
| | Whole size | Horizontal size | 16 | 2Byte | Standard of the coordinates |
| | | Vertical size | 18 | 2Byte | |
| | Display area size | Horizontal size | 20 | 2Byte | The whole size is equal to the display area size when the image is not enlarged. |
| | | Vertical size | 22 | 2Byte | |
| | Display center coordinates | Horizontal size | 24 | 2Byte | |
| | | Vertical size | 26 | 2Byte | |
| | Live view image Image size | Horizontal size | 28 | 2Byte | The size set in the LiveViewImageSize property becomes the image size. |
| | | Vertical size | 30 | 2Byte | |
| | Live view image Quality | | 32 | 1Byte | The quality is set in the LiveViewImageCompression property. |
| | Reserved | | 33 | 7Byte | |
| | AF driving enabled status | | 40 | 1Byte | 0: AF driving disabled, 1: AF driving enabled |
| | Focus driving status | | 41 | 1Byte | 0: Not driving, 1: Driving |
| | Focusing judgement result | | 42 | 1Byte | 0: No information, 1: Not focused,<br>2: Focused |
| | AF mode state | | 43 | 1Byte | 0: Other<br>1: Subject detection AF<br>2: Auto-area AF / Target tracking AF / 3D-Tracking AF |
| | AF area number | | 44 | 1Byte | From 0 to 64 (The number of subjects when using subject detection) |
| | Selected subject index | | 45 | 1Byte | From 0 to 34 |
| | Tracking state | | 46 | 1Byte | 0: Waiting<br>2: Tracking |
| | Reserved | | 47 | 1Byte | |
| | From 0 to 95 | AF frame size | 48 | 2Byte | Region of the AF frame size and the AF frame center coordinates for 96 areas. |
| | | Horizontal size | | | |
| | | Vertical size | 50 | 2Byte | |
| | | AF frame center coordinates | 52 | 2Byte | |
| | | Horizontal | | | |
| | | Vertical | 54 | 2Byte | |
| | Remaining time of video recording | | 816 | 4Byte | From 0 to 7500000 [msec]<br>* It is valid during the video recording state. |
| | Elapsed video recording time | | 820 | 4Byte | From 0 to 7500000 [msec]<br>* It is valid during the video recording state. |
| | Sound indicator (peak value) | L | 824 | 1Byte | From 0 to 14 |
| | | R | 825 | 1Byte | From 0 to 14 |
| | Sound indicator (current value) | L | 826 | 1Byte | From 0 to 14 |
| | | R | 827 | 1Byte | From 0 to 14 |
| | Video recording information | | 828 | 1Byte | 0: During LV execution<br>1: During video recording |

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| | | | | | |
|---|---|---|---|---|---|
| | State of synchronous recording by the external recorder (NOTE1) | | 829 | 1Byte | 0: Synchronous recording cannot be performed<br>1: Synchronous recording standby (Synchronous recording can be performed but does not start)<br>2: Synchronous recording is being performed |
| | Reserved | | 830 | 1Byte | |
| | Time code status | | 831 | 1Byte | 0: Off, 1: On |
| | Time code count | | 832 | 1Byte | Hour |
| | | | 833 | 1Byte | Minute |
| | | | 834 | 1Byte | Second |
| | | | 835 | 1Byte | Frame |
| | Countdown time | | 836 | 2Byte | Countdown every one second starting from 3600 (one hour) ; countdown starting from thirty seconds with a rise in temperature |
| | Spot WB condition | | 838 | 1Byte | 0: Off<br>1: On standby (not acquired)<br>2: During acquisition operation<br>3: On standby (acquisition success)<br>4: On standby (acquisition failure) |
| | Rotation | | 839 | 1Byte | 0: Off, 1: Rotate counterclockwise,<br>2: Rotate clockwise, 3: Upside down |
| | Level angle information | Rolling | 840 | 4Byte | |
| | | Pitching | 844 | 4Byte | |
| | | Yawing | 848 | 4Byte | |
| | Using white balance for photo Lv | | 852 | 1Byte | 0: Off<br>1: On |
| | Internal temperature condition | | 853 | 1Byte | 0: Normal<br>1: Medium temperature (low)<br>2: Medium temperature (high)<br>3: High temperature<br>4: Countdown display |
| | Reserved | | 854 | 170Byte | |
| Live view image | Image data | | 1024 | | |

The data specifications for LiveViewObject (With Version) are shown below.

| No. | Description |
|---|---|
| 1 | The size of the display information is 1024Byte. |
| 2 | When the electronic VR is ON, the image is 90% of angle of view. |

The values of AF frame size and AF frame center coordinates are shown below.

| No. | Description |
|---|---|
| 1 | The AF frame size and the AF frame center coordinates for the number of [AF area number] are valid. |

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Refer to the following property about the value for virtual horizon.

| No. | Description |
|---|---|
| 1 | AngleLevel property (section 6.5.6.19) |
| 2 | AngleLevelPitching property (section 6.5.6.20) |
| 3 | AngleLevelYawing property (section 6.5.6.21) |

The details of the AF mode state are shown below.

| No. | Description |
|---|---|
| 1 | The AF mode state becomes [1:Subject detect AF] under the following conditions.<br>- The camera detects subjects when the AF area mode is [Wide-area AF (S)], [Wide-area AF (L)], [Auto-area AF], [Target tracking AF], [3-D Tracking], [Wide area AF(C1)] or [Wide area AF(C2)]. |
| 2 | The AF mode state becomes [2:Auto-area AF/ Target tracking AF/3-D Tracking] under the following conditions.<br>- The camera doesn't detect subjects when the AF area mode is [Auto-area AF], [Target tracking AF] or [3-D Tracking]. |
| 3 | The AF mode state becomes [0:Other] under the following conditions.<br>- When the conditions of No1 and No2 are not met. |

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## 9.7.2 DataSet without Version

| | Field | | | Size | Description |
|---|---|---|---|---|---|
| | Display information area size | | | 4Byte | |
| | Live view image area size | | | 4Byte | |
| Display information | Live view image Image size | | Horizontal size | 2Byte | The size set in the LiveViewImageSize property becomes the image size. |
| | | | Vertical size | 2Byte | |
| | Whole size | | Horizontal size | 2Byte | Standard of the coordinates |
| | | | Vertical size | 2Byte | |
| | Display area size | | Horizontal size | 2Byte | The whole size is equal to the display area size when the image is not enlarged. |
| | | | Vertical size | 2Byte | |
| | Display center coordinates | | Horizontal size | 2Byte | |
| | | | Vertical size | 2Byte | |
| | AF frame size | | Horizontal size | 2Byte | |
| | | | Vertical size | 2Byte | |
| | AF frame center coordinates | | Horizontal size | 2Byte | |
| | | | Vertical size | 2Byte | |
| | Reserved | | | 4Byte | |
| | Selected focus area | | | 1Byte | From 0 to 51 |
| | Rotation | | | 1Byte | 0: Off, 1: Rotate counterclockwise,<br>2: Rotate clockwise, 3: Upside down |
| | Focus driving status | | | 1Byte | 0: Not driving, 1: Driving |
| | Reserved | | | 1Byte | |
| | Reserved | | | 4Byte | |
| | Reserved | | | 2Byte | |
| | Countdown time | | | 2Byte | Countdown every one second starting from 3600 (one hour) ; countdown starting from thirty seconds with a rise in temperature |
| | Focusing judgement result | | | 1Byte | 0: No information, 1: Not focused,<br>2: Focused |
| | AF driving enabled status | | | 1Byte | 0: AF driving disabled, 1: AF driving enabled |
| | Reserved | | | 2Byte | |
| | Level angle information | | Rolling | 4Byte | |
| | | | Pitching | 4Byte | |
| | | | Yawing | 4Byte | |
| | Remaining time of video recording | | | 4Byte | From 0 to 7500000 [msec]<br>* It is valid during the video recording state. |
| | Video recording information | | | 1Byte | 0: During LV execution<br>1: During video recording |
| | Subject detection AF mode state | | | 1Byte | 0: Subject detection AF is not set.<br>1: Subject detection AF is set. |
| | The number of detected subjects | | | 1Byte | From 0 to 35 |
| | Selected subject index | | | 1Byte | From 0 to 34 |
| | 0 to 34 | AF frame size | Horizontal size | 2Byte | Area of the AF frame size and the AF frame center coordinates for thirty-five persons<br>(4 Byte + 4 Byte) x 35 persons;<br>280 Byte in total |
| | | | Vertical size | 2Byte | |
| | | AF frame center coordinates | Horizontal position | 2Byte | |
| | | | Vertical position | 2Byte | |
| | Sound indicator | | L | 1Byte | From 0 to 14 |

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| | | | | |
|---|---|---|---|---|
| | (peak value) | R | 1Byte | From 0 to 14 |
| | Sound indicator (current value) | L | 1Byte | From 0 to 14 |
| | | R | 1Byte | From 0 to 14 |
| | Reserved | | 1Byte | |
| | Reserved | | 1Byte | |
| | Reserved | | 1Byte | |
| | Spot WB condition | | 1Byte | 0: Off<br>1: On standby (not acquired)<br>2: During acquisition operation<br>3: On standby (acquisition success)<br>4: On standby (acquisition failure) |
| | Reserved | | 24Byte | |
| Live view image | Image data | | | |

The data specifications for LiveViewObject (No Version) are shown below.

| No. | Description |
|---|---|
| 1 | The size of the display information is 376Byte. |
| 2 | When the electronic VR is ON, the image is 90% of angle of view. |

Refer to the following property about the value for virtual horizon.

| No. | Description |
|---|---|
| 1 | AngleLevel property (section 6.5.6.19) |
| 2 | AngleLevelPitching property (section 6.5.6.20) |
| 3 | AngleLevelYawing property (section 6.5.6.21) |

The details of the case in which the subject detection AF mode state is set to “1: Subject detection AF is set” are shown below.

| No. | Description |
|---|---|
| 1 | When the camera detects subject, the subject detection AF mode state is set to “1: Subject detection AF” is set”, and AF frame size and AF center coordinate for subject detection become valid. |

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# 9.8 IPTCPreset Data Set

The IPTCPreset data set is transmitted by the operation of the GetIPTCPresetInfo command or the SetIPTCPresetInfo command.

## 9.8.1 NMS-IPTC

NMS-IPTC deta is described below.

| Field | Size(Byte) | Data | DataType | Description |
|---|---|---|---|---|
| DatasetVersion | 2 | 0x00C9 | UINT16 | Version of this dataset<br>Version 2.01 |
| IPTC Type | 1 | 1 | UINT8 | IPTC data format<br>1:NMS-IPTC<br>2:XMP-IPTC |
| Reserved | 5 | - | - | |
| Profile | | | AUINT8 | Profile name<br>ASCII:Maximum 18Byte+NULL |
| NMS Data | | - | - | NMS format IPTC data |
| Caption | | | AUINT8 | Caption<br>UTF-8:Maximum 2000Byte+NULL |
| EventID | | | AUINT8 | Event ID<br>UTF-8:Maximum 64Byte+NULL |
| Headline | | | AUINT8 | Headline<br>UTF-8:Maximum 256Byte+NULL |
| ObjectName | | | AUINT8 | ObjectName<br>UTF-8:Maximum 256Byte+NULL |
| City | | | AUINT8 | City<br>UTF-8:Maximum 256Byte+NULL |
| State | | | AUINT8 | State<br>UTF-8:Maximum 256Byte+NULL |
| Country | | | AUINT8 | Country<br>UTF-8:Maximum 256Byte+NULL |
| Category | | | AUINT8 | Category<br>ASCII:Maximum 256Byte+NULL |
| SuppCat | | | AUINT8 | Supplemental Categories<br>UTF-8:Maximum 256Byte+NULL |
| Byline | | | AUINT8 | Byline<br>UTF-8:Maximum 256Byte+NULL |
| BylineTitle | | | AUINT8 | BylineTitle<br>UTF-8:Maximum 256Byte+NULL |
| WriterEditor | | | AUINT8 | Writer/Editor<br>UTF-8:Maximum 256Byte+NULL |
| Credit | | | AUINT8 | Credit<br>UTF-8:Maximum 256Byte+NULL |
| Source | | | AUINT8 | Source<br>UTF-8:Maximum 256Byte+NULL |

* Each element terminates with NULL. The maximum byte count equals the byte count that can be input plus 1.

The length of each element is variable. If the element is empty, the element count is 0x00000000 and no element exists.

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## 9.8.2 XMP-IPTC

XMP-IPTC deta is described below.

| Field | Size(Byte) | Data | DataType | Description |
|---|---|---|---|---|
| DatasetVersion | 2 | 0x00C9 | UINT16 | Version of this dataset<br>Version 2.01 |
| IPTC Type | 1 | 2 | UINT8 | IPTC data format<br>1:NMS-IPTC<br>2:XMP-IPTC |
| Reserved | 5 | - | - | |
| Profile | | | AUINT8 | Profile<br>ASCII:Maximum 18Byte+NULL |
| XMP Data | | | AUINT8 | XMP format IPTC data<br>Maximum 30KByte |

The length of each element is variable. If the element is empty, the element count is 0x00000000 and no element exists.

# 9.9 PictureControl DataSet

PictureControl DataSet is sent by operation of SetPicCtrlData command or GetPicCtrlData command.

PictureControl DataSet indicates information about a picture control.

| Field | Size | Reference |
|---|---|---|
| BasePictureControl | 2 | 9.9.1 |
| RegistrationName | 20 | 9.9.2 |
| ApplyLevel | 1 | 9.9.3 |
| QuickSharpFlag | 1 | 9.9.4 |
| QuickSharp | 1 | 9.9.5 |
| Sharpening | 1 | 9.9.6 |
| MiddleRangeSharpening | 1 | 9.9.7 |
| Clarity | 1 | 9.9.8 |
| Contrast | 1 | 9.9.9 |
| Brightness | 1 | 9.9.10 |
| Saturation | 1 | 9.9.11 |
| Hue | 1 | 9.9.12 |
| FilterEffects | 1 | 9.9.13 |
| Toning | 1 | 9.9.14 |
| ToningDensity | 1 | 9.9.15 |
| CustomCurveFlag | 1 | 9.9.16 |
| CustomCurveData | 0 or 578 | 9.9.17 |

The values for Sharpening, MiddleRangeSharpening, Clarity, Contrast, Brightness, Saturation, Hue and ToningDensity are shown below.

| Value | Description |
|---|---|
| -128 | Auto |
| -N | -N x 0.25 |
| ... | |
| -1 | -0.25 |
| 0 | 0 |
| 1 | 0.25 |
| ... | |
| N | N x 0.25 |

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## 9.9.1 BasePictureControl

These are the types of picture controls on which the custom picture control is based.
If the picture control is pre-installed, this indicates the picture control.
The value is zero for unused custom picture control.

| Value | Description |
|---|---|
| 1 | Standard |
| 2 | Neutral |
| 3 | Vivid |
| 4 | Monochrome |
| 5 | Portrait |
| 6 | Landscape |
| 7 | Flat |
| 8 | Auto |
| 9 | Flat Monochrome |
| 10 | Deep Tone Monochrome |
| 11 | Rich Tone Portrait |
| 101 | Dream |
| 102 | Morning |
| 103 | Pop |
| 104 | Sunday |
| 105 | Somber |
| 106 | Drama |
| 107 | Silence |
| 108 | Bleach |
| 109 | Melancholic |
| 110 | Pure |
| 111 | Denim |
| 112 | Toy |
| 113 | Sepia |
| 114 | Blue |
| 115 | Red |
| 116 | Pink |
| 117 | Charcoal |
| 118 | Graphite |
| 119 | Binary |
| 120 | Carbon |

## 9.9.2 RegistrationName

This is the registration name of picture control.
The size is fixed to 20Bytes (ASCII: 19Bytes + NULL).

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## 9.9.3 ApplyLevel

This is the degree of control for picture control.

| Value | Description |
|---|---|
| 0 | 0 |
| 10 | 10 |
| 20 | 20 |
| 30 | 30 |
| 40 | 40 |
| 50 | 50 |
| 60 | 60 |
| 70 | 70 |
| 80 | 80 |
| 90 | 90 |
| 100 | 100 |

Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone<br>Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| 0 | | | 0<br>10<br>20<br>30<br>40<br>50<br>60<br>70<br>80<br>90<br>100 | | | |

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## 9.9.4 QuickSharpFlag

This is the state of the quick sharpening control for a picture control.
If QuickSharpFlag is set to [Valid] when setting a picture control, the values of Sharpening, MiddleRangeSharpening and Clarity are ignored.

| Value | Description |
|---|---|
| 0 | Invalid |
| 1 | Valid |

## 9.9.5 QuickSharp

This is the quick sharpening control for a picture control.

| Value | Description |
|---|---|
| -128 | Auto |
| -2 | -2 |
| -1 | -1 |
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |

Available values for each picture control are shown below.
If QuickSharpFlag is set to [Invalid] when setting a picture control, [Auto] cannot be set.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| From -2 to 2 | -128<br>From -2 to 2 | | From -2 to 2 | | | |

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## 9.9.6 Sharpening

This is the sharpening control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| -8<br>-4<br>0<br>4<br>8 | -128 (When QuickSharp is set to [Auto])<br>From -12 to 36 | | | | | |

## 9.9.7 MiddleRangeSharpening

This is the middle-range sharpening control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| -8<br>-4<br>0<br>4<br>8 | -128 (When QuickSharp is set to [Auto])<br>From -20 to 20 | | | | | |

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## 9.9.8 Clarity

This is the clarity control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| -8<br>-4<br>0<br>4<br>8 | -128 (When QuickSharp is set to [Auto])<br>From -20 to 20 | | | | | |

## 9.9.9 Contrast

This is the contrast control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| -8<br>-4<br>0<br>4<br>8 | -128<br>From -12 to 12 | | From -12 to 12 | | | |

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## 9.9.10 Brightness

This is the brightness control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| 0 | From -6 to 6 | | 0 | | | |

## 9.9.11 Saturation

This is the saturation control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| -8<br>-4<br>0<br>4<br>8 | -128<br>From -12 to 12 | 0 | From -12 to 12 | | 0 | |

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## 9.9.12 Hue

This is the hue control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| 0 | From -12 to 12 | 0 | From -12 to 12 | 0 | | |

## 9.9.13 FilterEffects

This is the filter effect for a picture control.

| Value | Description |
|---|---|
| 0 | Off |
| 1 | Yellow |
| 2 | Orange |
| 3 | Red |
| 4 | Green |

Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| 0 | | From 0 to 4 | 0 | | From 0 to 4 | |

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## 9.9.14 Toning

This is the toning control for a picture control.
If toning can be selected and toning is set to [B&W] when setting a picture control, the value of ToningDensity is ignored.

| Value | Description |
|---|---|
| 0 | B&W |
| 1 | Sepia |
| 2 | Cyanotype |
| 3 | Red |
| 4 | Yellow |
| 5 | Green |
| 6 | Blue Green |
| 7 | Blue |
| 8 | Purple Blue |
| 9 | Red Purple |

Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| 0 | | From 0 to 9 | 0 | | | From 0 to 9 |

## 9.9.15 ToningDensity

This is the toning (density) control for a picture control.
Available values for each picture control are shown below.

| Auto | Standard<br>Vivid<br>Portrait<br>Rich Tone Portrait<br>Landscape<br>Neutral<br>Flat | Monochrome<br>Flat Monochrome<br>Deep Tone Monochrome | Dream<br>Morning<br>Pop<br>Sunday | Somber<br>Drama<br>Silence<br>Bleach<br>Melancholic<br>Pure<br>Denim<br>Toy | Sepia<br>Blue<br>Red<br>Pink | Charcoal<br>Graphite<br>Binary<br>Carbon |
|---|---|---|---|---|---|---|
| 0 | | From 0 to 28 | 0 | | From 0 to 28 | |

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## 9.9.16 CustomCurveFlag

This is the state of custom curve for a picture control.
[Valid] is available only for a custom picture control.

| Value | Description |
|---|---|
| 0 | Invalid |
| 1 | Valid |

## 9.9.17 CustomCurveData

This is the custom curve data for a picture control.
The data format for a custom curve is described in LUT Format (section 9.9.17.1).
If the CustomCurveFlag is set to [Invalid], this is not used.

### 9.9.17.1 LUT Format

The LUT data consists of the header (64 bytes) for use on the host and following 514 bytes (2 bytes x 257 points) lut data. The header format is specified by the host (The stored location for such as the line point of the LUT to be sent, the data for reproducing the LUT at the time of reading). The camera does not care about the header contents. However, since 2 bytes of the header is used for judging whether header data is present on the camera, it is necessary to set data in the header.

The LUT format is as follows.

| Byte | Description |
|---|---|
| From 0 to 63 | Lut Header |
| 64, 65 | Data0 |
| 66, 67 | Data1 |
| ... | |
| 576, 577 | Data256 |

As an example of the Lut header, the contents of the header set by the application made by Nikon are shown below.

| Byte | Description | Range |
|---|---|---|
| 1 | ID (Byte1) | 0x49 |
| 2 | ID (Byte2) | 0x30 |
| 3 | Input Minimum (Black Point) | 0-255 |
| 4 | Input Maximum | 0-255 |
| 5 | Output Minimum | 0-255 |
| 6 | Output Maximum | 0-255 |
| 7 | Gamma (integer portion) | 0-20 |
| 8 | Gamma (fractional portion) | 0-100 |
| 9 | Number of Spline Points | 2-20 |
| 10, 11 | Spline Point1 (x, y) | 0-255, 0-255 |
| 12, 13 | Spline Point2 (x, y) | 0-255, 0-255 |
| ... | | |
| 48, 49 | Spline Point20 (x, y) | 0-255, 0-255 |
| From 50 to 64 | Reserved | 0 |

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# 9.10 PictureControlCapability DataSet

The PictureControlCapability DataSet is sent by operation of GetPicCtrlCapability command.
The PictureControlCapability DataSet indicates information about the function of a picture control.

| Field | Size | Data | Description |
|---|---|---|---|
| ApplyLevelCapability | 1 | 0x00: Not selectable<br>0x80: Selectable<br>0x01: AUTO is available<br>0x81: Selectable and AUTO is available | Apply level: Selectability and availability of AUTO |
| QuickSharpCapability | 1 | | Quick sharpening: Selectability and availability of AUTO |
| SharpeningCapability | 1 | | Sharpening: Selectability and availability of AUTO |
| MiddleRangeSharpening Capability | 1 | | Middle range sharpening: Selectability and availability of AUTO |
| ClarityCapability | 1 | | Clarity: Selectability and availability of AUTO |
| ContrastCapability | 1 | | Contrast: Selectability and availability of AUTO |
| BrightnessCapability | 1 | | Brightness: Selectability and availability of AUTO |
| SaturationCapability | 1 | | Saturation: Selectability and availability of AUTO |
| HueCapability | 1 | | Hue: Selectability and availability of AUTO |
| FilterEffectsCapability | 1 | | Filter effect: Selectability and availability of AUTO |
| ToningCapability | 1 | | Toning: Selectability and availability of AUTO |
| ToningDensityCapability | 1 | | Toning (density): Selectability and availability of AUTO |
| DefaultQuickSharpLevel | 1 | QuickSharp (9.9.5) | The default value of QuickSharp |
| DefaultLevel[0] | 3 | DefaultLevel (9.10.1) | QuickSharp -2 |
| DefaultLevel[1] | 3 | | QuickSharp -1 |
| DefaultLevel[2] | 3 | | QuickSharp 0 |
| DefaultLevel[3] | 3 | | QuickSharp +1 |
| DefaultLevel[4] | 3 | | QuickSharp +2 |

## 9.10.1 DefaultLevel

The information for the default level used in the PictureControlCapability DataSet is shown below.

| Field | Size | Reference |
|---|---|---|
| Sharpening | 1 | 9.9.6 |
| MiddleRangeSharpening | 1 | 9.9.7 |
| Clarity | 1 | 9.9.8 |

Do Not Copy

# 9.11 PictureControlHlg DataSet

PictureControlHlg DataSet is transferred by the operation of SetPicCtrlHlgData command or GetPicCtrlHlgDataList command.

PictureControlHlg DataSet provides information about Picture Control (HLG).

| Field | Size | Section |
|---|---|---|
| BasePictureControl | 2 | 9.9.1 |
| RegistrationName | 20 | 9.9.2 |
| ApplyLevel | 1 | 9.9.3 |
| QuickSharpFlag | 1 | 9.9.4 |
| QuickSharp | 1 | 9.11.1 |
| Sharpening | 1 | 9.9.6 |
| MiddleRangeSharpening | 1 | 9.9.7 |
| Clarity | 1 | 9.9.8 |
| Contrast | 1 | 9.9.9 |
| Highlight | 1 | 9.11.3 |
| Shadow | 1 | 9.11.4 |
| Saturation | 1 | 9.11.5 |
| Hue | 1 | 9.9.12 |
| FilterEffects | 1 | 9.9.13 |
| Toning | 1 | 9.9.14 |
| ToningDensity | 1 | 9.9.15 |

Fields other than QuickSharp, Contrast, Highlight, Shadow, and Saturation have the same available values as PictureControl DataSet (Standard, Monochrome, Flat), so refer to PictureControl DataSet.

## 9.11.1 QuickSharp

Indicates a quick sharpening of Picture Control (HLG).

| Value | Description |
|---|---|
| -2 | -2 |
| -1 | -1 |
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |

The values available for each Picture Control (HLG) are shown below.

| Standard<br>Flat | Monochrome |
|---|---|
| From -2 to 2 | |

Do Not Copy

## 9.11.2 Contrast

Indicates the contrast of Picture Control (HLG).
The values available for each Picture Control (HLG) are shown below.

| Standard<br>Flat | Monochrome |
|---|---|
| From -12 to 12 | |

## 9.11.3 Highlight

Indicates the hilight of Picture Control (HLG).
The values available for each Picture Control (HLG) are shown below.

| Standard<br>Flat | Monochrome |
|---|---|
| From -12 to 12 | |

## 9.11.4 Shadow

Indicates the shadow of Picture Control (HLG).
The values available for each Picture Control (HLG) are shown below.

| Standard<br>Flat | Monochrome |
|---|---|
| From -12 to 12 | |

## 9.11.5 Saturation

Indicates the saturation of Picture Control (HLG).
The values available for each Picture Control (HLG) are shown below.

| Standard<br>Flat | Monochrome |
|---|---|
| From -12 to 12 | 0 |

Do Not Copy

# 9.12 PictureControlHlgCapability DataSet

PictureControlHlgCapability DataSet is transferred by the operation of GetPicCtrlHlgCapabilityList command.

PictureControlHlgCapability DataSet provides information about the functions of Picture Control (HLG).

| Field | Size | Data | Description |
|---|---|---|---|
| ApplyLevelCapability | 1 | 0x00 : Not selectable<br>0x80 : Selectable | Apply level: Selectability |
| QuickSharpCapability | 1 | | Quick sharpening: Selectability |
| SharpeningCapability | 1 | | Sharpening: Selectability |
| MiddleRangeSharpeningCapability | 1 | | Middle range sharpening: Selectability |
| ClarityCapability | 1 | | Clarity: Selectability |
| ContrastCapability | 1 | | Contrast: Selectability |
| Highlight | 1 | | Highlight: Selectability |
| Shadow | 1 | | Shadow: Selectability |
| SaturationCapability | 1 | | Saturation: Selectability |
| HueCapability | 1 | | Hue: Selectability |
| FilterEffectsCapability | 1 | | Filter effect: Selectability |
| ToningCapability | 1 | | Toning: Selectability |
| ToningDensityCapability | 1 | | Toning (density): Selectability |
| DefaultQuickSharpLevel | 1 | QuickSharp (section 9.9.5) | Default value of QuickSharp |
| DefaultLevel[0] | 3 | DefaultLevel (section 9.10.1) | QuickSharp -2 |
| DefaultLevel[1] | 3 | | QuickSharp -1 |
| DefaultLevel[2] | 3 | | QuickSharp 0 |
| DefaultLevel[3] | 3 | | QuickSharp +1 |
| DefaultLevel[4] | 3 | | QuickSharp +2 |

DefaultQuickSharpLevel and DefaultLevel have the same range of data as PictureControl DataSet, so refer to them.

Do Not Copy

# 9.13 HLGQuality DataSet

The HLGQuality DataSet provides information about HLG image quality adjustments sent by operating the SetHLGQualityData or GetHLGQualityData commands.

The details of the HLG Quality Data Set are shown below.

| Field | Size | Range | Description |
|---|---|---|---|
| QuickSharpFlag | 1 | 0: Invalid<br>1: Valid | Flag for quick sharpening<br>If this value is [valid] when setting HLG, the value of Sharpening, MiddleRangeSharpening and Clarityis ignored. |
| QuickSharp | 1 | -2 to +2 | Quick sharpening |
| Sharpening | 1 | -12 to +36 | Sharpening |
| MiddleRangeSharpening | 1 | -20 to +20 | Middle range sharpening |
| Clarity | 1 | -20 to +20 | Clarity |
| Contrast | 1 | -12 to +12 | Contrast |
| Saturation | 1 | -12 to +12 | Saturation (Color intensity) |
| Hue | 1 | -12 to +12 | Hue |

Sharpening, MiddleRangeSharpening, Clarity, Contrast, Saturation, and Hue have a resolution of 0.25, so the value set for the camera is in 0.25 increments. The values used are shown below.

| Value | Description |
|---|---|
| -N | -N x 0.25 |
| --- | |
| -1 | -0.25 |
| 0 | 0 |
| 1 | 0.25 |
| --- | |
| N | N x 0.25 |

Do Not Copy

# 9.14 SBAttributeDesc DataSet

The SBAttributeDesc data set is sent by the operation of the GetSBAttrDesc command or GetSBGroupAttrDesc command.

The SBAttributeDesc data set is used to indicate the information of the flash attribute.

The details of the SBAttributeDesc data set are shown below.

| Field | Size (Byte) | DataType | Description |
|---|---|---|---|
| AttributeID | 2 | UINT16 | SBAttrID or SBGroupAttrID |
| DataType | 2 | UINT16 | Data type of the attribute<br>The value varies depending on the attribute.<br>Refer to DataTypeCode (section 7.1). |
| GetSet | 1 | UINT8 | Availability of Get and Set operations.<br>0x00 : Only Get operation is available (Get)<br>0x01 : Get and Set operations are available (Get / Set) |
| DefaultValue | DTS | DTS | The default value in the camera<br>The value varies depending on the attribute. |
| CurrentValue | DTS | DTS | The current value in the camera<br>The value varies depending on the attribute. |
| FormFlag | 1 | UINT8 | Type of the data set descriptor<br>0x00 : None<br>0x01 : Range<br>0x02 : Enumeration |
| FORM | DTS | DTS | The value in the field depends on FormFlag field.<br>If FormFlag is 0, this field doesn't exist. |

Do Not Copy

# 9.15 SBAttributeDescList DataSet

The SBAttributeList data set is sent by the operation of the GetSBAttrValue command, the SetSBAttrValue command, the GetSBGroupAttrValue command or the SetSBGroupAttrValue command.

The SBAttributeList data set is used to indicate the list of the current values of the flash attribute.

The details of the SBAttributeList data set are shown below.

| Field | Size (Byte) | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of attributes in the list. |
| Element1 AttributeDesc | DTS | Refer to SBAttributeDesc (section 9.14) | SBAttributeDesc for element 1 |
| Element2 AttributeDesc | DTS | Refer to SBAttributeDesc (section 9.14) | SBAttributeDesc for element 2 |
| ... | | | |
| ElementN AttributeDesc | DTS | Refer to SBAttributeDesc (section 9.14) | SBAttributeDesc for element N |

Do Not Copy

# 9.16 SBAttribute DataSet

The SBAttribute data set is sent by the operation of the GetSBAttrValue command, the SetSBAttrValue command, the GetSBGroupAttrValue command or the SetSBGroupAttrValue command.

The SBAttribute data set is used to indicate the current values of the flash attribute.

The details of the SBAttribute data set are shown below.

| Field | Size (Byte) | DataType | Description |
|---|---|---|---|
| AttributeID | 2 | UINT16 | SBAttrID or SBGroupAttrID |
| DataType | 2 | UINT16 | Data type of the attribute<br>The value varies depending on the attribute.<br>Refer to DataTypeCode (section 7.1). |
| CurrentValue | DTS | DTS | The current value in the camera<br>The value varies depending on the attribute. |

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# 9.17 SBAttributeList DataSet

The SBAttributeList data set is sent by the operation of the GetSBAttrValue command, the SetSBAttrValue command, the GetSBGroupAttrValue command or the SetSBGroupAttrValue command.

The SBAttributeList data set is used to indicate the list of the current values of the flash attribute.

The details of the SBAttributeList data set are shown below.

| Field | Size (Byte) | DataType or DataSet | Description |
|---|---|---|---|
| NumberOfElements | 4 | UINT32 | The number of attributes in the list. |
| Element1 Attribute | DTS | Refer to SBAttribute (section 9.16) | SBAttribute for element 1 |
| Element2 Attribute | DTS | Refer to SBAttribute (section 9.16) | SBAttribute for element 2 |
| ... | | | |
| ElementN Attribute | DTS | Refer to SBAttribute (section 9.16) | SBAttribute for element N |

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# 9.18 PortraitImpressionBalance DataSet

PortraitImpressionBalance DataSet is transferred by the operation of SetPortraitImpressionBalanceData command or GetPortraitImpressionBalanceDataList command and contains information about portrait impression adjustment.

The details of PortraitImpressionBalance DataSet are shown below.

| Field | Size | Range | Description |
|---|---|---|---|
| Brightness | 1 | -6 to +6 | Brightness |
| hue | 1 | -6 to +6 | hue |

Since the setting resolution of Brightness and hue is 0.5, the value set for the camera is in 0.5 increments.

The values to be used are shown below.

| Value | Description |
|---|---|
| -N | -N x 0.5 |
| ・・・ | |
| -1 | -0.5 |
| 0 | 0 |
| 1 | 0.5 |
| ・・・ | |
| N | N x 0.5 |

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# 10 Data Format

## 10.1 ASCII Codes

For the property related to the comment of the camera, only the following 90 characters of ASCII codes can be input.

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SP | ! | " | # | $ | % | & | ' | ( | ) | * | + | , | - | . | / |
| : | ; | < | = | > | ? | @ | [ | ] | _ | { | } | | | | |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | | | | | | |
| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| Q | R | S | T | U | V | W | X | Y | Z | | | | | | |
| a | b | c | d | E | f | g | h | i | j | k | l | m | n | o | p |
| q | r | s | t | U | v | w | x | y | z | | | | | | |

In addition to the above characters, following five characters can be used in the ManualSettingLensName property (section 6.5.5.4), the CommentString property (section 6.5.5.7), ArtistV property (section 6.5.5.9) and CopyrightV property (section 6.5.5.10).

| | | | | |
|---|---|---|---|---|
| ¥ | ^ | ` | \| | ~ |

| 7-Bit ASCII Code Table (JIS Roman letter set: C0, GL) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 0x0? | 0x1? | 0x2? | 0x3? | 0x4? | 0x5? | 0x6? | 0x7? |
| 0x?0 | NUL | DLE | SP | 0 | @ | P | ` | p |
| 0x?1 | SOH | DC1 | ! | 1 | A | Q | a | q |
| 0x?2 | STX | DC2 | " | 2 | B | R | b | r |
| 0x?3 | ETX | DC3 | # | 3 | C | S | c | s |
| 0x?4 | EOT | DC4 | $ | 4 | D | T | d | t |
| 0x?5 | ENQ | NAK | % | 5 | E | U | e | u |
| 0x?6 | ACK | SYN | & | 6 | F | V | f | v |
| 0x?7 | BEL | ETB | ' | 7 | G | W | g | w |
| 0x?8 | BS | CAN | ( | 8 | H | X | h | x |
| 0x?9 | HT | EM | ) | 9 | I | Y | i | y |
| 0x?a | LF | SUB | * | : | J | Z | j | z |
| 0x?b | VT | ESC | + | ; | K | [ | k | { |
| 0x?c | FF | FS | , | < | L | ¥ | l | \| |
| 0x?d | CR | GS | - | = | M | ] | m | } |
| 0x?e | SO | RS | . | > | N | ^ | n | ~ |
| 0x?f | SI | US | / | ? | O | _ | o | DEL |

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# 11 Appendices

## 11.1 Properties Affected by Mounting the CPU Lens

The following properties are affected by mounting the CPU lens.

| Property | Reference |
|---|---|
| LensSort | 6.5.10.1 |
| Fnumber | 6.5.1.5 |
| FocalLength | 6.5.1.6 |
| LensTypeML | 6.5.10.2 |
| LensTypeF | 6.5.10.3 |
| LensID | 6.5.10.4 |
| LensFocalMin | 6.5.10.5 |
| LensFocalMax | 6.5.10.6 |
| LensApatureMin | 6.5.10.7 |
| LensApatureMax | 6.5.10.8 |
| MovieFnumber | 6.5.12.1 |

## 11.2 Properties Affected by Mounting the Speedlight

| Property | Speedlight mounted | | Speedlight not mounted |
|---|---|---|---|
| | New-type communication | Noncommunication | |
| ExternalSpeedLightExist | Mounted | Mounted | Not mounted |
| ExternalSpeedLightSort | New-type communication | Noncommunication | Not fixed |
| ExternalSpeedLightStatus | Valid | Valid | Not fixed |
| NewExternalSpeedLightMode | Valid | Not fixed | Not fixed |
| FlashCompensation | Valid/Not fixed | Not fixed | Not fixed |
| ExposureTime | Bulb ～ (*1) (*2) | Bulb ～ (*1) | Bulb ～ 1/8000 |
| ShutterSpeed | Bulb ～ (*1) (*2) | Bulb ～ (*1) | Bulb ～ 1/8000 |

*1: Flash shooting synchronization speed
*2: Speed limit of the speedlight

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# 11.3 Properties Affected by the Shooting Mode

| Property | Shooting mode | | | |
|---|---|---|---|---|
| | M | P | A | S |
| ExposureProgramMode | M | P | A | S |
| Fnumber | Get / Set (*1) | Get | Get / Set (*1) | Get |
| ExposureTime | Get / Set | Get | Get | Get / Set (*2) |
| ShutterSpeed | Get / Set | Get | Get | Get / Set |
| FlexibleProgram | Not fixed | Get / Set | Not fixed | Not fixed |

*1: Setting is disabled when a lens other than the CPU lens is mounted.
*2: Setting is disabled with Bulb.

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# 11.4 Properties Affected by Time Zone

The UTC is retained in the camera. When displaying the time on the menu or getting/setting the DateTime property, “Time zone” and “Daylight saving time” in the camera are considered.

When getting the DateTime property, the calculated value shown below is passed to the host.

Time in the camera + Time zone difference + Daylight saving time

When the camera settings are as shown in the table below, the calculation is “13:00:00 + 09:00 + 0:00” and the value passed to the host is “2020/06/01 22:00:00”.

| | |
|---|---|
| Time in the camera (UTC) | 2020/06/01 13:00:00 |
| Time zone | UTC+9 (Tokyo, Seoul) |
| Daylight saving time | Off |

When the DateTime property is set, the value calculated as shown below is set in the camera.

Time set by the host - Time zone difference - Daylight saving time

When the camera settings are as shown in the table below and the time set by the host is “2020/06/01 13:00:00”, the calculation is “13:00:00 - 09:00 - 0:00” and the time setting in the camera is “2020/06/01 04:00:00”.

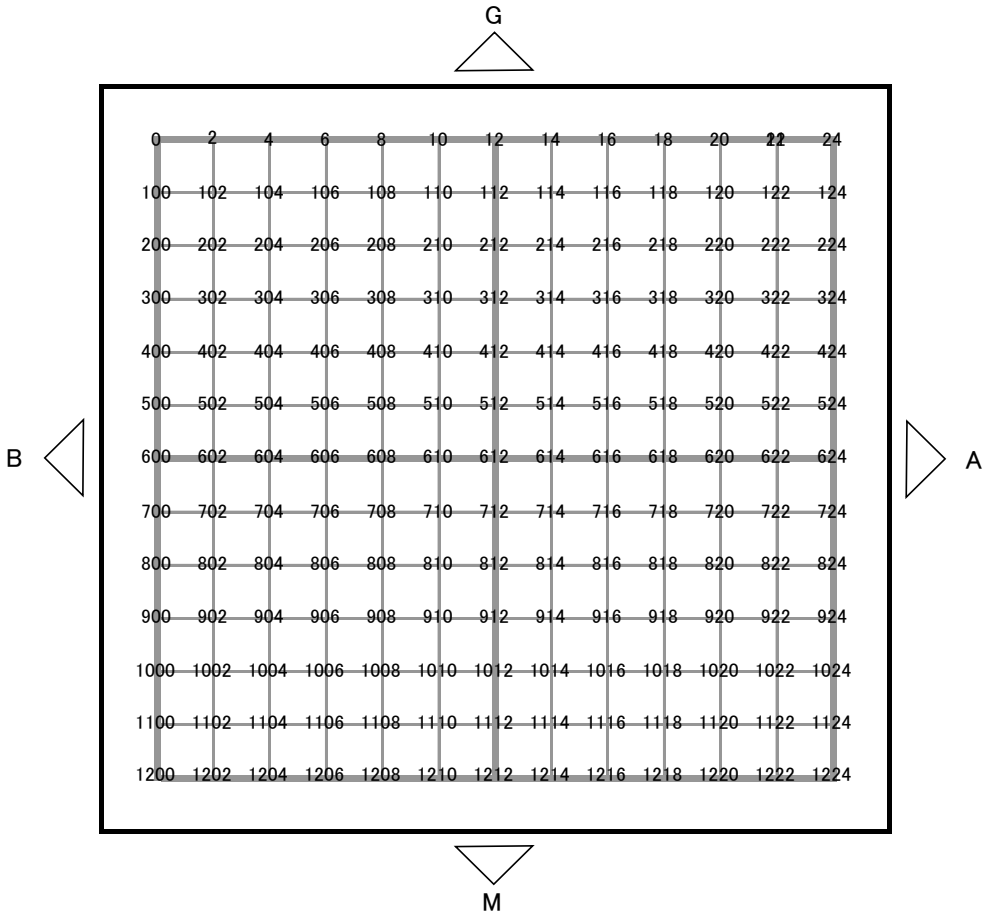
| | |
|---|---|
| Time zone | UTC+9 (Tokyo, Seoul) |
| Daylight saving time | Off |

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# 11.5 White Balance Fine-Tuning Coordinates

The relationship between the values of PropertyValue of the property affecting the white balance fine-tuning and the actual setting coordinates is shown below.

On the A-B axis: in steps of 0.5, on the G-M axis: in steps of 0.25




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# 11.6 Speedlight Types

The communication status types of the speedlight are shown below.

| CLS (New-type communication) with the operating and setting section | CLS (New-type communication) without the operating and setting section | Old-type communication | Noncommunication | Mounting not detected |
|---|---|---|---|---|
| SB-5000<br>SB-910<br>SB-900<br>SB-800<br>SB-700<br>SB-600<br>SU-800 | SB-500<br>SB-400<br>SB-300 | SB-80DX<br>SB-50DX<br>SB-28DX<br>SB-28D<br>SB-28<br>SB-27<br>SB-26<br>SB-25<br>SB-24 | SB-30<br>SB-29<br>SB-29S<br>SB-23<br>SB-22<br>SB-22S<br>SB-21A<br>SB-21B<br>SB-20<br>SB-19<br>SB-18<br>SB-17<br>SB-16A<br>SB-16B<br>SB-15<br>SB-14<br>SB-12<br>SB-11<br>SB-10<br>SB-E | SB-9<br>SB-8<br>SB-7<br>SB-6<br>SB-5<br>SB-4<br>SB-3<br>SB-2<br>SB-1 |

*For old-type communication and noncommunication, mounting is not detected in the camera.

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# 11.7 Command Release Operation

The operation by the command release is described in this section.

## 11.7.1 Continuous Shooting

The number of images that can be captured continuously by the continuous shooting is the smallest number among the following property combination.

| Property |
|---|
| BurstNumber (section 6.5.1.16) |
| ContinuousShootingCount (section 6.5.7.5) |

The availability of the continuous shooting in the release mode of StillCaptureMode (section 6.5.1.15) is described below.

| Release mode | BurstNumber | Description |
|---|---|---|
| Single frame | Invalid | Only one image can be captured. |
| Continuous high-speed shot<br>Continuous low-speed shot<br>High-speed frame capture | Valid | Among the BurstNumber setting value, the number of images that can be recorded in the SDRAM that is calculated in the camera, and the number of remaining images for recording while the bracketing is being performed, until the least number is reached, the acquisition of the new objects can be performed.<br>Only one image can be captured when the internal flash is enabled or the HDR mode is set. |
| Self-timer | Invalid | Only one image can be captured.<br>StillCaptureMode is temporarily set to the single frame when release. |

## 11.7.2 AF Operation

AF is performed in accordance with the setting of the FocusMode property (section 6.5.1.7).

## 11.7.3 Preset WB acquisition

During photo mode, the preset WB acquisition is performed with release sequence.

Refer to InitiateCaptureRecInMedia command about the preset measurement release.

During video mode, it is the same as the spot WB acquisition.

Refer to StartSpotWb command.

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## 11.7.4 Event on the Release Operation

When the release operation is started, one or more new objects are created. When the new objects are recorded, the camera generates the asynchronous interrupt event to inform the host of the addition of the new objects.

The new object addition event includes the ObjectHandle indicating the new object that is created. If two or more new objects are created, the new object addition event is issued two or more times.
When all the new objects that can be acquired are recorded, the camera issues the capturing completion event to the host to inform that the acquisition of all the new objects is completed.
The new object addition event and the capturing completion event vary depending on the recording destination. The following table shows the correspondence of the destination to the event.

| Recording destination | New object addition event | Capturing completion event |
|---|---|---|
| Card | ObjectAdded | CaptureComplete |
| SDRAM | ObjectAddedInSdram | CaptureCompleteInSdram |
| Card and SDRAM | ObjectAdded and<br>ObjectAddedInSdram | CaptureComplete and<br>CaptureCompleteInSdram |

When the recording destination is set to [Card and SDRAM], the order of issuing the new object addition event is not decided and the events are issued in order of completing image recording.
The capturing completion event is issued in the same way; the event of the recording destination in which acquisition of all the new objects is completed first, is issued first.

## 11.7.5 Recovery from the Sequence Error

When the Bit0 of the WarningStatus property (section 6.5.6.22) is set to 1 [Sequence error] and the release operation is performed by the command, the sequence error is cancelled if the sequence error can be recovered. When it is cancelled, the release is not started and the command is terminated.

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